STUDY GUIDE

CAREER, COMMUNICATION, SYSTEM DESIGN AND RESEARCH: BECOMING HIREABLE AND STAYING EXCELLENT

A practical mini-textbook for a working professional moving into data, ML and AI roles in India.

Taught the IIT Professor Way — Real Numbers. Real Problems. Real Understanding.

CHAPTER 1 The Career Map

CHAPTER 2 Learning How to Learn

CHAPTER 3 Spoken English for Technical People

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A Note From Your Professor

You already have a rare advantage. If you have spent time finding out why a process is slow, reading numbers, and explaining them to people who did not want to hear bad news, you have already done a version of data work. Here the object is a business or a model instead of a single process.

This document is about the parts of a career that nobody grades but everybody notices: how you learn, how you speak and write, how you solve problems, how you prepare for interviews, how you design systems, how you read research, how you present yourself, and how you look after your energy. These skills decide how far and how fast you grow, in every role.

I will be honest with you. There are no shortcuts here, and I will not invent salary numbers or promise a job in 90 days. I will give you methods, real numbers for planning, and templates that work. Some days you will feel behind. That feeling is common, and it is not proof that you are not good enough.

One rule above all: never move on until the current topic feels obvious. If a section is fuzzy, go back, do the worked problems again with the book closed, and teach it aloud to an imaginary beginner. A slow, solid step is faster than a fast, shaky one.

What Every Topic Contains

1. An **intuition** callout: the everyday problem the idea solves.
2. A **formal definition or method**, in plain words.
3. **Worked problems** with real numbers and every step.
4. **The picture**: a diagram, table or mental model.
5. **Connection to the real job**.
6. **Applications**: where it is used.
7. **Common mistake** callouts.
8. **Check yourself**: three short questions with answers.

Every chapter ends with a recap (with a pattern cheat-sheet), a practice set of 8 to 12 problems with answers, and a prompt pack of 5 prompts for any AI assistant.

Contents

1. **The Career Map** - six data roles, what each does day to day, and how to tell your own transition story.
2. **Learning How to Learn** - active recall, spaced repetition, interleaving, teaching back, avoiding resource hopping, and energy for a working learner.
3. **Spoken English for Technical People** - daily routine, shadowing, phrase bank, PREP and STAR, handling questions, pronunciation.
4. **Written Communication** - emails, README, design docs, ADRs, Slack etiquette, asking good questions, learning in public.
5. **Data Storytelling and Presenting Results** - headline, evidence, chart choice, executive summaries, Q&A.
6. **Problem Solving and Debugging Method** - the coding framework, issue trees, root-cause analysis, estimation.
7. **Technical Interview Preparation by Role** - SQL, Python and DSA, stats and ML, cases, eight STAR stories, a 12-week plan.
8. **System Design I: Software Systems** - framework, estimation, caching, sharding, queues; URL shortener, rate limiter, news feed.
9. **System Design II: Data and ML Systems** - data platform, recommender, fraud, RAG chatbot, feature pipeline.
10. **Research Skills** - three-pass paper reading, reproducing results, experimental design, ablations, writing, ethics.
11. **Portfolio, GitHub, LinkedIn and Resume** - project README template, resume bullets with impact, headline, first post.
12. **Job Search in India** - referrals, platforms, internal transfers, company types, salary research, notice period, rejection log.
13. **Domain Knowledge and Business Sense** - BFSI, e-commerce, SaaS, telecom, healthcare; unit economics worked examples.
14. **Certifications, Open Source, Kaggle and Long-Term Growth** - credential decisions, contribution path, IC vs manager, wellbeing and self-doubt.

An appendix (`99_cheatsheet.md`) collects the key patterns and ends with an honest closing note.

How To Study This Document

Use this learning loop in every chapter:

1. **Build first.** Try the worked problem or a similar one before reading the full explanation.
2. **Ask.** When stuck, ask an AI assistant with a prompt from the chapter's prompt pack. Ask for hints before answers.
3. **Just-in-time theory.** Read the explanation for exactly what blocked you.
4. **Rebuild.** Close the book and redo the problem from memory three days later.
5. **Teach.** Explain it aloud for two minutes, or write a short post (Chapter 4).

Suggested pace: one chapter per week at about one hour a day. Chapters 1 to 7 are the foundation; do them first. Chapters 8 and 9 (system design) are best done together with a project. Chapters 10 to 14 can be taken in order of need: if you have an interview in a month, read 11, 12 and 7 first.

Use the prompt packs like this: paste a prompt, add your own details where marked `[PASTE HERE]`, and answer the AI's questions in your own words. Never accept an answer you cannot explain back.

Prerequisites

- Basic school-level arithmetic (percentages, ratios).
- Some SQL and Python helps for the examples, but you can start without them. See the learner's SQL and Python study documents.
- Statistics ideas such as mean, median and percentile are used in a few examples; see the learner's Math PDF.
- Willingness to speak aloud and write short posts.

Tools you need

All free:

- A notebook or a folder of markdown files (Obsidian-style) for your learning log.
- Anki (flashcards) or any spaced-repetition app.
- A voice recorder on your phone.
- Python 3 and SQLite to run the examples.
- A GitHub account and a LinkedIn profile.
- A free grammar checker (LanguageTool or the free tier of Grammarly).

- A spreadsheet for tracking problems and applications.

CHAPTER 1

THE CAREER MAP

Six roles, what each one does all day, and why your existing skills are closer to them than you think.

You want a better job, but the words "data analyst", "data engineer", "ML engineer" and "AI engineer" sound like one big blur. If you pick the wrong target, you will study the wrong things for a year. This chapter gives you a clear map. You will learn what each role does in a normal working day, which of your current skills already count, and how to plan your move in honest hours instead of wishes. By the end, you will be able to read any job description and say which role it really is, and you will have a first draft of your own transition story.

1.1 The Six Roles and What They Do Day to Day

INTUITION

Think of a restaurant. Someone buys and stores the food (Data Engineer). Someone tastes the dishes and tells the owner what sells (Analyst). Someone invents a new recipe using science (Data Scientist). Someone makes sure the recipe can be cooked 500 times a night, the same way every time (ML Engineer). Someone builds the ordering robot that talks to customers (AI Engineer). Someone prepares clean ingredient tables so everyone cooks faster (Analytics Engineer). Job titles are messy, but these six jobs are real.

The six roles in plain words

- **Data Analyst** answers business questions using data. Main tools: SQL, Excel, a BI tool (a dashboard tool such as Power BI or Tableau), and some Python. Output: dashboards, reports, recommendations.
- **Data Engineer (DE)** builds the pipelines (automatic data movers) that bring data from many sources into one reliable place, usually a warehouse (a database designed for analysis). Main tools: SQL, Python, Airflow, Spark, cloud storage, Kafka.
- **Data Scientist (DS)** uses statistics and machine learning (ML, programs that learn patterns from data) to predict or explain things. Also designs experiments such as A/B tests. Output: models, analyses, experiment results.

- **ML Engineer (MLE)** takes a model and makes it a reliable product: training pipelines, serving APIs, monitoring, retraining. This is software engineering plus ML.
- **AI Engineer** builds applications on top of large language models (LLMs): prompts, RAG (retrieval-augmented generation, where the model reads your documents before answering), agents, evaluation, guardrails.
- **Analytics Engineer** sits between DE and Analyst. Uses SQL and tools like dbt to build clean, tested, documented tables that analysts use.

Titles differ by company. A "Data Scientist" at one company may do pure dashboards; at another, pure deep learning. Always read the job description (JD), not the title.

A typical day, by role

Role	Morning	Afternoon	Biggest pain
Analyst	Check dashboard numbers, answer a stakeholder ping	Write SQL for a new question, make 3 charts	Vague questions, dirty data
Data Engineer	Check overnight pipeline failures	Build or fix a pipeline, code review	Late or broken upstream data
Data Scientist	Explore data, plan features	Train and compare models, write findings	Unclear target, messy labels
ML Engineer	Check model monitoring, drift alerts	Improve serving code, deploy	Models that work in a notebook but fail in production
AI Engineer	Review failed chatbot answers	Improve retrieval, prompts, evaluation set	Hallucinations, cost, latency
Analytics Engineer	Review dbt test failures	Model new tables, write docs	Definitions that differ by team

Skill weight by role

Rate 0 (none) to 3 (heavy). These are rough, from the skill maps in this course.

Skill	Analyst	DE	DS	MLE	AI Eng	Analytics Eng
SQL	3	3	2	1	1	3
Python	1	3	2	3	3	1
Statistics	2	0	3	2	1	1
ML	0	0	3	3	2	0
System design	0	3	1	3	3	1
Business sense	3	1	3	1	2	2

Problem 1.1.1 You enjoy SQL and pipelines, dislike heavy statistics, and like systems thinking. Using the table, which two roles fit best?

Step 1	Pick the skills you like: SQL, system design. Dislike: statistics.
Step 2	Add scores for SQL + system design, and subtract nothing for statistics except to remove roles where it scores 3. DE: 3 + 3 = 6, statistics 0. MLE: 1 + 3 = 4, statistics 2. Analytics Eng: 3 + 1 = 4. DS: 2 + 1 = 3, statistics 3 (bad fit).
Answer	Data Engineer first (6 points, no statistics), Analytics Engineer or ML Engineer second. This is a rough guide, not a law.
Check	Add the DE row again: 3 (SQL) + 3 (system design) = 6, and the statistics score is 0. Correct.
Why it matters	A simple score table stops you from chasing a role only because it is popular.

Problem 1.1.2 A JD says "Data Scientist" but lists: Airflow, dbt, Snowflake, SQL, dashboards. Which role is it really?

Step 1	Count the keywords. Airflow, dbt and Snowflake are pipeline and warehouse tools. SQL and dashboards are analytics tools.
Step 2	No model training or statistics keywords appear.
Answer	It is an Analytics Engineer or Data Engineer job with a misleading title. Decide whether you want that job before you apply.
Check	Look at the skill table: the keywords match the SQL = 3 and DE/Analytics Eng columns, not the ML column. In plain words: you will build tables and pipelines, not models.

Real-world mapping

Role	Example output	Where it appears in the job
Analyst	Weekly sales dashboard with 6 charts	Monday business review meeting
Data Engineer	Pipeline that loads 10 million rows nightly by 6 am	A bank's morning risk report
Data Scientist	Churn model with 0.80 AUC	Telecom retention campaign
ML Engineer	Model API answering in under 100 ms	Payment fraud check at checkout
AI Engineer	Chatbot answering from 5,000 HR documents	Employee help desk
Analytics Engineer	A tested `orders` table everyone trusts	Finance and marketing use the same revenue number

The picture

The diagram below shows one flow. Read it top to bottom. Data starts raw at the top. Each role takes what the role above produced and makes it more useful. The Analyst and the ML/AI side are both customers of the clean tables.

Career map: how the six roles connect

```
Raw data sources
    |
[Data Engineer]  --> Warehouse / Lake
    |                     |
[Analytics Engineer] --> clean tables --> [Analyst] --> dashboards
                          |
                 [Data Scientist] --> model idea
                          |
                 [ML Engineer] --> production model
                          |
                 [AI Engineer] --> LLM apps using the same data
```

Connection to the real job

Small companies hire one "data person" who does all of this. Large companies split it into many roles. Your first job will probably touch two or three of these boxes. That is good: it shows you what you like.

Applications

- Banks: DE moves transaction data, DS builds credit models, Analyst tracks collections.
- E-commerce: Analytics Engineer defines "orders"; MLE serves search ranking.
- SaaS: AI Engineer builds a support chatbot over help articles.

COMMON MISTAKE

Choosing a role from its title or from social media hype. Read five real JDs for each role and list the tools. The tools show the real work.

Check yourself

1. Which role builds pipelines that move data? Answer: Data Engineer.
2. Which role mainly uses dbt to model clean tables? Answer: Analytics Engineer.
3. Why read the JD and not the title? Answer: Titles are inconsistent between companies; the tool list shows the real work.

1.2 Your Transition Story: From IT to Data

INTUITION

Think of a school office that records every pupil's late arrival for a year, finds that most lateness happens on Mondays, and tells the head teacher what to change. That is data work already, and anyone who has collected measurements, found the slow part and explained it to a team has done a version of it. You are not starting from zero. You are changing the object you study, from a system or a process to a business or a model.

The learner of this document is a working professional moving from a technical IT or testing role into data roles. That is the only background the document assumes; every example below is general.

What you already do, and its data twin

Skill many working professionals already have	Data/ML equivalent
Reading median and 95th-percentile waiting times (a hospital's appointment waits)	Percentiles, distributions, outliers in EDA
Planning for a busy day (customers per hour, time per customer)	Simulation, data generation, capacity plans
Finding why a queue at the bank is slow	Root-cause of a metric drop, model error analysis
Weekly reports for managers	Dashboards, executive summaries
Comparing this month's results with last month's	A/B tests, regression detection
Throughput, queues, Little's law	Pipeline throughput, streaming lag, autoscaling
Monitoring, SLAs	Data freshness SLAs, model monitoring

The story (use this in interviews)

Write your story in four parts: **where I was, what I noticed, what I built, where I am going.**

Example in simple English:

> "For five years I worked in IT support for a retail company. I noticed that every week the help desk produced thousands of ticket rows, but we only looked at averages. I built a small pipeline that loads the tickets into a database and draws the 95th-percentile resolution time per week. It showed a slow trend two weeks before customers complained. I enjoyed the data part more than the ticket-closing part, so I studied SQL, Python and statistics for a year. Now I want to work on data platforms where reliability and speed matter."

Notice what this story does: it is true, it has one result, and it explains why you want to change.

Problem 1.2.1 Turn this weak bullet into a transition-story sentence: "I worked on reports for a company's help desk."

Step 1	Add object and tool: "I analysed help-desk tickets using SQL."
Step 2	Add scale: "for a team handling up to 1,000 tickets a day." (Use your own true number.)
Step 3	Add result and data angle: "I analysed 2 million ticket rows and found that one approval step was the bottleneck, which cut the 95th-percentile resolution time from 24 hours to 15 hours."
Answer	"I analysed help-desk tickets for a team handling up to 1,000 a day, studied 2 million rows, and found an approval bottleneck that cut the 95th-percentile resolution time from 24 hours to 15 hours." Check: (24 - 15) / 24 = 37.5% faster. Use only numbers that are true for you.
Why it matters	A sentence with scale, method and result is believable. "I worked on reports" is not.

Problem 1.2.2 Which target role has the smallest gap for a working professional with strong SQL and scripting but no ML?

Step 1	List gaps. Analyst: BI tool and business sense. DE: pipelines, warehouse, orchestration. DS: statistics and ML (big gap). MLE: ML plus software engineering.
Step 2	Your strengths (systems, scripting, reporting) match DE, Data Platform, and Observability-analytics best.
Answer	Data Engineer or Data Platform roles have the smallest gap. ML roles come after.
Check	Count gap items: DE has 3 gaps, DS has 2 big ones (statistics, ML) that each take months. A short gap list of familiar skills is a quicker climb.

Niches that use your advantage

Operations Analytics, Data Platform, ML Platform, Observability/AIOps (using data to find and predict system problems). Few data people can answer "will this system survive 10 times the usual number of customers?" If you can reason about capacity and queues, you have an edge.

Connection to the real job

Interviewers love a clear story with one memorable project. See Chapter 11 for the project template and Chapter 7 for behavioral stories.

Applications

- Data Platform teams at product companies and GCCs (Global Capability Centers, offices that large foreign companies run in India).
- Observability tools companies.
- Any bank or telecom with large pipelines.

COMMON MISTAKE

Apologising for where you start ("I am only a support person"). Say what you did, with numbers. Your starting point is your difference, not your weakness.

Check yourself

1. Name two data-world twins of everyday work skills. Answer: waiting-time percentiles and EDA; build comparison and A/B tests (others are fine).

2. What four parts make the transition story? Answer: where I was, what I noticed, what I built, where I am going.
3. Which niche suits you best? Answer: Data platform, observability or operations analytics.

1.3 A Realistic Path and Timeline

INTUITION

Climbing a mountain at night with a torch: you do not see the top, only the next 10 metres. You need a plan for the next stage, and honest checkpoints.

The course plan stages: Analyst skills (SQL, Python) first, then Data Engineering, then ML, then AI Engineering. Switching to a data role does not need all stages. Many people move after SQL + Python + one pipeline project + one dashboard project.

Rough effort

At 1 hour a day, you get about 365 hours a year. Check: 1 x 365 = 365. Most people need roughly 6 to 12 months of steady work to be interview-ready for a first data role. This is a rough guide; it varies a lot with background and target.

Problem 1.3.1 You study 1 hour on 5 weekdays and 3 hours on one weekend day. How many hours in 26 weeks?

Step 1	Weekly hours = 5 x 1 + 3 = 8.
Step 2	26 x 8 = 208.
Answer	208 hours in half a year. If you miss 20% of weeks, expect about 166 hours. Plan for misses.
Check	208 x 0.8 = 166.4. In plain words: about 6 months of effort is only about 4 full-time work weeks (166 / 40 = 4.2).

Problem 1.3.2 You want 3 projects done in 26 weeks, each needing about 50 hours of work, plus 60 hours of theory. Is the plan feasible with 208 hours?

Step 1	Projects: 3 x 50 = 150. Theory: 60. Total 210.
Step 2	210 is more than 208, and you will miss some days.
Answer	Not feasible as written. Reduce to 2 projects (100 + 60 = 160 hours) or extend to 32 weeks. Honest planning prevents guilt.
Check	210 / 8 hours per week = 26.25 weeks, so you need at least 27 weeks with zero missed days. With misses, 32 weeks is a fair choice.

The picture

```
Month 0–3   SQL + Python basics, 1 small analysis project
Month 3–6   Pipelines, warehouse, dashboard project, start applying to
analytics/DE roles
Month 6–9   Cloud basics, ML basics, operations–analytics project
Month 9–12  Mocks, interviews, internal transfer talks, referrals
(Stay in your current role until you hold a signed offer.)
```

Connection to the real job

Start applying before you feel ready. Interview calls are themselves the best teacher. Chapter 12 gives the job-search system.

Applications

- Planning a quarterly review of your own progress.
- Explaining to a manager how you will transition.

COMMON MISTAKE

Waiting until you know "everything." The skill maps are maps, not to-do lists. No working professional knows all of it.

Check yourself

1. How many hours is 1 hour/day for a year? Answer: 365.
2. Why plan for missed days? Answer: Life happens; plans that assume perfection create guilt and quitting.

3. When should you resign from your current job? Answer: After a signed offer in a data role.

Chapter recap

- There are six main data roles; the tool list in the JD tells the real job.
- Analyst answers questions, DE builds pipelines, DS models, MLE productionises, AI Engineer builds LLM apps, Analytics Engineer builds clean tables.
- Technical work already gives you percentiles, root-cause thinking, reporting and capacity sense.
- Tell your transition story in four parts with one true number.
- Niches such as data platform and observability use your existing strengths.
- 1 hour a day is about 365 hours a year; plan for missed days.
- Apply before you feel fully ready; stay in your current role until a signed offer.
- Compare yourself with last month's you.

Pattern cheat-sheet

Need	Pattern
Read a job description	Count tools, ignore the title
Percent improvement	`(old – new) / old x 100`
Study hours per year	`hours per week x weeks x (1 – missed share)`
Transition story	Where I was, what I noticed, what I built, where I am going

Practice set

1. (Easy) Which role mainly uses dbt? **Answer:** Analytics Engineer.
2. (Easy) A JD lists Spark, Kafka, Airflow. Which role? **Answer:** Data Engineer.
3. (Easy) 1 hour/day for 90 days is how many hours? **Answer:** 90.
4. (Medium) Convert "improved report turnaround" into a quantified bullet using the 95th-percentile turnaround from 30 minutes to 18 minutes. **Answer:** "Cut 95th-percentile report turnaround from 30 minutes to 18 minutes (40%) by batching the approvals." Check: 1.2 / 3.0 = 40%.

5. (Medium) What is the output? You work 7 h/week for 30 weeks, miss 4 weeks. Hours? **Answer:** 7 x 26 = 182 hours.
6. (Medium) Find the weakness: "I am just a support person, but I want to try data." **Answer:** It apologises and gives no evidence. Replace with what you built and measured.
7. (Medium) Scenario: your manager asks why you want to move from your current role into data. Give a 3-sentence answer. **Answer:** "I enjoy analysing results more than the routine tasks. I built a small pipeline that spotted a slowdown early. I want to work where data and reliability meet."
8. (Hard) Interview: "Why should we hire a former IT-support or testing professional as a data engineer?" **Answer:** Mention pipelines are systems: throughput, bottlenecks, SLAs, monitoring. Give one example where you found a bottleneck with data. Then name your SQL/Python/pipeline project.
9. (Hard) Interview: "Walk me through your career in two minutes." **Answer:** Use the four-part story: past work with scale, noticed problem, what you built, target role. About 260 words at 130 words per minute is 2 minutes; aim for 200 to leave room.
10. (Hard) Scenario: you have three offers of study time, 150 hours: a bootcamp, a project, or a course. Which first? **Answer:** The project; it creates proof. Add a course only for a gap the project exposes.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in
India moving to data roles; English is my second language, so use simple
words. Teach me the differences between Data Analyst, Data Engineer, Data
Scientist, ML Engineer, AI Engineer and Analytics Engineer. Explain step
by step with a typical day for each, then ask me five questions one at a
time to check I understand. Correct me kindly and finish with a table of
skills by role.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in
India moving into data roles; English is my second language. Help me write
my transition story in four parts: where I was, what I noticed, what I
built, where I am going. Ask me questions first, one at a time. Then draft
a 150-word version in simple English and a one-line version. Do not invent
```

numbers; ask me for true ones. Here is my background: [PASTE HERE].

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles in India. Topic: reading job descriptions to find the real role. I will paste 3 job descriptions. For each, tell me the real role, the tools that matter, my gap, and how many weeks of study the gap needs at 1 hour a day. Explain your reasoning in simple English. JDs: [PASTE HERE].

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional in India learning data skills, 1 hour a day. Topic: planning a 6-month career transition. Give me 8 graded exercises on estimating study hours, setting stage goals and planning for missed days. Do not give solutions yet. After I answer each, give feedback and fix my arithmetic.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data. Run a mock interview for a Data Platform role: ask "Tell me about yourself" and "Why data?" and three follow-ups, one at a time. After each answer give feedback on clarity, evidence and numbers, then show a better version in simple English. My first answer: [PASTE HERE].

CHAPTER 2

LEARNING HOW TO LEARN

You do not have an information problem. You have an energy and method problem. Fix the method and one hour a day goes much further.

You will study for years while you work full time, so the way you study matters more than the number of courses you buy. Many learners watch ten hours of video and remember almost nothing a week later. That is not a lack of talent. It is a method problem, and methods can be changed. In this chapter you will learn the study techniques that research supports, how to set up a one-hour session, how to stop jumping between courses, and how to protect your energy. By the end you will have a weekly routine you can actually keep.

2.1 Active Recall and Spaced Repetition

INTUITION

Imagine a school that gives its pupils one exam in September and never checks again. You would not trust that the pupils still know the material in June. Memory is the same. A fact you met once was "checked once". Each time you pull it out of your head on purpose, the memory gets stronger. Watching a video again is like re-reading the old exam paper. Recalling is like sitting a fresh quiz.

How it works

- **Active recall** means you try to produce the answer from memory before you look. Example: close the notes and write what a LEFT JOIN does.
- **Spaced repetition** means you repeat the recall with growing gaps: 1 day, 3 days, 7 days, 14 days, 30 days. You review just when you are about to forget.
- **Desirable difficulty** means a little struggle helps. If recall feels easy, you may be repeating too soon. If you cannot recall at all, the gap was too long or the first learning was thin.
- **Anki** is a free flashcard app that schedules the gaps for you. Use your own words on the card, one idea per card.

Good cards and bad cards

Bad card	Good card
"Explain gradient descent" (too big)	"In gradient descent, which direction do we move the weights?" Answer: against the gradient (downhill)
"p-value"	"A p-value of 0.03 means what, in one sentence?"
Copy of a paragraph	A small cloze: "A window function does not {{collapse}} rows"

Problem 2.1.1 You learn "what is an index in SQL" on Monday 9 November 2026. Use gaps of 1, 3, 7, 14, 30 days. When are the reviews?

Step 1	Day 0 is Monday 9 Nov. Add 1 day: Tue 10 Nov.
Step 2	Add 3 days: Fri 13 Nov. Add 7: Fri 20 Nov. Add 14: Fri 4 Dec. Add 30: Sun 3 Jan 2027. (Computed with Python.)
Answer	Reviews on 10 Nov, 13 Nov, 20 Nov, 4 Dec and 3 Jan. That is 5 reviews, about 30 seconds each, so 2.5 minutes of work per card for a lasting memory.
Check	Total days from learning to the last review: 1 + 3 + 7 + 14 + 30 = 55. From 9 Nov plus 55 days is 3 Jan. Correct. In plain words: 2.5 minutes spread over 55 days buys months of memory.

Problem 2.1.2 You add 5 new cards a day. How many cards after 60 days, and is it affordable at 10 minutes of Anki per day?

Step 1	5 x 60 = 300 cards.
Step 2	Each card needs 5 reviews x 0.5 min = 2.5 min over its life. 300 x 2.5 = 750 minutes in total across 60+ days, about 12.5 minutes per day averaged over 60 days.
Answer	It is slightly above 10 minutes. Reduce to 3 or 4 new cards a day, and write only cards for things you really need. Quality over quantity.
Check	750 / 60 days = 12.5 min per day. At 4 new cards a day: 240 cards x 2.5 = 600 min; 600 / 60 = 10 min per day. Matches the budget.

Real-world mapping

What you learn	Card example	Where it appears in the job
SQL syntax	"Which clause filters groups after GROUP BY?" HAVING	SQL screen in interviews
Statistic definition	"What does p95 mean?"	Every business report and A/B test
System numbers	"Seconds in a day?" 86,400	System design estimation (Chapter 8)
Vocabulary	"What is idempotent?"	Pipeline design discussions

The picture

Read the curve like this. The vertical axis is how strongly you remember. The horizontal axis is days. Without review, the line falls fast. Each review lifts it back up, and after each review it falls more slowly, so the next review can wait longer.



Connection to the real job

SQL syntax, window functions, Python idioms, metrics definitions and system design numbers are all recall skills in an interview. Put the "numbers to remember" (e.g. 1 day = 86,400 seconds) on cards.

Applications

- Interview prep (SQL patterns, statistics definitions).
- Cloud exam certifications.
- Learning a domain's vocabulary (Chapter 13).

COMMON MISTAKE

Making 50 cards from one video. Cards pile up, reviews become a burden and you quit. Cap new cards at 3 to 5 a day and delete cards you do not need.

Check yourself

1. What is the difference between re-reading and recall? Answer: Recall produces the answer from memory; re-reading only recognises it.
2. Name the review gaps. Answer: about 1, 3, 7, 14, 30 days.
3. Why one idea per card? Answer: You can tell exactly what you forgot.

2.2 Interleaving, Teaching Back and Deliberate Practice

INTUITION

A cricket batter who only practises one shot against one bowler is surprised in the match. Practising mixed shots is harder in the nets and better in the game. This is interleaving.

Three methods

- **Interleaving**: mix problem types in one session instead of 10 of the same kind. Example: one SQL join, one window function, one Python dictionary problem, one probability question.
- **Teaching back**: explain the idea aloud to an imaginary beginner (or write a short post). Gaps in your explanation show gaps in your understanding. This is also the Feynman technique.
- **Deliberate practice**: work at the edge of your ability on one specific weakness, with fast feedback. Example: "I fail window-function questions" leads to 10 window problems, check each answer, write why you missed.

A one-hour session shape

Minutes	Activity
0-10	Recall: Anki reviews and "what did I learn yesterday" from memory

10-40	Build or solve problems on today's topic (no copy-paste)
40-50	Interleave: 2 old-topic problems
50-60	Teach back aloud, then write 2 cards and 1 log line

Problem 2.2.1 You have 8 SQL practice problems: 4 joins, 4 window functions. Show an interleaved order versus a blocked order, and say which gives better learning.

Step 1	Blocked: J J J J W W W W. You know the next problem type, so you skip the decision "which tool?".
Step 2	Interleaved: J W J W W J W J. You must first recognise the pattern, as in a real interview.
Answer	Interleaved. It feels harder and slower, but it trains the skill the interview tests: recognising which pattern to use.
Check	In an interview nobody tells you "this is a join question". Recognising the pattern is half the marks.

Problem 2.2.2 Your weak spot is "GROUP BY with HAVING". Design deliberate practice for one week.

Step 1	Specific goal: write correct GROUP BY / HAVING queries in under 5 minutes.
Step 2	Daily 3 problems of rising difficulty; check against a known answer or run on sample data.
Step 3	After each miss, write the cause on one line ("used WHERE instead of HAVING").
Answer	After 7 days you have about 21 attempts and a list of 3 to 4 repeated errors. Turn each into a flashcard.
Check	3 problems x 7 days = 21. If the error list is empty, the problems were too easy: raise the difficulty.

Connection to the real job

Teaching back is what you will do in stand-ups, code reviews and interviews. If you cannot explain a model to a manager, you do not fully own it yet.

Applications

- Writing a LinkedIn post per week (Chapter 4).
- Mock interviews (Chapter 7).
- Mentoring juniors later in your career.

COMMON MISTAKE

Calling "I watched the course" practice. Rule: for every 20 minutes of video, 40 minutes of doing. If you could not do it with the video closed, you have not learned it.

Check yourself

1. What is interleaving? Answer: mixing different problem types in a session.
2. What does a gap in teach-back show? Answer: a gap in understanding.
3. What is deliberate practice? Answer: practising a specific weakness at the edge of ability with fast feedback.

2.3 Avoiding Resource Hopping and Beating Overwhelm

INTUITION

Digging five holes 1 metre deep never gives water. One hole 5 metres deep does. Starting a new course every time the current one gets hard is digging many shallow holes.

The rules

- **One main resource per topic.** Choose it in 15 minutes, using a short list of 2 or 3. Finish it.
- **One active project at a time.** A project forces you to use what you learn.
- **Finish before starting.** Define "finished": all exercises done, one project built, notes in your log.
- **Parking lot.** When a new course tempts you, write it in a "later" list with a date. Decide again in 4 weeks.
- **Note system.** One notebook or a folder of markdown files (an Obsidian-style vault). Keep a log: "what I learned" and "what confused me". The confused list is gold: it tells you what to study next and what to ask.

Problem 2.3.1 You start 5 courses, each 30 hours, and finish 20% of each. How many hours spent and how many courses completed? What if you finished one?

Step 1	5 x 30 x 0.20 = 30 hours spent, 0 courses complete.
Step 2	Finishing one course: 30 hours, 1 course complete, 1 certificate-worthy skill.
Answer	The same 30 hours buys either 0 finished courses or 1. Hopping has the same cost and zero finished output.
Check	5 x 6 hours (20% of 30) = 30 hours. Same hours, different result.

Problem 2.3.2 Your parking lot has 12 courses. You can finish 1 course every 6 weeks at 1 hour/day. How long does the whole list take, and what do you do?

Step 1	12 x 6 = 72 weeks, about 16.6 months.
Step 2	Most courses overlap in content.
Answer	You will never finish all. Keep the 2 that fix your biggest current gap, delete the rest, and re-check the list in a month.
Check	72 / 4.33 weeks per month = 16.6 months. Correct.

Connection to the real job

Companies pay for finished things: a pipeline running every day, a deployed model. Finishing is a professional habit.

Applications

- Choosing between courses on Coursera, YouTube, and Zoomcamps.
- Picking a single book per subject.
- Setting up your study repo.

COMMON MISTAKE

Searching "best course" for hours. The best course is the one you finish. Spend at most 15 minutes choosing.

Check yourself

1. What is resource hopping? Answer: changing courses before finishing the current one.
2. What is a parking lot? Answer: a list of tempting courses you decide on later.
3. How many projects at a time? Answer: one.

2.4 Energy, Consistency and Wellbeing for a Working Learner

INTUITION

A phone with 5% battery cannot run the best app. If you sleep five hours and then study, the hour is wasted. Charging the battery is part of studying.

The rules that work

- **Fixed slot.** Same time daily. After you have eaten, before you are exhausted: perhaps 6:30 am, or 9 pm. Pick the slot you actually keep.
- **Sleep 7+ hours.** Memory is stored during sleep. Skipping sleep to study is a bad trade.
- **Pomodoro 25/5.** 25 minutes focused, 5 minutes break. Two rounds is an hour.
- **Movement and a weekly rest day.** One day with no study, guilt-free.
- **Weekends for projects**; weekdays for small steps.
- **1 hour beats 6 hours occasionally.** Seven hours a week as 1 x 7 days is better for memory than 7 hours on Sunday because of spacing.
- **Financial resilience.** Keep an emergency fund, avoid loans for courses, stay in your current role until a signed offer.
- **When to seek help.** If self-criticism becomes heavy or constant, talk to a trusted person or a counselor. This is part of a sustainable career.

Problem 2.4.1 You have 10 free hours a week. Option A: 10 hours on Sunday. Option B: 1.5 hours a day for 7 days (10.5 h). Compare using spacing.

Step 1	In A you meet each topic once a week: one exposure, a gap of 7 days.
Step 2	In B you meet topics daily; Anki gaps 1, 3, 7 become possible.
Answer	B gives about the same hours but much better retention, and is less tiring. Use A only for big project days on top of B.
Check	1.5 x 7 = 10.5 hours. In plain words: spreading the same hours across the week is the same cost with a better result.

Problem 2.4.2 You miss 3 days in a 30-day month at 1 hour/day. How much did you lose and what do you do?

Step 1	27 hours studied instead of 30, so you studied 90%.
Step 2	Never "make up" 3 hours in one night; just continue tomorrow.
Answer	You lost 3 hours (10%). A 90% month is excellent. Restart without guilt.
Check	27 / 30 = 0.90. You still studied 27 hours, which is more than most people study in a year of good intentions.

The picture

```
Energy budget (a 24-hour day)
Sleep 7–8h | Job 9h | Commute/chores 4h | Family/rest 3h | STUDY 1h
Study fits only if sleep is protected.
```

Connection to the real job

Data careers take years. People who stay consistent for 3 years beat people who sprint for 3 months. Burnout stops your career; rest prolongs it.

Applications

- Planning your daily slot and weekly review.
- Setting monthly goals (Chapter 14).
- Talking to your manager about workload.

COMMON MISTAKE

Thinking missed days mean you failed. A streak is a tool, not a judge. Missing one day is nothing; missing two days in a row is the real rule to avoid.

Check yourself

1. Why is 1 h daily better than 7 h once? Answer: spacing and less fatigue.
2. How long is a Pomodoro? Answer: 25 minutes work, 5 minutes break.
3. When should you talk to a counselor? Answer: when self-criticism is heavy or constant.

Chapter recap

- Recall beats re-reading. Close the notes, then try.
- Space your reviews: 1, 3, 7, 14, 30 days. Anki does this for you.
- One idea per card; 3 to 5 new cards a day maximum.
- Mix problem types (interleaving) because interviews mix them.
- Explain aloud (teach back) to find gaps.
- Practise your specific weak spot with fast feedback.
- One resource, one project, finish before starting something new.
- Sleep, a fixed slot and a rest day are part of the study plan.
- Heavy self-criticism is a reason to talk to someone, not to push harder.

Pattern cheat-sheet

Need	Pattern
Review schedule	Gaps of 1, 3, 7, 14, 30 days
New cards per day	3 to 5 maximum
One-hour session	10 recall, 30 build, 10 interleave, 10 teach back
Video to practice ratio	20 minutes video : 40 minutes doing
Focus blocks	Pomodoro 25 on, 5 off

Practice set

1. (Easy) Learned on day 0, first review is on which day? **Answer:** Day 1.
2. (Easy) How many Pomodoro rounds fit in one hour? **Answer:** 2 (25+5, 25+5).
3. (Easy) Active recall or re-reading? Which is stronger? **Answer:** Active recall.
4. (Medium) You add 4 cards/day for 50 days. How many cards, and lifetime review minutes at 2.5 min each? **Answer:** 200 cards; 500 minutes.
5. (Medium) What is wrong with this card: "Explain the entire ML pipeline"? **Answer:** It is too big. Break into one-idea cards.
6. (Medium) Find the bug in this plan: "Sunday 8 hours study, nothing else all week." **Answer:** No spacing, high fatigue, one failed Sunday ruins the week. Spread into daily slots.
7. (Medium) Scenario: you keep switching courses every two weeks. What do you do? **Answer:** Pick one for the topic, write down finish criteria, park others for 4 weeks.
8. (Hard) Interview-style: "How do you keep your skills current while working full time?" **Answer:** Fixed one-hour slot, one resource and one project at a time, Anki for recall, a weekly post to teach back, monthly review.
9. (Hard) Interview-style: "Tell me about a time you learned something fast." **Answer:** Use STAR (Chapter 3): Situation (new tool), Task (deadline), Action (project-first plan, daily recall, asked an expert), Result (delivered in N days).
10. (Hard) Compute: 3 pomodoro rounds of 25 min is how many focus minutes, and total time with breaks? **Answer:** 75 focus minutes. With a 5-minute break after each round it is 90 minutes; if you skip the last break it is 85.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in India learning data skills for 1 hour a day; English is my second language. Teach me active recall and spaced repetition for technical topics. Explain step by step with a worked example for SQL, then show me how to write 5 good Anki cards from a topic I give. Then quiz me one question at a time to check I understand the difference between a good and a bad card.
```

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working learner in India moving into data roles. Topic: interleaving and deliberate practice. Create a 1-week practice schedule (1 hour a day) that mixes SQL, Python and statistics. Show which problems to interleave and why. Then ask me what my weakest topic is and design a deliberate practice plan for it with fast feedback. My weak topic: [PASTE HERE].

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional learning data skills; I keep switching courses. Topic: avoiding resource hopping. Ask me about my current courses and projects, then help me choose one resource and one project, define "finished" for each, and build a parking lot list. Give me a simple weekly review template. My current list: [PASTE HERE].

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I study 1 hour a day while working full time in India. Topic: energy, sleep and consistency. Review my weekly routine and suggest changes that protect sleep and keep a fixed study slot. Be realistic and kind. Also tell me signs that I should talk to a counselor. My routine: [PASTE HERE].

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving to data. Topic: teaching back. I will explain a concept to you as if you are a beginner. Find gaps, ask follow-up questions a beginner would ask, and grade my explanation on clarity from 1 to 5. Then show a better version in simple English. The concept I will explain: [PASTE HERE].

CHAPTER 3

SPOKEN ENGLISH FOR TECHNICAL PEOPLE

Clear, simple, confident English beats fancy English. Your accent is not the problem. Clarity is the goal.

In every data job you will speak: in stand-ups, in meetings with business people, in interviews, and when you explain a result to a manager. Many strong engineers lose offers, not because they lack skill, but because the interviewer could not follow their explanation. This chapter gives you a daily 10-minute routine, a phrase bank you can copy, two simple structures for almost every answer, and a method for handling questions you cannot answer. By the end you will be able to give a 60-second status update and a clear two-minute explanation of a technical idea, and you will know how to practise alone.

3.1 The Daily Speaking Routine, Shadowing and Vocabulary

INTUITION

Speaking is a physical skill, like cycling or batting. You cannot learn it by reading about it. Your mouth and brain need repeated practice, and you need feedback. The good news: 10 minutes a day is enough to improve visibly in 3 months, because the practice is daily.

Words you need

Word	Meaning
Shadowing	Playing a recording and repeating each sentence right after the speaker, copying rhythm and sound
Rhythm	Which words are strong and which are soft in a sentence
Filler	Sounds like "umm", "you know" used while thinking
Phrase bank	Your own notebook of ready-made sentences

The 10-minute routine

1. **Pick one concept** (for example: "What is an index in a database?").
2. **Explain it aloud** for 2 minutes as if to a beginner. Use your phone voice recorder.
3. **Listen** to the recording once. Do not judge. Note only ONE thing to fix (too fast? many "umm"? a word you mispronounced?).
4. **Next day**, explain a new concept and fix that one thing.
5. Once a week, re-record an old topic and compare. You will hear the progress.

Shadowing (5 minutes, separate from the routine)

1. Choose a short tech talk of 5 to 10 minutes with clear speech and subtitles.
2. Listen to one sentence. Pause.
3. Repeat it at the same speed, copying the rhythm and the stressed words.
4. Do 10 sentences. Next day, shadow the same clip again, because repeating the same clip is useful.

Vocabulary: 5 phrases a week, in context

Do not memorise word lists. Take 5 phrases from your real work or a tech talk, write them in a sentence of your own, and put each on an Anki card (Chapter 2). Example: "We are blocked on access" (you cannot proceed because you lack permission).

Where to practise with people

- Toastmasters clubs (many cities in India have them) or a local speaking group.
- A language-exchange partner or a study buddy, with a weekly 20-minute call.
- Optional paid apps and tutors (ELSA Speak, Cambly). Check current prices yourself; they change.
- YouTube channels on English for tech professionals.

Problem 3.1.1 You speak at 110 words per minute (wpm) and a good explanation target is 130 wpm. Your 2-minute recording has 190 words. Is the problem speed or length?

Formula	words per minute = words / minutes
Plug in	190 / 2
Compute	190 / 2 = 95 wpm
Answer	You speak at 95 wpm, slower than the 110 you thought, and well below 130. A slow pace is fine for clarity; the real issue is long pauses and fillers. Next day, fix one thing: remove long pauses by planning your first sentence.
Check	At 130 wpm, 2 minutes is 260 words. You said 190, which is 190 / 260 = 73% of that. Matches "slower than target".
Why it matters	Numbers turn a vague feeling ("my English is bad") into one fixable thing.

Problem 3.1.2 You do the 10-minute routine for 5 days a week for 12 weeks. How many hours is that?

Formula	total minutes = minutes per day x days per week x weeks
Plug in	10 x 5 x 12
Compute	10 x 5 = 50, and 50 x 12 = 600 minutes
Answer	600 minutes = 10 hours. Ten hours of recorded practice in a quarter is more speaking practice than most people do in a year.
Check	600 / 60 = 10 hours. Correct.

The picture



Read it as a loop. The arrow back from "pick ONE fix" to the next day is the whole method. One fix per day, not ten.

Connection to the real job

Stand-ups, demos and interviews are short spoken tests. A learner who has recorded 40 explanations is calm in them.

Applications

- Data Analyst: explaining a dashboard to a sales head.
- Data Engineer: explaining a pipeline failure to a manager in 30 seconds.
- ML/AI Engineer: explaining model accuracy to a non-technical product owner.

COMMON MISTAKE

Trying to copy an American or British accent. Your goal is to be understood. Clear stress on the key word and a calm pace help more than any accent.

Check yourself

1. How long is the routine? Answer: 10 minutes, one concept, one fix.
2. What is shadowing? Answer: repeating a speaker's sentences right after them to copy rhythm.
3. How many new phrases per week? Answer: 5, learned in context.

3.2 The Meeting Phrase Bank

INTUITION

A chef does not invent a knife every time. He keeps a few good ones sharp. A phrase bank is your set of sharp knives: a few polite sentences you can use without thinking, so your brain is free for the technical part.

The phrase bank

Situation	Say this
Agree	"I agree with that." / "That makes sense to me."
Disagree politely	"I see your point. Can I share a different view?" / "I am not sure that will work, because..."

Ask for clarification	"Could you explain that a little more?" / "When you say 'users', do you mean all users or active users?"
Confirm understanding	"So to confirm, you want X by Friday, is that right?"
Buy time	"Let me check and get back to you." / "That is a good question. Give me a moment to think."
Say you do not know	"I do not know yet. I will find out and reply by tomorrow."
Interrupt politely	"Sorry to jump in, can I add one point?"
Ask for repetition	"Sorry, could you say that again, a little slower?"
Give a status	"I have finished A, I am working on B, I am blocked on C."
Raise a risk	"One risk I see is... The impact would be... I suggest..."
Close a meeting	"Let me summarise the actions: Ravi will do X by Monday, I will do Y by Wednesday."

Stand-up and status structure

A stand-up is a short daily meeting where each person says what they did. Use this order: **what I did, the result, what is next, blockers.** A blocker is anything that stops your work.

Sample 60-second stand-up (68 words, checked by code; at 130 wpm this is about 31 seconds):

> "Yesterday I finished the SQL query for the weekly sales report and tested it on one month of data. The result matches the old report. Today I will add the region filter and send it for review. I have one blocker: I do not have access to the returns table. I have asked the data owner, and I will follow up this afternoon if there is no reply."

Problem 3.2.1 Turn this vague status into the four-part structure: "Working on the pipeline, some issues."

Step 1	What I did: "I fixed the null-value error in the orders load."
Step 2	Result: "The nightly run now finishes in 22 minutes instead of failing."
Step 3	Next: "Today I will add a row-count check."
Step 4	Blockers: "I need read access to the staging bucket; I raised a ticket yesterday."
Answer	"I fixed the null-value error in the orders load. The nightly run now finishes in 22 minutes instead of failing. Today I will add a row-count check. I need read access to the staging bucket, and I raised a ticket yesterday."
Check	The new status has 4 parts, 1 number, and a clear ask. The old one had none of these. (Use only real numbers.)
Why it matters	Managers who hear "some issues" must ask more questions; managers who hear the structure can help in 10 seconds.

Problem 3.2.2 A colleague asks, "Can you finish the report today?" You are not sure. Use the phrase bank to answer in two sentences, without promising or refusing.

Step 1	Buy time: "Let me check my tasks and get back to you."
Step 2	Give a time: "I will reply by 2 pm."
Answer	"Let me check my tasks and get back to you. I will reply by 2 pm with a clear time."
Check	The answer is honest (no false promise), polite and has a deadline. That matters more than a quick "yes" you cannot keep.

The picture

```
Meeting flow          Phrase to use
Listen             -> (nod, take notes)
Not clear?         -> "Could you explain that a little more?"
Understand?        -> "So to confirm, ..."
Need to disagree?  -> "I see your point. Can I share a different view?"
End                -> "Let me summarise the actions..."
```

Connection to the real job

Business people often use words loosely ("customer", "active"). The phrase "When you say X, do you mean...?" prevents weeks of wrong work. This is the first step of the problem-solving method in Chapter 6.

Applications

- Analyst: confirming the definition of a metric before building a dashboard.
- Data Engineer: negotiating deadlines with a data owner.
- ML Engineer: raising a risk about an unrealistic accuracy target.

COMMON MISTAKE

Saying "yes" to everything to avoid conflict, then missing the deadline. A polite "Let me check and reply by 2 pm" builds more trust than a wrong yes.

Check yourself

1. Give the four parts of a status. Answer: what I did, result, next, blockers.
2. How do you buy time politely? Answer: "Let me check and get back to you."
3. What does "So to confirm" do? Answer: It repeats the request to avoid misunderstanding.

3.3 Structuring Answers: PREP and STAR

INTUITION

When you give directions in a city, you say "go straight, turn left at the temple, it is the blue gate." You do not start with the history of the road. A structure is a route: the listener always knows where you are and where you are going.

PREP for explanations

Point, **R**eason, **E**xample, **P**oint again. Say your answer first, then why, then a real case, then repeat the answer.

Sample (55 words):

> "I would choose a median here. The reason is that salaries have a few very large values, and those pull the average up. For example, if five people earn 30, 32, 35, 36 and 400

> thousand, the average is 106.6 thousand but the median is 35 thousand. So the median describes a typical person better."

STAR for behavioral questions

A behavioral question asks about your past ("Tell me about a time you..."). The company wants to see how you act. **S**ituation, **T**ask, **A**ction, **R**esult. Spend about 15% on Situation and Task, 60% on Action, 25% on Result. Say "I", not only "we", because they hire you.

Sample (66 words):

> "At my company a new ordering page was due on Friday. On Wednesday, our timing check showed the checkout page taking 4 seconds at the 95th percentile. My task was to find the cause before Friday. I checked the database logs and saw one query without an index. I added the index and measured again. The 95th percentile fell to 1.6 seconds, and we released on time."

(Your own story must be true. Chapter 7 gives eight story templates.)

Problem 3.3.1 Check the sample numbers. The median of 30, 32, 35, 36, 400 and the average?

Step 1	Sort: 30, 32, 35, 36, 400. Five values, so the median is the 3rd value.
Step 2	Median = 35.
Step 3	Sum = 30 + 32 + 35 + 36 + 400 = 533. Average = 533 / 5 = 106.6.
Answer	Median is 35 thousand; average is 106.6 thousand. The one big value of 400 pulls the average to three times the median.
Check	Ran in Python: (30+32+35+36+400)/5 gives 106.6.
Why it matters	The example in the "E" of PREP must have real numbers or the listener will not believe it.

Problem 3.3.2 Rewrite this rambling answer to "Why SQL over Excel for 5 million rows?" in PREP: "Well, Excel is good, but there are limits, and I think databases are used in companies, and also..."

Step 1	Point: "I would use SQL."
Step 2	Reason: "Excel allows about 1 million rows per sheet, so 5 million rows will not even open."
Step 3	Example: "In my last report, a query on 5 million rows ran in a few seconds in the database."
Step 4	Point: "So for this size, SQL is the right tool."
Answer	"I would use SQL. Excel allows about 1 million rows per sheet, so 5 million will not fit. For example, I ran a report on 5 million rows in a database and it finished in seconds. So SQL is the right tool here." (Excel's limit is 1,048,576 rows per sheet.)
Check	5,000,000 / 1,048,576 = 4.8, so you would need five sheets. The reason holds.

Time and length

At about 130 words per minute: a 60-second answer is about 130 words, 2 minutes is about 260 words. A good interview answer is 1 to 2 minutes. A STAR story is 2 minutes.

Answer type	Target	Words at 130 wpm
"Tell me about yourself"	90 seconds	about 195
Technical explanation (PREP)	60 seconds	about 130
STAR story	2 minutes	about 260
Stand-up update	30 to 45 seconds	65 to 100

The picture

```
PREP:  [Point] -> [Reason] -> [Example] -> [Point]
STAR:  [Situation 15%] [Task] -> [Action 60%] -> [Result 25%]
Status:[Did] -> [Result] -> [Next] -> [Blockers]
```

Read left to right. Each box is one or two sentences. If you cannot say what box you are in, you are rambling.

Connection to the real job

Interviewers give you 20 seconds of patience before they decide whether you are clear. Starting with the answer (Point) wins that 20 seconds.

Applications

- Interviews: PREP for technical, STAR for behavioral.
- Written updates: the same status structure works in Slack.
- Stakeholder meetings: lead with the recommendation.

COMMON MISTAKE

Telling the whole backstory before the answer. Say the answer first. The listener can ask for the story if they want it.

Check yourself

1. What does PREP stand for? Answer: Point, Reason, Example, Point.
2. Which STAR part gets the most time? Answer: Action, about 60%.
3. How many words is a 2-minute answer? Answer: about 260 at 130 wpm.

3.4 Handling Questions, Presenting, and Pronouncing Tech Words

INTUITION

A question is not an attack. It is the other person asking for a map because the road was not clear. When you stay calm, you turn a hard moment into a conversation.

The Q&A routine: pause, restate, answer briefly, be honest

1. **Pause** for 2 seconds. It looks calm and gives your brain time.
2. **Restate**: "So you are asking how we handle late data, is that right?" This gives you more time and confirms the question.
3. **Answer briefly** in PREP.
4. **If you do not know**, say it honestly: "I do not know yet. My guess would be X because Y. I

will find out and send you the answer." A reasoned guess plus honesty is stronger than silence or a bluff.

Presenting results: problem, approach, result, recommendation

Slide plan for a 5-minute talk (one message per slide):

Slide	Message	Time
1	The problem and why it matters (with a number)	45 s
2	Approach in 3 steps	60 s
3	Result: the one key chart	90 s
4	Recommendation and next step	45 s
5	Risks and questions	60 s

Practise with a timer. If you take 7 minutes at home, you will take 8 on stage. Cut a slide.

Pronouncing common tech words

English has regional variations; these are the commonly accepted forms, and pronunciations in India and abroad can differ slightly. Capital letters show the strong syllable.

Word	Say it like
cache	"cash"
queue	"kyoo"
latency	"LAY-ten-see"
median	"MEE-dee-un"
parameter	"puh-RAM-uh-ter"
algorithm	"AL-guh-rith-um"
Kubernetes	"koo-ber-NET-eez"
nginx	"engine-X"

regex	"REJ-eks"
async	"AY-sink"
SaaS	"sass"
SQL	"S-Q-L" or "sequel" (both are used)
Jira	"JEE-ruh"
PyTorch	"PIE-torch"
scikit-learn	"SY-kit learn"
Hadoop	"huh-DOOP"
GUI	"GOO-ee"
dbt	"D-B-T" (say each letter)

Problem 3.4.1 A senior engineer asks in a design review: "What happens if the upstream file arrives two hours late?" You do not know. Write the reply.

Step 1	Pause and restate: "So you are asking what happens to the downstream tables if the file is late."
Step 2	Give what you do know: "The pipeline starts at 2 am and has no retry after 4 am."
Step 3	Be honest: "I have not tested a two-hour delay. My guess is that the job fails and the dashboard shows yesterday's data."
Step 4	Promise follow-up: "I will test it and share the result by Thursday."
Answer	A reply of four short sentences that restates the question, shares known facts, gives a labelled guess, and promises a date.
Check	It has no bluff and it gives a next action. A bluff can be caught; honesty builds trust.

Problem 3.4.2 Plan a 5-minute presentation: you have 7 slides. Each slide takes 1 minute to say. Fix it.

Step 1	Total time = 7 x 1 = 7 minutes, so you are 2 minutes over.
Step 2	Remove two slides: merge approach details into one slide and move extra charts to an appendix.
Answer	Keep 5 slides at 1 minute each (or the table above). Extra material goes in a backup slide for Q&A.
Check	5 x 1 = 5 minutes. In plain words: a short talk you finish is better than a long one you rush.

The picture

```
Question arrives -> [PAUSE 2 s] -> [RESTATE] -> [ANSWER brief]
                                                      |
                                            know it? yes -> PREP
                                                    no  -> honest guess +
promise
```

Connection to the real job

In interviews, how you handle a question you cannot answer is often scored more than a perfect answer. In real work, saying "I will find out" is a professional habit.

Applications

- Interview rounds (technical and behavioral).
- Sprint demos and design reviews.
- Customer or stakeholder calls where you represent your team.

COMMON MISTAKE

Guessing confidently without saying it is a guess. If a senior person finds the error, your trust drops. Say "my guess is..." clearly.

Check yourself

1. What do you do first when you get a hard question? Answer: pause, then restate.
2. What do you say when you do not know? Answer: "I do not know yet; here is my guess; I

will find out."

3. How many messages per slide? Answer: one.

Chapter recap

- Practise speaking 10 minutes a day: explain, record, listen, fix one thing.
- Shadow a short tech talk to improve rhythm and sound.
- Learn 5 phrases a week in context, not word lists.
- Use a phrase bank for agreeing, disagreeing, clarifying and buying time.
- Use PREP for explanations and STAR for behavioral stories.
- Status updates: what I did, result, next, blockers.
- On a hard question: pause, restate, answer briefly, be honest.
- Rough speaking speed is about 130 words per minute; plan answer length from that.

Pattern cheat-sheet

Need	Pattern
Explain a concept	PREP: Point, Reason, Example, Point
Behavioral answer	STAR with 15 / 60 / 25 time split
Stand-up	Did, result, next, blockers
Buy time	"Let me check and get back to you"
Do not know	"I do not know yet. My guess is... I will find out by..."
Answer length	words = minutes x 130

Practice set

1. (Easy) How many words is a 90-second answer at 130 wpm? **Answer:** 1.5 x 130 = 195 words.
2. (Easy) Expand PREP. **Answer:** Point, Reason, Example, Point.
3. (Easy) How do you say "cache"? **Answer:** "cash".

4. (Medium) What is wrong with this status: "Working on stuff"? **Answer:** No result, no next step, no blockers, no number. Use the four-part status.
5. (Medium) What is the output? A speaker uses 260 words in 2 minutes. Words per minute? **Answer:** 260 / 2 = 130 wpm.
6. (Medium) Find the bug in this STAR answer: "We, as a team, decided..., we did..., we solved it." **Answer:** No "I". The interviewer cannot tell what you did. State your own actions.
7. (Medium) Scenario: a colleague talks fast and you did not understand. What do you say? **Answer:** "Sorry, could you say that again, a little slower?"
8. (Hard) Interview: "Tell me about yourself" in 90 seconds. **Answer:** Present (role and one number), past (how you got here: a technical role with some data analysis), future (target role and why). About 195 words. Practise aloud with a timer.
9. (Hard) Interview: "What is your biggest weakness?" **Answer:** Name a real, fixable one, show action and progress. Example: "I used to avoid speaking in large meetings. I now record 10 minutes daily and give a weekly update to my team. I am more comfortable, and I still prepare my first sentence."
10. (Hard) Scenario: in a meeting the VP asks a question outside your area. What do you say? **Answer:** "That is outside my area, so I do not want to guess. Anita on the platform team would know. I will ask her and reply by tomorrow."

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in
India moving into data roles; English is my second language. Teach me the
PREP structure (Point, Reason, Example, Point) for explaining technical
ideas. Show 3 worked examples (SQL, statistics, pipelines), then give me 5
concepts one at a time. I will answer in PREP, and you will correct my
structure and grammar kindly and show a better version in simple English.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in
India moving to data roles; English is my second language. Topic: the STAR
method for behavioral interview answers. Interview me with 8 behavioral
questions one at a time (conflict, failure, ambiguity, deadline, mistake,
influence, leadership, learning fast). After each answer, score Situation,
Task, Action and Result and show a 2-minute improved version. My story
```

notes: [PASTE HERE].

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional in India; English is my second language. Topic: a phrase bank for meetings. Give me 10 realistic meeting scenarios in a data team (late data, unclear metric, disagreement on deadline). For each, I will reply and you will check politeness, clarity and grammar, and offer a better sentence. Start with scenario 1 now.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data. Topic: handling Q&A when I do not know the answer. Ask me 8 technical questions that are slightly beyond my level, one at a time. I will answer; you will assess whether I paused, restated, answered briefly, and was honest, then show how a calm professional would reply in simple English.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional in India; English is my second language. Topic: my daily 10-minute speaking routine and pronunciation of tech words. Build me a 4-week plan with one concept per day and one thing to fix per week. Include a list of 30 tech words with simple respellings, flagging any where pronunciation varies. Then ask me to say a 60-second explanation in text and review it.

CHAPTER 4

WRITTEN COMMUNICATION

Most of your work life is written: emails, tickets, docs, messages. Good writing is how people trust you when you are not in the room.

In a data job, your SQL or model may be perfect and still be ignored if your message is confusing. Remote and hybrid teams, and global companies with teams in many time zones, depend on writing. This chapter shows you how to write an email that gets a reply, a README that lets a stranger run your project, a design record that explains why you chose something, a Slack question that gets a fast answer, and a weekly post that builds your name. By the end you will have four templates you can copy and use this week.

4.1 Emails and Executive Summaries

INTUITION

Your manager reads 80 emails a day. An email is like a lift ride: you have about 10 seconds. If the ask is not in the first line, the ride is over. So write like a newspaper: the headline first, details after.

The rules

- **Subject line contains the ask.** Bad: "Query". Good: "Decision needed by Fri: use March or April data for churn report?"
- **Lead with the conclusion.** First sentence = the answer or the request.
- **Use bullets** for details (3 to 5 bullets).
- **End with clear next steps**: who does what by when.
- **Keep it short.** Under 150 words for most work emails.

An **executive summary** is a short top section of a report for busy leaders. Format: one line with the headline result and a number, then 3 bullets of evidence, then a recommendation, then risks.

Words you need

Word	Meaning
Stakeholder	A person who cares about or depends on your work
Ask	The thing you want the reader to do
BLUF	"Bottom line up front": conclusion first
Executive summary	One page, numbers first

Problem 4.1.1 Rewrite this email. Subject: "Hi". Body: "I was checking the pipeline and there were some things, the load took long, and I think maybe we need more memory, let me know your thoughts."

Step 1	Make the subject carry the ask: "Approval needed: raise pipeline memory from 8 GB to 16 GB".
Step 2	Lead with the conclusion: "I recommend doubling the pipeline memory to 16 GB."
Step 3	Add evidence in bullets (use true numbers; these are an example): "Nightly load now takes 3 h 10 min, past the 2 h target. Memory is at 95% for the last 40 minutes. A test with 16 GB finished in 1 h 20 min."
Step 4	Next steps: "Please approve by Friday. I will make the change Saturday night."
Answer	A subject, a one-line recommendation, 3 evidence bullets and an approval request with dates, about 70 words.
Check	Time saved: 3 h 10 min = 190 min and 1 h 20 min = 80 min, so (190 - 80) / 190 = 57.9% faster. The email can be understood in 10 seconds.
Why it matters	The reader can approve without a meeting.

Problem 4.1.2 Write a one-line headline for an executive summary. Facts: in the south region the price rose from 100 to 110 rupees, and units sold fell by 12%. What happened to revenue?

Formula	revenue = price x units
Plug in	price change factor = 110 / 100 = 1.10; units change factor = 1 - 0.12 = 0.88
Compute	revenue factor = 1.10 x 0.88 = 0.968, so revenue changed by 0.968 - 1 = -0.032, which is -3.2%
Answer	"In the south region, a 10% price rise cut units by 12%, so revenue fell about 3.2%."
Check	Take 1,000 units at 100 rupees = 100,000. After the change: 880 units x 110 = 96,800. (100,000 - 96,800) / 100,000 = 3.2%. Matches.
Why it matters	In plain words: the price rise did not pay for itself. Writing numbers forces you to calculate them, and a summary with wrong numbers destroys trust.

The picture

Read it from top to bottom: it is the order of lines in the email.

```
Subject: [ASK + DEADLINE]
Line 1:  Conclusion / request
Bullets: evidence (3–5)
Last:    who does what by when
```

Connection to the real job

Weekly status mails, incident reports (what broke, impact, cause, fix, prevention) and stakeholder updates all follow this shape. A monthly sales report is already an executive summary if you put the numbers first.

Applications

- Analyst: weekly KPI email.
- Data Engineer: incident summary after a pipeline failure.
- Data Scientist / ML: model result email with accuracy and a clear recommendation.

COMMON MISTAKE

A long email with the ask hidden in paragraph four. If the reader must search for the ask, they will reply "what do you need?" and you lose a day.

Check yourself

1. Where does the ask go? Answer: in the subject line and the first sentence.
2. What is an executive summary? Answer: a one-page summary with numbers first.
3. What ends a good email? Answer: clear next steps with owner and date.

4.2 Documentation: README, Design Docs, ADRs, Runbooks

INTUITION

Imagine you go on leave for a month. A new person must run your pipeline at 2 am when it breaks. Documentation is the note you leave on the kitchen table. If it is clear, they cook without calling you.

Words you need

Document	Purpose	Reader
README	How to understand and run a project	New user or hiring manager
Design doc	What we will build and why, before building	Team and reviewers
ADR (Architecture Decision Record)	One decision, its reasons and alternatives	Future team members
Runbook	Step-by-step fix for a known problem	On-call engineer at 2 am
Data dictionary	Each column's meaning, type and source	Analysts and engineers

README template

```
# Project name
One sentence: what problem it solves.

## Result
One headline number or screenshot.

## How to run
1. git clone ...
2. pip install -r requirements.txt
3. python main.py

## How it works
Short diagram + 5 bullets.

## Data
Source, size, license.

## Decisions and limits
What I chose and why; what does not work yet.

## Next steps
```

ADR template

```
# ADR 003: Use Postgres for the orders store
Status: Accepted (2026-11-07)
Context: We need transactions and queries by customer; 5 million
rows/year.
Decision: Use Postgres.
Alternatives considered: MongoDB (flexible, but we need joins), DynamoDB
(cost model unclear).
Consequences: Plan for read replicas after 50 million rows. Easy for the
team (SQL skills).
```

An ADR is short (half a page). Its value is the **reasons**: in two years nobody remembers why. Write what was decided, when, and what you would revisit.

Design doc outline

Problem, goals and non-goals, proposed design (diagram), alternatives, risks, rollout plan, open questions. Share it before building. Reviewers find problems when they are cheap to fix.

Runbook example

```
Alert: nightly_orders_load failed
1. Open Airflow, find the failed task, read the last 50 log lines.
2. If error is "connection timeout": re-run the task once.
3. If it fails again: check the source DB status page; message #data-
oncall.
4. If row count < 90% of yesterday: do NOT publish; notify finance.
Owner: Data Platform team. Last tested: 2026-10-15.
```

Problem 4.2.1 Write a one-sentence README opening for a project that loads a school's exam results into a database and charts the average mark by class.

Step 1	State the problem: engineers cannot compare p95 across releases.
Step 2	State the solution and tools: a pipeline in Python and SQLite with a chart per release.
Answer	"This project loads exam result files into a database and draws the average mark for every class, so a drop can be seen in seconds."
Check	Does it say what, for whom, and why? What: loads and draws. Who: engineers (implied). Why: find a regression. It is one sentence of 25 words.
Why it matters	Hiring managers read the first line only. Make it count.

Problem 4.2.2 Write the key line of an ADR for choosing batch (nightly) over streaming for a daily sales report.

Step 1	Facts: the report is read once at 9 am; data is available at 1 am; streaming needs more infrastructure.
Step 2	Decision: batch.
Answer	"Decision: use a nightly batch job at 2 am, because the report is read once a day at 9 am and streaming adds cost and complexity without a user benefit. Revisit if the business needs hourly numbers."
Check	The ADR includes the reason and the condition to revisit. Both matter. In plain words: a decision without a reason cannot be challenged or improved.

The picture

```
Idea -> Design doc (before) -> Build -> README (how to run) -> Runbook
(how to fix)
              |
            ADRs record each important decision along the way
```

Read left to right: documents are written at different moments for different readers.

Connection to the real job

Interview take-homes and portfolio projects are judged on README quality. In companies, a good runbook reduces 2 am calls.

Applications

- Data Engineering: data dictionary for each table, runbook per pipeline.
- ML: model card (what the model does, data, metrics, limits).
- Analytics: a page of metric definitions.

COMMON MISTAKE

Writing docs once and never updating them. Add the date and owner, and update when you change the code.

Check yourself

1. What does an ADR record? Answer: one decision with reasons and alternatives.

2. Who reads a runbook? Answer: the on-call engineer fixing a known issue.
3. What is the first line of a README? Answer: one sentence on the problem solved.

4.3 Slack Etiquette and Asking Good Questions

INTUITION

A doctor cannot help if you say "I feel bad." She asks: where, since when, what did you try? A good question gives the answerer everything needed, so they can answer in one reply.

Slack and Teams etiquette

- Put the question in the first message; do not send just "Hi" and wait.
- Use threads, so the channel stays readable.
- Use the right channel; do not send private messages for questions others also have.
- Respect time zones and working hours; do not expect an instant answer.
- Say thanks and post the solution when you solve it. The next person will find it.
- Mark urgency honestly. Not everything is urgent.

The four-part question

1. **What I am trying to do** (goal).
2. **What I tried** (steps, commands).
3. **Expected vs actual** (with the exact error message).
4. **Minimal example** (the smallest code or data that shows the problem).

Good question example

> "Hi team, I am loading a CSV of 2 million rows into Postgres with Python. Goal: finish in under 10 minutes. I tried `pandas.to_sql` with default settings; it took 47 minutes and used 6 GB RAM. I also tried `chunksize=50000` : 21 minutes. A minimal example and the schema are in the thread. Any suggestion on bulk loading?"

Problem 4.3.1 Rewrite this message: "My code is not working, please help."

Step 1	Goal: "I am joining orders and customers in pandas."
Step 2	Tried: "I used `orders.merge(customers, on='cust_id')`."
Step 3	Expected vs actual: "Expected 1,000 rows, got 1,200."
Step 4	Find the clue and add a minimal example: "`customers.cust_id` has duplicate ids; here are 5 sample rows from each table."
Answer	"I am joining orders (1,000 rows) with customers using `orders.merge(customers, on='cust_id')`. I expected 1,000 rows but got 1,200. I checked and 200 orders match a duplicated `cust_id`. Here are 5 sample rows from each table. Should I drop duplicates first, or is there a better key?"
Check	The numbers explain the bug: 1,000 + 200 extra matches = 1,200. A reader can answer at once: deduplicate customers.
Why it matters	Writing the question often answers it. Many engineers say "while writing the question I found the bug."

Problem 4.3.2 A colleague is in a UTC team. It is 5 pm IST for you. What time is it in UTC, and when might a reply come?

Formula	UTC = IST - 5 h 30 min (IST is the Indian time zone; it is 5 h 30 min ahead of UTC)
Plug in	17:00 - 5:30
Compute	11:30
Answer	It is 11:30 UTC. A UTC colleague's day ends around 17:00 UTC, which is 22:30 IST, so a reply could come late at night or the next morning your time. Send a complete question so they can answer in one go.
Check	11:30 + 5:30 = 17:00 IST. Correct.
Why it matters	A half-written question across time zones costs a full day per round trip.

The picture

```
Bad:   "Help"  -> reply: "What is the problem?"  -> 1 day lost
Good:  goal + tried + expected/actual + example -> reply: the answer
```

Connection to the real job

Senior people judge junior people by their questions. A clean question shows you tried first and respect their time. It is also how you get answers from open-source communities (Chapter 14).

Applications

- Team chat and code reviews.
- Stack Overflow and GitHub issues.
- AI assistants: the same four parts give much better AI answers.

COMMON MISTAKE

Pasting a screenshot of an error instead of the text. Text can be searched and copied. Paste the full error as text in a code block.

Check yourself

1. What are the four parts of a good question? Answer: goal, tried, expected vs actual, minimal example.
2. Why use threads? Answer: they keep the channel readable.
3. Should you paste errors as screenshots? Answer: no, use text.

4.4 Learning in Public: Weekly Posts and Tools

INTUITION

A shop with a closed door gets no customers, even if the shop is excellent. Posting what you learn opens the door a little. One short post a week for a year makes 52 posts. Hiring managers, referrers and mentors find you through these.

What to post

- **TIL ("Today I learned")**: 80 to 150 words, one small lesson.

- **Project write-up**: problem, approach, result with number, mistakes.
- **Paper summary** (Chapter 10).
- **Comparison with a benchmark**: for example, two ways to load data, with timings.

A simple template for a weekly post

```
Title: What I learned: window functions in 20 minutes
1 line: the problem I had
3–5 lines: what I did (with a small code snippet or number)
1 line: the mistake I made
1 line: what I will try next
```

Tools

- **Grammarly** or **LanguageTool** (free tiers exist) to catch grammar errors. Learn from the corrections.
- Read your post aloud once. If you run out of breath, the sentence is too long.
- Keep sentences under 20 words. One idea per sentence.

Ethics of posting

Never post confidential company data, client names, internal numbers or screenshots from work. Use public or simulated data. Check your employment contract if unsure.

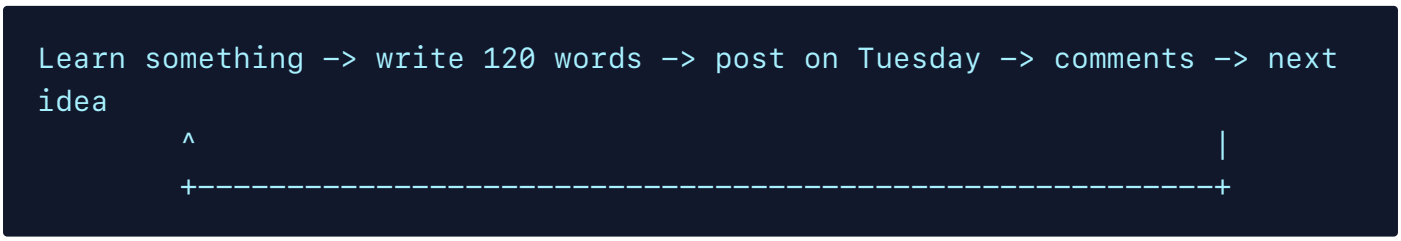
Problem 4.4.1 You plan 52 short posts (45 minutes each) and 24 blog posts (3 hours each) a year. How many hours is that, and is it realistic?

Formula	time = posts x hours per post + blogs x hours per blog
Plug in	52 x 0.75 + 24 x 3
Compute	52 x 0.75 = 39 hours; 24 x 3 = 72 hours; total = 111 hours
Answer	About 111 hours a year, or 2.1 hours a week. That is too heavy if you only study 7 hours a week.
Check	111 / 52 = 2.13. With 12 blogs instead (36 hours) the total is 75 hours, 1.4 hours a week. The skill map says 2 posts a month is plenty; here 12 blogs a year is even lighter, and you can still do it.
Why it matters	Plan the writing load honestly or you will drop it after 3 weeks.

Problem 4.4.2 Make this sentence shorter: "In this particular post I would like to try to explain in some detail about how it is that joins work."

Step 1	Remove filler: "In this particular post", "I would like to try to", "in some detail", "how it is that".
Step 2	Keep subject, verb, object.
Answer	"This post explains how joins work." That is 6 words instead of 21.
Check	The original has 21 words (count them). 21 - 6 = 15 words saved, about 71% shorter, with the same meaning.

The picture



Connection to the real job

Chapter 11 turns your posts into a portfolio. Recruiters and referrers read your last 5 posts to judge if you are real.

Applications

- LinkedIn posts, dev.to, Hashnode, Medium, a personal blog.
- Internal company wiki pages (visibility for internal transfers).

COMMON MISTAKE

Waiting for a perfect, long post. Short and honest beats long and never published. Post what you learned this week.

Check yourself

1. What is a TIL post? Answer: a short post on one thing you learned today.
2. Name a tool for grammar. Answer: Grammarly or LanguageTool.
3. What should you never post? Answer: confidential company data.

Chapter recap

- Put the ask in the email subject and the conclusion in the first line.
- A one-page executive summary has numbers first.
- READMEs say what, how to run, result and limits.
- ADRs record decisions with reasons; runbooks help the on-call person.
- Ask questions with goal, tried, expected vs actual, minimal example.
- Use threads, channels and time-zone respect in chat.
- Post one short thing a week; keep sentences under 20 words.
- Never share confidential data.

Pattern cheat-sheet

Need	Pattern
Email	Subject = ask; line 1 = conclusion; bullets; next steps
README	One sentence, result, how to run, how it works, limits
ADR	Status, context, decision, alternatives, consequences
Question	Goal, tried, expected vs actual, minimal example

Weekly post	Problem, what I did, mistake, next
Time zone	UTC = IST - 5 h 30 min

Practice set

1. (Easy) What is BLUF? **Answer:** Bottom line up front: put the conclusion first.
2. (Easy) Name two items in a README. **Answer:** How to run, result, data, limits (any two).
3. (Easy) 15:00 IST is what time in UTC? **Answer:** 09:30.
4. (Medium) Rewrite the subject "Update" for a request to approve 2 extra days. **Answer:** "Approval needed by Wed: extend data migration by 2 days".
5. (Medium) What is wrong with "Hi, are you there?" in Slack? **Answer:** It forces a wait. Send the full question in the first message.
6. (Medium) What is the result? Price 200 rises 5% and units fall 4%. Revenue change? **Answer:** 1.05 x 0.96 = 1.008, so revenue rises 0.8%.
7. (Medium) Find the bug: an ADR says only "We chose Kafka." **Answer:** No reasons, no alternatives. Add context, why, alternatives, consequences.
8. (Hard) Interview: "How do you document a data pipeline?" **Answer:** README (purpose, run steps), data dictionary, ADRs for key choices, runbook for failures, with owners and dates; keep it near the code.
9. (Hard) Interview: "How do you ask for help when stuck?" **Answer:** I define the goal, list what I tried with timings, give expected vs actual and a minimal example, then ask in the team thread after 30 to 60 minutes of trying.
10. (Hard) Scenario: you wrote a post using real company numbers. What do you do? **Answer:** Remove it, replace the numbers with simulated or public data, and check the company policy before posting again.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in
India moving into data roles; English is my second language. Teach me how
to write work emails: subject with the ask, conclusion first, bullets,
next steps. Show 3 bad emails and fix each step by step. Then give me 5
```

scenarios one at a time (approval request, delay notice, incident summary, status, clarification). Review each of my drafts and show a shorter version.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving to data engineering; English is my second language. Topic: README files, design docs and ADRs. Explain the purpose of each with a tiny example, then review my README for a project: [PASTE HERE]. Point out missing parts, vague sentences and what a hiring manager would not understand in 10 seconds. Give a rewritten version.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional in India moving to data. Topic: asking good questions (goal, what I tried, expected vs actual, minimal example). I will paste a messy question or bug. Rewrite it using the four parts, tell me what information is missing, and ask me for it. Then give me 5 more messy questions to fix myself, one at a time, without solutions first. My first question: [PASTE HERE].

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional in India learning data skills; English is my second language. Topic: learning in public. Help me plan 12 weekly LinkedIn posts from my study notes. Give a template, 12 topic ideas from SQL, Python and statistics, and tips for short sentences. Remind me not to post confidential company data. Then review my first draft: [PASTE HERE].

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data roles in India. Topic: executive summaries and incident reports. Show a one-page summary template with numbers first. Then give me a short scenario with data (numbers provided by you), ask me to write the summary, and grade it on clarity, correct numbers and recommendation. Check my arithmetic too.

CHAPTER 5

DATA STORYTELLING AND PRESENTING RESULTS

A number does nothing until someone understands it and acts on it. Your task is to move the person from the table to the decision.

You will often find the right answer and still lose the room. A manager sees 12 charts and asks "so what?". This chapter teaches you to build a short story around your data: who is listening, what the one headline is, which chart shows it, and what you recommend. By the end you will be able to take a messy table, find the single insight, show it in one clean chart, and present it in five minutes with answers ready for the questions that usually follow. We use one running example: monthly sales by region for a company with three regions, in lakh rupees (1 lakh = 100,000).

5.1 Audience First: Headline, Evidence, Recommendation

INTUITION

Think of a doctor reading your blood report. She does not read you 40 numbers. She says: "Your sugar is high. Here are two numbers that show it. Please reduce sweets and test again in a month." Headline, evidence, recommendation. Your data story is the same.

The storyline

1. **Audience**: who decides, what do they care about, how much time do they have, how technical are they?
2. **Headline insight**: one sentence with a number and a direction.
3. **Evidence**: the 1 to 3 charts or numbers that prove it.
4. **Recommendation**: what should we do, and what happens if we do nothing?
5. **Next step**: who does what by when.

Words you need

Word	Meaning

Insight	A finding that can change a decision (not just a fact)
Headline	The one-sentence version of the insight
Takeaway	The message the audience should remember
Clutter	Anything on a chart that does not help the message (extra gridlines, 3D, many colours)

Running example: the data

Monthly sales (lakh rupees):

Month	North	West	South	Total
Jan	120	95	80	295
Feb	125	97	60	282
Mar	130	99	55	284

Problem 5.1.1 Find the headline insight from the table.

Formula	change % = (new - old) / old x 100
Plug in	Total: (284 - 295) / 295; North: (130 - 120) / 120; West: (99 - 95) / 95; South: (55 - 80) / 80
Compute	Total = -11 / 295 = -3.7%. North = +8.3%. West = +4.2%. South = -25 / 80 = -31.2%. (Run in Python, the same values print.)
Answer	Total sales fell 3.7% from January to March, but only because South fell 31.2%. North and West are growing. Headline: "Sales are down 3.7% and the whole decline is in the South region, which lost 25 lakh per month."
Check	North + West gained 10 + 4 = 14 lakh. South lost 25 lakh. Net = 14 - 25 = -11 lakh, which equals 284 - 295 = -11. The story adds up.
Why it matters	In plain words: the problem is not "sales", it is "one region". That changes the action from a company-wide discount to a South investigation.

Problem 5.1.2 Write the recommendation and next step for the South drop, given you do not yet know the cause.

Step 1	Say what you know: South lost 25 lakh per month (31.2%).
Step 2	Say what you do not know: the cause (price, stock, competitor, data error?).
Step 3	Recommend the cheapest next test: check the South data for a data-loading problem first, then stock-outs, then price changes.
Answer	"Recommendation: before any discount, spend one week to check South for (1) missing data, (2) stock-outs, (3) price changes. I will report back on Friday 14 November."
Check	The recommendation names an owner (I), a deadline and a cheap order of checks. It does not claim a cause we cannot prove.
Why it matters	Honest, small next steps beat a confident wrong answer.

The picture

```
Table of 9 numbers
        |
  find ONE insight  ->  "South down 31.2%, rest growing"
        |
  one chart, one message
        |
  recommendation + next step
```

Read it top to bottom: your work is to shrink, not to expand. A nine-number table becomes one sentence.

Real-world mapping

Story part	Example	Where it appears in the job
Audience	Sales head, 10 minutes, non-technical	Monday business review
Headline	"South lost 25 lakh/month"	Slide title, email subject

Evidence	Chart of regions over 3 months	Dashboard page
Recommendation	Check data and stock in South	Action item in minutes

Connection to the real job

Analysts spend half their time on this skill. Data scientists must "sell" model results. Data engineers use it when they report pipeline reliability to managers. Chapter 3 gave you the speaking side; Chapter 4 the written side.

Applications

- Analytics: weekly business review decks.
- Data Science: A/B test results ("variant B lifts conversion by 0.4 percentage points").
- ML/AI: explaining accuracy and error cost to a product owner.
- Data Engineering: monthly pipeline reliability report.

COMMON MISTAKE

Presenting everything you analysed. The audience does not need your journey. Give them the destination. Move the rest to an appendix.

Check yourself

1. What are the five parts of the storyline? Answer: audience, headline, evidence, recommendation, next step.
2. What is an insight? Answer: a finding that can change a decision.
3. Why is the South headline better than "sales fell"? Answer: it points to where to act.

5.2 Choosing the Right Chart and Removing Clutter

INTUITION

A chart is a road sign. A sign that needs 30 seconds to read causes accidents. The best chart is understood in 3 seconds, and the viewer's eye goes straight to the one thing you want them to see.

Rule: one message per chart

Write the message as the chart title, as a full sentence: "South sales fell 31% in two months" instead of "Sales by region". Then choose the chart that makes that message easiest.

Chart choice table

Question	Use	Avoid
How does one thing change over time?	Line chart	Pie charts
Compare a few categories	Bar chart (sorted, horizontal if long labels)	3D bars
Parts of a whole, 2 to 4 parts	Stacked bar or a simple bar of shares	Pie with 8 slices
Relationship between two numbers	Scatter plot	Dual axes (confusing)
Distribution (for example response times)	Histogram or box plot	Only the average
Funnel stages	Horizontal bars with percentages	Tables only
Exact values needed	Table	Chart

Removing clutter

- Remove gridlines and borders unless needed.
- Label lines directly instead of a legend.
- Use one highlight colour (the message) and grey for the rest.
- Start bar charts at zero. Line charts may start higher but say so.
- Axis labels with units ("lakh rupees per month").
- Use the same scale when comparing charts.

Words you need

Word	Meaning
Axis	The horizontal (x) or vertical (y) line with numbers
Legend	The key that explains colours
Annotation	A short text on the chart pointing at the key point
Truncated axis	An axis that starts above zero and can make small changes look huge

Problem 5.2.1 Why does a bar chart that starts at 270 instead of 0 mislead? Use the totals 295, 282, 284.

Formula	apparent ratio of bar heights = (value - baseline) / (other - baseline)
Plug in	Baseline 0: 284 / 295. Baseline 270: (284 - 270) / (295 - 270) = 14 / 25.
Compute	284 / 295 = 0.963 (the bars look 96% as tall). 14 / 25 = 0.56 (the bars look only 56% as tall).
Answer	With a zero baseline, March is 96% as tall as January, which is true (a 3.7% fall). With a baseline of 270, March looks 56% as tall as January, as if sales nearly halved.
Check	The true change is -3.7%. A chart showing -44% (1 - 0.56) is a visual lie.
Why it matters	A manager who feels tricked will not trust your next chart.

Problem 5.2.2 Choose a chart for each message and say why: (a) "South fell while North and West grew over 3 months", (b) "Our funnel loses most people between visit and add-to-cart: of 10,000 visits, 3,200 add to cart, 900 start checkout, 270 pay".

Step 1	(a) Time plus 3 categories: a line chart with 3 lines, South in a strong colour, others grey, and direct labels.
Step 2	(b) Funnel stages: horizontal bars. Compute step rates: 3,200 / 10,000 = 32.0%; 900 / 3,200 = 28.1%; 270 / 900 = 30.0%; overall 270 / 10,000 = 2.7%.
Step 3	Look at the rates. The step losses are similar (about 30% continue at each step), so the first-line claim "most are lost between visit and cart" is not special: each step loses about 70%.
Answer	(a) line chart with South highlighted. (b) horizontal bars with step rates; the message should be corrected to "Only 2.7% of visits end in payment; each step loses about 70% of the people".
Check	0.32 x 0.281 x 0.30 = 0.027. Matches 2.7%.
Why it matters	The data corrected the first idea. If you had written the story before computing, you would have said something false.

The picture

```
Bad chart                          Good chart
Title: "Sales"                     Title: "South sales fell 31% in 2
months"
3D colourful bars, legend          Lines: South red, others grey, labelled
Axis from 270 (hidden)             Axis from 0, unit: lakh rupees / month
Gridlines everywhere               No gridlines; one annotation "Jan ->
Mar: -25"
```

Read left to right. The right chart takes about 3 seconds because the eye goes only to the red line.

Connection to the real job

Dashboard tools (Power BI, Tableau, Metabase, plain matplotlib) all let you break these rules. Your discipline matters more than the tool. Books to read: *Storytelling with Data* (Knaflic) and *Good Charts* (Berinato).

Applications

- Analytics: executive dashboards with a handful of tiles.
- Data Science: model-performance charts (ROC curve, precision-recall).
- Operations and service teams: median and 95th-percentile waiting time per week, with a target line.
- Data Engineering: pipeline freshness chart with a threshold line.

COMMON MISTAKE

Pie charts with many slices, dual-axis charts and 3D effects. They look impressive and communicate badly. Use bars and lines.

Check yourself

1. Which chart for change over time? Answer: line chart.
2. Why start bars at zero? Answer: bar length is read as size; truncating exaggerates differences.
3. What goes in the chart title? Answer: the message as a full sentence.

5.3 Executive Summaries and Handling Q&A

INTUITION

A senior leader is like a train passenger with five minutes before their stop. Give them a one-page ticket: the result, the evidence and what to do. The rest goes in the luggage (appendix).

One-page executive summary

```
Headline:   Sales down 3.7% (Jan to Mar); all of the decline is South
(-31%).
Evidence:   - North +8.3%, West +4.2%
            - South 80 -> 55 lakh/month (-25 lakh)
            - Decline began in February
Recommendation: One-week check of South: data load, stock, price.
Risk:       If not fixed, South loss of 25 lakh/month = 300 lakh/year.
Ask / next: Approve one analyst for one week. Report on Friday 14 Nov.
```

Check the risk number: 25 x 12 = 300 lakh per year (3 crore, since 100 lakh = 1 crore).

Handling Q&A in a data presentation

Question type	What to do
"How did you get this number?"	Show the definition and the source table in the appendix
"Is this significant or noise?"	State the sample size and the uncertainty, honestly
"What if it is a data error?"	Say what you checked (row counts, duplicates); say what you could not check
"Why not do X?"	Say what you considered and why you ranked it lower
A question you cannot answer	"I do not know yet. I will check and reply by tomorrow."

Prepare an appendix with definitions, data sources and 2 backup charts. Prepare answers for the 3 hardest questions before the meeting.

Presenting a result in 5 minutes

Use the plan from Chapter 3: problem 45 s, approach 60 s, result 90 s, recommendation 45 s, questions 60 s. Say the headline in the first 15 seconds and again at the end.

Problem 5.3.1 An A/B test: control 10,000 users, 400 converted. Variant 10,000 users, 450 converted. Write the headline, and state what you must still check.

Formula	conversion rate = converted / users
Plug in	Control 400 / 10,000; Variant 450 / 10,000
Compute	Control 4.0%, variant 4.5%. Difference = 0.5 percentage points. Relative lift = 0.5 / 4.0 = 12.5%.
Answer	"Variant B converted at 4.5% against 4.0%: a lift of 0.5 percentage points (12.5% relative)." You must still check whether the difference is bigger than random noise (a significance test, see the learner's statistics chapters) and the sample size.
Check	4.0% x 1.125 = 4.5%. Correct. A quick estimate of noise: the standard error of the difference is about sqrt(0.04 x 0.96 / 10000 + 0.045 x 0.955 / 10000) = sqrt(0.00000384 + 0.0000042975) = sqrt(0.0000081375) = 0.002853, so about 0.29 percentage points. The 0.5 point lift is about 1.75 standard errors: promising but not conclusive at the usual 95% level (which needs about 1.96).
Why it matters	In plain words: "it looks better, but the evidence is not yet strong." Saying this honestly is part of good storytelling.

Problem 5.3.2 A director asks, "Could the South drop be a data error?" Write the reply.

Step 1	Restate: "You are asking if South numbers could be wrong in the data."
Step 2	Say what you checked: "I compared row counts for South: January 41,200 orders, March 28,900 orders; no missing days; no duplicate order ids."
Step 3	Say what is unchecked: "I have not yet compared with the finance ledger."
Step 4	Promise: "I will compare with finance by Thursday."
Answer	A four-sentence reply that separates checked from unchecked.
Check	28,900 / 41,200 = 0.70, a 30% fall in orders, close to the 31% fall in sales, which supports a real drop in orders and not a pricing artefact (these order counts are made-up illustration numbers, so replace them with your own).
Why it matters	Knowing what you checked builds trust even when you cannot answer fully.

The picture

```
Appendix backup:  definitions | data sources | 2 backup charts | checks
done
Main story:       headline -> evidence -> recommendation -> next step
Q&A:              pause -> restate -> brief answer -> honest "not yet" if
needed
```

Connection to the real job

Stakeholders care about decisions and risk. You will often be asked "what should we do?". Always have a recommendation, even a small one.

Applications

- Analytics: monthly business review.
- Data Science: model go/no-go meetings.
- AI Engineering: showing evaluation results of a chatbot (accuracy on a test set, cost per question).
- Platform: reliability reviews.

COMMON MISTAKE

Ending with "any questions?" and no recommendation. Always end with your recommendation, then take questions.

Check yourself

1. What goes in the appendix? Answer: definitions, sources, backup charts, checks.
2. What do you do if asked about data errors? Answer: say what you checked and what you did not.
3. How do you compute a relative lift? Answer: difference divided by the control rate.

Chapter recap

- Know your audience: decision-maker, time, technical level.
- Story: headline, evidence, recommendation, next step.
- Shrink a table to one insight with a number.
- One message per chart; the title is the message.

- Line for time, bar for categories, scatter for relationships, table for exact values.
- Start bar charts at zero; remove clutter; highlight one thing.
- An executive summary has headline, evidence, recommendation, risk and ask.
- In Q&A, separate what you checked from what you did not; be honest.
- Compute numbers before writing the story; the data may correct your first idea.

Pattern cheat-sheet

Need	Pattern
Percent change	`(new − old) / old x 100`
Conversion rate	`converted / users`
Relative lift	`(variant − control) / control`
Chart title	One full sentence stating the message
Executive summary	Headline, evidence, recommendation, risk, ask
Q&A	Pause, restate, brief answer, honest "not yet"

Practice set

1. (Easy) 80 to 55: percent change? **Answer:** -31.25%, about -31.2%.
2. (Easy) Which chart for sales over 12 months? **Answer:** Line chart.
3. (Easy) What is a truncated axis? **Answer:** An axis not starting at zero, which can exaggerate changes.
4. (Medium) Compute: 5,000 visits, 150 purchases. Conversion rate? **Answer:** 150 / 5,000 = 3.0%.
5. (Medium) A chart title says "Revenue by month". Rewrite as a message for numbers 100, 102, 120. **Answer:** "Revenue jumped 17.6% in the third month" (120 / 102 = 1.176).
6. (Medium) Find the bug: pie chart of 9 regions. **Answer:** Too many slices; use a sorted bar chart.
7. (Medium) Scenario: your manager says "just show me everything." What do you do? **Answer:** Give a one-page summary with the main message and put the rest in an appendix, offering to go deeper on request.

8. (Hard) Interview: "Explain a time you influenced a decision with data." **Answer:** STAR: situation (problem), task (answer a question), action (analysis, one chart, recommendation), result (decision made, with a number).
9. (Hard) Interview: "A stakeholder disagrees with your analysis. What do you do?" **Answer:** Listen, restate their concern, check the data definition and sources together, show what you checked, agree on what test would settle it.
10. (Hard) Scenario: A/B test control 2,000 users with 80 conversions, variant 2,000 users with 96 conversions. The PM wants to launch. **Answer:** Rates 4.0% and 4.8%, lift 0.8 points (20% relative). Standard error = sqrt(0.04 x 0.96 / 2000 + 0.048 x 0.952 / 2000) = sqrt(0.0000192 + 0.00002285) = 0.00648, so the lift is 0.008 / 0.00648 = 1.23 standard errors, not significant. Say: "promising, but too early; we need more users."

Prompt pack

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional in India moving into data roles; English is my second language. Topic: data storytelling (audience, headline, evidence, recommendation, next step). Give me a small table of numbers with a hidden story, ask me to find the headline insight and compute the percent changes, then check my arithmetic and teach me how to turn it into a one-sentence headline and a recommendation.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving to data. Topic: choosing the right chart and removing clutter. Explain with 6 messages (change over time, ranking, parts of a whole, relationship, distribution, funnel) and which chart for each. Then show me 5 bad chart descriptions and ask me to fix each one at a time. Include one truncated-axis example with numbers so I can compute the visual exaggeration.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional in India; English is my second language. Topic: writing a one-page executive summary with numbers first. I will paste my analysis results. Turn them into headline, evidence (3 bullets), recommendation, risk and next step,
```

in simple English. Point out any numbers that look inconsistent. My results: [PASTE HERE].

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving to data in India. Topic: handling Q&A after presenting results. Act as a sceptical director. Ask me 8 hard questions about my analysis one at a time (data errors, significance, why not another option, cost). Give feedback after each answer on honesty, brevity and structure. My analysis summary: [PASTE HERE].

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional in India moving to data. Topic: presenting an A/B test result. Give me 3 A/B test scenarios with user counts and conversions. For each, I will compute rates, lift and standard error, and decide whether to launch. Check my steps, explain the standard error in plain words, and show how to say the result honestly to a non-technical manager.

CHAPTER 6

PROBLEM SOLVING AND DEBUGGING METHOD

Strong engineers are not people who know every answer. They are people who have a method for finding answers when they do not know.

Most of your time in a data job is spent on problems nobody has solved before in your company: "why did the dashboard number drop?", "why is this pipeline slow?", "how big should this system be?". Beginners guess or panic. Professionals follow a short method. You already own one of the best methods, because a working professional hunts root causes every day. This chapter writes that method down: a framework for coding problems, issue trees for business problems, a debugging loop, root-cause analysis, and estimation (back-of-envelope) problems. By the end you will have a repeatable approach for coding interviews, for real bugs, and for "how many..." questions.

6.1 The Coding Problem Framework

INTUITION

A good doctor does not operate before diagnosis. She asks questions, repeats back what she heard, tests simple ideas first, and only then treats. In a coding problem, the code is the treatment. Everything before it is diagnosis.

The eight steps

1. **Clarify**: ask about inputs, outputs, size, edge cases ("Can the list be empty? Are there duplicates? How large?").
2. **Restate** the problem in your own words.
3. **Examples**: write 2 small examples and 1 edge case with the expected answers.
4. **Brute force**: say the simplest solution aloud, even if slow, and its cost.
5. **Optimize**: find what is repeated and remove it (a hash map, sorting, two pointers, a better data structure).
6. **Code** it, with clear names.
7. **Test** on your examples, by hand.
8. **Analyze complexity**: time and memory in Big-O terms.

Big-O describes how the work grows as the input size n grows. `O(n)` means work grows in direct proportion. `O(n²)` means doubling n makes the work 4 times as much.

Running example: Two Sum

Problem: given a list of numbers and a target, return the positions of two numbers that add up to the target.

Problem 6.1.1 Walk the framework on the list `[2, 7, 11, 15]` with target 9.

Step 1	Clarify: exactly one answer exists; the same element cannot be used twice; return indices.
Step 2	Restate: find i and j, i not equal to j, with nums[i] + nums[j] = 9.
Step 3	Example: `[2, 7, 11, 15]`, 9 gives `(0, 1)` because 2 + 7 = 9. Edge: `[3, 3]`, 6 gives `(0, 1)`.
Step 4	Brute force: try every pair. For n numbers there are n(n-1)/2 pairs. For n = 4000 that is 4000 x 3999 / 2 = 7,998,000 checks. This is `O(n²)`.
Step 5	Optimize: for each number x, we need target - x. Store numbers already seen in a dictionary (a hash map, which finds a key in about one step). Then each number needs one lookup: `O(n)`.
Answer	The code below (both versions run and agree).

```
def two_sum_brute(nums, target):
    for i in range(len(nums)):
        for j in range(i + 1, len(nums)):
            if nums[i] + nums[j] == target:
                return (i, j)
    return None

def two_sum_fast(nums, target):
    seen = {}                        # value -> index
    for i, x in enumerate(nums):
        if target - x in seen:
            return (seen[target - x], i)
        seen[x] = i
    return None

print(two_sum_brute([2, 7, 11, 15], 9), two_sum_fast([2, 7, 11, 15], 9))
```

Before you run it, what do you think prints? Both functions should return the same pair. Real output:

```
(0, 1) (0, 1)
```

- **Check** In the fast version at i = 0, x = 2: 9 - 2 = 7 is not in `seen` (empty), so store `{2: 0}`. At i = 1, x = 7: 9 - 7 = 2 is in `seen` at index 0, so return `(0, 1)`. Same as brute force. I also ran both on 4,000 random numbers and they returned the same pair `(2088, 2332)`.
- **Why it matters** In plain words: the fast version does about 4,000 steps; the brute version up to 7,998,000. In an interview, saying the brute force first and then the improvement is exactly what is scored.

Problem 6.1.2 What if the input is a sorted list, and memory must be tiny?

Step 1	Clarify: sorted ascending; extra memory should be O(1).
Step 2	Idea: two pointers, one at each end. If the sum is too small, move the left pointer right; if too big, move the right pointer left.
Step 3	Example `[2, 7, 11, 15]`, target 9: left = 2, right = 15, sum 17 > 9, so move right. Right = 11, sum 13 > 9, move right. Right = 7, sum 9. Found.
Answer	Two pointers solve it in `O(n)` time and `O(1)` memory when the list is sorted.
Check	It took 3 comparisons for 4 numbers, at most n - 1.
Why it matters	A trade-off: hash map uses more memory; two pointers needs sorted input. Naming trade-offs earns marks.

Picture

```
Clarify -> Restate -> Examples -> Brute force -> Optimize -> Code -> Test
-> Complexity
   ^ ask       ^ say it     ^ write 3     ^ say cost     ^ find the
repeated work
```

Real-world mapping

Step	Interview behaviour	Job behaviour
Clarify	"How large is n? Duplicates?"	"Which orders count as 'active'?"
Brute force	State the simple way and its cost	Make it work first
Optimize	Hash map, sort, two pointers	Add an index, partition, cache
Test	Walk examples by hand	Unit tests, small sample data

Connection to the real job

SQL and pandas problems follow the same method: clarify the grain (one row means what?), test on a tiny table, then scale. Chapter 7 uses this in interviews.

Applications

- Data Analyst: SQL challenges (clarify the question, test on 5 rows).
- Data Engineer: finding why a join explodes (brute force thinking about row counts).
- ML Engineer: optimizing feature computation from $O(n^2)$ to $O(n)$.

COMMON MISTAKE

Starting to code in the first 30 seconds. You then solve the wrong problem, and the interviewer sees no thinking. Spend the first 3 to 5 minutes on steps 1 to 4.

Check yourself

1. Why state the brute force first? Answer: it proves you understand the problem and gives a base to improve.
2. What does $O(n^2)$ mean? Answer: doubling the input makes work 4 times larger.
3. Why use a hash map in Two Sum? Answer: it finds a needed value in about one step.

6.2 Structured Thinking: Issue Trees, MECE and Hypotheses

INTUITION

Imagine your phone bill doubled. You do not stare at the total. You split it: calls, data, roaming, extra services. Then you split the biggest piece again. Splitting a big problem into non-overlapping parts is how you find where to look.

Words you need

Term	Meaning
Issue tree	A diagram that splits a problem into smaller parts, branch by branch
MECE	"Mutually Exclusive, Collectively Exhaustive": parts do not overlap and together cover everything
Hypothesis	A guess you can test ("The drop is caused by a data load failure")
First principles	Break a problem to basic facts and rebuild, instead of copying what others do

Example issue tree: "Revenue fell 10%"

Revenue = number of orders x average order value (AOV). This is MECE: a change must come from orders, from AOV, or both.

```
Revenue fell 10%
|-- Orders fell?
|    |-- Fewer visitors (traffic)?
|    |    |-- by channel: search, ads, email, direct
|    |-- Lower conversion (visitor -> order)?
|         |-- by device, by product category, by new vs returning
|-- AOV fell?
|    |-- fewer items per order?
|    |-- lower prices or more discounts?
|-- Data problem? (always check first)
     |-- missing days, duplicates, definition change
```

Hypothesis-driven analysis

Do not open ten dashboards hoping to see something. Write a hypothesis, decide which one

cheap query would disprove it, run it, then move on.

Problem 6.2.1 Revenue fell from 295 to 284 lakh. Orders went from 59,000 to 52,000 and AOV is the rest. Which branch of the tree explains the drop?

Formula	revenue = orders x AOV, so AOV = revenue / orders
Plug in	Jan AOV = 295 lakh / 59,000 = 29,500,000 / 59,000 rupees; March AOV = 28,400,000 / 52,000 rupees
Compute	Jan = 500 rupees. March = 546.15 rupees (28,400,000 / 52,000 = 546.15).
Answer	Orders fell 11.9% ((52,000 - 59,000) / 59,000), AOV rose 9.2% (546.15 / 500 = 1.0923). So the drop comes from fewer orders, not from smaller baskets. Go down the Orders branch (traffic or conversion).
Check	52,000 x 546.15 = 28,399,800, close to 28,400,000 (rounding). Also 0.881 x 1.0923 = 0.962, a 3.8% fall, near the actual -3.7%.
Why it matters	In plain words: customers are buying fewer times, but spending more when they buy. A discount on price would be the wrong fix.

Problem 6.2.2 Which of these splits is MECE for "why did the pipeline fail": (a) network, database, bug; (b) bug in my code, bug in someone else's code, human error?

Step 1	Test exclusive: in (a) "network" and "database" can overlap (a database network timeout). In (b) a human error can be the cause of a bug in the code: overlap.
Step 2	Test exhaustive: in (a) missing "infrastructure, data, configuration"; in (b) missing "input data problem".
Answer	Neither is perfectly MECE. A better split: by stage (ingest, transform, load), because every failure happens in exactly one stage.
Check	Take any failure and ask "which stage?": there is exactly one answer.
Why it matters	A good split means you never forget a branch and never check the same thing twice.

Picture

The tree above reads left to right in time: you check the top branches first, then go down only

the branch that holds the cause. At each node ask: is the cause here? One cheap check closes or opens the branch.

Connection to the real job

This is how analysts handle "why did the KPI move?" questions. It is also how a restaurant manager splits a customer's total wait into ordering, cooking, plating and serving time.

Applications

- Analytics: KPI drop analysis, funnel diagnosis.
- Data Engineering: pipeline failure triage by stage.
- Data Science: error analysis by segment.
- Consulting-style case interviews (Chapter 7).

COMMON MISTAKE

Checking random things because they "feel" likely. Write the tree, so you can see what you have not yet checked.

Check yourself

1. What does MECE stand for? Answer: mutually exclusive, collectively exhaustive.
2. Revenue splits into what two factors? Answer: orders and average order value.
3. What do you always check first in a KPI drop? Answer: a data problem.

6.3 Debugging and Root-Cause Analysis

INTUITION

Your house lights go out. You do not replace all the bulbs. You check the fuse box first, then the room. You halve the problem each time. Debugging is a search, and good searches cut the space in half.

The debugging loop

Reproduce -> Isolate -> Hypothesize -> Test -> Fix -> Prevent.

- **Reproduce**: make it fail on demand. If you cannot reproduce it, you cannot know you fixed it.
- **Isolate**: shrink the failing case. Fewer rows, fewer lines, fewer services.
- **Hypothesize**: write one specific guess.
- **Test**: change one thing; observe.
- **Fix** the cause, not the symptom.
- **Prevent**: add a test, an alert, or a check, and write a short note.

Bisecting

If something worked in build 1 and fails in build 1000, test build 500. If it fails, the bug is in 1 to 500. Repeat. The number of tests needed is $\log_2(n)$ rounded up. For 1,000 builds that is 10 tests.

```
lo, hi, steps = 1, 1000, 0
first_bad = 637
while lo < hi:
    mid = (lo + hi) // 2
    steps += 1
    if mid >= first_bad:     # build 'mid' is bad
        hi = mid
    else:
        lo = mid + 1
print("first bad build", lo, "after", steps, "tests")
```

What do you think prints? Real output:

```
first bad build 637 after 10 tests
```

5 Whys and root cause

The 5 Whys technique: ask "why?" until you reach a cause you can fix and prevent. Stop when the answer is a process or a design, not a person.

Problem 6.3.1 Apply 5 Whys. A dashboard showed yesterday's revenue as 0.

Step 1	Why was revenue 0? The orders table had no rows for yesterday.
Step 2	Why no rows? The nightly load job failed at 2 am.
Step 3	Why did it fail? The source file arrived with a new column.
Step 4	Why did a new column break the job? The code used fixed column positions.
Step 5	Why fixed positions with no alert? We had no schema check and no alert on zero-row loads.
Answer	Root cause: no schema validation and no zero-row alert. Fix: read columns by name, add a schema check, alert if row count is below 50% of the 7-day average. A person was not the cause; the process was.
Check	If the check had existed, would the failure have been caught at 2 am? Yes: the new column would have triggered the schema alert.
Why it matters	Fixing only "re-run the job" leaves the same bug for next week.

Problem 6.3.2 A query that took 4 seconds now takes 40. You have 10 recent changes. How many tests do you need with bisecting versus testing one by one?

Step 1	One by one: up to 10 tests.
Step 2	Bisecting: $\log_2(10) = 3.32$, so 4 tests.
Answer	At most 4 tests against up to 10.
Check	$2^4 = 16$ which is at least 10, but $2^3 = 8$ is less than 10. So 4 is right.
Why it matters	The gap grows quickly: for 1,000 changes it is 10 tests against 1,000.

Reading logs

Read from the first error, not the last. The first error is usually the cause; later errors are consequences. Note timestamps and compare with deployment times. Search for the request ID across services.

Decisions under uncertainty

- Ask whether the decision is **reversible** (easy to undo, like a feature flag) or **irreversible** (like deleting data). Decide reversible things fast; slow down for irreversible ones.
- Write the trade-off: options, cost, benefit, risk, and why you chose.
- Do the cheapest test that can change your mind.

Picture

```
Reproduce -> Isolate -> Hypothesize -> Test -> Fix -> Prevent
    ^                                           |
    +------------ if not fixed, loop ------------+
```

Connection to the real job

A restaurant kitchen shows this well. Orders are coming out slowly: you reproduce it at the Friday rush, isolate the slow station by timing each station, hypothesize (only one grill cook?), and test by changing one thing. This is root-cause analysis.

Applications

- Data Engineering: broken pipelines, slow Spark jobs, duplicate rows.
- ML: model accuracy dropped (data drift? feature bug? label delay?).
- Analytics: wrong numbers after a definition change.

COMMON MISTAKE

Changing three things at once. If it works, you do not know which fix did it; if it breaks, you do not know which one broke it. Change one thing at a time.

Check yourself

1. Name the debugging loop. Answer: reproduce, isolate, hypothesize, test, fix, prevent.
2. How many tests to bisect 1,000 builds? Answer: 10.
3. When do you stop the 5 Whys? Answer: at a process or design cause you can fix.

6.4 Estimation: Fermi Problems and Back-of-the-Envelope Math

INTUITION

Before you drive to another city you estimate: 300 km at 60 km per hour is 5 hours. You do not need exact numbers; you need the right order of magnitude. A Fermi problem (named after the physicist Enrico Fermi) is an estimate built from simple guesses that are probably within a factor of 2 or 3.

The method

1. State the question and what unit the answer has.
2. Break it into factors you can estimate.
3. Write each assumption clearly and label it as an assumption.
4. Multiply or add.
5. Sanity-check against something you know.

Numbers worth remembering

Fact	Value
Seconds in a day	86,400 (about 10^5)
Seconds in a month	about 2.6 million
1 million per day	about 12 per second
1 billion per day	about 11,600 per second
1 KB, 1 MB, 1 GB, 1 TB	10^3, 10^6, 10^9, 10^{12} bytes
1 lakh, 1 crore	10^5, 10^7

Problem 6.4.1 (Assumptions are made up for practice.) Estimate cab rides per second in a city with 13 million people, if 3% take a ride per day and rides happen mostly in 16 hours.

Formula	rides per second = rides per day / seconds of activity
Plug in	Rides per day = 13,000,000 x 0.03; seconds = 16 x 3,600
Compute	13,000,000 x 0.03 = 390,000 rides per day. 16 x 3,600 = 57,600 seconds. 390,000 / 57,600 = 6.77 rides per second.
Answer	About 7 rides per second on average. If the peak is 3 times the average, plan for about 20 rides per second.
Check	390,000 rides over 86,400 seconds would be 4.5 per second; concentrating them into 16 hours gives a higher number, 6.8. The direction is correct.
Why it matters	In plain words: this system needs a modest server, not a giant one; the important traffic is the location pings, which are far more frequent. Estimation tells you where the real load is.

Problem 6.4.2 A bank has 5 million app users. Each opens the app about 3 times a day, and each open makes about 8 API calls. Estimate average and peak API calls per second.

Formula	calls per second = users x opens x calls per open / 86,400
Plug in	5,000,000 x 3 x 8 / 86,400
Compute	5,000,000 x 3 = 15,000,000 opens; x 8 = 120,000,000 calls per day; / 86,400 = 1,388.9 per second.
Answer	About 1,400 calls per second on average. If peak is 3 times average, about 4,200 per second.
Check	By the shortcut "1 million per day is about 12 per second": 120 million per day gives 120 x 12 = 1,440 per second. Close to 1,389.
Why it matters	A capacity planner would turn this directly into a staffing target and compare with Little's law. If each call takes 0.2 s, concurrent calls = 4,200 x 0.2 = 840.

Picture

```
Big unknown -> factors -> each factor: assumption (labelled) -> multiply -
> sanity check
```

Connection to the real job

System design interviews begin with this (Chapter 8). On the job, you estimate storage and cost before you build, and the estimate can save weeks.

Applications

- Data Engineering: data volume per day, storage cost, cluster size.
- ML/AI: GPU hours, tokens per month, cost per 1,000 predictions.
- Analytics: market sizing, campaign reach.

COMMON MISTAKE

Hiding assumptions or giving one false-precise number. Say the assumptions aloud and give a range ("between 1,000 and 2,000 per second").

Check yourself

1. How many seconds are in a day? Answer: 86,400.
2. How do you check a Fermi estimate? Answer: compare with a known fact or a quick shortcut.
3. What do you do with assumptions? Answer: state them clearly and label them.

Chapter recap

- For coding problems: clarify, restate, examples, brute force, optimize, code, test, complexity.
- State the brute force first; it shows understanding.
- Split problems into MECE branches and test one hypothesis at a time.
- Revenue = orders x average order value is a good first split.
- Debug by reproducing, isolating, hypothesizing, testing, fixing and preventing.
- Bisecting finds the first bad change in $\log_2(n)$ tests.
- 5 Whys ends at a process cause, not a person.
- Estimate with labelled assumptions and sanity-check the result.

Pattern cheat-sheet

Need	Pattern

Coding steps	Clarify, restate, examples, brute, optimize, code, test, Big-O
KPI drop	Check data first, then revenue = orders x AOV
Bisect tests	`ceil(log2(n))`
Events per second	`events per day / 86,400`
Peak load	average x 2 to 3 (state the assumption)
Concurrent users	`throughput x response time` (Little's law)

Practice set

1. (Easy) How many tests to bisect 16 builds? **Answer:** $\log_2(16) = 4$.
2. (Easy) 2 million events per day is about how many per second? **Answer:** 2,000,000 / 86,400 = 23.1.
3. (Easy) Expand MECE. **Answer:** Mutually exclusive, collectively exhaustive.
4. (Medium) Find the output: `two_sum_fast([3, 3], 6)`. **Answer:** At i = 0, store {3: 0}. At i = 1, 6 - 3 = 3 is in seen, return `(0, 1)`.
5. (Medium) Orders fell from 1,000 to 900, AOV rose from 200 to 210. Revenue change? **Answer:** 0.9 x 1.05 = 0.945, a 5.5% fall.
6. (Medium) Find the bug in the plan: "The job failed, I re-ran it, it passed, done." **Answer:** No root cause and no prevention; it will fail again. Do 5 Whys and add a check.
7. (Medium) Scenario: a dashboard number is 20% lower than finance's number. What do you check first? **Answer:** Definitions (what counts as revenue), date range and time zone, duplicates, missing partitions.
8. (Hard) Interview: "How many bytes does an app store if 10 million users each send 5 messages of 200 bytes per day for a year?" **Answer:** 10,000,000 x 5 x 200 = 10,000,000,000 bytes (10 GB) per day; x 365 = 3.65 TB per year before replication.
9. (Hard) Interview: "Walk me through how you would debug a sudden latency spike at 3 pm." **Answer:** Reproduce or find the time window; compare p50/p95 by endpoint; check deployments, traffic, GC, DB slow queries and connection pool; test one hypothesis at a time; fix and add an alert.
10. (Hard) Scenario: you must choose between a quick fix that deletes old data (irreversible) and a slower reversible one. **Answer:** Choose the reversible one unless the cost of waiting

is higher than the cost of a mistake; if you must delete, take a backup first and record the decision.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in India moving into data roles; English is my second language. Teach me the coding problem framework: clarify, restate, examples, brute force, optimize, code, test, complexity. Walk through Two Sum and one more problem step by step. Then give me 5 problems, one at a time. For each I must follow the framework in words before writing code; you review my process.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving to data. Topic: issue trees and MECE for business problems. Give me 3 KPI-drop scenarios with numbers (revenue, signups, churn). For each, I will draw an issue tree in text, and then find the branch that explains the change by computing. Check my arithmetic and tell me where my tree overlaps or leaves gaps.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in India moving to data. Topic: debugging method and root-cause analysis (reproduce, isolate, hypothesize, test, fix, prevent; bisecting; 5 Whys). Give me 4 realistic data-pipeline failures with logs. I will diagnose each, one at a time. Grade my root cause, my prevention step and whether I changed one thing at a time.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in India moving to data. Topic: estimation (Fermi) problems and back-of-the-envelope math. Give me 8 graded estimation problems: users per second, storage per year, cost per month, GPU hours. Do not give answers first. After I answer, show your own calculation with labelled assumptions, compare, and teach me sanity checks.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving to data. Topic: decision-making under uncertainty (reversible vs
irreversible choices, trade-off documentation). Describe 5 work scenarios
in a data team. For each I will say which decision type it is, list
options with cost, benefit and risk, and choose. Then you critique my
reasoning and show how to write a short decision record.
```

CHAPTER 7

TECHNICAL INTERVIEW PREPARATION BY ROLE

An interview is a skill, separate from your day-to-day work skills. You can train it with a plan, a question bank and mock practice.

Many capable engineers fail interviews because they never practised the interview format: 45 minutes, a stranger, a time limit, someone watching. This chapter shows what each role is tested on, how to prepare each round (SQL, Python and DSA, statistics and machine learning, system design, case and product questions, behavioral), how to build eight reusable STAR stories from your own work experience, and a 12-week plan with 96 hours of practice. By the end you will know what to practise first, how to measure readiness, and how to run a mock interview.

7.1 What Each Role Tests

INTUITION

Interview rounds are exam papers. A Data Engineer exam has heavy SQL and design; a Data Scientist exam has heavy statistics and ML. Do not study for all papers equally; study for the paper you will actually write.

Words you need

Term	Meaning
Online assessment (OA)	A timed test on a website, often SQL, Python or multiple choice
Screen	A short first call (recruiter or engineer) to check basic fit
Loop	The full set of interview rounds for one job
DSA	Data structures and algorithms (lists, dictionaries, trees, sorting, graphs)
Case round	A business problem you reason through aloud

Behavioral round	Questions about your past actions (use STAR from Chapter 3)

The round map (weight by role)

Round	Analyst	Data Eng	Data Scientist	ML/AI Eng
SQL	Heavy	Heavy	Medium-Heavy	Medium
Python/coding	Light	Medium-Heavy	Medium	Heavy (+DSA)
Statistics/probability	Medium	Light	Heavy	Medium
ML	none	none	Heavy	Heavy
System design	none	Heavy (data systems)	Medium (ML design)	Heavy (ML systems)
Case/product	Heavy	Light	Heavy	Medium
Behavioral	Always	Always	Always	Always

A typical loop

Typical interview timeline: 5 to 8 weeks from application to offer

Read the picture left to right. Each box is a stage. The week labels above the boxes are typical, not promised. Many companies are faster; some take longer. The last stage is the offer, and your notice period starts after you accept.

Running example: you target Data Engineer roles

Your weights: SQL (heavy), Python (medium-heavy), system design (heavy), behavioral (always), stats (light).

Problem 7.1.1 You have 96 hours of preparation. Allocate hours for the Data Engineer target using weights Heavy = 3, Medium-Heavy = 2, Light = 1, Always = 2 for behavioral.

Formula	hours for a round = total hours x (round weight / sum of weights)
Plug in	Weights: SQL 3, Python 2, system design 3, behavioral 2, stats 1. Sum = 11.
Compute	SQL = 96 x 3/11 = 26.2; Python = 96 x 2/11 = 17.5; system design = 26.2; behavioral = 17.5; stats = 96 x 1/11 = 8.7. Total = 96.1 (rounding).
Answer	About 26 hours SQL, 18 Python, 26 system design, 18 behavioral, 9 stats.
Check	26.2 + 17.5 + 26.2 + 17.5 + 8.7 = 96.1, within rounding of 96.
Why it matters	In plain words: you spend your hours where the marks are, not where your hobby is.

Problem 7.1.2 Same total for a Data Scientist target: SQL 2, Python 2, stats 3, ML 3, ML design 2, case 3, behavioral 2.

Step 1	Sum of weights = 2 + 2 + 3 + 3 + 2 + 3 + 2 = 17.
Step 2	Hours: SQL 96 x 2/17 = 11.3; Python 11.3; stats 96 x 3/17 = 16.9; ML 16.9; ML design 11.3; case 16.9; behavioral 11.3.
Answer	Stats, ML and case each get about 17 hours; the others about 11.
Check	11.3 x 4 + 16.9 x 3 = 45.2 + 50.7 = 95.9. Rounds to 96.

Connection to the real job

Read the JD (Chapter 1) and ask the recruiter: "What are the interview rounds and what does each cover?" Most recruiters tell you. Then you prepare for the actual loop.

Applications

- Every role target: use the map to set study hours.
- Choosing between two applications: prepare for the one with the nearest loop first.

COMMON MISTAKE

Preparing equally for everything, or only for what you enjoy. Use the weights.

Check yourself

1. What is an OA? Answer: a timed online assessment.
2. Which role has heavy statistics? Answer: Data Scientist.
3. What always appears in every role? Answer: the behavioral round.

7.2 SQL, Python/DSA and Stats/ML Preparation

INTUITION

Practice problems are the gym. You do not get strong by watching someone lift. You get strong by lifting a little more, with a spotter who corrects your form.

SQL and Python practice

- Use LeetCode (SQL and Python), plus other sites, in a pattern-based way, not random order. For SQL: joins, GROUP BY / HAVING, window functions (ROW_NUMBER, RANK, DENSE_RANK, LAG), self-joins, dates, NULL handling. For Python/DSA: arrays and hashing, two pointers, sliding window, stacks, binary search, trees, graphs, simple dynamic programming.
- 2 to 3 problems a week during general study; 5 to 7 per week in the last 6 weeks. For each problem: try 25 minutes alone, then study a solution, then redo it from memory 3 days later.
- Track in a spreadsheet: problem, pattern, date, solved alone (yes/no), mistake.

A real SQL trap: the "second highest" with ties

Table `emp(id, name, dept, salary)`. Question: for each department, find the employee with the second-highest salary. Before you run it, what do you think `LIMIT 1 OFFSET 1` returns for the Data department, where two people share the top salary of 120?

```
create table emp(id int, name text, dept text, salary int);
insert into emp values (1,'Asha','Data',90),(2,'Ben','Data',120),
(3,'Chitra','Data',120),
 (4,'Dev','Ops',70),(5,'Esha','Ops',80),(6,'Farid','Ops',95);

-- Correct: DENSE_RANK handles ties
select dept, name, salary from (
  select dept, name, salary,
         dense_rank() over (partition by dept order by salary desc) r
  from emp) where r = 2 order by dept;

-- Wrong: OFFSET ignores ties
select name, salary from emp where dept='Data' order by salary desc limit
1 offset 1;
```

Real output (SQLite):

```
dept  name  salary
----  ----  ------
Data  Asha  90
Ops   Esha  80

name    salary
------  ------
Chitra  120
```

Explanation line by line. `partition by dept` restarts ranking inside each department. `order by salary desc` ranks highest first. `dense_rank()` gives tied rows the same rank and does not skip numbers, so both 120 salaries are rank 1 and 90 is rank 2. `LIMIT 1 OFFSET 1` just skips the first row, so it returns the second row, Chitra with 120, which is still the highest salary. That is wrong when ties exist.

Problem 7.2.1 What would `rank()` instead of `dense_rank()` return for Data, and why?

Step 1	`rank()` gives ties the same rank, then skips. Salaries 120, 120, 90 get ranks 1, 1, 3.
Step 2	`r = 2` matches no row in Data.
Answer	`rank()` returns no row for Data; `dense_rank()` returns Asha. Ask the interviewer: "do you want the second highest value or the second row?" That is the clarify step.
Check	Ops salaries 95, 80, 70 are distinct, so both functions return Esha. Only ties show the difference.
Why it matters	In plain words: ties are the favourite hidden trap in SQL interviews.

Statistics and ML Q&A bank

Write your own answers to 100 common questions (do not copy). Group them:

Group	Sample questions	Count
Probability	Bayes rule, conditional probability, expected value	15
Statistics	p-value, confidence interval, Type I and II errors, A/B test design, power	20
ML basics	bias-variance, overfitting, regularization, cross-validation	20
Models	linear/logistic regression, trees, random forest, boosting	15
Metrics	precision, recall, F1, ROC-AUC, class imbalance	15
Practice	data leakage, feature engineering, monitoring, drift	15

Answer each in PREP form (Chapter 3), 60 seconds. Record yourself. 100 questions at 15 minutes each (write, then speak) is 25 hours.

Problem 7.2.2 Explain precision and recall with numbers: a fraud model flags 100 transactions; 40 are really fraud; the data has 50 frauds in total.

Formula	precision = true positives / flagged; recall = true positives / actual frauds
Plug in	Precision = 40 / 100; recall = 40 / 50
Compute	Precision = 0.40; recall = 0.80
Answer	Of the flagged transactions, 40% are real fraud (precision). The model catches 80% of all fraud (recall). F1 = 2 x 0.4 x 0.8 / (0.4 + 0.8) = 0.64 / 1.2 = 0.533.
Check	40 / 100 = 0.4, 40 / 50 = 0.8. Misses = 50 - 40 = 10 frauds, false alarms = 100 - 40 = 60.
Why it matters	In plain words: the model finds most fraud but 6 of 10 alerts are false alarms; the business decides whether that cost is acceptable. Always connect the metric to a cost.

Picture

```
Weekly plan:  Mon SQL x1 | Tue Python x1 | Wed stats Q&A (5) | Thu SQL x1
| Fri review |
              Sat project / mock (3 h) | Sun rest
```

Connection to the real job

SQL and Python are daily tools; statistics Q&A is a day-to-day skill for experiment review. Practising explanations makes you a better team member, not only a better candidate.

Applications

- Data Analyst: SQL window functions, A/B test questions.
- Data Engineer: SQL, Python scripting, DSA basics.
- ML Engineer: DSA, ML concepts, metrics.

COMMON MISTAKE

Reading solutions without solving. Try for 25 minutes first; the struggle is where learning happens (Chapter 2).

Check yourself

1. What does DENSE_RANK do with ties? Answer: gives tied rows the same rank without skipping numbers.
2. What is precision? Answer: true positives divided by everything flagged.
3. How long do you try before looking at a solution? Answer: about 25 minutes.

7.3 Case, Product and System-Design Rounds; Behavioral Stories

INTUITION

A case round is a conversation where you act as the analyst. The interviewer does not want the right answer; they want to watch you think: clarify, structure, calculate, recommend.

The case framework

1. **Clarify** the goal and the metric ("What does 'drop' mean? Which period?").
2. **Structure** with an issue tree (Chapter 6).
3. **Prioritize** branches by size and likelihood; ask for data.
4. **Calculate** with the numbers they give.
5. **Recommend** with a next step and a risk.

Problem 7.3.1 Case: "Daily active users (DAU) fell from 200,000 to 170,000. What do you do?" The interviewer gives: Android 100,000 to 99,000; iOS 60,000 to 59,000; web 40,000 to 12,000.

Formula	change % = (new - old) / old x 100
Compute	Total: (170,000 - 200,000) / 200,000 = -15%. Android: -1.0%. iOS: -1.7%. Web: (12,000 - 40,000) / 40,000 = -70%.
Step	Share of the loss: total loss 30,000; web lost 28,000, which is 28,000 / 30,000 = 93.3%.
Answer	The drop is a web problem: 93% of the lost users are on web, and web fell 70%. First check data and tracking on web (a broken tag or login bug), then recent web releases, then SEO or ad traffic changes. Recommendation: investigate web today, and confirm the tracking before any marketing spend.
Check	1,000 + 1,000 + 28,000 = 30,000. Matches the total loss.
Why it matters	In plain words: one platform explains almost everything, so you ignore the others for now.

System design rounds

For Data Engineers and ML/AI Engineers, system design is a heavy round. Chapters 8 and 9 give the framework and worked designs. For Data Scientists, an ML design round asks you to design a model system (metric, data, features, model, evaluation, monitoring). You will use the same frameworks.

Eight STAR story templates from your work experience

Prepare eight stories. Write each in about 260 words (2 minutes) in your own true words. Fill the brackets with your real events and numbers; do not invent.

#	Theme	Template prompt (fill with a true event)
1	Conflict	A colleague disagreed that [slow report turnaround] was a problem. Action: showed the 95th-percentile data, agreed on a threshold. Result: [fix, number]
2	Failure	A check missed [a defect or a wrong assumption]. Action: told the team early, fixed the plan, added a checklist. Result: [what changed]
3	Leadership	Led [a release cycle or a project] with [N people]. Action: split work, daily check-in. Result: [on time]

4	Ambiguity	Requirement was "make it quicker". Action: asked for a service target, defined a 95th-percentile goal. Result: [clear target met]
5	Deadline	Release moved up by [N days]. Action: prioritized critical flows. Result: [shipped, risk accepted]
6	Mistake	You ran a month-end report on the wrong data file. Action: owned it, fixed it, added a pre-run check. Result: [no repeat]
7	Influence	Convinced a manager to [invest in an index or capacity] using data. Result: [number]
8	Learning fast	Learned [a new tool or SQL] in [N days] to deliver [project]. Result: [outcome]

Problem 7.3.2 Draft story 6 (mistake) in STAR in about 6 sentences, with placeholders where facts must be yours.

Step 1	Situation: "I produced a month-end sales report last quarter, but the results looked too good."
Step 2	Task: "I needed to confirm the results before reporting."
Step 3	Action: "I found I had used a file from a smaller test branch of the shop, not the full company. I told my lead the same day, re-ran the report on the correct file, and added a file check to our reporting checklist."
Step 4	Result: "The correct report showed [your number] instead of [wrong number]. The checklist stopped this error from repeating over the next [N] reports."
Answer	A STAR story that owns the mistake, shows speed in reporting, and ends with prevention. Use only what truly happened.
Check	Count words aloud: aim for 200 to 260. The Action part should take about 60% of the time.
Why it matters	Interviewers use the mistake question to see honesty and learning. Owning it clearly is a strength.

The picture

```
Case:      Clarify -> Structure (tree) -> Prioritize -> Calculate ->
Recommend
Behavioral: 8 stories, each tagged with 2–3 themes (one story can answer
many questions)
```

Connection to the real job

Cases are real business conversations. Behavioral stories are how managers decide if you are safe to work with.

Applications

- Analyst loops: heavy on cases and SQL.
- Data Scientist loops: product sense, experiments.
- Engineer loops: design plus behavioral.

COMMON MISTAKE

Telling team stories only ("we"). Use "I" for your actions. Also avoid blaming people.

Check yourself

1. Steps of the case framework? Answer: clarify, structure, prioritize, calculate, recommend.
2. How many STAR stories to prepare? Answer: eight.
3. What do you say in a mistake story? Answer: own it, fix it, prevent it.

7.4 The 12-Week Plan, Mocks and Resume Alignment

INTUITION

A marathon runner trains with a weekly plan and a few practice races. Mock interviews are the practice races: they show what practice alone hides.

The 12-week plan (96 hours)

The 12-week interview plan

Each week: five 1-hour weekday sessions plus one 3-hour weekend block, 8 hours total. Twelve weeks give 96 hours.

Weeks	Focus	Output
1-2	SQL patterns, DSA basics (arrays, hashing)	10 SQL and 6 DSA problems logged
3-4	Python patterns, two pointers, sliding window	12 problems, 1 mini project cleanup
5-6	Statistics and ML Q&A (50 questions)	50 written and recorded answers
7-8	System design frameworks (Chapters 8, 9)	4 designs explained aloud
9-10	Case practice, 8 STAR stories	stories written and timed
11-12	Mock interviews (4 or more), apply in batches	mock log, 20 applications

Mock interviews

- Use peers, Pramp and interviewing.io (check what is free now), or a friend with a list of questions.
- Record yourself (with permission) and review: structure, filler words, pace, clarity.
- Start monthly from month 6 of your plan; weekly in the last 4 weeks.
- After each mock write 3 lines: what went well, one fix, one topic to study.

Problem 7.4.1 You do 4 mocks in weeks 11 to 12 and fail 3 of them for the same reason: you code before clarifying. What is the plan?

Step 1	Diagnose: one repeated error, so one deliberate practice target.
Step 2	Rule: before coding, speak 4 sentences (clarify, restate, example, brute force). Put them on a sticky note.
Step 3	Practise on 10 problems with a timer: 4 minutes of thinking aloud first.
Answer	After 10 practice problems, run 2 more mocks and count how many times you started coding before step 4. Target: zero.
Check	If you fixed it, the next mocks show your first 4 minutes are speaking. A measurable behaviour is better than "improve".
Why it matters	In plain words: fix the one thing that repeats; ignore the ten you imagine.

Problem 7.4.2 You send 60 applications and get 15 recruiter screens. What is your screen rate, and how many offers do you expect if 40% pass the technical round, 33% pass the final round and 50% of finalists receive an offer?

Formula	rate = next stage / previous stage
Compute	Screen rate = 15 / 60 = 25%. Technical: 15 x 0.40 = 6. Final: 6 x 0.33 = 2. Offers: 2 x 0.50 = 1.
Answer	About 1 offer from 60 applications, using these planning numbers. They are illustrative; track your own.
Check	0.25 x 0.40 x 0.33 x 0.50 = 0.0165; 60 x 0.0165 = 0.99, about 1. Matches.
Why it matters	In plain words: job search is a numbers game, but referrals and good targeting change the rates (Chapter 12).

Resume alignment

Chapter 11 covers the resume. For interviews, make sure each bullet is something you can talk about for 3 minutes. If you cannot explain a technology, remove it.

Connection to the real job

The plan builds habits that also help at work: regular practice, feedback, and review.

Applications

- Interview season: weekly plan.
- Between seasons: keep one problem a week to stay sharp.

COMMON MISTAKE

Starting mocks only after you "feel ready". You will never feel ready. Start mocks in week 6 even if you do badly.

Check yourself

1. How many hours is the 12-week plan? Answer: 96.
2. What do you write after each mock? Answer: what went well, one fix, one topic.
3. When should mocks start? Answer: early, not after feeling ready.

Chapter recap

- Prepare for your actual role: use the round-weight table.
- Check the loop by asking the recruiter; typical loops take 5 to 8 weeks.
- Practise SQL and Python by pattern, try 25 minutes before reading solutions.
- Watch for ties, NULLs and duplicates in SQL.
- Write answers to 100 stats and ML questions in your own words.
- Cases: clarify, structure, prioritize, calculate, recommend.
- Prepare 8 STAR stories from real work; use "I".
- Run mocks early and track one fix at a time.

Pattern cheat-sheet

Need	Pattern
Time split	`hours x weight / sum of weights`
Second highest per group	`dense_rank() over (partition by g order by v desc)` then `r = 2`
Precision / recall	`TP / flagged`, `TP / actual`

F1	`2PR / (P + R)`
Funnel rate	`next / previous`
Case steps	Clarify, structure, prioritize, calculate, recommend

Practice set

1. (Easy) Weights SQL 2, Python 1: 30 hours. SQL hours? **Answer:** 30 x 2/3 = 20.
2. (Easy) 15 screens out of 60 applications is what rate? **Answer:** 25%.
3. (Easy) What does `rank()` do with ties? **Answer:** gives ties the same rank and skips the next numbers.
4. (Medium) Precision 0.5 and recall 0.5: F1? **Answer:** 2 x 0.25 / 1.0 = 0.5.
5. (Medium) What is the output? Values 100, 100, 90, `dense_rank` of 90? **Answer:** 2.
6. (Medium) Find the bug: `select max(salary) from emp where salary < max(salary)`. **Answer:** An aggregate cannot be used in WHERE; use a subquery: `where salary < (select max(salary) from emp)`.
7. (Medium) DAU fell 200,000 to 150,000; web lost 40,000 of that 50,000. What share? **Answer:** 40,000 / 50,000 = 80%.
8. (Hard) Interview: "Tell me about a time you made a mistake." **Answer:** Use story 6: situation, own it, report early, fix, add a prevention step, give the result.
9. (Hard) Interview: "How would you decide between logistic regression and gradient boosting for churn?" **Answer:** Start simple as a baseline (interpretable, fast); try boosting; compare on the same cross-validation using the business metric; consider explainability, latency and maintenance; pick the simplest that meets the target.
10. (Hard) Scenario: you have 48 hours before a Data Engineer loop with SQL, Python, design, behavioral, stats weights 3, 2, 3, 2, 1. Allocate. **Answer:** Sum 11. SQL 13.1, Python 8.7, design 13.1, behavioral 8.7, stats 4.4 hours.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in
India moving to data roles; English is my second language. Topic:
```

technical interview preparation by role. I target [Data Engineer / Analyst / Data Scientist / ML Engineer]. Tell me what each round tests, build a weighted preparation plan for 96 hours, and ask me 5 questions one at a time to check I understand the weights and the typical loop timeline.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional in India moving to data. Topic: SQL interview practice (joins, GROUP BY, window functions, ties, NULLs). Give me 8 graded SQL exercises on a small schema you provide, without solutions. After I answer each one, test my query logic with sample data, point out tie and NULL traps, and show a cleaner version.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional in India moving to data. Topic: statistics and ML interview Q&A. Quiz me one question at a time from this list: p-value, confidence interval, bias-variance, regularization, precision/recall, A/B test design. After each answer, give feedback on correctness, clarity and a 60-second PREP version. Track my weak topics and end with a study list.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional in India moving to data. Topic: case interviews. Act as interviewer. Give me a case with numbers (a KPI drop or a conversion problem) and reveal data only when I ask. Judge my clarify, structure, prioritization, calculation and recommendation. After the case, give a score for each and one fix for next time.

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional moving into data. Topic: eight STAR stories (conflict, failure, leadership, ambiguity, deadline, mistake, influence, learning fast). I will paste my raw notes for one story. Ask me clarifying questions, shape it into a 2-minute STAR answer in simple English, remove blame, keep only true facts and check there is a result with a number. Notes: [PASTE HERE].

CHAPTER 8

SYSTEM DESIGN I: SOFTWARE SYSTEMS

How to design a service for millions of users, with real numbers, in the language of a working professional.

In a system design interview someone says "design a URL shortener" and watches how you think for 45 minutes. At work, the same skill decides whether your service survives launch day. You already know the hard part: load, bottlenecks, percentiles and capacity. What you may lack is a vocabulary (cache, shard, queue) and a framework that turns a vague request into a clear design. By the end of this chapter you will estimate load from a few assumptions, choose between a cache, a shard and a queue with a reason, and walk through three classic designs: a URL shortener, a rate limiter and a news feed. Chapter 6 taught estimation as a thinking tool; here we apply it to machines.

8.1 The Design Framework and the Words You Need

INTUITION

Designing a system is like planning a restaurant before it opens. You do not start by buying stoves. You ask how many guests come, what they order, how fast they expect food, and what happens if the chef is ill. Weak answers draw boxes in minute one. Strong answers spend the first five minutes on questions and numbers.

The eight steps

Use the same order every time.

1. **Requirements.** Functional (what the system does) and non-functional (how well: latency, availability, scale, cost, consistency).
2. **Estimates.** Requests per second, storage, bandwidth, memory. Say your assumptions aloud.
3. **API.** The few calls a client makes.
4. **Data model.** What is stored, the lookup key, the read and write pattern.
5. **High-level design.** Boxes and arrows for one request, end to end.
6. **Deep dives.** The two or three hardest parts.

7. **Bottlenecks.** What breaks first at 10 times the load? Capacity thinking lives here.
8. **Trade-offs.** What you gave up and why.

Words you need

Word	Plain meaning	Where it appears
REST / gRPC	Web-style named calls (`GET /urls/abc`) / faster binary calls between services	APIs
DNS	The phone book that turns `example.com` into an IP address	Before the first request
CDN	Content Delivery Network: copies of files near the user	Images, scripts
Load balancer	A traffic officer that spreads requests over servers	In front of app servers
Cache	A small fast copy of slow data	Redis, browser, CDN
Sharding	Splitting data over machines by key	Big tables
Replication	Copies of data on several machines	Reads and failover
Queue	A waiting line between producer and consumer	Kafka, SQS
Idempotent	Doing it twice has the same effect as once	Payments, retries
Stateless	A server keeps nothing between requests, so any server can serve any request	App servers

SQL or NoSQL? SQL gives rules, joins and transactions (all-or-nothing changes). NoSQL stores (key-value, document, wide-column) usually trade some of that for easier scaling. Choose by access pattern: "find one row by key, billions of rows" leans NoSQL; "many questions with joins" leans SQL.

CAP in one paragraph. When the network between machines breaks (a *partition*), a system

must choose: stay *consistent* (everyone sees the same data, some requests fail) or stay *available* (every request is answered, some answers are stale). PACELC adds that even without a partition you trade latency against consistency. Partitions do happen, so the real choice is C or A during a failure. Observability (logs, metrics, traces) is how you see all of this at run time; it is your home ground.

Problem 8.1.1 Interviewer: "Design profile-photo upload." Write the requirements you confirm in two minutes.

Step 1	Functional: upload one photo, replace it, view anyone's photo, delete.
Step 2	Non-functional: max size 5 MB, view latency (reads far exceed writes), durability (never lose a photo), availability.
Step 3	Out of scope: editing, albums, face detection.
Answer	You design upload, storage and delivery, and you state that editing is excluded. A CDN, a load balancer with several servers, and an idempotent upload (a client-made request ID so a retry cannot create two photos) follow from these lines.
Check	Each line says what or how well, never how it is built. Requirements must not contain the solution.

```
Client -> DNS -> CDN (static files)
              \-> Load balancer -> App servers (stateless)
                                     |-> Cache (Redis)
                                     |-> Database (primary + replicas)
                                     |-> Queue -> Workers (slow jobs)
```

Where it is used

Data engineers design ingestion services, ML engineers prediction APIs, AI engineers chat backends. All start with these eight steps.

COMMON MISTAKE

Saying "Kafka and microservices" in minute one. If you cannot justify a box with a number or a requirement, remove it.

INTERVIEW TIP

Say your method aloud. Interviewers score structure.

Check yourself

1. First two steps? Answer: requirements, then estimates.
2. Is "latency under 200 ms" functional? Answer: no, non-functional.
3. Idempotent means? Answer: doing it twice equals doing it once.

8.2 Estimation With Little's Law

INTUITION

A bank manager sets the number of counters by asking how many customers arrive per minute and how long each takes. If 60 arrive per minute and each needs 30 seconds, about 30 customers are being served at any moment, so you need about 30 counters. That is Little's law, which planners use to size counters, tills and servers.

Formal definition

Little's law: $L = \lambda \times W$. L is the average number of requests inside the system, λ (lambda) is the arrival rate per second, W is the average seconds each request spends inside. It holds for any stable system. For a system with think time (the pause between a user's actions): $\text{users} = \text{throughput} \times (\text{response time} + \text{think time})$.

Peak traffic is not average. A common assumption is peak = 2 to 3 times the average; state it as an assumption. A month has about 2.6 million seconds, a day 86,400.

Problem 8.2.1 100 requests per second, each in the system 0.2 s. How many in flight?

Formula	$L = \lambda \times W$
Compute	$100 \times 0.2 = 20$
Answer	About 20 requests in flight, so you need at least 20 worker slots.
Check	One slot serves 1 / 0.2 = 5 per second; 20 slots serve 100 per second.

Problem 8.2.2 A shortener team says: 100 million new short URLs per month, each read 100 times. Estimate writes and reads per second, peak, and 5-year storage at 500 bytes per record. Predict first: are writes nearer 4, 40 or 400 per second?

```
new_per_month = 100_000_000
read_write = 100
secs_month = 30 * 86400
writes = new_per_month / secs_month
reads = writes * read_write
print("writes/s      :", round(writes, 1))
print("reads/s avg   :", round(reads))
print("reads/s peak  :", round(reads * 3))
records = new_per_month * 12 * 5
print("records 5y    :", f"{records:,}")
print("storage TB    :", records * 500 / 1e12)
print("7-char base62 :", f"{62**7:,}", "ratio", round(62**7 / records))
print("bandwidth MB/s:", round(reads * 500 / 1e6, 2))
```

```
writes/s      : 38.6
reads/s avg   : 3858
reads/s peak  : 11574
records 5y    : 6,000,000,000
storage TB    : 3.0
7-char base62 : 3,521,614,606,208 ratio 587
bandwidth MB/s: 1.93
```

- **Step 1** Seconds per month: 30 × 86,400 = 2,592,000. Writes: 100,000,000 / 2,592,000 = 38.6 per second.
- **Step 2** Reads are 100 times writes: 3,858 per second; peak at 3 times: 11,574.
- **Step 3** Records: 100 M × 12 × 5 = 6 billion. Storage: 6×10^9 × 500 bytes = 3 TB. Seven base62 characters give 587 times more codes than we need.
- **Answer** About 39 writes and 3,900 reads per second (11,600 at peak), 3 TB in five years, 2 MB per second of bandwidth.
- **Check** 3,858 / 100 gives back 38.6. 3 TB fits on one modern disk, so size alone does not force sharding.
- **Why it matters** This is a read-heavy, small-data system, so caching matters more than sharding. The numbers shape the design.

Problem 8.2.3 If each read takes 50 ms end to end, how many requests are in flight at average and peak, and how many servers if one handles 100 concurrent requests?

Compute	average `3,858 × 0.05 = 192.9`; peak `11,574 × 0.05 = 578.7`
Answer	About 193 in flight on average and 579 at peak: at least 6 servers at peak, so plan 8 with spare for failures.
Check	578.7 / 100 = 5.8, rounded up to 6.

```
arrival rate λ ----> [ system: W seconds each ] ----> done
                     requests inside = L = λ × W
```

Where it is used

Pipelines: events per second × seconds per event = busy workers, which sizes a Kafka consumer group. ML serving: replicas = requests per second × latency per request. Prices change, so look up current cloud prices before costing.

COMMON MISTAKE

Sizing for the average, or mixing ms and seconds in `λ × W`. Write units beside every number.

Check yourself

1. 500 per second, 0.4 s each. In flight? Answer: 200.
2. 8 jobs run at once, 0.5 s each. Max throughput? Answer: λ = L / W = 16 per second.
3. Why design for peak? Answer: the system must work at its busiest moment.

8.3 Caching

INTUITION

You keep a water bottle on your desk instead of walking to the kitchen. The kitchen is the database: complete but slow. The bottle is the cache: small, near, quick, and sometimes out of date.

How it works

A **cache** keeps popular results in fast memory. A read asks the cache first: found is a **hit**; not found is a **miss**, so read the database, store the result, return it (the **cache-aside** pattern). The share of hits is the **hit ratio** `h`.

```
average latency = h × cache_time + (1 − h) × database_time
```

Levels: the browser, the **CDN**, and an application cache such as **Redis** (an in-memory key-value store). **Invalidation** (removing stale data) is the hard part: use a **TTL** (time to live, the entry expires), or delete the entry when the row changes. **LRU** (least recently used) evicts the item untouched longest when the cache is full.

Problem 8.3.1 Cache 1 ms, database 10 ms, hit ratio 80%. Average latency?

Compute	`0.8 × 1 + 0.2 × 10 = 0.8 + 2.0 = 2.8`
Answer	2.8 ms, against 10 ms without a cache.
Check	h = 0 gives 10 and h = 1 gives 1; 2.8 is between.

Problem 8.3.2 The shortener has 3,858 reads per second. What happens to the database at hit ratios 0, 50, 90, 99%, and how big is the cache for the hottest 20% of 100 million URLs?

```
reads = 3858
for h in (0.0, 0.5, 0.9, 0.99):
    avg_ms = h * 1 + (1 - h) * 10
    db = reads * (1 - h)
    print(f"hit {h:4.2f}: avg {avg_ms:5.2f} ms, DB reads/s {db:7.1f}")
hot = 100_000_000 * 0.2 * 500 / 1e9
print("cache GB for hottest 20% of 100M urls:", hot)
```

```
hit 0.00: avg 10.00 ms, DB reads/s  3858.0
hit 0.50: avg  5.50 ms, DB reads/s  1929.0
hit 0.90: avg  1.90 ms, DB reads/s   385.8
hit 0.99: avg  1.09 ms, DB reads/s    38.6
cache GB for hottest 20% of 100M urls: 10.0
```

- **Step 1** Database reads = reads × miss ratio = 3,858 × (1 − h).
- **Step 2** At h = 0.9 the database sees 386 per second (a 10 times cut); at 0.99, 38.6.
- **Step 3** Cache: 20,000,000 URLs × 500 bytes = 10 GB, one Redis machine.
- **Answer** A 90% hit ratio cuts database reads from 3,858 to 386 per second and latency from

10 to 1.9 ms, using 10 GB.

- **Check** "20% of URLs get 80% of reads" is an assumption; measure your real hit ratio.
- **Why it matters** The first 90% is cheap. Going to 99% needs a much bigger cache, so the return per GB falls.

A failure mode: stampede. If a viral key expires while 50,000 requests per second ask for it, all miss together and crush the database. Fixes: let one request refill while others wait, add random jitter to TTLs, refresh popular keys early. A traffic spike right after a cache flush (a test you can run with a tool such as Locust or k6) reproduces it.

Where it is used

Feature stores cache features for model serving. RAG systems cache embeddings and repeat answers (saves tokens). BI tools cache query results; pre-aggregated tables are a cache of an expensive query.

COMMON MISTAKE

Caching without an invalidation plan. For each cached item write down how stale is acceptable. A profile photo can be minutes old; an account balance cannot.

Check yourself

1. h = 95%, cache 2 ms, database 20 ms. Average? Answer: 1.9 + 1.0 = 2.9 ms.
2. TTL means? Answer: how long an entry lives before it expires.
3. Stampede means? Answer: many requests miss at once and all hit the database.

8.4 Sharding, Replication and Consistent Hashing

INTUITION

A library with one librarian has long queues. Copy popular books to more branches (replication: more readers served). Or split the collection, A to M in one building and N to Z in another (sharding: more data and writes fit). One helps reading, the other helps size.

Definitions

- **Replication:** the same data on several machines. A **primary** takes writes, **replicas** copy it and serve reads. If the primary dies, a replica takes over (**failover**). Replicas lag a little, so a replica read can be slightly old.
- **Sharding:** each machine owns a slice of the keys. The **shard key** picks the slice. A skewed key makes a **hot shard**.
- **Choosing the shard:** `hash(key) mod N` is easy, but when N changes nearly all keys move. **Consistent hashing** puts machines (many virtual points each) on a circle; a key belongs to the next machine clockwise, so adding a machine takes keys only from its neighbours.

Problem 8.4.1 Shard = `id mod 4`. Ids 10 to 14. Where do they go, and what if we move to 5 shards?

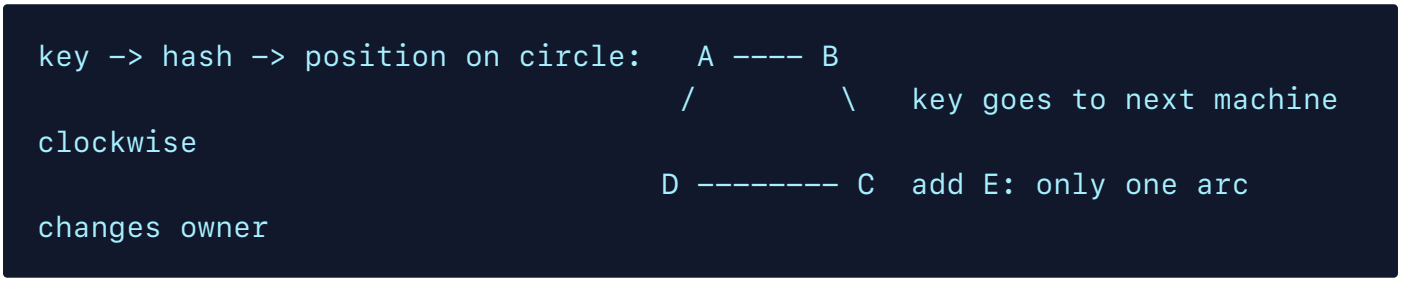
Step 1	Mod 4: 10→2, 11→3, 12→0, 13→1, 14→2.
Step 2	Mod 5: 10→0, 11→1, 12→2, 13→3, 14→4.
Answer	All five ids change shard when N goes from 4 to 5.
Check	A key stays only if `k mod 4 = k mod 5`; in general about 80% of keys move (next problem).

Problem 8.4.2 Compare mod and consistent hashing on 100,000 keys, 4 to 5 machines.

```
import hashlib, bisect
def h(s): return int(hashlib.md5(s.encode()).hexdigest(), 16)
keys = [f"key{i}" for i in range(100000)]
moved = sum(1 for k in keys if h(k) % 4 != h(k) % 5)
print("mod 4 -> mod 5 moved:", moved / len(keys))
def ring(nodes, vn=100):
    pts = sorted((h(f"{n}#{v}"), n) for n in nodes for v in range(vn))
    return [p for p, _ in pts], [n for _, n in pts]
def owner(r, k):
    ps, ns = r
    i = bisect.bisect(ps, h(k)) % len(ps)
    return ns[i]
r4 = ring(["A","B","C","D"]); r5 = ring(["A","B","C","D","E"])
moved = sum(1 for k in keys if owner(r4, k) != owner(r5, k))
print("ring 4 -> 5 moved   :", moved / len(keys))
from collections import Counter
print(sorted(Counter(owner(r5, k) for k in keys).items()))
```

```
mod 4 -> mod 5 moved: 0.80045
ring 4 -> 5 moved   : 0.19343
[('A', 20388), ('B', 22592), ('C', 19712), ('D', 17965), ('E', 19343)]
```

- **Step 1** `h` turns text into a big number. `ring` puts 100 points per machine on a sorted list (the circle). `owner` finds the first point after the key's hash, wrapping with `% len(ps)`.
- **Step 2** Mod moved 80.0% of keys; the ring moved 19.3%. The ideal is 1/5 = 20%.
- **Step 3** The last line shows each machine holds about 20,000 keys: balanced.
- **Answer** Consistent hashing moves about 19% of keys instead of 80% when adding a fifth machine.
- **Check** The moved keys all go to the new machine E, so only neighbours gave up keys.
- **Why it matters** Moving 80% of cached data means an empty cache and a crushed database; 19% is survivable. For the shortener (3 TB, about 1 TB per machine, writes only 39 per second) we shard because of size, not write speed: 6 shards with replicas leave room to grow.



Where it is used

Kafka topics split into partitions by key. Spark and warehouses shuffle by key; one customer producing 40% of events gives the "slow last task" you know from performance work. Vector indexes and embedding tables are sharded too.

COMMON MISTAKE

Sharding by a skewed key (country, or date, where today's shard takes all writes). Check the key's distribution first.

Check yourself

1. Which helps write scale, replicas or shards? Answer: shards.
2. Why is mod N bad when N changes? Answer: almost all keys move.
3. What is a hot shard? Answer: one shard getting far more load than the rest.

8.5 Queues, Retries and Reliability

INTUITION

A restaurant counter takes orders and sends tickets to the kitchen on a rail. In the rush the rail fills, but the counter does not collapse. The kitchen works at its own speed. A queue separates how fast work arrives from how fast it is done.

How it works

A **queue** sits between a **producer** and **consumers**. It smooths bursts, lets slow work happen later, and lets you add consumers. A **stream** (Kafka) is a queue that keeps a log so many readers can replay it. *At-least-once* delivery never loses a message but may repeat it, so consumers must be idempotent (store a message ID, skip repeats).

The utilization curve. In a simple random queue (the M/M/1 model), time in the system is `service_time / (1 − utilization)`. Utilization is the share of time a worker is busy. Near 100% small increases cause huge waits: the hockey-stick you see in any latency-versus-load graph. It is a simplified model; the shape is right, real numbers differ.

Retries help brief faults but can crush a struggling service. Use **exponential backoff** (0.1 s, 0.2, 0.4, ...), a cap, and **jitter** (randomness, so clients do not retry together). A **circuit breaker** stops calling a failing service and fails fast; **graceful degradation** shows a simpler result instead of an error.

Problem 8.5.1 1,000 jobs per second, 20 ms of one worker each. Workers for 60% utilization? Time in system at 50, 80, 95%?

```
jobs = 1000; svc = 0.020
busy = jobs * svc
print("busy workers (Little):", busy)
for u in (0.5, 0.6, 0.8, 0.9, 0.95):
    print(f"util {u:.2f} -> workers {busy/u:5.1f}, time in system =
service x {1/(1-u):4.1f} = {svc/(1-u)*1000:6.1f} ms")
```

```
busy workers (Little): 20.0
util 0.50 -> workers  40.0, time in system = service x  2.0 =   40.0 ms
util 0.60 -> workers  33.3, time in system = service x  2.5 =   50.0 ms
util 0.80 -> workers  25.0, time in system = service x  5.0 =  100.0 ms
util 0.90 -> workers  22.2, time in system = service x 10.0 =  200.0 ms
util 0.95 -> workers  21.1, time in system = service x 20.0 =  400.0 ms
```

- **Step 1** Little's law: 1,000 × 0.020 = 20 busy workers. That would be 100% utilization, which is unstable.
- **Step 2** For utilization u, workers = 20 / u. At 60%: 33.3, so 34.
- **Step 3** Time = 20 ms / (1 − u). At 95%: 20 / 0.05 = 400 ms.
- **Answer** Use about 34 workers; time in system is about 50 ms. Running at 95% saves 12 workers (34 versus 22) but multiplies latency by 8 (400 versus 50 ms).
- **Check** At u = 0 the formula gives 20 ms, the bare service time.
- **Why it matters** This is the capacity argument for keeping CPU near 60 to 70%: headroom buys latency.

Problem 8.5.2 Three services in a chain, each 99.9% available. Chain availability? Two 99% replicas in parallel? Downtime of 99.9% per month?

```
for a in (0.99, 0.999, 0.9999):
    print(a, "downtime min/month:", round((1-a) * 30 * 1440, 1))
print("3 services in series at 99.9% each:", round(0.999**3, 6))
print("2 replicas each 99%, parallel:", 1 - 0.01**2)
base, cap = 0.1, 5
print([min(cap, base * 2**i) for i in range(8)])
```

```
0.99 downtime min/month: 432.0
0.999 downtime min/month: 43.2
0.9999 downtime min/month: 4.3
3 services in series at 99.9% each: 0.997003
2 replicas each 99%, parallel: 0.9999
[0.1, 0.2, 0.4, 0.8, 1.6, 3.2, 5, 5]
```

- **Step 1** Series (all must work): $0.999^3 = 0.997003$.
- **Step 2** Parallel (fails only if both fail): 1 − 0.01 × 0.01 = 0.9999, assuming independent failures.
- **Step 3** 0.001 × 30 × 1,440 = 43.2 minutes per month. The last line is backoff: 0.1 s doubling

to the 5 s cap.

- **Answer** The chain gives 99.70% (about 130 minutes of downtime per month), the replicas 99.99%, and 99.9% allows 43 minutes.
- **Check** 0.003 × 43,200 = 129.6 minutes. Series lowers availability; parallel raises it.

```
Producer --> [ queue: waiting jobs ] --> Consumer 1
 (bursty)                              --> Consumer 2   (add consumers to
drain faster)
```

Where it is used

Consumer lag (messages waiting) is the data engineer's key health metric: it is `L` of Little's law. ML batch scoring and GenAI request queues in front of rate-limited models use the same pattern.

COMMON MISTAKE

Retrying instantly in a loop. Add backoff, a cap and jitter, and make the operation idempotent first.

Check yourself

1. 200 jobs per second, 50 ms each. Busy workers? Answer: 10.
2. Why can at-least-once duplicate? Answer: a lost acknowledgement makes the queue resend.
3. Two services in series, each 99%? Answer: 98.01%.

8.6 Worked Design 1: URL Shortener

INTUITION

A shortener is a giant phone book: short code in, long address out. Few writes, many reads. The clever parts are making unique short codes and answering reads very fast.

URL shortener architecture

Read it left to right. A request passes a load balancer to stateless app servers. A read asks Redis first and falls back to the database on a miss. A write asks the ID generator (counter ranges) for a number, turns it into a code and stores it.

The eight steps

1. **Requirements.** Create a short URL; open it and be redirected; optional alias and expiry. Redirect p99 under about 100 ms (assumed), high availability, links never change.
2. **Estimates.** From 8.2: 39 writes and 3,858 reads per second (11,574 peak), 3 TB, 10 GB hot cache.
3. **API.** `POST /urls` with `{long_url}` returns `{short_code}`. `GET /{short_code}` redirects: 301 (permanent, browsers may cache) or 302 (temporary, you count every click). Choose by whether analytics matter.
4. **Data model.** One table: `short_code` (key), `long_url`, `created_at`, `expires_at`, `user_id`. Lookup by key only.
5. **High-level design.** The picture above.
6. **Deep dive: codes.** (a) Hash the long URL and take 7 characters: needs collision checks. (b) A counter converted to base62 (digits `0–9a–zA–Z`): no collisions, but guessable. (c) Random 7 characters with a retry on collision.
7. **Bottlenecks.** At 10 times load the cache still absorbs 90%; the database sees about 11,500 reads per second, so shard and add replicas. The counter is a single point of failure: give each app server a **range** of numbers (say 1,000,000 at a time).
8. **Trade-offs.** Counter codes are short and collision-free but predictable; random codes hide order but need a collision check. 302 gives analytics but more traffic.

Problem 8.6.1 Write counter 125 in base62 (`0–9`, then `a–z` for 10 to 35, then `A–Z` for 36 to 61). At 39 writes per second, when do we need more than 6 characters?

Step 1	125 / 62 = 2 remainder 1, so the digits are 2 and 1: `21`.
Step 2	Six characters hold 62^6 = 56,800,235,584 codes.
Step 3	Time to use them: 56.8×10^9 / 38.6 per second / 86,400 / 365 = about 47 years.
Answer	125 becomes `21`, and 6 characters would already last decades. Seven characters are generous.
Check	2 × 62 + 1 = 125.

Problem 8.6.2 Chance a random 7-character code collides when 6 billion of 3.52 trillion are used?

Compute	$6 \times 10^9 / 3.52 \times 10^{12} = 0.0017$
Answer	About 0.17% at the end of 5 years, so one retry almost always fixes it.
Check	1 / 587 = 0.0017, the ratio from 8.2.

Where it is used

Click analytics is a data pipeline: redirect logs flow through a queue into a warehouse. It makes a good portfolio project (Chapter 11).

COMMON MISTAKE

Designing for write scale. The numbers say writes are only 39 per second. Read the estimate before choosing the architecture.

Check yourself

1. Reads or writes bigger here? Answer: reads, by 100 times.
2. Why counter ranges? Answer: no call to the counter on every write, and no single point of failure.
3. 301 or 302 for click counting? Answer: 302.

8.7 Worked Design 2: Rate Limiter

INTUITION

A nightclub lets in 5 people a minute, and a group of 10 at once if it has been quiet. The bouncer holds a bucket of tokens; each guest takes one; the bucket refills slowly; when empty, guests wait.

The problem

A **rate limiter** rejects requests from a client above an allowed speed. It protects servers, controls cost and stops abuse. The reply is HTTP status 429 (too many requests), often with a

`Retry-After` header.

Algorithm	Idea	Good	Weak
Fixed window	Count per minute, reset at the minute	Simple	Edge burst up to 2x
Sliding window log	Store timestamps	Exact	Memory per request
Sliding window counter	Blend of two windows	Cheap, close	Approximate
Token bucket	Tokens refill at a rate; capacity = burst	Allows bursts, tiny memory	Two parameters
Leaky bucket	Requests drain at a fixed speed	Smooth output	No burst

A **token bucket** keeps two numbers per client: `tokens` and `last_time`. On each request, add `(now − last) × rate` tokens (up to `capacity`), then spend one if at least one is there.

Problem 8.7.1 Rate 5 per second, capacity 10, bucket starts full. 15 requests arrive at t = 0. Then 16 requests arrive 0.125 s apart from t = 1. Predict which pass, then run.

```
class TokenBucket:
    def __init__(self, rate, capacity):
        self.rate, self.capacity = rate, capacity
        self.tokens, self.last = capacity, 0.0
    def allow(self, now):
        self.tokens = min(self.capacity, self.tokens + (now - self.last) *
self.rate)
        self.last = now
        if self.tokens >= 1:
            self.tokens -= 1
            return True
        return False

b = TokenBucket(rate=5, capacity=10)
out = [b.allow(0.0) for _ in range(15)]
print("burst of 15 at t=0:", "".join("Y" if x else "N" for x in out))
out = [b.allow(1.0 + i * 0.125) for i in range(16)]
print("8 req/s for 2 s   :", "".join("Y" if x else "N" for x in out))
```

```
burst of 15 at t=0: YYYYYYYYYYNNNNN
8 req/s for 2 s   : YYYYYYYYYYNYYNY
```

- **Step 1** At t = 0 there are 10 tokens and no time passes: 10 pass, 5 are refused.
- **Step 2** Tokens are 0. At t = 1 the refill is 5. Each next request is 0.125 s later: +0.625 tokens, then −1.
- **Step 3** Levels after each allowed request: 4, 3.625, 3.25, 2.875, 2.5, 2.125, 1.75, 1.375, 1.0, 0.625, 0.25: eleven in a row.
- **Step 4** Request 12 sees 0.875: refused. Request 13 sees 1.5: allowed (0.5 left). Request 14 sees 1.125: allowed (0.125). Request 15 sees 0.75: refused. Request 16 sees 1.375: allowed.
- **Answer** 14 of 16 pass: the saved burst credit drains, then traffic settles toward 5 per second.
- **Check** Tokens supplied after t = 1: 5 + 1.875 × 5 = 14.375, so at most 14 can pass. We got exactly 14.
- **Why it matters** A limiter must give a predictable answer you can verify by hand.

Problem 8.7.2 Memory for 10 million users at 64 bytes each (tokens 8, time 8, key and overhead about 48, an assumption)?

Compute	`10,000,000 × 64 = 640,000,000 bytes`
Answer	About 640 MB: one Redis instance. At 50,000 API requests per second with two Redis operations each, Redis sees 100,000 operations per second, which you must confirm with a traffic test.
Check	6.4×10^8 bytes = 640 MB.

Deep dive: many servers

The counters must be shared, so use Redis. "Read then write" is a **race** (two servers read the same value), so use an atomic operation (`INCR` or a script). If Redis is down, **fail open** (allow all, risk overload) or **fail closed** (block all, risk an outage). A payment API leans closed; a search page leans open.

```
Request -> App server -> Redis: [user42: tokens=3.2, last=...]
              | allowed? yes -> backend      no -> 429
```

Where it is used

LLM providers limit tokens per minute, so a RAG client needs its own limiter plus backoff (8.5). Data extraction and scrapers must respect limits and site terms.

COMMON MISTAKE

A fixed window: a client sends the full limit at 11:59:59 and again at 12:00:00, doubling the rate. Use a bucket or sliding window when it matters.

Check yourself

1. Which status code? Answer: 429.
2. Rate 2 per second, capacity 4, full: instant requests passed? Answer: 4.
3. Fail open means? Answer: allow requests when the limiter store is down.

8.8 Worked Design 3: News Feed

INTUITION

Each person has a notice board. When a friend posts, a helper copies the post onto each follower's board (push): reading is instant, posting costs more. Or you walk to each friend's board when you open the app (pull): posting is cheap, reading is slow.

News feed fan-out

A post goes to a post service and a queue. Workers fan it out into each follower's feed cache (a list of post IDs) for normal users. Accounts with millions of followers skip fan-out; their posts are merged in when a follower opens the feed.

Assume 50 million daily active users (DAU), 10 feed opens per user per day, 20% of users post once a day, 200 followers on average, 16 bytes per feed entry.

```
dau = 50_000_000
opens = 10
print("feed reads/s avg :", round(dau * opens / 86400))
print("feed reads/s peak:", round(dau * opens / 86400 * 3))
posts = dau * 0.2 / 86400
print("posts/s          :", round(posts, 1))
fan = 200
print("fanout writes/s  :", round(posts * fan))
print("entries/day      :", dau * 0.2 * fan, "-> GB at 16 B:", dau * 0.2 *
fan * 16 / 1e9)
print("one celebrity post to 10M followers:", 10_000_000, "writes")
print("at 20k writes/s that takes", 10_000_000 / 20000, "s")
```

```
feed reads/s avg : 5787
feed reads/s peak: 17361
posts/s          : 115.7
fanout writes/s  : 23148
entries/day      : 2000000000.0 -> GB at 16 B: 32.0
one celebrity post to 10M followers: 10000000 writes
at 20k writes/s that takes 500.0 s
```

Problem 8.8.1 What is the read-to-post ratio, and does it favour push or pull?

Step 1	5,787 / 115.7 = 50 reads per post.
Step 2	Push: each post becomes 200 writes (23,148 per second); each read is one cache lookup.
Step 3	Pull: each read gathers from about 200 sources: 5,787 × 200 = 1,157,400 lookups per second.
Answer	Reads dominate 50 to 1, so push is cheaper: 23 thousand writes per second versus 1.16 million lookups per second for pull.
Check	1,157,400 / 23,148 = 50, the read-to-post ratio.

Problem 8.8.2 A celebrity with 10 million followers posts. What goes wrong with push, and what is the fix?

Step 1	Push means 10,000,000 feed writes: 500 seconds of the whole fan-out capacity at 20,000 per second, so followers wait.
Step 2	**Hybrid:** push for accounts under a threshold (say 100,000 followers; set it by measurement). Larger accounts store the post once and are merged in at read time.
Step 3	Read path: take the precomputed list, add recent posts of the few celebrities this user follows, sort, return the top 20.
Answer	Push for normal users, pull for celebrities. It caps write cost and adds only a few lookups per read.
Check	Storage: 2 billion entries per day at 16 bytes is 32 GB per day, so keep only the latest 500 entries per user.

Other deep dives: ranking by a model (Chapter 9), cursor pagination ("posts older than ID X", since new posts shift page numbers), and deletes (mark deleted and filter at read time).

Where it is used

Fan-out is the pattern behind notification services and event distribution. E-commerce uses it for personalised home pages; ML systems precompute recommendations per user.

COMMON MISTAKE

Presenting push or pull as the answer. Show the numbers and the hybrid, with a threshold and a reason.

Check yourself

1. Which is cheaper for readers? Answer: push.
2. Why skip fan-out for celebrities? Answer: one post would cause millions of writes.
3. What is cursor pagination? Answer: asking for posts after a given ID instead of a page number.

Chapter recap

- Use the eight steps in order; tools come last.
- Little's law `L = λ × W` turns rates and latency into concurrency and servers.
- Estimate first: numbers tell you read-heavy or write-heavy, large or small.
- A cache cuts database load by the miss ratio; plan invalidation; beware stampedes.
- Replicas help reads and failover; shards help size and writes; consistent hashing moves about 1/N of keys.
- Queues smooth bursts; keep utilization near 60 to 70%, since waits grow like `1 / (1 − u)`.
- Retries need backoff, a cap, jitter and idempotency.
- Shortener: read-heavy, cache plus ID ranges. Limiter: token bucket in Redis, atomic. Feed: hybrid push and pull.

Pattern cheat-sheet

Need	Pattern
Requests in flight	`L = λ × W`
Servers needed	`L / concurrency per server`, plus spare
Latency with cache	`h × t_cache + (1 − h) × t_db`
DB load with cache	`reads × (1 − h)`

Queue wait	`service / (1 − utilization)`
Availability	series: multiply; parallel: $1 - (1 - a)^n$
Backoff	$\min(cap, base \times 2^n)$ plus jitter

Practice set

1. (Easy) 300 requests per second, 0.1 s each. In flight? **Answer:** 30.
2. (Easy) Hit ratio 80%, cache 2 ms, database 20 ms. Average? **Answer:** 1.6 + 4 = 5.6 ms.
3. (Easy) Downtime of 99.9% in a 30-day month? **Answer:** 43.2 minutes.
4. (Medium) 20 million new records per month, read:write 50:1. Reads per second? **Answer:** writes 20,000,000 / 2,592,000 = 7.72 per second; reads about 386.
5. (Medium) What is `min(5, 0.1 * 2**6)` ? **Answer:** 6.4 is above the cap, so 5.
6. (Medium) Find the bug: `count = redis.get(k); redis.set(k, count + 1)` on 20 servers. **Answer:** a race; two servers read the same count and one increment is lost. Use atomic `INCR` .
7. (Medium) 95% hits; the database handles 500 reads per second. Max total reads? **Answer:** 500 / 0.05 = 10,000 per second.
8. (Hard) 20 workers, 20 ms jobs, 900 arrivals per second. Utilization, time, workers for 60%? **Answer:** busy 18, utilization 0.9, time 20 ms / 0.1 = 200 ms; 18 / 0.6 = 30 workers.
9. (Hard, interview) "Design a notification service." **Answer:** Requirements: push, email, SMS, priorities, preferences, deduplication. Estimate: 100 million per day is 1,157 per second, with spikes. A queue per channel, workers with provider rate limits and backoff, an idempotency key against duplicates, cached preferences. Trade-off: guaranteed delivery versus duplicates.
10. (Hard, interview) "p99 doubled after a launch; the average did not." **Answer:** Look for a slow subset: a hot key or shard, cache misses for a new data type, GC pauses, a slow dependency, retries. Compare percentiles per instance and endpoint, check queue depth and utilization, reproduce with a traffic test.

Prompt pack

✨ AI prompt: copy and paste into any AI

You are my patient, practical professor. I am a working professional in India moving into data and AI roles; English is my second language. Topic: the system design framework (requirements, estimates, API, data model, high-level design, deep dives, bottlenecks, trade-offs). Act as interviewer for "design a pastebin service". Ask one step at a time, wait for my answer, score each step, and correct my numbers.

✨ **AI prompt: copy and paste into any AI**

You are my patient, practical professor. I am a working professional in India moving into data and AI roles. Topic: back-of-envelope estimation for system design (requests per second, storage, bandwidth, Little's law, peak versus average). Give me 8 graded exercises with real numbers, without solutions. After each answer, check my arithmetic, units and assumptions, and give a Python snippet to verify.

✨ **AI prompt: copy and paste into any AI**

You are my patient, practical professor. I am a working professional moving into data. Topic: caching, sharding, replication and consistent hashing. Explain each with an everyday analogy, then give a product scenario with numbers and ask me to choose between cache, replica and shard with reasons. Quiz me one question at a time on hit ratio, stampede, hot shard, and keys moved when a machine is added.

✨ **AI prompt: copy and paste into any AI**

You are my patient, practical professor. I am a working professional moving into data. Topic: queues, retries, idempotency and availability maths, including the utilization curve service / (1 − utilization). Teach step by step, then give me 6 exercises one at a time, including a "find the bug" about retries without backoff. Review each answer and ask me to explain it back in two sentences.

✨ **AI prompt: copy and paste into any AI**

You are my patient, practical professor. I am a working professional in India moving to data roles. Topic: worked designs (URL shortener, rate limiter, news feed). Review my own design for [CHOOSE ONE] with the eight-step framework: check my estimates, find the first bottleneck at 10x load, challenge one trade-off, and tell me what a strong answer adds. My design: [PASTE HERE].

CHAPTER 9

SYSTEM DESIGN II: DATA AND ML SYSTEMS

From "build me a recommender" to a diagram, a number for every box, and a list of what can go wrong.

Chapter 8 designed software services. Data and ML systems add three new difficulties: the data keeps changing, the model can be silently wrong while every server is healthy, and the answer is a probability, not a fact. In interviews for data engineer, ML engineer and AI engineer roles, this is the round that decides the offer. At work it is the difference between a model that lives in a notebook and one that earns money. By the end of this chapter you will have one framework for ML design, and five worked designs: a data platform, a feature pipeline, a recommender, a fraud detector and a RAG chatbot. Every one is sized with real numbers, as a restaurant owner sizes a kitchen.

9.1 The ML Design Framework and the Data Platform

INTUITION

A model is the chef; the data platform is the whole supply chain. A brilliant chef with late, dirty or missing ingredients cooks bad meals. Most ML failures in companies are supply-chain failures, not model failures.

The framework (eleven steps)

Use it for any ML or AI design question.

1. **Business objective.** What outcome do we want (fewer chargebacks, more orders)?
2. **Metrics.** *Offline* (measured on past data: precision, recall, NDCG) and *online* (measured live: conversion, revenue, complaints). Add **guardrail metrics** that must not get worse (latency, complaints).
3. **Data.** Sources, volume, labels, freshness, privacy.
4. **Features.** Inputs the model uses, and how they are computed.
5. **Model options.** Start with a simple baseline, then something stronger.
6. **Training.** How often, on what, with what split.

7. **Evaluation.** Offline tests, then an A/B test (Chapter 7 and the learner's statistics document).
8. **Serving.** Batch (scores computed ahead) or online (scores computed per request); latency and cost.
9. **Monitoring.** Data drift, model quality, system health.
10. **Iteration.** How you ship version 2 safely.
11. **Risks.** Bias, privacy, abuse, failure modes.

The data platform

A **data platform** moves data from where it is produced to where it is used, reliably. Words you need:

Word	Plain meaning
Ingestion	Bringing data in: batch (files on a schedule) or streaming (events as they happen)
CDC	Change Data Capture: copying each database change as an event
Raw zone / data lake	Cheap storage (object storage) keeping data exactly as received
Transform	Cleaning and joining (Spark, dbt)
Warehouse	A database built for analysis; clean tables
Orchestration	A scheduler that runs steps in order and retries (Airflow)
Lineage	A record of where each table came from
Freshness	How old the newest data is

Data platform

Left to right: sources (apps, databases, logs) enter through ingestion (CDC, Kafka or batch), land untouched in the raw zone, are cleaned by transforms into the warehouse, then feed dashboards, APIs, the feature store and model training. The bottom band says orchestration, quality tests, lineage and access control touch every box.

Problem 9.1.1 An app produces 1 billion events per day, 1 KB each (raw JSON). Size ingestion and storage. Assume peak is 3 times average, columnar files compress 5 times (an assumption: measure on your own data), and one Kafka partition handles about 10 MB per second (also an assumption).

```
events = 1_000_000_000      # per day
size = 1000                 # bytes per event, raw JSON
print("events/s avg :", round(events / 86400))
print("events/s peak:", round(events / 86400 * 3))
raw_tb = events * size / 1e12
print("raw TB/day   :", raw_tb, " per year:", raw_tb * 365)
comp = 5
print("columnar TB/year at 5x:", raw_tb * 365 / comp)
mbps_peak = events / 86400 * 3 * size / 1e6
print("peak MB/s:", round(mbps_peak, 1), "-> partitions at 10 MB/s each:",
-(-mbps_peak // 10))
batch_hour = 3600
print("worst-case staleness hourly batch + 10 min job:", (batch_hour +
600) / 60, "min")
```

```
events/s avg : 11574
events/s peak: 34722
raw TB/day   : 1.0  per year: 365.0
columnar TB/year at 5x: 73.0
peak MB/s: 34.7 -> partitions at 10 MB/s each: 4.0
worst-case staleness hourly batch + 10 min job: 70.0 min
```

- **Step 1** Average: 1,000,000,000 / 86,400 = 11,574 events per second. Peak: 3 × 11,574 = 34,722.
- **Step 2** Raw volume: 10^9 × 1,000 bytes = 10^{12} bytes = 1 TB per day, 365 TB per year.
- **Step 3** Columnar compression at 5 times: 73 TB per year.
- **Step 4** Peak bandwidth: 34,722 × 1 KB = 34.7 MB per second. At 10 MB per second per partition that is 3.5, so at least 4 partitions; real designs add headroom and parallelism for consumers, so plan around 12.
- **Answer** About 11,600 events per second (35,000 at peak), 1 TB per day raw, about 73 TB per year compressed, at least 4 and about 12 partitions.
- **Check** The average rate is 11,574 events × 1 KB = 11.6 MB per second, and 11.6 MB × 86,400 s = 1.0 TB per day. This agrees with Step 2.
- **Why it matters** Storage is cheap and ingestion is moderate; the hard parts are freshness,

quality and cost of compute.

Problem 9.1.2 The business wants a dashboard "fresh within 15 minutes". Your design is an hourly batch plus a 10-minute job. Does it meet the need?

Step 1	Worst case: an event arrives just after a batch starts, waits 60 minutes for the next batch, then 10 minutes to process: 70 minutes.
Step 2	70 is greater than 15, so no.
Step 3	Options: micro-batch every 5 minutes (about 5 + 3 = 8 minutes worst case) or streaming.
Answer	The hourly design fails; a 5-minute micro-batch meets the target.
Check	5-minute interval + 3-minute job = 8 minutes, under 15.

Where it is used

Data engineers build this picture daily. Analysts consume the warehouse. ML engineers add the feature store. Quality tests (nulls, duplicates, row counts) and freshness alerts are the platform's equivalent of a shop's daily till check.

COMMON MISTAKE

Drawing the model box first. Ask for the metric and the data freshness before the model.

Check yourself

1. Offline versus online metric? Answer: offline is measured on past data, online on live traffic.
2. What does CDC copy? Answer: each database change, as an event.
3. What is a guardrail metric? Answer: one that must not get worse, such as latency.

9.2 Feature Pipelines and Training-Serving Skew

INTUITION

A student practises with a past paper and the answers revealed would score 100% in practice and fail the real exam. Leaking the future into training does the same to a model. A **feature** is an input column the model reads, such as "average spend in the last 30 days".

How it works

A **feature store** keeps feature values so that training and serving use the same definitions. It has an **offline store** (history, for training) and an **online store** (latest values, for fast lookup). Two dangers:

- **Training-serving skew:** the feature is computed one way in training (SQL) and another in serving (Python), so the model sees different data live than it learned on.
- **Leakage:** training rows use information that did not exist at prediction time.

The cure for leakage is a **point-in-time join**: for each label at time `t`, use the latest feature value at or before `t`.

Problem 9.2.1 Table `feat` holds `avg_spend` per user over time; table `label` says whether the user churned (left) at a time. Compare the wrong and the right join.

```
import sqlite3
c = sqlite3.connect(":memory:")
c.executescript("""
create table feat(user_id int, ts text, avg_spend real);
insert into feat values (1,'2026-01-01',100),(1,'2026-01-10',500),
(2,'2026-01-05',80);
create table label(user_id int, ts text, churned int);
insert into label values (1,'2026-01-08',0),(1,'2026-01-12',1),(2,'2026-
01-04',0);
""")
print("WRONG (latest feature, leaks future):")
for r in c.execute("""select l.user_id,l.ts,l.churned,
  (select avg_spend from feat f where f.user_id=l.user_id order by ts desc
limit 1) from label l order by 1,2"""): print(r)
print("RIGHT (latest feature at or before label time):")
for r in c.execute("""select l.user_id,l.ts,l.churned,
  (select avg_spend from feat f where f.user_id=l.user_id and f.ts<=l.ts
order by ts desc limit 1) from label l order by 1,2"""): print(r)
```

```
WRONG (latest feature, leaks future):
(1, '2026-01-08', 0, 500.0)
(1, '2026-01-12', 1, 500.0)
(2, '2026-01-04', 0, 80.0)
RIGHT (latest feature at or before label time):
(1, '2026-01-08', 0, 100.0)
(1, '2026-01-12', 1, 500.0)
(2, '2026-01-04', 0, None)
```

- **Step 1** Wrong query: the sub-select takes the newest feature for the user, whatever its date. User 1 at 2026-01-08 gets 500, which was only computed on 2026-01-10, two days in the future. User 2 at 2026-01-04 gets 80, from 2026-01-05, one day in the future.
- **Step 2** Right query adds `f.ts <= l.ts`. User 1 at 01-08 gets 100 (from 01-01). User 1 at 01-12 gets 500 (from 01-10). User 2 at 01-04 gets `None`: no feature existed yet.
- **Answer** The right join shows what the model could really have known; the wrong join gave 2 of 3 rows information from the future.
- **Check** For every row in the right output, the feature date is at or before the label date. For user 2 the honest answer is "missing", not 80.
- **Why it matters** Leakage makes offline scores look great and online results disappointing. It is the commonest cause of "the model worked in the notebook".

Problem 9.2.2 Your fraud model uses `avg_txn_30d`. Training computes it in SQL with calendar days; serving computes it in code with the last 720 hours. What is the risk and the fix?

Step 1	The two definitions differ for a user active near a day boundary, so training and serving feature values disagree.
Step 2	The model sees a distribution at serving time it never trained on: skew.
Step 3	Fix: define the feature once (one code path or one feature-store definition) and log the served feature values, then compare them to training values regularly.
Answer	One definition, shared by training and serving, plus a logged comparison.
Check	If the logged and recomputed values differ for the same user and time, the skew exists.

Where it is used

ML engineers own this daily. Data engineers build the pipelines that fill the stores. AI engineers

have the same problem for retrieval metadata.

COMMON MISTAKE

Random train-test split on time data. Use a time split (train on the past, test on later weeks), or you leak the future.

Check yourself

1. What does a point-in-time join do? Answer: uses the latest feature at or before the label time.
2. What is skew? Answer: features computed differently in training and serving.
3. Offline versus online store? Answer: history for training versus latest values for fast serving.

9.3 Worked Design: Recommender System

INTUITION

A shop assistant cannot show you all 1 million products. She first picks a few hundred that might suit you (fast, rough), then ranks them with more care (slow, precise), then arranges the shelf. Recommenders work in these stages because a heavy model on every item is too slow.

The funnel

A catalogue of 1 million items and a budget of about 100 ms per request. The standard design is a **funnel**:

1. **Candidate retrieval:** an approximate nearest neighbour (ANN) search over embeddings (vectors that place similar things close together) gives about 500 items.
2. **Light ranker:** a fast model cuts 500 to 100.
3. **Heavy ranker:** a richer model scores 100 and keeps 20.
4. **Rules and diversity:** remove out-of-stock and duplicates, mix categories, return 10.

```
stages = [("retrieval ANN", 1_000_000, 500, 15), ("light ranker", 500,
100, 10),
          ("heavy ranker", 100, 20, 40), ("rules + diversity", 20, 10, 5)]
total = 0
for name, n_in, n_out, ms in stages:
    total += ms
    print(f"{name:18s} {n_in:>9,} -> {n_out:>4}  {ms:3d} ms  cumulative
{total} ms")
relevant = {3, 7, 9}
ranked = [7, 1, 3, 4, 5, 9, 2]
for k in (2, 4, 6):
    hits = len(relevant & set(ranked[:k]))
    print(f"precision@{k} = {hits}/{k} = {hits/k:.2f}   recall@{k} =
{hits}/{len(relevant)} = {hits/len(relevant):.2f}")
items = 1_000_000; d = 64
print("item embeddings MB (float32, d=64):", items * d * 4 / 1e6)
```

```
retrieval ANN      1,000,000 ->  500   15 ms  cumulative 15 ms
light ranker             500 ->  100   10 ms  cumulative 25 ms
heavy ranker             100 ->   20   40 ms  cumulative 65 ms
rules + diversity         20 ->   10    5 ms  cumulative 70 ms
precision@2 = 1/2 = 0.50   recall@2 = 1/3 = 0.33
precision@4 = 2/4 = 0.50   recall@4 = 2/3 = 0.67
precision@6 = 3/6 = 0.50   recall@6 = 3/3 = 1.00
item embeddings MB (float32, d=64): 256.0
```

The stage times are assumed budgets, not measurements; you would set them by testing.

Problem 9.3.1 Read the funnel table. Total latency, and the heavy ranker's share?

Step 1	Add: 15 + 10 + 40 + 5 = 70 ms (the cumulative column ends at 70).
Step 2	Heavy share: 40 / 70 = 0.571.
Answer	70 ms in total; the heavy ranker uses 57% of the time budget, so it is the first thing to optimise or cap (score fewer than 100 candidates).
Check	If the heavy ranker scored 200 candidates and time doubled, the total would be 110 ms, over a 100 ms budget. This is a capacity trade-off you can state in an interview.
Why it matters	Each stage trades quality for speed; the funnel exists only because of the latency budget.

Problem 9.3.2 Items ranked in order are 7, 1, 3, 4, 5, 9, 2; the user really liked items 3, 7, 9. Compute precision and recall at 2, 4, 6.

Formula	precision@k = relevant items in the top k / k; recall@k = relevant items in the top k / all relevant items.
Plug in	Top 2 is {7, 1}: one hit. Top 4 is {7, 1, 3, 4}: two hits. Top 6 adds 5 and 9: three hits.
Compute	precision: 1/2, 2/4, 3/6, which is 0.50 each time. Recall: 1/3, 2/3, 3/3.
Answer	Precision stays at 0.50 while recall climbs to 1.00 by position 6.
Check	The output above prints the same fractions.

Two more points an interviewer expects. **Cold start:** a new user or item has no history, so use popularity, content features and exploration. **Online metrics and bias:** the model only learns from items it showed, so keep a small share of random exploration and run A/B tests with guardrails (return rate, latency). Chapter 8.8 gave the feed side of delivery.

Where it is used

E-commerce home pages, video and music queues, job and course suggestions, "similar items" on product pages. An ML engineer owns the ranker, a data engineer the feature and event pipelines.

COMMON MISTAKE

Optimising an offline metric that does not match the business goal (for example clicks when you care about purchases). Define the online metric first.

Check yourself

1. Why a funnel? Answer: a heavy model on every item would exceed the latency budget.
2. precision@3 if 2 of the top 3 are relevant? Answer: 2/3 = 0.67.
3. What is cold start? Answer: no history for a new user or item.

9.4 Worked Design: Fraud Detection

INTUITION

A bank sees 999 genuine transactions for every fraud. If your alarm is even slightly jumpy, it rings mostly for innocent people. The fight is not "is my model accurate"; it is "how many false alarms per real catch can the team handle".

Why accuracy lies

Fraud is rare. If 0.1% of transactions are fraud, a model that says "never fraud" is 99.9% accurate and catches nothing. Use **recall** (the share of frauds caught), **false positive rate** (FPR: the share of genuine transactions flagged) and **precision** (the share of flags that are really fraud). By Bayes' rule, with base rate `b`:

```
precision = recall × b / (recall × b + FPR × (1 − b))
```

Problem 9.4.1 5,000 transactions per second, 0.1% fraud. For three models, find precision and the flags per second.

```
tps = 5000; fraud_rate = 0.001
print("fraud tx/s:", tps * fraud_rate, " per day:", tps * fraud_rate *
86400)
for recall, fpr in ((0.8, 0.005), (0.8, 0.001), (0.9, 0.02)):
    tp = recall * fraud_rate
    fp = fpr * (1 - fraud_rate)
    prec = tp / (tp + fp)
    flagged_per_s = tps * (tp + fp)
    print(f"recall {recall} fpr {fpr}: precision {prec:.3f}, flagged/s
{flagged_per_s:.1f}, false alarms/s {tps*fp:.1f}")
```

```
fraud tx/s: 5.0  per day: 432000.0
recall 0.8 fpr 0.005: precision 0.138, flagged/s 29.0, false alarms/s 25.0
recall 0.8 fpr 0.001: precision 0.445, flagged/s 9.0, false alarms/s 5.0
recall 0.9 fpr 0.02: precision 0.043, flagged/s 104.4, false alarms/s 99.9
```

- **Step 1** Model A: true positives per transaction = 0.8 × 0.001 = 0.0008. False positives = 0.005 × 0.999 = 0.004995.
- **Step 2** Precision = 0.0008 / (0.0008 + 0.004995) = 0.0008 / 0.005795 = 0.138.
- **Step 3** Flags per second = 5,000 × 0.005795 = 29; of these 25 are false alarms.
- **Answer** A catches 80% of fraud but only 14% of its flags are real: about 6 false alarms for

every real fraud. Model C catches 90% but only 4% of flags are real, with 100 flags per second.

- **Check** Real frauds caught per second for A = 5 × 0.8 = 4; flags 29; 4 / 29 = 0.138. Matches.
- **Why it matters** A model with a 0.5% false-positive rate sounds excellent and still gives 25 false alarms per second. Base rates decide.

Problem 9.4.2 Pick the threshold by money. Per 1 million transactions: a missed fraud costs 8,000 (rupees, assumed), reviewing one flag costs 40 (assumed). Compare model A (recall 0.8, FPR 0.005) and model B (recall 0.9, FPR 0.02).

```
loss_per_fraud = 8000; review_cost = 40
n = 1_000_000
frauds = n * fraud_rate
for name, recall, fpr in (("A", 0.8, 0.005), ("B", 0.9, 0.02)):
    caught = frauds * recall
    fa = (n - frauds) * fpr
    net = caught * loss_per_fraud - (caught + fa) * review_cost
    print(name, "caught", caught, "false alarms", round(fa), "net saving",
round(net))
```

```
A caught 800.0 false alarms 4995 net saving 6168200
B caught 900.0 false alarms 19980 net saving 6364800
```

- **Step 1** Frauds: 1,000,000 × 0.001 = 1,000. Model A catches 800; model B catches 900.
- **Step 2** A: savings 800 × 8,000 = 6,400,000; review cost (800 + 4,995) × 40 = 231,800; net 6,168,200.
- **Step 3** B: savings 900 × 8,000 = 7,200,000; review cost (900 + 19,980) × 40 = 835,200; net 6,364,800.
- **Answer** B is worth 196,600 more per million transactions, even with four times the false alarms, because each caught fraud is worth 200 reviews.
- **Check** 6,364,800 − 6,168,200 = 196,600. This assumes every flag is reviewed and every caught fraud is fully saved; real analyst capacity, customer annoyance and review time would change the answer.

The design in one view

- **Data:** transactions, device, merchant, history. **Labels** arrive late (chargebacks take weeks), so train on older data and expect label delay.

- **Features:** velocity (count in the last 1, 10, 60 minutes), amount versus the user's usual, new device, geography. Use streaming aggregates and a point-in-time join (9.2).
- **Serving:** online scoring inside the payment path with a strict budget (for example under 100 ms, a business choice). Fallback rules if the model service is slow: apply rules and fail safe.
- **Decision:** approve, step-up verification (one-time password), or block. Threshold by cost (9.4.2), not by 0.5.
- **Monitoring:** drift, flag rate, review outcomes; fraudsters adapt, so retrain often.
- **Risks:** unfair bias against groups, privacy rules (India's data protection law and your company's policy; check with your compliance team).

Where it is used

Banks, UPI apps, card networks, insurance claims, e-commerce returns, ad click fraud.

COMMON MISTAKE

Judging by accuracy or by ROC curve alone. Always compute precision and the alert volume at the base rate, and the cost at the chosen threshold.

Check yourself

1. 0.2% fraud, recall 0.5, FPR 0.01. Precision? Answer: 0.001 / (0.001 + 0.00998) = 0.091.
2. Why are labels delayed? Answer: chargebacks and confirmations arrive weeks later.
3. Why not threshold 0.5? Answer: costs of misses and false alarms are unequal; pick by expected cost.

9.5 Worked Design: RAG Chatbot for Company Documents

INTUITION

An open-book exam. The model does not memorise your policies. When a question arrives, a librarian fetches the 3 most relevant pages and the model answers from them, quoting the page. That is **RAG** (retrieval-augmented generation).

RAG architecture

Offline (top): documents are cleaned and cut into chunks, each chunk is turned into an embedding by an embedding model, and stored with metadata in a vector database. Online (bottom): the user's question is embedded, a top-k search finds similar chunks (optionally reranked), and the large language model (LLM) answers from them with citations. A band says guardrails and evaluation wrap the whole path: log every question, chunk and answer.

First: prompt, RAG, fine-tune or agent?

Need	Choose	Why
Change tone or format, general knowledge is enough	Prompting	Cheapest, fastest
Answers must use private or changing documents	RAG	Update the index, not the model
Consistent style or a narrow task at scale	Fine-tuning	Bakes behaviour into weights; needs data and evaluation
Multi-step actions with tools (look up, calculate, book)	Agent	Loop of plan and tool calls; higher risk and cost

Start with prompting, add RAG, fine-tune only with evidence it is needed, add agents only when the task needs actions.

Problem 9.5.1 50,000 documents, 20 chunks each, 768-number embeddings (float32). How big is the vector index? And with int8 quantization (a smaller number format)?

```
chunks = 50_000 * 20; d = 768
print("chunks:", f"{chunks:,}")
print("vectors GB float32:", chunks * d * 4 / 1e9)
print("vectors GB int8   :", chunks * d / 1e9)
```

```
chunks: 1,000,000
vectors GB float32: 3.072
vectors GB int8   : 0.768
```

- **Step 1** Chunks: 50,000 × 20 = 1,000,000.
- **Step 2** float32 uses 4 bytes per number: 1,000,000 × 768 × 4 = 3,072,000,000 bytes = 3.07

GB.

- **Step 3** int8 uses 1 byte: 0.77 GB.
- **Answer** About 3 GB, which fits in memory on a single machine; no sharding is needed at this size. (Index overhead and metadata add more; measure.)
- **Check** Doubling the documents doubles the size, which is linear. At 100 million chunks you would need about 307 GB and sharding (Chapter 8.4).

Problem 9.5.2 2,000 questions per day over an 8-hour working day. Each uses 3,000 input tokens (question plus chunks) and 500 output tokens. Latency parts: embed 30 ms, search 40 ms, rerank 120 ms, LLM first token 700 ms, then output streams at 50 tokens per second. Find tokens per month, time to first token, full answer time, and streams open at a peak 5 times average.

```
q, tin, tout = 2000, 3000, 500
print("tokens/day:", q * (tin + tout), "in", q * tin, "out", q * tout)
print("tokens/month (M):", q * (tin + tout) * 30 / 1e6)
parts = {"embed query": 30, "vector search": 40, "rerank": 120, "LLM first
token": 700}
first = sum(parts.values())
stream = tout / 50 * 1000
print("first token ms:", first, "| full answer s:", (first + stream) /
1000)
qps = q / (8 * 3600)
print("avg qps:", round(qps, 3), "peak x5:", round(qps * 5, 2), "streams
open at peak (Little):", round(qps * 5 * (first + stream) / 1000, 1))
```

```
tokens/day: 7000000 in 6000000 out 1000000
tokens/month (M): 210.0
first token ms: 890 | full answer s: 10.89
avg qps: 0.069 peak x5: 0.35 streams open at peak (Little): 3.8
```

- **Step 1** Tokens per day: 2,000 × 3,500 = 7,000,000; per month: 210 million (30 days). Multiply by the provider's current price per million tokens (look it up; prices change) to get cost.
- **Step 2** First token: 30 + 40 + 120 + 700 = 890 ms. Streaming 500 tokens at 50 per second takes 10 s, so the full answer is 10.89 s.
- **Step 3** Average rate: 2,000 / 28,800 = 0.069 per second; peak 0.35 per second. Little's law: 0.35 × 10.89 = 3.8 answers streaming at once.

- **Answer** 210 million tokens per month, 0.9 s to the first word, 10.9 s for a full answer, and fewer than 4 concurrent streams at peak. The load is tiny; the cost and quality matter more than throughput.
- **Check** 7,000,000 × 30 = 210,000,000. The bulk of tokens (6 of 7 million) are input, so trimming retrieved context cuts cost most.
- **Why it matters** Perceived speed depends on time to first token and streaming, not total time.

Evaluation, guardrails, failures

You cannot improve what you do not measure. Build a **golden set**: 50 to 200 real questions with the correct source chunk. Measure retrieval first (hit@k: is the right chunk in the top k?) and answer quality separately (is it correct, grounded in the chunks, with citations?).

Problem 9.5.3 Four questions have gold chunks d4, d9, d2, d7. The retriever returned: q1 [d4, d1, d3], q2 [d5, d6, d9], q3 [d8, d1, d5], q4 [d7, d2, d3]. Compute hit@1 and hit@3.

Step 1	hit@1 looks at the first item only: q1 yes (d4), q2 no, q3 no, q4 yes (d7). That is 2 of 4.
Step 2	hit@3 looks at all three: q1 yes, q2 yes (d9 is third), q3 no, q4 yes. That is 3 of 4.
Answer	hit@1 = 0.50 and hit@3 = 0.75. Question q3 never finds its chunk, so inspect q3's chunks (perhaps the document was split badly).
Check	The code printed `hit@1 = 2/4 = 0.50` and `hit@3 = 3/4 = 0.75`.

Design points for the interview: **chunk size** (too small loses context, too large dilutes meaning; test 300 to 800 tokens), **metadata filters** (department, date, access rights: never show a user a document they may not read), **hybrid search** (keyword plus vectors helps with names and codes), **rerank**, **refusal** ("I could not find this in the documents"), **prompt injection** (text inside a document that tries to give the model orders; treat retrieved text as data, never as instructions), **human in the loop** for risky answers, and **logging** every question, chunk and answer for review. Chapter 4 covers the README and design doc to write this up.

Where it is used

Internal helpdesks, policy and HR assistants, customer-support copilots, search over contracts and manuals. AI engineers own it, data engineers feed and refresh the index.

COMMON MISTAKE

Judging the chatbot by a few good demos. Without a golden set you cannot tell whether a change helped, and you will ship hallucinations (confident wrong answers).

Check yourself

1. RAG or fine-tuning for changing policies? Answer: RAG; update the index.
2. What is hit@k? Answer: the share of questions whose correct chunk is in the top k results.
3. What is prompt injection? Answer: text in the data that tries to command the model.

Chapter recap

- Use the eleven-step ML design framework; metric and data come before the model.
- A data platform ingests, stores raw data, transforms into a warehouse and serves features; test freshness and quality.
- Avoid leakage with point-in-time joins; avoid skew with one feature definition.
- Recommenders use a funnel of retrieval, light ranking, heavy ranking and rules because of the latency budget.
- Fraud is a base-rate problem: compute precision and alert volume, and choose the threshold by cost.
- RAG: chunk, embed, store, retrieve, rerank, answer with citations; measure retrieval and answers separately.
- Choose prompting, then RAG, then fine-tuning, then agents.
- Always size the system: events per second, bytes, memory, tokens, concurrent users (Little's law).

Pattern cheat-sheet

Need	Pattern
Precision from base rate	`r·b / (r·b + FPR·(1 − b))`
Net value of a model	`caught × loss − flags × review cost`
Point-in-time join	`feature.ts <= label.ts`, latest one

Vector index size	`chunks × dims × bytes per number`
Latency budget	sum stage times; first token plus streaming
precision@k, recall@k	hits in top k over `k`, over all relevant

Practice set

1. (Easy) 200 million events per day. Events per second? **Answer:** 200,000,000 / 86,400 = 2,315.
2. (Easy) 2 million chunks, 384 dimensions, float32. Size? **Answer:** 2,000,000 × 384 × 4 = 3.07 GB.
3. (Easy) Top 5 results, 2 relevant of 4 total. precision@5 and recall@5? **Answer:** 2/5 = 0.40 and 2/4 = 0.50.
4. (Medium) 1% fraud, recall 0.9, FPR 0.02. Precision? **Answer:** 0.009 / (0.009 + 0.0198) = 0.3125.
5. (Medium) What is the output of the point-in-time join if a feature is dated after the label? **Answer:** it is not picked; the earlier feature, or `None` if there is none.
6. (Medium) Find the bug: a model is trained with a random split on one year of daily data, then fails in production. **Answer:** time leakage; use a time-based split (train earlier, test later).
7. (Medium) A funnel has stages of 12, 8, 35 and 5 ms. Budget 70 ms. Does it fit? **Answer:** 12 + 8 + 35 + 5 = 60 ms, yes, with 10 ms spare.
8. (Hard) A chatbot gets 10,000 questions per day, 4,000 tokens each in total. Tokens per month? And if 60% of questions are repeats, how many tokens can a cache save? **Answer:** 10,000 × 4,000 × 30 = 1.2 billion; repeats save 60%, about 720 million tokens (if answers can safely be reused and nothing in them goes stale).
9. (Hard, interview) "Design fraud detection for UPI payments." **Answer:** Follow the framework: objective (loss prevented with low friction), metrics (recall at fixed false-alarm volume, review cost, customer friction guardrail), delayed labels, streaming velocity features, online scoring under a strict budget with fallback rules, threshold by cost, step-up verification for middle scores, drift and retraining, bias and privacy risks.
10. (Hard, interview) "Our RAG bot gives wrong answers. How do you debug?" **Answer:** Build a golden set; measure hit@k. If retrieval fails, change chunking, add hybrid search or reranking, check metadata filters. If retrieval is fine, fix the prompt (answer only from

context, cite, refuse), check context length and ordering, then model choice. Log failures into the golden set.

Prompt pack

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional in India moving into data and AI roles; English is my second language. Topic: the ML system design framework (objective, metrics, data, features, model, training, evaluation, serving, monitoring, risks). Act as interviewer for "design a product recommendation system". Ask one framework step at a time, score my answer, and push me to give numbers.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data engineering. Topic: data platform design (ingestion, CDC, raw zone, warehouse, orchestration, freshness, quality). Give me a scenario with event volumes, ask me to size ingestion, storage and partitions, and check my arithmetic. Then ask me three questions about freshness targets and what to test. Keep it step by step.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into ML engineering. Topic: feature pipelines, point-in-time joins, leakage and training-serving skew. Give me 6 graded exercises with small SQL tables I can run in SQLite, without solutions. After my answers, show a corrected query, explain the leak, and ask me to describe skew in my own words.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into ML roles. Topic: fraud detection design, including base rates, precision and recall, alert volume and cost-based thresholds. Give me a transaction volume and fraud rate, then ask me one calculation at a time. Challenge my threshold choice with analyst capacity and customer friction, and end with a 5-minute interview answer outline.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into AI engineering. Topic: RAG chatbot design (chunking,
embeddings, vector search, rerank, guardrails, evaluation with a golden
set). Here is my design: [PASTE HERE]. Review it, estimate index size and
token cost with me step by step, list the three most likely failure modes,
and tell me how I would measure each.
```

CHAPTER 10

RESEARCH SKILLS

How to read a paper without drowning, check whether a claim is true, run a fair experiment, and write down what you learned.

A new paper appears and your manager asks, "Can we use this?" A team shows a chart where the new model is 0.5 points better and asks for budget. A recruiter for an applied scientist role asks about your last experiment. In all three cases the skill is the same: reading claims critically and producing evidence honestly. You already do this in performance work: you do not trust one test run, you check the baseline, you look for what else changed. This chapter turns that instinct into a method for papers and ML experiments. By the end you will read a paper in three passes within a time budget, reproduce a result and judge it against run-to-run noise, design an experiment and an ablation, write a paper-style summary or a blog post, and know the ethical rules that keep your work honest. A running example appears throughout: a pretend paper called "Paper X" that claims a retrieval trick improves a question-answering score.

10.1 Reading Papers: The Three-Pass Method

INTUITION

You do not read a map by walking every street. First you look at the whole city, then the district you need, then the one street you must drive. A paper is the same. Most papers need only the first look. A few deserve the whole street.

The method

The three-pass method comes from S. Keshav's short article "How to Read a Paper". Each pass has a goal and a stop rule.

Pass	What you read	Time (rough guide)	Question it answers	Output
1	Title, abstract, headings, figures, conclusion	5 to 10 minutes	Is this relevant and worth more time?	Yes or no, plus a 3-line summary

2	Introduction, method, experiments; skip proofs	up to about 1 hour	What exactly did they do and show?	You can explain the main figure and the claim
3	Everything; rebuild the method in your head or code	several hours, longer for beginners	Would I get the same result? Where is it weak?	Notes, doubts, a list of tests you would run

Times are rough and differ by reader and field.

Ten questions to ask of every paper (write the answers in your log):

1. What problem do they solve, and why does it matter?
2. What is new compared with earlier work (the *novelty*)?
3. What do they assume (data, scale, compute)?
4. What are the baselines (the older methods they compare against), and are those fair and well tuned?
5. On which datasets, and how big?
6. Which metric, and does it match the real goal?
7. Do they report variance (several runs, error bars)?
8. Do the ablations (removing parts one by one) support the story?
9. What are the limitations, stated and unstated?
10. Would it work on my data and budget?

Where to find papers

arXiv (categories cs.LG, cs.CL, cs.CV, stat.ML) is where most preprints (papers shared before formal review) appear. Others: Papers with Code (papers linked to code), Semantic Scholar (search and citations), the ACL Anthology (language papers), conference proceedings (NeurIPS, ICML, ICLR, CVPR, KDD), Hugging Face Daily Papers, and Distill-style explanatory blogs. A preprint has not been checked by reviewers, so read it with extra care.

A starter reading list

Attention Is All You Need (the Transformer), BERT, ResNet, Adam, Dropout, GPT-3, LoRA, InstructGPT (RLHF), the Chinchilla scaling paper, the RAG paper (Lewis et al.), ReAct, Chain-of-Thought, CLIP, Latent Diffusion, XGBoost, and the classics of your chosen specialisation. Read one a month early on; one a week from the stage where you build ML projects (the study

plan in Chapter 7).

Problem 10.1.1 You have 7 hours per week for reading. You scan 5 papers at pass 1 (10 minutes each) and take one to pass 2 and a half of pass 3 (3 hours total). Does it fit?

Formula	total minutes = passes × minutes each
Plug in	5 × 10 = 50 minutes for pass 1; plus 180 minutes for the deep paper
Compute	50 + 180 = 230 minutes = 3 hours 50 minutes
Answer	It uses 230 of your 420 minutes, about 55%. You still have 190 minutes for coding and writing the summary.
Check	7 hours × 60 = 420 minutes; 420 − 230 = 190.
Why it matters	Without a time box, one paper can eat a week. Pass 1 is a filter, and most papers should die there.

Problem 10.1.2 Pass 1 of Paper X (pretend). Abstract: "We add a reranking step to retrieval and improve answer F1 from 79.5 to 84.2 on one benchmark." Figure 1 is a bar chart with no error bars. Conclusion: "reranking is essential." What is your pass-1 decision and what do you note?

Step 1	Relevant? Yes, you are building RAG (Chapter 9.5).
Step 2	Red flags from questions 5 and 7: one benchmark, no error bars, and a strong word ("essential").
Step 3	Three-line summary: "Adds reranker to retrieval. Claims 4.7 F1 points on one dataset. No variance reported."
Answer	Go on to pass 2, with a doubt list: does the baseline use hybrid search? how many seeds? is the gain real beyond noise (see 10.2)?
Check	84.2 − 79.5 = 4.7, the claimed gain.

The picture: a paper log row

```
Title | Link | 3-line summary | Key idea | What I would test | Date |
Shared?
Paper X | arXiv:... | rerank step; +4.7 F1; no variance | rerank top 50 ->
5 | try on my docs | 2026-11-07 | post draft
```

Where it is used

Applied scientists and research engineers read weekly. ML and AI engineers use papers to choose methods (for example whether to add reranking). Data scientists read papers on experiment design and causal methods.

Applications

- **Machine learning:** choosing a loss, an optimizer or an architecture.
- **GenAI:** retrieval, fine-tuning and evaluation methods.
- **Data engineering:** system papers (stream processing, storage formats).
- **Interviews:** "tell me about a paper you read recently" is common for ML roles.

COMMON MISTAKE

Reading every paper start to finish, or taking the abstract as truth. The abstract is advertising. Check the experiments table before you believe it.

Check yourself

1. What is the goal of pass 1? Answer: decide in 5 to 10 minutes whether the paper is relevant.
2. What is a baseline? Answer: an older or simpler method the new method is compared against.
3. Why is a preprint treated carefully? Answer: reviewers have not checked it yet.

10.2 Reproducing Results and Reading Noise

INTUITION

If a bus takes 48 minutes on Monday and 52 minutes on Tuesday with the same route, you know a single trip is not the truth. ML training has the same noise: random starting weights and random data order. A 0.5-point win from one run may be luck.

Reproducibility checklist

To reproduce a result (get the same outcome from the same method), version everything:

- **Code:** a Git commit hash.
- **Data:** a dataset version or checksum, and the exact split.
- **Config:** every hyperparameter (settings you choose, such as learning rate).
- **Environment:** library versions (a lock file or container image), hardware, compute time.
- **Seeds:** the random-number seeds. Fixing a seed makes one run repeatable, but results should still be shown over several seeds.
- **Report:** compute used, number of runs, mean and spread.

Tools: Git, DVC or similar for data, `wandb` or MLflow-style trackers for runs, Hugging Face Hub for models and datasets, `lm-evaluation-harness` for standard language-model evaluations. Cheap compute: Colab, Kaggle notebooks, Lightning AI, cloud credits and spot (discounted, interruptible) machines. Check each service's current free limits, which change.

Problem 10.2.1 (simulation) Two models are truly identical: accuracy has a mean of 91.0% and a standard deviation of 0.4 points from run to run. Train 5 seeds of each and compare.

```
import random, statistics as st, math
random.seed(0)
base = [random.gauss(91.0, 0.4) for _ in range(5)]
new  = [random.gauss(91.0, 0.4) for _ in range(5)]
print("baseline runs:", [round(x, 2) for x in base])
print("new runs     :", [round(x, 2) for x in new])
mb, mn = st.mean(base), st.mean(new)
sb, sn = st.stdev(base), st.stdev(new)
print(f"baseline {mb:.2f} +- {sb:.2f}   new {mn:.2f} +- {sn:.2f}   diff
{mn-mb:+.2f}")
se = math.sqrt(sb**2/5 + sn**2/5)
t = (mn - mb) / se
print(f"Welch t = {t:.2f}  (|t| below about 2.3 for 5 runs each means no
clear difference)")
wins = 0; N = 100000
for _ in range(N):
    if random.gauss(91, 0.4) - random.gauss(91, 0.4) >= 0.5: wins += 1
print("single-run luck: P(new beats base by >= 0.5 with no real gain) =",
round(wins / N, 3))
```

```
baseline runs: [91.38, 90.44, 90.73, 91.15, 90.59]
new runs     : [90.97, 91.07, 90.67, 90.48, 91.08]
baseline 90.86 +- 0.39   new 90.85 +- 0.27   diff -0.00
Welch t = -0.02  (|t| below about 2.3 for 5 runs each means no clear
difference)
single-run luck: P(new beats base by >= 0.5 with no real gain) = 0.189
```

- **Step 1** Look at the baseline runs alone: they range from 90.44 to 91.38, almost a full point, with no change in the method at all.
- **Step 2** The means are 90.86 and 90.85, a difference of about zero. The standard error of the difference is $\sqrt{(0.39^2/5 + 0.27^2/5)} = \sqrt{(0.0450)} = 0.21$, so $t = diff / se$, about −0.02.
- **Step 3** The last line runs 100,000 pretend single-run comparisons. In 18.9% of them the "new" model beat the baseline by 0.5 or more with no real improvement.
- **Answer** One run each would have a nearly 1-in-5 chance of showing a fake 0.5-point win. With 5 runs each, the means agree and no gain is visible.
- **Check** The difference of two runs has a spread of $0.4 \times \sqrt{2} = 0.566$. A gain of 0.5 is $0.5 / 0.566 = 0.88$ spreads above zero, and the chance of exceeding that is about 0.19 on the normal curve. It matches 0.189.
- **Why it matters** A gain smaller than the run-to-run spread is not evidence. Report the mean

and standard deviation over several seeds, and use a test or confidence interval (see the learner's statistics document).

Problem 10.2.2 You try to reproduce Paper X. They report 84.2 F1. You get 83.1, 83.6, 82.9, 83.4, 83.0 over 5 seeds. Is this a successful reproduction?

Step 1	Mean = (83.1 + 83.6 + 82.9 + 83.4 + 83.0) / 5 = 416.0 / 5 = 83.2.
Step 2	Deviations from the mean: −0.1, 0.4, −0.3, 0.2, −0.2. Squares: 0.01, 0.16, 0.09, 0.04, 0.04; sum 0.34. Sample variance 0.34 / 4 = 0.085, standard deviation about 0.29.
Step 3	The paper's 84.2 is `(84.2 − 83.2) / 0.29 = 3.4` standard deviations above your mean. That is far outside your run-to-run spread.
Answer	You reproduced the direction of the result but not the size: 83.2 versus 84.2. Report it honestly as "partial reproduction, about 1 point lower", then look for differences (data version, preprocessing, tuning, tokenizer).
Check	Compute the mean twice; 83.2 × 5 = 416.0, matching the sum.

The picture: how to write a reproduction report

```
Claim: +4.7 F1 on benchmark B          Paper: 84.2
My setup: commit abc123, data v2, 5 seeds, 1 GPU-hour each
My result: 83.2 +- 0.3                  Gap: -1.0
Possible causes: baseline tuning, preprocessing, undisclosed trick
Verdict: partially reproduced
```

Where it is used

ML engineers reproduce before adopting a method. Research roles reproduce as the first step of every project. Engineers already do the same when they re-check a benchmark.

Applications

- **Model selection:** avoid shipping a change that is only noise.
- **MLOps:** run tracking makes experiments repeatable.
- **Data science:** check that an analysis can be rerun on fresh data.

COMMON MISTAKE

Comparing single runs, or tuning your own method for hours while leaving the baseline at default settings. Give the baseline the same tuning budget.

Check yourself

1. Why use several seeds? Answer: results vary from run to run, so you need the spread.
2. What goes into a reproducibility record? Answer: code version, data version, config, environment, seeds, compute.
3. A gain of 0.2 with run spread 0.4: evidence? Answer: no; it is smaller than the noise.

10.3 Experimental Design

INTUITION

A school testing a new teaching method keeps the class size, the syllabus and the exam the same, changes one thing, and repeats. If you change the method and the class size together, you cannot say which one caused the improvement. An experiment is a fair race with one difference.

The parts of a good experiment

- **Question and hypothesis.** A question: "Does reranking help?" A **falsifiable hypothesis** (one that data could prove wrong): "Adding a reranker raises F1 by at least 1 point on our test set." Decide the success threshold *before* running.
- **Baseline.** The method to beat. Include a simple baseline (keyword search) and the current best you can run.
- **Controls.** Everything except the change stays the same: data, split, budget, tuning effort.
- **Confounders.** Hidden causes that move the result: a bigger model, more training time, a different data version. List them and neutralise them.
- **Randomization.** Random seeds, random assignment to groups in online tests, shuffled data.
- **Power and sample size.** *Power* is the chance of detecting a real effect. Small test sets and few runs give weak power. A tiny test set makes a 1-point gain meaningless.
- **Multiple comparisons.** Each extra test is another chance of a false positive.

- **Reporting.** Mean and spread over seeds, the exact numbers, the negative results too.

Multiple comparisons, with numbers

If you test one idea at the 5% level, the chance of a false positive is 5%. If you test 20 ideas and keep the best, the chance that *at least one* looks significant is $1 - 0.95^{20}$. One fix is the **Bonferroni correction**: divide the threshold by the number of tests.

```
for m in (1, 5, 20, 100):
    print(m, "tests: P(at least one false positive at 0.05) =", round(1-0.95**m,3), " Bonferroni alpha =", 0.05/m)
```

```
1 tests: P(at least one false positive at 0.05) = 0.05  Bonferroni alpha = 0.05
5 tests: P(at least one false positive at 0.05) = 0.226  Bonferroni alpha = 0.01
20 tests: P(at least one false positive at 0.05) = 0.642  Bonferroni alpha = 0.0025
100 tests: P(at least one false positive at 0.05) = 0.994  Bonferroni alpha = 0.0005
```

Problem 10.3.1 A teammate tried 20 feature ideas on one validation set and reports the best, which "improves AUC significantly (p = 0.03)". Should you believe it?

Step 1	With 20 tries and a 0.05 threshold, the chance that at least one looks significant by luck is 0.642 (see the output).
Step 2	Bonferroni for 20 tests needs $p < 0.05 / 20 = 0.0025$. The reported p = 0.03 does not pass.
Step 3	The honest fix: choose a final candidate, then confirm it on a **held-out test set** that was never used for choosing.
Answer	Do not believe it yet. A p-value of 0.03 after 20 attempts is weak evidence; test the winner once on fresh data.
Check	$0.95^{20} = 0.358$, and $1 - 0.358 = 0.642$.
Why it matters	This is called "p-hacking" or "researcher degrees of freedom", and it explains many results that fail to replicate.

Problem 10.3.2 You must decide how many seeds you need. The run-to-run standard deviation is 0.4 and you want to detect a real gain of 0.5. Using the rough rule that the standard error of a mean difference is $0.4 \times \sqrt{(2 / n)}$, find n so that 0.5 is about 2 standard errors.

Formula	$se = 0.4 \times \sqrt{(2 / n)}$; we need $0.5 / se \geq 2$, so $se \leq 0.25$.
Plug in	$0.4 \times \sqrt{(2 / n)} \leq 0.25$ gives $\sqrt{(2 / n)} \leq 0.625$, so $2 / n \leq 0.3906$.
Compute	$n \geq 2 / 0.3906 = 5.12$, so n = 6.
Answer	About 6 seeds per method. With 5 the margin is thin (se = 0.253).
Check	For n = 6: $0.4 \times \sqrt{(1/3)} = 0.231$, and 0.5 / 0.231 = 2.2. For n = 5: $0.4 \times \sqrt{0.4} = 0.253$, and 0.5 / 0.253 = 1.98, just under 2.
Why it matters	Smaller gains need many more runs: halving the gain to 0.25 needs four times as many seeds. This is a rule of thumb, not a full power analysis.

Where it is used

Data scientists design A/B tests (the learner's statistics document), ML engineers compare models, applied scientists design studies. The same discipline applies to any measurement: control the environment, change one variable, repeat.

Applications

- **A/B testing:** randomised assignment and guardrail metrics.
- **Model comparison:** same splits, same tuning budget, several seeds.
- **Offline evaluation:** held-out test sets used once.
- **Engineering benchmarks:** one change per test, repeated runs.

COMMON MISTAKE

Tuning on the test set, then reporting the same set as your result. Keep a train set, a validation set for choices, and a test set touched once.

Check yourself

1. What is a falsifiable hypothesis? Answer: a claim that data could show to be wrong, with a threshold set beforehand.

2. Why do 20 tests inflate false positives? Answer: each test is a new chance of luck; the total chance adds up to 64%.
3. What is a confounder? Answer: a hidden cause that changes the result along with the thing you test.

10.4 Ablations and Applied Research Projects

INTUITION

To find which ingredient makes the soup taste good, you cook it again leaving out one ingredient at a time. If the soup is still good without the herbs, the herbs were not the point. An **ablation** removes one part of a system at a time and measures what is lost.

How to run an ablation

1. Fix the full system and measure it (mean over seeds).
2. Remove or replace *one* component, keep everything else, remeasure.
3. Also measure the baseline with all components removed.
4. Compare the drop of each component to the noise from 10.2. A drop smaller than the noise is not a finding.

The numbers below are illustrative (made up for teaching), as the mean of 5 seeds.

```
rows = [("full model", 84.2), ("− reranker", 82.9), ("− hybrid search",
83.1), ("− query rewrite", 84.0), ("− all three (baseline)", 79.5)]
full = rows[0][1]
for name, f1 in rows:
    print(f"{name:24s} F1 {f1:5.1f}   drop vs full {full − f1:4.1f}")
print("sum of single drops:", round(sum(full − f for _, f in rows[1:4]),
1), " vs total gain over baseline:", round(full − rows[4][1], 1))
```

```
full model               F1  84.2   drop vs full  0.0
− reranker               F1  82.9   drop vs full  1.3
− hybrid search          F1  83.1   drop vs full  1.1
− query rewrite          F1  84.0   drop vs full  0.2
− all three (baseline)   F1  79.5   drop vs full  4.7
sum of single drops: 2.6  vs total gain over baseline: 4.7
```

Problem 10.4.1 Read the table. Which components matter, given a run-to-run standard deviation of 0.3?

Step 1	Query rewrite: drop 0.2, below the noise of 0.3. Not shown to matter.
Step 2	Reranker: 1.3 is about 4 standard deviations (1.3 / 0.3 = 4.3). Hybrid search: 1.1 is about 3.7. Both matter.
Step 3	The single drops add to 2.6 but the total gain is 4.7, so the parts *interact* (together they give more than the sum), or other parts of the pipeline matter too.
Answer	Keep the reranker and hybrid search; drop the query rewrite unless a larger test shows value. Report the interaction honestly.
Check	1.3 + 1.1 + 0.2 = 2.6, and 84.2 − 79.5 = 4.7.
Why it matters	Ablations tell you where to spend engineering time and where to cut cost. A component that adds latency and no accuracy should go.

Applied research projects for you

An applied project follows the same steps as a paper: question, baseline, experiment, ablation, write-up. Ideas that match the skill maps: improve retrieval quality for a chatbot; evaluate a small domain fine-tune; Indian-language NLP; efficient inference (latency and cost per 1,000 requests); better evaluation methods for LLM answers; and **anomaly detection on operational data**, such as detecting a delivery-time slowdown across weeks with a statistical test and reporting precision and recall. That last one is a signature project because it joins real operational data with ML. Reproduce a paper at small scale first, run ablations, try one improvement, and document honestly even if the improvement fails.

Problem 10.4.2 Plan a reproduction project in one table: "reproduce a small retrieval-reranking paper in 4 weeks at 1 hour per day". Allocate hours.

Step 1	Total hours: 4 weeks × 7 = 28.
Step 2	Split: reading and setup 6; baseline reproduction 8; main method 6; ablations 4; write-up 4. Total 6 + 8 + 6 + 4 + 4 = 28.
Answer	28 hours: 6 reading and setup, 8 baseline, 6 method, 4 ablations, 4 write-up.
Check	The sum is 28, equal to the budget.

Algorithm design in one paragraph

Research also includes designing algorithms: paradigms (divide and conquer, greedy, dynamic programming, randomization, approximation), proving correctness (induction, invariants), and analysing complexity. Resources: CLRS, Skiena's *Algorithm Design Manual*, and the MIT 6.006 and 6.046 lectures. Use them after the DSA practice in Chapter 7.

Where it is used

Applied scientist roles, ML platform teams and any team that must justify a cost with evidence.

Applications

- **GenAI:** which of retrieval, reranking and prompt changes earn their latency.
- **ML engineering:** which features justify their pipeline cost.
- **Performance and AIOps:** which detector settings give acceptable false alarms.

COMMON MISTAKE

Reporting only the full model and not the ablation, or claiming a component "helps" when its effect is inside the noise.

Check yourself

1. What does an ablation do? Answer: removes one part at a time and measures the loss.
2. A component's drop is 0.2 and noise is 0.3. Conclusion? Answer: not shown to matter.
3. Why can the single drops not add to the total? Answer: components interact.

10.5 Writing Papers and Technical Blog Posts

INTUITION

A good paper is a report you would hand to a busy colleague: the problem, the idea, the proof, the limits, in that order. A good blog post is the same story told to a friend. Writing is how you find out whether you really understood. Chapter 4 gave the general rules of clear writing; here are the shapes specific to research.

Structure of a research paper

Section	Job	Typical content
Title	Say what and for whom	Plain claim, not a slogan
Abstract	Full story in about 150 to 250 words	Problem, gap, method, result with a number, limit
Introduction	Why this matters	Problem, what is missing, **contributions** as a short list
Related work	Where you sit	What others did and how you differ
Method	What you did	Enough detail to repeat it; define notation
Experiments	Evidence	Data, baselines, setup, metrics, seeds
Results	What happened	Tables, plots, ablations, error bars
Limitations	Honesty	Where it fails, what you did not test
Conclusion	Close	One paragraph, no new claims

Tools: LaTeX (a typesetting language) on Overleaf (an online editor), Zotero (a reference manager), clear figures with labelled axes, and consistent notation. Writing rules: short sentences, define every term, state contributions in plain words ("We show X. We release code and data."), and ask a colleague to read it for feedback. Learn the review process: reviewers judge novelty, correctness, clarity and reproducibility; many venues use a reproducibility checklist.

Venues, easiest start first: a **workshop** paper (shorter, friendlier), an **arXiv** preprint, a conference, a journal. A technical blog post with code is a perfectly good start.

Technical blog posts

Post types: tutorials, "what I learned" retrospectives, project write-ups, paper summaries, and benchmark comparisons. Platforms: Medium, dev.to, Hashnode or a personal blog. A schedule of 2 posts a month is plenty. Upgrade over time to local meetup talks, workshop papers, open-source docs and Kaggle write-ups.

Craft in five steps: (1) name the reader ("a data analyst who knows SQL"), (2) outline in 5 headings, (3) code that runs, with the output pasted, (4) one diagram, (5) edit ruthlessly, then pick a clear title.

Problem 10.5.1 Rewrite this weak abstract in four sentences: "We propose a novel approach for improving chatbots. Results are good. Our method is better than others on many datasets."

Step 1	Find the missing parts: problem (which chatbot, what failure), method, result with a number, and limits.
Step 2	Write: "Company-document chatbots often fail because the right passage is not retrieved. We add a reranking step that rescores the top 50 search results and keeps 5. On one internal question set of 200 questions, answer F1 rises from 79.5 to 84.2 (mean of 5 seeds, standard deviation 0.3). We have not tested other languages or documents longer than 20 pages."
Answer	The rewrite names the problem, method, a number with its spread, and a limit, in four sentences. (Numbers belong to the pretend Paper X.)
Check	Each of the four jobs of an abstract is present: problem, method, result, limit.
Why it matters	Readers and recruiters decide in the first 30 seconds; vague words ("novel", "good") are not evidence.

Problem 10.5.2 A blog post draft is 1,800 words: introduction 600, method 200, code 100, results 100, history of the field 800. What is wrong, and what is the budget fix?

Step 1	The history takes 800 / 1,800 = 44% and the results take 100 / 1,800 = 5.6%. The reader came for the method and the results.
Step 2	Target: introduction 250, method 500, code (with output) 300, results 350, lessons and limits 250. Total 250 + 500 + 300 + 350 + 250 = 1,650 words.
Answer	Cut history to a short paragraph in the introduction and move the space to method and results; the post becomes 1,650 words with results at 21% of the text.
Check	350 / 1,650 = 0.212.

Where it is used

Applied scientists write reports and papers; ML and data engineers write design docs and blog

posts that win interviews (Chapter 11 uses blog posts as portfolio pieces).

Applications

- **Hiring:** a write-up of a reproduction is stronger than a certificate.
- **Teams:** design docs and decision records (Chapter 4).
- **Community:** meetup talks, Kaggle write-ups, open-source docs.

COMMON MISTAKE

Hiding limitations. Reviewers and readers trust you more when you say where the method fails.

Check yourself

1. What goes in the abstract? Answer: problem, method, result with a number, and a limit.
2. What is the easiest first venue? Answer: a workshop paper, an arXiv preprint or a blog post.
3. Why paste real output in a post? Answer: so readers can trust and rerun it.

10.6 Ethics and Honest Reporting

INTUITION

A hospital would never hide a failed lab test because the report looks better without it. Research ethics is that same rule applied to data, people and results.

The rules

- **Consent and data rights.** Use data you are allowed to use. For personal data, follow the consent given, your company's policy and the law (India's Digital Personal Data Protection Act, 2023, and GDPR for European users; check current text and your compliance team). Do not scrape or use personal data without checking terms.
- **Privacy.** Remove or hash identifiers, keep only what you need, and watch for re-identification (joining "anonymous" data to find a person).
- **Bias and fairness.** Check performance across groups (for example by region, gender, language). A model with a high average can be poor for a smaller group.
- **Dual use.** Some methods help attackers (deepfakes, surveillance, malware). Think about

misuse before publishing and what safeguards you can add.

- **Responsible disclosure.** If you find a security flaw or a harmful model behaviour, tell the owner privately and give time to fix it before publishing details.
- **Honest reporting.** Report negative results, all seeds, all tuned baselines, failed ideas, and the limits. Do not drop bad runs. Do not cherry-pick the benchmark. Credit others' work and cite sources.
- **Applied versus fundamental research.** Fundamental research asks "how does this work?" and may take years; applied research asks "does this solve our problem under our constraints?" Industry research is usually applied, academic often fundamental, and the best work in either is honest.

Problem 10.6.1 Your anomaly detector catches 90% of regressions on the 5 services where you tuned it and 55% on the other 15 services. The report headline says "90% recall". What is the honest number?

Step 1	Services: 5 + 15 = 20. Weighted recall = (5 × 0.90 + 15 × 0.55) / 20.
Step 2	Numerator: 4.5 + 8.25 = 12.75. Divide by 20: 0.6375.
Step 3	The 90% was measured where you tuned; the fair number is overall 64% (and you should say so).
Answer	Report 64% overall (assuming equal numbers of regressions per service), and give the split (90% tuned services, 55% others).
Check	0.6375 lies between 0.55 and 0.90, and is closer to 0.55 because 15 of 20 services have it.
Why it matters	Headlines from the best subset are a quiet form of dishonesty. You would be the one on call when the detector misses.

Problem 10.6.2 You find that a public chatbot leaks other users' chat titles when you vary a URL parameter. What do you do?

Step 1	Stop testing as soon as you have confirmed the problem; do not read or save other people's data.
Step 2	Report privately to the owner through their security contact, with steps to reproduce and no real user data.
Step 3	Give reasonable time to fix before any public write-up; publish details only after the fix, or not at all.
Answer	Private report with a reproducible description, minimal access, and delayed disclosure.
Check	You never published user data, and the owner could act on your report.

Where it is used

Every data and ML role handles personal data. Interviews for ML roles include fairness and privacy questions (Chapter 7). Regulated domains (banking, health) audit these practices.

Applications

- **Banking and insurance:** fairness checks on credit and claims models.
- **Healthcare:** consent and de-identification.
- **GenAI:** data licensing, prompt-injection and leakage risks.
- **Open source:** licences and attribution.

COMMON MISTAKE

Treating ethics as a final checkbox. Raise data rights, bias and misuse at the design stage (Chapter 9, step 11), when changes are cheap.

Check yourself

1. Name two honest-reporting habits. Answer: report all seeds and negative results; keep the baseline fairly tuned.
2. What is responsible disclosure? Answer: telling the owner privately and allowing time to fix before publishing.
3. Why check fairness by group? Answer: a good average can hide poor results for a smaller group.

Chapter recap

- Read in three passes: filter in 10 minutes, understand in about an hour, rebuild only the few that matter.
- Keep a paper log: title, 3-line summary, key idea, what you would test.
- Reproduce by versioning code, data, config, environment and seeds, and report mean and spread over several seeds.
- A gain smaller than run-to-run noise is not evidence; single runs fooled us about 19% of the time in our simulation.
- A good experiment has a falsifiable hypothesis, a fair baseline, controls, randomization and a final untouched test set.
- Many tests inflate false positives (64% at 20 tests); correct or confirm on fresh data.
- Ablations remove one part at a time; compare each drop to the noise.
- Write papers and posts in a clear structure with numbers and limits, and keep ethics in the design from the start.

Pattern cheat-sheet

Need	Pattern
Three passes	10 min scan, 1 h understand, hours to rebuild
Standard error of a mean difference	$\sqrt{(s_1^2/n + s_2^2/n)}$
Seeds needed	$se = s \times \sqrt{(2/n)}$; want gain ≥ 2 × se
False positives over m tests	$1 - 0.95^m$; Bonferroni $0.05 / m$
Ablation drop	`full − without part` vs noise
Reproduction report	claim, setup, result ± spread, gap, causes, verdict
Abstract	problem, method, result with number, limit

Practice set

1. (Easy) Pass 1 takes 10 minutes. How many papers can you scan in 2 hours? **Answer:** 120 / 10 = 12.

2. (Easy) What do you note in the paper log? **Answer:** title, link, 3-line summary, key idea, what you would test, date.
3. (Easy) Five runs: 90, 91, 92, 91, 91. Mean? **Answer:** 455 / 5 = 91.
4. (Medium) Five runs above, sample standard deviation? **Answer:** deviations −1, 0, 1, 0, 0; squares 1, 0, 1, 0, 0 sum 2; variance 2 / 4 = 0.5; standard deviation 0.707.
5. (Medium) False-positive chance for 10 tests at 0.05? **Answer:** $1 - 0.95^{10} = 1 - 0.599 = 0.401$.
6. (Medium) What is the output? Ablation: full 88.0, without part A 87.9, noise 0.4. **Answer:** The drop is 0.1, below the noise; no evidence that A matters.
7. (Medium) Find the bug: "We tuned the learning rate on the test set and report test accuracy 92%." **Answer:** The test set was used for choices, so the number is optimistic. Tune on validation data and test once.
8. (Hard) A paper claims +1.0 over baseline with one run each. You measure a run-to-run standard deviation of 0.6. How likely is a gap of at least 1.0 from luck alone? **Answer:** Spread of the difference $0.6 \times \sqrt{2} = 0.849$; 1.0 / 0.849 = 1.18; the chance of exceeding is about 0.12 (one tail), so luck alone could do it about 1 time in 8.
9. (Hard, interview) "Tell me about a paper you read recently." **Answer:** Use problem, idea, evidence, doubt, use: "Problem X; they do Y; they show Z on one dataset without variance; I doubted the baseline tuning; I reproduced a small version and saw about 1 point less; I would use it only with testing on our data."
10. (Hard, interview) "Your offline metric improved 2% but the online A/B test shows no change. Why?" **Answer:** Possible reasons: the offline metric does not match the business goal, leakage in the offline set, a time or population shift, noise in either test, or too little power online. Check logs for feature skew and run the test longer.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in
India moving into data and ML roles; English is my second language. Topic:
reading research papers with the three-pass method. I will paste the
abstract and figure captions of a paper: [PASTE HERE]. Walk me through
pass 1, ask the ten critical questions one at a time, check my answers,
and help me write a 3-line summary for my paper log.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into ML. Topic: reproducing a published result and judging it
against seed-to-seed noise. Give me 6 graded exercises with small sets of
run results, without solutions. Ask me to compute mean, standard
deviation, a standard error of the difference, and decide if a claimed
gain is real. Then check my reasoning step by step.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data science. Topic: experimental design (hypothesis,
baseline, controls, confounders, power, multiple comparisons, held-out
test set). I will describe an experiment I plan: [PASTE HERE]. Find the
confounders, name what I should fix, estimate how many runs I need, and
write the final plan in a short checklist.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into applied AI. Topic: ablation studies and applied research
projects. Help me design a 4-week project that uses data I can legally
access (for example anomaly detection on delivery-time data). Ask about my
data, propose a baseline, an ablation table and a risk list, and give me
an hour-by-hour plan for 1 hour a day.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data. Topic: writing a short research-style summary or
technical blog post, and research ethics. Here is my draft: [PASTE HERE].
Check the structure (problem, method, result with a number, limits),
simplify my English, flag any claim without evidence, and list any ethics
concerns such as data rights, privacy or bias that I should address.
```

CHAPTER 11

PORTFOLIO, GITHUB, LINKEDIN AND RESUME

How strangers decide in 30 seconds whether to talk to you, and how to make that decision easy.

A recruiter opens your resume for about the time it takes to drink a sip of tea. A hiring manager opens your GitHub link if the resume made them curious. A referrer checks your LinkedIn before putting their name on you. These three places tell one story or three confusing ones. This chapter shows you how to build proof (a small set of strong projects), how to present the proof (README, GitHub, resume bullets with numbers, LinkedIn headline) and how to write your first public post. You already own good raw material: years of finding bottlenecks, explaining bad news and reading percentiles. By the end you will have a project plan with hours, a README that a stranger can run, five resume bullets with impact numbers, a headline, and a first post draft. Chapter 4 gave the README and post templates; Chapter 7 covered interview stories. Here we assemble them into a hireable profile.

11.1 Portfolio Strategy: Fewer, Stronger Projects

INTUITION

A restaurant with 30 dishes done badly loses to a restaurant with 6 dishes done well. A hiring manager has little time. Six solid projects show range, depth and finishing power. Thirty half-done tutorials show only that you started things.

The rule and the mix

Aim for **6 to 8 strong projects**, not 30. A project is *strong* when it has:

- a business or problem statement in one sentence;
- a named data source (public or simulated; never company data);
- an approach, and results **with numbers**;
- "what I would do next";
- an architecture diagram (Chapter 8 and 9 style);
- clear steps to run it, so anyone can reproduce it;

- some tests;
- a demo where possible (a hosted app, or a 2 to 3 minute video);
- a short blog post: problem, approach, results, mistakes (Chapter 10.5).

A good mix for a working professional moving to data:

Type	How many	Example idea
Analytics	2	Sales dashboard with a written recommendation (Chapter 5)
Data engineering	2	Event pipeline with orchestration and data-quality tests
ML or deep learning	2	Churn model with a proper time split and error analysis
GenAI or agents	1 to 2	RAG chatbot with a golden evaluation set (Chapter 9.5)
MLOps or operations	1	Traffic test of an ML endpoint with a cost model

Your signature pieces

Most applicants build the same Titanic and house-price projects. You can build four things that stand out:

1. **Daily order files to warehouse to dashboard** (a food-delivery or shop example), with slowdown detection across weeks.
2. **Anomaly detection on real monitoring metrics**, with documented precision and recall.
3. **Traffic test of a data pipeline or ML endpoint** (for example with Locust), with bottleneck analysis and a cost model.
4. **A blog series**: "What my first job taught me about data engineering."

Use public or simulated data only. Never publish company logs, client names or internal numbers. If a project is inspired by your work, rebuild it from scratch on open data, and check your employment contract and policy first.

Problem 11.1.1 Plan 8 projects (2 + 2 + 2 + 1 + 1) at about 25 hours each. At 7 hours a week, how long do they take? How long do the first 3 take?

Formula	weeks = total hours / hours per week
Plug in	8 × 25 = 200 hours; 200 / 7. First 3: 3 × 25 = 75; 75 / 7.
Compute	200 / 7 = 28.6 weeks; 75 / 7 = 10.7 weeks.

```
projects = {"analytics": 2, "data engineering": 2, "ML/DL": 2, "GenAI": 1,
"MLOps/operations": 1}
n = sum(projects.values())
hours_each = 25
total = n * hours_each
print("projects:", n, " hours:", total, " weeks at 7 h/week:", round(total
/ 7, 1))
first3 = 3 * hours_each
print("first 3 projects hours:", first3, " weeks:", round(first3 / 7, 1))
```

```
projects: 8  hours: 200  weeks at 7 h/week: 28.6
first 3 projects hours: 75  weeks: 10.7
```

- **Answer** All 8 projects take about 29 weeks (7 months) of study time. The first three take about 11 weeks, which fits the 12-week plan of Chapter 7. Start applying when 3 strong projects are ready; keep building the rest.
- **Check** 28.6 × 7 = 200.2 hours, matching 200.
- **Why it matters** An honest time plan stops you from promising yourself 8 projects in 2 months and quitting at project 2.

Problem 11.1.2 You have one free weekend for a project and a target role of Data Engineer. Pick from: (a) a Kaggle Titanic notebook, (b) a pipeline that loads daily open weather data into SQLite with tests and a schedule, (c) a sentiment model on movie reviews. Which one, and why?

Step 1	Role: Data Engineer, so the proof needed is ingestion, quality and scheduling.
Step 2	(a) and (c) show modelling, which is a different role. (b) shows the pipeline skills, data tests and a run-it-yourself setup.
Step 3	Scope it to fit a weekend: one source, one table, three tests (no nulls, no duplicates, row count above zero), a scheduled run, a README.
Answer	Choose (b). Match the project to the role in the job description.
Check	Count the JD keywords it covers (pipeline, scheduling, SQL, testing): four.

The picture

```
Resume bullet -> "see GitHub: repo link"  -> README (30 s) -> demo or
diagram -> blog post (details)
                      ^                                              |
                      +------------ LinkedIn Featured section --------+
```

Each layer lets a reader stop when they are convinced or go deeper when curious.

Connection to the real job

Interviewers pick questions from your projects ("why SQLite and not Postgres?"). A project you cannot explain for 3 minutes is a liability; remove it from your resume (Chapter 7.4).

Applications

- **Data analyst:** dashboards with a clear recommendation.
- **Data engineer:** pipelines with tests and scheduling.
- **ML engineer:** a model plus serving and monitoring.
- **AI engineer:** RAG or agent with evaluation and guardrails.

COMMON MISTAKE

Copying a tutorial and listing it as a project. Change the data, add your own question, measure something, and write what went wrong.

REAL JOB TIP

Write the blog post while you build, not after. Notes from the first week make the best "mistakes I made" section.

Check yourself

1. How many strong projects do you need? Answer: 6 to 8; start applying after about 3.
2. Which data may you use? Answer: public or simulated data, never company data.
3. Name one signature project. Answer: daily order files into a warehouse with slowdown detection.

11.2 The Project README and Your GitHub Profile

INTUITION

Your GitHub page is a shop window. A visitor looks for 30 seconds. If the window is clean, labelled and shows the best items, they step in. If it shows 40 forked repos with no description, they walk on. The README is the label on each item.

The README template for a portfolio project

Chapter 4.2 gave the basic README. A portfolio project adds proof. Copy this and fill it in.

```
# Project name: one-line promise
What problem it solves, for whom, in one or two sentences.

## Result
One headline number or screenshot.
Example: "Detects delivery-time slowdowns with 92% precision and 85%
recall on 40 simulated weeks."

## Demo
Link to hosted app or a 2–3 minute video. Screenshot if neither exists.

## Architecture
Diagram (image) and 5 bullets: what each box does and why it is there.

## How to run (reproducible)
1. git clone <repo>
2. python -m venv .venv && source .venv/bin/activate
3. pip install -r requirements.txt
4. python -m pytest        # tests pass
5. python run.py           # prints the result above

## Data
Source, size, license, how to download. Never real secrets or private
data.

## Method and results
Baseline, what I tried, table of numbers, what worked and what failed.

## Decisions and limits
What I chose and why (link to ADR); what does not work yet.

## Next steps
Three bullets.
```

The numbers in the `Result` example are an example for the template, not a real result. Your README must contain only numbers you measured.

GitHub hygiene checklist

- **Pin 6 repositories** (the profile allows 6): your best, one per type.
- **Profile README:** a repository with the same name as your username holds a short intro. Write 5 lines: who you are, what you build, your top 3 projects with links, how to reach you.
- **Each repo:** a description, topics (labels), a licence (a file stating how others may use the

code), a `.gitignore`, and no large data files.

- **Secrets:** never commit passwords, keys or tokens. Use environment variables and add `.env` to `.gitignore`. If a secret is committed, revoke it at once (deleting the file does not remove it from history).
- **Commit messages:** short and clear ("Add null-check test for orders"), not "final2".
- **History:** steady small commits show real work; one giant commit looks pasted.
- **Tests and CI:** a few tests and a simple automated test run give confidence (GitHub Actions can run tests on every push; check current free limits).
- **Tidy:** delete or archive dead repos; forks you never changed add noise.

Problem 11.2.1 A stranger lands on your repository. Within 30 seconds they read the title, the first paragraph and the first screen of the README. Using the template, which three sections must be on the first screen?

Step 1	The reader asks: what is it, does it work, can I see it? That is Title and promise, Result, Demo.
Step 2	Architecture and run steps can sit below; they serve the reader who decided to stay.
Answer	Title with a one-line promise, Result with one number, and Demo (link, video or screenshot) on the first screen.
Check	Cover the page below the third heading: can you still say what the project does and how well it works? If yes, the order is right.

Problem 11.2.2 (Predict, run, explain) A quick script checks that a repository folder has the needed files. Before you run it, which of the three files do you think will be flagged missing?

```
import os, tempfile
d = tempfile.mkdtemp()
for name in ["README.md", "requirements.txt", ".gitignore"]:
    if name != ".gitignore":
        open(os.path.join(d, name), "w").close()
need = ["README.md", "requirements.txt", ".gitignore", "LICENSE"]
missing = [n for n in need if not os.path.exists(os.path.join(d, n))]
print("missing:", missing)
```

Run it with `python3 file.py`; the real output is:

```
missing: ['.gitignore', 'LICENSE']
```

- **Step 1** `tempfile.mkdtemp()` makes an empty temporary folder.
- **Step 2** The loop creates `README.md` and `requirements.txt` as empty files and skips `.gitignore` on purpose.
- **Step 3** The list comprehension keeps every needed name whose path does not exist: `.gitignore` and `LICENSE`.
- **Answer** The script prints `.gitignore` and `LICENSE` as missing.
- **Check** Four names needed, two created, so two missing.
- **Why it matters** A ten-line script like this can run before every portfolio push; it is the same habit as a smoke test (a quick test that the basics work).

Connection to the real job

Teams read READMEs daily; a clear one saves onboarding days. Hiring managers for data and ML roles often open a pinned repo before the interview.

Applications

- **Hiring:** a clean repo is proof of working style.
- **Open source:** the same README and tests earn pull-request acceptance (Chapter 14).
- **Teamwork:** every internal tool needs a run-it-in-5-minutes README.

COMMON MISTAKE

A README that says "This project does X" with no way to run it and no numbers. Add a result line and a command that prints it.

COMMON MISTAKE

Committing a secret or a dataset with personal data. Check `git status` and your `.gitignore` before every push.

Check yourself

1. How many repositories can you pin? Answer: 6.
2. What should the first screen of a README show? Answer: promise, result, demo.

3. What do you do if you commit a password? Answer: revoke it immediately; deleting the file does not remove it from history.

11.3 Resume Bullets With Impact Numbers

INTUITION

A resume bullet is an advertisement of one line. "Responsible for delivery operations" tells the reader what your role title was. "Cut delivery time 40% by finding a slow dispatch step" tells the reader what you *achieved*. People hire results, not job descriptions.

The bullet formula

`Action verb + what you did + how (tool or method) + result with a number + scale`

Examples of numbers you can use (all must be true and explainable): percent improvement, time saved, users or requests, volume of data, money saved or protected, error rate, number of teams or releases. Percent improvement is `(old − new) / old × 100`, from the learner's math work.

Rules:

- **One page.** Reverse chronological order. Skills and projects near the top for a career changer.
- **Start with a strong verb:** built, reduced, automated, designed, found, wrote, led, shipped.
- **Avoid weak openings:** "responsible for", "worked on", "helped", "involved in".
- **One idea per bullet, about 12 to 28 words.**
- **Keywords from the job description**, but only for skills you actually have. If you cannot talk about it for 3 minutes in an interview, delete it.
- **Truth:** every number must be real and defensible. If you do not have an exact number, give a range or a careful estimate and be ready to explain how you got it. Never invent.
- **Confidential details:** no client names, internal system names or secret numbers if your contract forbids; say "a large Indian bank" if allowed, or "a retail banking application".

Problem 11.3.1 Compute the percent improvements for four real-looking results and write each as a bullet fragment.

```
def pct(old, new): return round((old - new) / old * 100, 1)
print("delivery time 50 -> 30 min:", pct(50, 30), "% lower")
print("report preparation 6 h -> 90 min:", pct(360, 90), "% shorter")
print("orders handled 450 -> 900 per hour:", round((900 - 450) / 450 *
100), "% higher")
print("cost 40000 -> 31000 per month:", pct(40000, 31000), "% lower")
```

```
delivery time 50 -> 30 min: 40.0 % lower
report preparation 6 h -> 90 min: 75.0 % shorter
orders handled 450 -> 900 per hour: 100 % higher
cost 40000 -> 31000 per month: 22.5 % lower
```

- **Step 1** Delivery time: `(50 - 30) / 50 = 20 / 50 = 0.40` , so 40% lower.
- **Step 2** Report preparation: the same units are needed. 6 hours is 360 minutes: `(360 - 90) / 360 = 0.75` , so 75% shorter.
- **Step 3** Orders handled per hour go *up*, so the base is the old value: `(900 - 450) / 450 = 1.0` , which is 100% higher (it doubled).
- **Step 4** Cost: `(40000 - 31000) / 40000 = 0.225` , so 22.5% lower.
- **Answer** 40% lower delivery time, 75% shorter report preparation, orders per hour doubled, 22.5% lower monthly cost.
- **Check** Reverse the first: 50 × (1 − 0.40) = 30. Correct. (These numbers are for practice; use your real ones.)
- **Why it matters** Unit mistakes (hours versus minutes) are the commonest error in impact numbers, and an interviewer will recompute them.

Problem 11.3.2 (Predict, run, explain) A small "bullet checker" scores a bullet: 2 points for a digit, 1 point for not starting weakly, 1 point for 12 to 28 words. Predict which of the four bullets scores 4 out of 4.

```
import re
bullets = [
 "Responsible for operations of a food-delivery app.",
 "Worked on reports and dashboards for many projects.",
 "Cut 95th-percentile delivery time from 50 to 30 minutes (40%) by finding
a slow dispatch step in an app serving 5,000 orders a day.",
 "Built a Python tool that loads daily order files into a warehouse and
flags late-delivery spikes; used by 3 teams on 12 weekly reports.",
]
weak_starts = ("responsible for", "worked on", "helped", "involved in",
"assisted")
for b in bullets:
    words = len(b.split())
    has_num = bool(re.search(r"\d", b))
    weak = b.lower().startswith(weak_starts)
    score = (2 if has_num else 0) + (0 if weak else 1) + (1 if 12 <= words
<= 28 else 0)
    print(f"score {score}/4 | words {words:2d} | number {has_num!s:5} |
weak start {weak!s:5} | {b[:48]}...")
```

```
score 0/4 | words  7 | number False | weak start True  | Responsible for
operations of a food-delivery ap...
score 0/4 | words  8 | number False | weak start True  | Worked on reports
and dashboards for many projec...
score 4/4 | words 24 | number True  | weak start False | Cut 95th-
percentile delivery time from 50 to 30 ...
score 4/4 | words 24 | number True  | weak start False | Built a Python
tool that loads daily order files...
```

- **Step 1** `re.search(r"\d", b)` looks for any digit. `startswith(weak_starts)` accepts a tuple of prefixes.
- **Step 2** Bullets 1 and 2 start weakly, have no digit and are short, so score 0.
- **Step 3** Bullets 3 and 4 have numbers, strong verbs (Cut, Built) and 21 to 23 words, so score 4.
- **Answer** Bullets 3 and 4 score 4 of 4.
- **Check** The checker is only a rough filter. It cannot tell whether a number is true. You are the final judge.
- **Why it matters** Writing a rule down makes it visible. Run your own bullets through the check and rewrite any below 3.

Problem 11.3.3 Rewrite the weak bullet "Worked on reports and dashboards for many projects"

using these true facts: you built dashboards for 14 cities, you found a dispatch limit that caused late deliveries at 800 orders an hour, and raising the limit cut the late-delivery rate from 6% to 0.4%.

Step 1	Action and result: found the cause, fixed it, late-delivery rate improved.
Step 2	Scale: 14 cities for context; the dispatch finding is the highlight.
Step 3	Write: "Found a dispatch limit that caused 6% late deliveries at 800 orders an hour; the fix cut them to 0.4% (93% lower), across dashboards for 14 cities."
Answer	The bullet above (27 words, inside the 12 to 28 word range).
Check	`(6 - 0.4) / 6 = 5.6 / 6 = 0.933`, so 93% lower. Every claim comes from the facts you gave.

A one-page structure for a career changer

```
Name | city | email | LinkedIn | GitHub
Headline line: IT professional moving into data engineering (same as
LinkedIn)
Skills: SQL, Python, Airflow, Git, ... (only what you can explain)
Projects (3 strong, each 2 bullets with numbers and a link)
Experience (current role: 4 to 5 bullets that show data and systems
thinking)
Education and certifications
```

Chapter 12 covers applications, referrals and the job-search funnel. Its illustrative planning numbers (60 applications giving about 15 recruiter screens) show why the first screen of your resume matters.

Connection to the real job

The same formula appears in performance reviews and promotion documents: result, number, scale. Writing bullets now trains the habit.

Applications

- **All data roles:** bullets with metrics get past the first screen.
- **Internal transfers:** a one-page "impact summary" helps managers.
- **Performance reviews:** the same bullets feed your self-review.

COMMON MISTAKE

Inventing or inflating a number. Interviewers ask "how did you measure that?" Keep the evidence (a report, a graph) in your own notes.

INTERVIEW TIP

For each bullet, prepare a 2-minute STAR story (Chapter 7.3). The bullet is the headline; the story is the article.

Check yourself

1. Percent change formula? Answer: `(old − new) / old × 100` for a reduction; use the old value as the base.
2. Name two weak bullet openings. Answer: "responsible for", "worked on".
3. Can you list a tool you cannot explain? Answer: no; remove it.

11.4 LinkedIn: Headline, About and Featured

INTUITION

Recruiters search LinkedIn the way you search a database: by keywords and filters. The headline is the title of your row. A headline that says only your current job title is found by people looking for that job, not the one you want.

The parts that matter

- **Headline:** the line under your name, which also appears in search results and comments. It is limited in length (about 220 characters at the time of writing; check the current limit). Use the formula: *what you are now moving to* + *key skills* + *proof or niche*.
- **About:** 3 short paragraphs: who you are, what you built (with 1 or 2 numbers), what you want next.
- **Featured:** pin your best repository, blog post and demo video.
- **Experience:** reuse your resume bullets (shorter lines).
- **Skills:** 10 to 20 real skills; ask a colleague to endorse the main ones.
- **Recommendations:** ask 2 or 3 colleagues or managers who saw your work; give them 3

facts to mention.

- **Photo, location, links:** a clear face photo, your city, and your GitHub link.
- **Open-to-work signals and privacy:** LinkedIn has settings to tell only recruiters you are looking, hiding it from your current employer's company page. Check the current settings; they change.
- **Confidentiality:** do not post client names, internal numbers or screenshots from work.

Problem 11.4.1 Compare three headlines by length and by search value.

```
heads = [
 "IT Support Analyst at XYZ",
 "IT professional moving into data engineering | SQL, Python, Airflow |
Reliable data pipelines",
 "Aspiring Data Scientist | Passionate | Hardworking | Seeking
opportunities",
]
for h in heads: print(len(h), "chars:", h)
post = "Last month I loaded 2 million delivery records into SQLite and
found that late deliveries doubled on every Friday evening. Here is the
12-line query. What would you check next?"
print(len(post.split()), "words,", len(post), "chars")
```

```
25 chars: IT Support Analyst at XYZ
93 chars: IT professional moving into data engineering | SQL, Python,
Airflow | Reliable data pipelines
74 chars: Aspiring Data Scientist | Passionate | Hardworking | Seeking
opportunities
30 words, 176 chars
```

- **Step 1** Headline 1 (25 characters) wastes space and a recruiter searching "data engineer" will not find it.
- **Step 2** Headline 3 (74 characters) has empty words ("Passionate", "Hardworking", "Seeking opportunities") and no skill keywords. Every candidate says these.
- **Step 3** Headline 2 (93 characters, well below 220) names the target role, three skills and your niche. The last output line is a different check: a three-sentence mini post is 30 words and 176 characters, which is a good size for a post opening.
- **Answer** Headline 2 is best: target role, searchable skills and a story in 93 characters.
- **Check** Count the keywords a recruiter might search: data engineering, SQL, Python, Airflow, data pipelines. Headline 2 contains all five.

- **Why it matters** Some views cut a long headline short, so put the target role first.

Problem 11.4.2 Write your About section in three short paragraphs using this skeleton. Fill the brackets with your own true facts.

```
Paragraph 1 (who): I am a working professional with [N] years in [domain],
moving into data engineering.
Paragraph 2 (proof): I built [project] that [result with number]. I use
[tools].
Paragraph 3 (next): I am looking for [role] work where [what you want to
do]. I write about [topic] at [link].
```

- **Step 1** Say who you are in one sentence with years and domain, then the move you are making.
- **Step 2** Pick one project and one number, and name 3 tools.
- **Step 3** State the role you want in plain words; end with where readers can see more.
- **Answer** A 70 to 120 word About section that a recruiter can read in 20 seconds.
- **Check** Does it contain the target role, one real number and a link? If yes, it is complete.

The picture: what a recruiter sees first

```
[photo]  Your name
         Headline (put the target role first; some views cut it short)
         City | Contact | GitHub
         About (first 2–3 lines before "see more")
         Featured: repo | blog | demo video
```

Connection to the real job

Your profile is searchable data. Referrals (Chapter 12) arrive faster when a stranger can verify you in a minute. Many companies check LinkedIn and GitHub together.

Applications

- **Job search:** recruiters and referrers find you by keywords.
- **Networking:** your headline appears with every comment you leave.
- **Learning in public:** posts bring people to a profile that makes sense.

COMMON MISTAKE

Keeping the headline as your current job title only, or filling it with adjectives. Put the target role and the skills.

COMMON MISTAKE

Claiming skills or titles you do not hold. Say "moving into" and show projects.

Check yourself

1. Which three things go in a headline? Answer: target role, key skills, proof or niche.
2. Name two things to pin in Featured. Answer: best repository, blog post or demo video.
3. Can you post client names? Answer: no.

11.5 Writing Your First Post (and Keeping a Rhythm)

INTUITION

Your first post is like the first day of a new restaurant: it will not be perfect, and its purpose is to find out how things really run. Nobody remembers it. What they notice is whether you kept going.

What to write

Chapter 4.4 gave the weekly-post template and the ethics of posting. Your first post should be a small project write-up or a "what I learned" note with **one real number**. Good first topics:

- "What 2 million delivery records taught me about percentiles."
- "I rebuilt a rate limiter in Python (token bucket) and tested it with a small script that sends many requests."
- "My first data pipeline: three bugs I made."

Structure in six lines:

1. **Hook:** one sentence with a number or a surprise.
2. **Problem:** what you tried to do.
3. **What I did:** 3 to 5 lines, with a small code snippet or screenshot.

4. **Result:** the number.
5. **Mistake or lesson:** one honest line.
6. **Question:** invite a reply.

Problem 11.5.1 Draft your first post from this true experience (replace with yours): you loaded 2 million delivery records into SQLite and noticed that late deliveries doubled on every Friday evening.

Step 1	Hook: "Last month I loaded 2 million delivery records into SQLite and found that late deliveries doubled on every Friday evening."
Step 2	Problem: "I wanted to see trends across weeks, not single reports."
Step 3	What I did: "I wrote a 12-line SQL query with a window function (a function that looks at neighbouring rows) to compare each week to the previous one."
Step 4	Mistake: "My first query used the average, which hid the spike. The 95th percentile showed it."
Step 5	Question: "What would you check next?"
Answer	A post of about 120 words once you add your details. The short three-sentence version (hook, query, question) measured in 11.4.1 is 29 words and 160 characters.
Check	The draft contains one number (2 million), one surprise (doubled on Fridays), one mistake and one question. It contains no company or client name.
Why it matters	Posts with a concrete number and an honest mistake are read and remembered; vague advice is scrolled past.

Problem 11.5.2 Plan a rhythm for the next 3 months, at about 2 posts per month plus five new connections per month (a number from the skill map). How many posts and connections is that, and how much time at 45 minutes per post?

Formula	posts = 2 × 3; connections = 5 × 3; time = posts × 0.75 hours
Plug in	2 × 3 = 6; 5 × 3 = 15; 6 × 0.75
Compute	6 posts, 15 connections, 4.5 hours
Answer	6 posts and 15 genuine connections in 3 months, costing about 4.5 hours of writing. Over a 7-hour week that is small.
Check	4.5 hours / 13 weeks = 0.35 hours per week (about 21 minutes).
Why it matters	A tiny, steady habit beats a big burst that stops after two weeks (Chapter 2).

Connecting well, not loudly

Send a connection request with a short note ("I read your post on rate limiting; I am a working professional moving into data and would like to follow your work"). Comment with something useful, not "great post". Never ask strangers only for a job; ask a specific question, offer thanks, and share what you made. Chapter 14 covers mentors and communities.

Connection to the real job

Writing is a daily part of data work (Chapter 4). A public post is also a small proof that you can explain things, which every team values.

Applications

- **Referrals:** people who know your work refer you more readily.
- **Interviews:** a post can start the interviewer's first question.
- **Learning:** teaching forces understanding (Chapter 2).

COMMON MISTAKE

Waiting until the post is perfect. Publish at "good and honest"; improve the next one.

REAL JOB TIP

Keep a running file of "things I fixed this week". Each line is a future post or resume bullet, and a ready story for the interview.

Check yourself

1. What must a first post contain? Answer: a hook with a number, the problem, what you did, the result, a lesson, and a question.
2. Can you post work screenshots? Answer: no, not confidential ones; use public or simulated data.
3. How many connections per month does the plan suggest? Answer: five genuine ones.

Chapter recap

- Build 6 to 8 strong projects, not 30; start applying after about 3 are solid.
- Use signature projects: order-files-to-warehouse, anomaly detection, traffic tests of pipelines and endpoints.
- Every project needs a problem, data source, numbers, a diagram, run steps, tests and a short blog post.
- A README shows promise, result and demo on the first screen; GitHub needs pinned repos, a profile README, licences and no secrets.
- Resume bullets follow: action, what, how, result with number, scale; use percent change `(old − new) / old × 100` and keep units consistent.
- Never invent a number; never list a skill you cannot explain for 3 minutes.
- The LinkedIn headline names the target role, key skills and your niche; About and Featured carry proof.
- Publish a first post with one real number and one honest mistake, then keep a light rhythm.

Pattern cheat-sheet

Need	Pattern
Bullet	`Verb + what + how + result with number + scale`
Percent reduction	`(old − new) / old × 100`
Percent increase	`(new − old) / old × 100`
Headline	`target role + skills + niche` (about 220 characters max; check)
README order	promise, result, demo, architecture, run, data, method, limits, next

First post	hook, problem, what I did, result, lesson, question
Time plan	`projects × hours / hours per week`

Practice set

1. (Easy) Latency fell from 500 ms to 350 ms. Percent reduction? **Answer:** 150 / 500 = 30%.
2. (Easy) Throughput rose from 200 to 260 tps. Percent increase? **Answer:** 60 / 200 = 30%.
3. (Easy) How many repositories can you pin on GitHub? **Answer:** 6.
4. (Medium) A job takes 2 hours and falls to 45 minutes. Percent shorter? **Answer:** 2 hours = 120 minutes; (120 − 45) / 120 = 62.5%.
5. (Medium) What does the bullet checker (11.3.2) score for "Built a tool for testing"? **Answer:** no digit (0), strong start (1), only 5 words (0), so 1 out of 4.
6. (Medium) Find the bug: a README says "pip install requirements.txt". **Answer:** it should be `pip install -r requirements.txt` ; without `-r` , pip looks for a package named requirements.txt.
7. (Medium) Which is a better headline: "Data Enthusiast" or "IT professional moving into data engineering | SQL, Python | reliable pipelines"? **Answer:** the second; it has a target role, searchable skills and a niche.
8. (Hard) Rewrite "Helped the team improve release quality": you wrote 120 automated checks for 4 releases and caught 7 defects before production. **Answer:** "Wrote 120 automated checks across 4 releases and caught 7 defects before production." (Starts with a verb, has 3 numbers, 13 words.)
9. (Hard, interview) "Walk me through your portfolio project." **Answer:** Use a 2-minute structure: problem in one sentence; data and why; architecture in three boxes; one result with a number; one mistake and its fix; what you would do next; why you chose the main tool over an alternative.
10. (Hard, interview) "Your resume says 40% delivery-time improvement. How did you measure it?" **Answer:** Name the setup (same period, same order volume, several weeks), the metric (95th percentile), the before and after values (50 to 30 minutes), and the cause (a slow dispatch step). Keep the report to show.

Prompt pack

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional in India moving into data roles; English is my second language. Topic: planning a portfolio of 6 to 8 strong projects. My target role is [Data Engineer / Data Scientist / ML Engineer / AI Engineer]. Ask me five questions about my background, then propose a project list with the mix of types, hours per project at 7 hours a week, and one signature project that uses my performance-testing experience.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data. Topic: writing a portfolio project README and cleaning my GitHub profile. Here is my README: [PASTE HERE]. Review it against this order: promise, result, demo, architecture, how to run, data, method, limits, next steps. Tell me what the first screen shows, list any missing pieces, check for secrets or private data, and rewrite my first paragraph in simple English.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data. Topic: resume bullets with impact numbers. I will paste five bullets and the true facts behind each: [PASTE HERE]. Rewrite each as: strong verb, what, how, result with a number, scale, in 12 to 28 words. Check every percent calculation with me step by step. Do not invent facts; ask me when a number is missing.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional in India moving into data. Topic: LinkedIn headline, About and Featured section. My target role is [ROLE]. Here is my current profile text: [PASTE HERE]. Give me three headline options with character counts, rewrite my About in three short paragraphs, and list what to pin in Featured. Warn me about confidentiality and claims I cannot support.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data. Topic: writing my first public post. Ask me for one small project or lesson I learned, with one real number. Then help me
```

```
write a post of about 120 words: hook, problem, what I did, result,
mistake, question. Simplify my English, remove anything confidential, and
suggest a title and a posting plan for the next four weeks.
```

CHAPTER 12

JOB SEARCH IN INDIA

A job search is a funnel you can measure, improve and run calmly. It is not a lottery.

You have a resume (Chapter 11) and interview skills (Chapter 7). Now you must put them in front of the right people and turn interviews into an offer you are happy with. This chapter shows how to treat the search as a pipeline with numbers, where referrals beat cold applications, which type of company fits which goal, how to move inside your own company, how to research salary without guessing, how to negotiate with facts, how to plan your notice period, and how to keep a rejection log that makes each failure useful. By the end you will have a weekly routine and a tracking sheet. All money figures in this chapter are **made-up practice numbers**. Salaries change with city, year and company, so I will show you how to find real data, not give you a number to quote.

12.1 The Search as a Funnel

INTUITION

Think of a shop with several stages: enter, browse, fill a basket, pay. Not every visitor reaches the till. If 1,000 walk in and 16 pay, you do not blame the till alone. You look at where visitors drop. A job search is the same: applications drop at the resume screen, the technical round, the final round.

Words you need

Term	Meaning
Funnel	A chain of stages where only some candidates pass each stage
Conversion rate	The share that passes a stage: passed / entered
ATS	Applicant Tracking System, software that stores and filters resumes
Referral	A current employee recommends you; your resume goes to a person, not only a pile

Pipeline	All your live applications at all stages right now

Job search funnel: 60 applications to 1 offer, with illustrative stage rates

Read the picture top to bottom. Each bar is a stage, and the number is how many candidates are still alive. The percentages are the share that passed the previous stage. The note at the bottom says the rates are illustrative. Your own rates will differ, and you should measure them.

Problem 12.1.1 Start with 60 applications. Stage pass rates: screen 25%, technical 40%, final 33%, offer 50%. How many offers do you expect?

Formula	count after a stage = count before x pass rate
Plug in	60 x 0.25 x 0.40 x 0.33 x 0.50
Compute	60 x 0.25 = 15; 15 x 0.40 = 6; 6 x 0.33 = 1.98 (about 2); 2 x 0.50 = 1
Answer	About 1 offer from 60 applications.
Check	Overall rate = 0.25 x 0.40 x 0.33 x 0.50 = 0.0165. Then 60 x 0.0165 = 0.99, about 1. The code below prints the same.
Why it matters	Plain words: if you send 10 applications and hear nothing, that is normal. It is not a verdict on you.

```
stages=[("applied",1.0),("screen",0.25),("technical",0.40),("final",0.33),
("offer",0.50)]
n=60
for name,r in stages:
    n*= r if name!="applied" else 1
    print(f"{name:10s} rate={r:.2f} count={n:.1f}")
p=0.25*0.40*0.33*0.50
print("overall rate",round(p,4),"apps per offer",round(1/p,1))
```

Before you run it, what do you think the last line prints? Here is the real output:

```
applied    rate=1.00 count=60.0
screen     rate=0.25 count=15.0
technical  rate=0.40 count=6.0
final      rate=0.33 count=2.0
offer      rate=0.50 count=1.0
overall rate 0.0165 apps per offer 60.6
```

Line by line: each loop step multiplies the running count by that stage's rate. The last line uses `1/p` because if 1.65 in 100 applications become offers, you need 100 / 1.65, about 60.6, applications per offer.

Problem 12.1.2 Improve the first stage. A referral raises the screen rate from 25% to 60% (a practice number, not a promise). Later stages stay the same. How many applications do you need for one offer?

Plug in	overall rate = 0.60 x 0.40 x 0.33 x 0.50
Compute	0.60 x 0.40 = 0.24; 0.24 x 0.33 = 0.0792; 0.0792 x 0.50 = 0.0396
Answer	1 / 0.0396 = 25.3 applications, against 60.6 without referrals.
Check	Screen rate rose 2.4 times (0.60 / 0.25), so applications needed should fall by about 2.4 times: 60.6 / 2.4 = 25.3. Yes.
Why it matters	The first stage is the cheapest to improve with a relationship. Interview skill lifts later stages, but that takes weeks of practice.

The picture: where to fix what

```
Low screen rate        -> resume, headline, channel (get referrals)
Good screen, no tech   -> technical prep (Chapter 7)
Good tech, no final    -> system design, communication, stories
Good final, no offer   -> negotiation, fit, compensation expectations
```

Connection to the real job

Treat your search like a small experiment: define a goal (for example 5 quality applications a week), measure each stage, and change one thing at a time. A weak funnel stage tells you what to practise.

Applications

- Data Analyst search: resume keyword fit matters most at the first stage.
- Data Engineer and ML Engineer search: technical-round pass rate often limits you.
- Internal transfer: the funnel has fewer stages but the same logic.

COMMON MISTAKE

Judging the search after 8 applications. With rates like these the result is mostly noise at small numbers. Look at 30 or more applications before drawing a conclusion, and look at stage rates, not the final result only.

Check yourself

1. If 40 of 100 pass screen and 10 of those 40 pass the technical round, what is the technical conversion? Answer: 10 / 40 = 25%.
2. Which stage is cheapest to improve with a person you know? Answer: the first screen, through a referral.
3. Why measure stage rates and not only offers? Answer: they show which skill to fix.

12.2 Channels: Referrals, Platforms and Outreach

INTUITION

A cold application is a letter dropped into a big postbox. A referral is a letter handed to the manager by a colleague he trusts. Both can work, but the second is read first.

The channels

Channel	How it works	Your effort
Referral	An employee submits you through the company's referral system	Highest value; needs relationships
LinkedIn	Jobs, recruiters, and direct messages to engineers and managers	Daily 15 minutes
Company career pages	Direct apply; some roles are posted	Weekly check of 10

	only here	target companies
Naukri, Instahyre, Hirist, Cutshort, Wellfound	Job and recruiter platforms; Wellfound leans to startups	Keep profile current; respond to recruiters
Communities	DataTalks.Club Slack, local meetups, Kaggle and GitHub circles	Give help first
Recruiters and consultants	Third-party recruiters hold roles for client companies	Reply politely; never pay a fee to get a job

Platform names and features change. Check each site yourself and do not assume what was true last year is true now.

REAL JOB TIP

A genuine recruiter never asks you to pay money for a job, an interview or a "registration". If someone does, stop and report it. Also be careful with unknown links and requests for identity documents before an official offer.

How to ask for a referral (a template)

A referral request must be easy to say yes to. Give the person the job link, two lines about you, and a way out.

```
Hi Priya, I saw you work on the data platform team at <Company>.
I am a working professional with 6 years of experience in IT moving into
data engineering.
I built <one project, one number>. There is an open role: <link>.
Would you be comfortable referring me? If it is not a good fit, no problem
at all.
I have attached my resume.
```

Cold outreach to a hiring manager

```
Hi Arjun, I read your post about reducing pipeline delay. I did a similar
analysis on test data
and reduced report time from 40 to 6 minutes (repo: <link>). I am applying
for the Data Engineer
role on your team. May I send you a 3-line summary of how my background
fits?
```

Problem 12.2.1 You send 20 cold messages and get 4 replies, and 2 of those become referrals. What are the reply rate and the referral rate?

Formula	rate = successes / attempts
Plug in	reply 4 / 20; referral 2 / 20
Compute	4 / 20 = 0.20; 2 / 20 = 0.10
Answer	20% reply and 10% referral.
Check	20 x 0.10 = 2 referrals. Yes.
Why it matters	Even 10% is good value: 2 referrals from 20 short messages.

Problem 12.2.2 You have 4 hours a week for the search. Plan it using 3 hours of applications and outreach and 1 hour of tracking. Split the 3 hours: 40% referral outreach, 40% tailored applications, 20% recruiter replies and profile upkeep.

Step 1	3 hours = 180 minutes.
Step 2	40% of 180 = 72 minutes; 40% = 72 minutes; 20% = 36 minutes.
Answer	72 minutes outreach, 72 minutes applications, 36 minutes upkeep, plus 60 minutes tracking.
Check	72 + 72 + 36 = 180. Yes.

Connection to the real job

Your network grows from the habits in Chapters 4 and 5: public posts, small projects, helpful answers. People refer those whose work they have seen.

Applications

- Moving into a new role family: referrals reduce the "no direct experience" filter.
- Switching city: recruiters and communities help when you lack a local network.

COMMON MISTAKE

Messaging a stranger "Please refer me" with no context. State the role, your proof, and make it easy to decline.

Check yourself

1. Name two things a referral message must contain. Answer: the job link and a short proof of fit (resume or project).
2. Should you pay a recruiter to get a job? Answer: no.
3. Why give help in communities first? Answer: it builds trust that later makes referrals possible.

12.3 Company Types: Product, GCC, Service and Startup

INTUITION

Choosing a company type is like choosing a gym. All of them can make you fit, but the equipment, coaching and fees differ. Decide what you want first (learning, stability, money, speed), then choose.

Words you need

Term	Meaning
Product company	Builds and sells its own software or platform (for example an online marketplace, a payments app, a SaaS tool)
GCC	Global Capability Center, an India team owned by a foreign company; also called captive centre
Service company or IT services MNC	Builds software for client companies; you work on client projects

Startup	A young company, small team, fast change, less structure
Consulting firm	Advises clients on strategy and technology, often project and deadline driven

Comparison (general tendencies, not rules)

	Product	GCC	Service MNC	Startup
Work type	Own product, long-term ownership	Global team's work, often real product work	Client projects, varies widely	Broad, many hats
Learning	Deep, if team is strong	Often strong; depends on mandate	Depends on project and client	Fast, but self-driven
Stability	Medium to high	Medium to high, tied to parent's plans	Medium	Lower; funding risk
Pay	Varies; often higher with stock	Varies; check per company	Often lower at the same level, but varies	Varies; stock may be worth nothing
Interview style	Coding, design, case, bar-raiser rounds	Similar to parent's style	Skills screen, project-based	Practical task, founder chat

These are tendencies. A small team inside a large company can behave like a startup, and a service company can have an excellent AI team. Judge the **team**, not only the logo.

Problem 12.3.1 Choose using a weighted score. Your weights (sum 10): learning 4, pay 3, stability 2, flexibility 1. You score each option from 1 to 5 using your own research. Product: learning 5, pay 4, stability 3, flexibility 2. GCC: 4, 4, 4, 3. Startup: 5, 3, 2, 3. Compute each total.

Formula	total = sum of weight x score
Product	4x5 + 3x4 + 2x3 + 1x2 = 20 + 12 + 6 + 2 = 40
GCC	4x4 + 3x4 + 2x4 + 1x3 = 16 + 12 + 8 + 3 = 39
Startup	4x5 + 3x3 + 2x2 + 1x3 = 20 + 9 + 4 + 3 = 36
Answer	Product 40, GCC 39, Startup 36. Product and GCC are almost tied, so the decision depends on the specific teams.
Check	Maximum possible = 10 x 5 = 50; all totals are below 50 and above 10. Yes.
Why it matters	Plain words: the method shows what you value; a gap of 1 point means "look closer", not "decided".

How to research a company (all free)

- Read engineer posts and tech blogs of the company.
- Look for the team on LinkedIn: who is there, how long they stay, what they did before.
- Read employee reviews on AmbitionBox and Glassdoor with a calm mind: reviews are opinions, look for repeated points.
- Ask your interviewers: "What does a good first year look like in this role?" and "How is success measured?"
- Check for recent layoffs, funding news and leadership changes in public news.

Connection to the real job

Your first data job sets how fast you learn, so choose a team with real data and a mentor, even if pay is slightly less. A strong first move often beats a higher number in a weak team.

Applications

- Career changers with a technical IT or testing background: platform, observability and data infrastructure teams in product firms and GCCs value your skill.
- Early moves: service MNC to GCC or product is a common path.

COMMON MISTAKE

Treating a company type as a rule ("startups pay less", "GCCs are safe"). It is a tendency. Check the team.

Check yourself

1. What does GCC stand for? Answer: Global Capability Center.
2. Which one tends to have the highest funding risk? Answer: startup.
3. Should you judge only by company logo? Answer: no, judge the team and the work.

12.4 The Internal Transfer Playbook

INTUITION

Your own company already trusts you. A new employer must guess. An internal move swaps "unknown outsider with a promise" for "known colleague with a track record".

The five steps

1. **Talk.** Have coffee chats with data or analytics managers in your company. Ask what they need.
2. **Volunteer.** Offer to help with small data tasks: a dashboard, a cleanup script, an analysis.
3. **Build one visible project.** Example: an operations dashboard built from your own work data, with a one-page note on what it saved.
4. **Ask for a shadow assignment.** A few hours a week on the data team's work, agreed with your manager.
5. **Formalise.** Ask your manager and HR how internal job postings and transfers work, and apply through the official route.

REAL JOB TIP

Tell your current manager early, but frame it as growth: "I want to apply my current analytical work to data. Can I run a small project for our team?" Do not hide your plans, but never make threats about leaving.

Problem 12.4.1 Your project reduces report preparation from 3 hours to 20 minutes, 4 times a month, for a team of 1. How many hours per month and per year does it save?

Plug in	old 3 h = 180 min; new 20 min; saved per run = 180 - 20 = 160 min
Compute	per month 160 x 4 = 640 min = 10.67 h; per year 10.67 x 12 = 128 h
Answer	About 10.7 hours a month and 128 hours a year.
Check	640 / 60 = 10.67; 640 x 12 = 7,680 min; 7,680 / 60 = 128. Yes.
Why it matters	A number like "128 hours a year" persuades a manager better than "I built a dashboard".

Problem 12.4.2 Compare the cost of the new-employer path and the internal path in months: new path needs 2 months of search + 2 months of notice = 4 months before joining. Internal path: 3 months of project + 1 month of transfer process = 4 months, but you keep salary and benefits and test the work first. Which has less risk?

Step 1	Time is the same, 4 months.
Step 2	Internal path: no notice period, no background verification, no probation reset (check your company policy), and you can stop if you dislike the work.
Answer	The internal path has lower risk, even if the time is equal. It may give lower salary growth, so research both.
Check	Risk list is longer for the external path: new probation, new team, unknown culture.

Connection to the real job

Even if you later change company, the internal project becomes your first data resume bullet with a real number.

Applications

- Large IT firms and GCCs usually have an internal job portal; ask HR how it works.
- Small firms: propose a part-time hybrid role.

COMMON MISTAKE

Waiting to be "noticed". Managers are busy. Show a small finished project and ask.

Check yourself

1. What is the first step of the playbook? Answer: talk to data or analytics managers.
2. Name one advantage of an internal move. Answer: your track record is known, and risk is lower.
3. How should you describe it to your manager? Answer: as growth, and by offering a project for the team.

12.5 Salary Research and Negotiation

INTUITION

Negotiation is like pricing a used car. You do not guess the value; you measure it. You collect data from several sources, then state a range you can defend.

Words you need

Term	Meaning
CTC	Cost to Company, the total yearly amount the company spends on you; not the same as money in hand
Fixed pay	The part paid regularly, in your salary
Variable pay	A bonus that depends on performance; may be paid at less than 100%
In-hand	Money that arrives in your bank account after deductions
ESOP / RSU	Employee stock options / Restricted stock units, a right to company shares; value depends on the company
Joining bonus	A one-time payment; often with a clause to repay it if you leave early
Total compensation	Fixed + expected variable + stock value + bonuses, per year

Where to find data

- **AmbitionBox, Glassdoor, levels.fyi**: self-reported salaries. Treat each as a rough range. Samples are small, and users report in different ways (fixed or CTC).
- **Recruiters**: ask "What is the budget range for this role?" It is a normal question.
- **Peers** in your network: ask ranges, not exact figures.
- **Job posts**: some postings show ranges.

Use at least three sources, discard extreme values and write a range in a sheet. Always ask: what exact role, how many years of experience, which city, and which year?

The negotiation script

1. Let the company give the first number when possible. If asked early, say: "I am looking for a range that fits the role and market. Could you share the budget range?"
2. If you must answer, give a **range** with a lower bound you accept, based on your research.
3. When an offer arrives, say thank you, ask for the breakdown in writing (fixed, variable, stock, bonus), and ask for time to review.
4. Ask once, politely, with evidence: "Based on market data and my experience, I was expecting something closer to X. Is there flexibility?"
5. Negotiate more than the base: joining date, joining bonus, learning budget, level or title, remote days, review cycle.

INTERVIEW TIP

Never lie about your current salary. Companies can verify through documents or background checks. Instead say what you expect and why. Check local laws and the offer terms; if unsure, ask a trusted adviser.

Problem 12.5.1 Compare three offers (practice numbers, in lakh per year). A (service): fixed 14, variable target 1 paid at 80%. B (product): fixed 18, variable target 2 at 90%, joining bonus 2 (one time), stock worth 3 per year. C (startup): fixed 16, no variable, stock worth 4 per year. Compute year 1 total compensation.

Formula	year-1 total = fixed + (variable target x payout) + joining + stock
A	14 + 1 x 0.80 = 14.8
B	18 + 2 x 0.90 + 2 + 3 = 18 + 1.8 + 2 + 3 = 24.8
C	16 + 0 + 0 + 4 = 20.0
Answer	A 14.8, B 24.8, C 20.0 lakh. But B includes a one-time 2, and C's stock is uncertain.
Check	Real output below. Run it yourself.
Why it matters	The headline fixed numbers (14, 18, 16) hide a gap of 10 lakh between A and B once everything is counted.

```
offers={
 "A_service":
{"fixed":14.0,"variable_target":1.0,"variable_pay_pct":0.8,"joining":0.0,"
stock_per_year":0.0},
 "B_product":
{"fixed":18.0,"variable_target":2.0,"variable_pay_pct":0.9,"joining":2.0,"
stock_per_year":3.0},
 "C_startup":
{"fixed":16.0,"variable_target":0.0,"variable_pay_pct":1.0,"joining":0.0,"
stock_per_year":4.0},
}
for k,o in offers.items():
    var=o["variable_target"]*o["variable_pay_pct"]
    cash=o["fixed"]+var
    y1=cash+o["joining"]+o["stock_per_year"]
    print(k,"fixed",o["fixed"],"expected
variable",round(var,2),"cash",round(cash,2),"yr1 total",round(y1,2))
```

```
A_service fixed 14.0 expected variable 0.8 cash 14.8 yr1 total 14.8
B_product fixed 18.0 expected variable 1.8 cash 19.8 yr1 total 24.8
C_startup fixed 16.0 expected variable 0.0 cash 16.0 yr1 total 20.0
```

Read it: `cash` is money you can plan monthly; `yr1 total` adds one-time and stock items. For B, the **recurring** value after year 1 is 19.8 cash + 3 stock = 22.8, because the joining bonus does not repeat.

Problem 12.5.2 Stock risk. For offer C, value the stock at 100%, 50% and 0% of the stated value. What is the total each time?

Compute	16 + 1.0 x 4 = 20.0; 16 + 0.5 x 4 = 18.0; 16 + 0 x 4 = 16.0
Answer	20.0, 18.0 or 16.0 lakh.
Check	The difference between best and worst is exactly the stock value, 4. Yes.
Why it matters	Judge startup offers on the fixed part first and treat stock as a bonus you may never see.

Problem 12.5.3 Your current fixed pay is 12 lakh. An offer gives 15, another 16.8, another 18. What is each hike in percent?

Formula	hike % = (new - old) / old x 100
Compute	(15 - 12)/12 = 0.25; (16.8 - 12)/12 = 0.40; (18 - 12)/12 = 0.50
Answer	25%, 40% and 50%.
Check	12 x 1.25 = 15; 12 x 1.40 = 16.8; 12 x 1.5 = 18. Yes.
Why it matters	Plain words: a big percentage on a low base may still be below market. Compare to market range, not only your past salary.

Offer-letter checklist

- Fixed, variable and stock in writing; vesting schedule for stock.
- Notice period and buyout rules; non-compete and confidentiality clauses.
- Joining bonus repayment clause and its length.
- Probation length, location and work-from-office rules.
- Background verification: what they check and when.

Connection to the real job

The same habit helps in yearly reviews: bring evidence and a market range, not feelings.

Applications

- Job switch, internal promotion, freelance rate setting, and counter-offer decisions.

COMMON MISTAKE

Comparing CTC with CTC. One company counts insurance and gratuity inside CTC, another does not. Compare fixed pay, expected variable and stock separately. Consult a qualified adviser for tax questions.

Check yourself

1. Why is fixed pay safer to compare? Answer: it is regular and guaranteed.
2. What should you do when you receive an offer? Answer: thank them, ask for it in writing, and take time to review.
3. Name two non-salary items you can negotiate. Answer: joining date and learning budget (or remote days, level).

12.6 Notice-Period Strategy

INTUITION

A notice period is like a handover window in an incident rotation. A good exit leaves the team safe and keeps your name clean, because your next reference may come from this team.

Words you need

Term	Meaning
Notice period	Time between resignation and last working day, set in your contract (often 30, 60 or 90 days in India)
Buyout	Paying the company to shorten the notice period, if policy allows
Garden leave	You are paid but not required to work during notice (rare)
Relieving letter	Official document confirming you left properly; new employers often ask for it
Counter-	Your current employer offers more money or a new role to keep you

offer

Rules of thumb

1. **Read your contract** before you apply. Know the exact days and buyout rules.
2. **Resign only after** you have a signed offer letter and have cleared the background verification terms.
3. Tell new employers your notice period honestly in the first call. Some prefer immediate joiners, and some will wait for the right candidate.
4. Ask the new company: can the joining date be adjusted? Can they pay a buyout through a clause? (Many say no; ask politely, once.)
5. Do a **handover document**: running tests, dashboards, scripts, contacts, open issues.
6. Treat counter-offers carefully. Ask: why did it take my resignation to fix pay or growth?

Problem 12.6.1 The offer date is 1 December 2026. Compute the joining date for notice periods of 30, 60 and 90 days if you resign that day.

Plug in	date + days
Compute	(by code below)

```
from datetime import date, timedelta
offer=date(2026,12,1)
for notice in (30,60,90):
    start=offer+timedelta(days=notice)
    print("notice",notice,"days -> joining",start, start.strftime("%A"))
```

```
notice 30 days -> joining 2026-12-31 Thursday
notice 60 days -> joining 2027-01-30 Saturday
notice 90 days -> joining 2027-03-01 Monday
```

- **Answer** 31 December 2026, 30 January 2027 (a Saturday, so the real start may move to Monday 1 February) and 1 March 2027.
- **Check** December has 31 days, so 1 Dec + 30 = 31 Dec. Yes. January 2027 + 29 more days gives 30 Jan. Yes.
- **Why it matters** A 90-day notice means the new company waits three months, so plan the search months ahead and discuss timing.

Problem 12.6.2 A buyout costs the fixed monthly pay for each month bought out, prorated by days (illustrative; your policy may differ). Fixed monthly pay is 1,00,000 rupees. What does 45 days cost?

Formula	cost = monthly pay / 30 x days
Compute	1,00,000 / 30 x 45 = 3,333.33 x 45 = 1,50,000
Answer	1,50,000 rupees under this rule.
Check	45 days is 1.5 months, and 1.5 x 1,00,000 = 1,50,000. Yes.
Why it matters	Decide if saving 45 days is worth the cost, or whether the new company will cover it.

Notice period tactics for the search

- Start interviewing before you are tired, because a long notice means a long total timeline.
- Keep a clean record. Do not criticise your employer in interviews.
- Use notice for rest, not only for work: a gap of a few days before joining protects your energy.

Connection to the real job

References from your last team often matter more than the salary. A professional handover is a signal that you will be a good teammate.

Applications

- Anyone with 60 or 90 days' notice, planning when to start applying.

COMMON MISTAKE

Resigning on a verbal "we will send the offer". Wait for the signed offer letter and for any background-check conditions in writing.

Check yourself

1. When should you resign? Answer: after a signed offer letter.
2. What is a buyout? Answer: paying to shorten the notice period.
3. Why write a handover document? Answer: to leave the team safe and protect your

reputation.

12.7 The Rejection Log and Improvement Cycle

INTUITION

A failed driving-test attempt is not only a failure; it is data. If you throw the data away, you will repeat the failure. A rejection log turns every "no" into the next study topic.

The log

Use a spreadsheet or a small database. Columns: company, role, date, channel (referral, portal, recruiter), stage reached, outcome, **reason or question you struggled with**, action, and date you fixed it.

Problem 12.7.1 Seven records are in the log. Count the rejection reasons.

```
import sqlite3
con=sqlite3.connect(":memory:")
con.execute("create table log(company text, role text, stage text, outcome
text, reason text)")
rows=[("A","DE","screen","reject","no Spark exp"),
("B","DE","tech","reject","SQL window func"),
("C","DA","tech","reject","SQL window func"),
("D","DE","final","reject","system design"),
("E","DE","tech","reject","SQL window func"),
("F","DE","screen","pass",""),("G","DE","tech","reject","system design")]
con.executemany("insert into log values(?,?,?,?,?)",rows)
for r in con.execute("select reason,count(*) c from log where
outcome='reject' group by reason order by c desc"):print(r)
for r in con.execute("select stage,count(*) from log group by stage order
by 2 desc"):print(r)
```

Predict first: which reason appears most? Real output:

```
('SQL window func', 3)
('system design', 2)
('no Spark exp', 1)
('tech', 4)
('screen', 2)
('final', 1)
```

- **Step 1** The first query groups rejections by reason and sorts by count.
- **Step 2** The second counts how many times each stage appears in the log.
- **Answer** "SQL window functions" caused 3 of 6 rejections, so it is the first thing to fix. Most records reach the technical round (4 of 7).
- **Check** 3 + 2 + 1 = 6 rejections, and the table has 7 rows with 1 pass. Yes.
- **Why it matters** Plain words: half your rejections are one fixable skill. Study SQL window functions for two weeks before your next technical round.

The improvement cycle

```
Apply -> Interview -> Log the questions you struggled with (same day)
   ^                                                          |
   |                                                          v
Retest yourself <- Study one weakness for 1–2 weeks <- Rank by count
```

Rules for the cycle:

1. Write the log the same day, while you remember the questions.
2. Rank reasons by count. Fix the top one first.
3. Retest with a mock interview (Chapter 7).
4. Ask recruiters for feedback; some give it, many do not. Do not argue.
5. Review every 2 weeks and update your funnel numbers from 12.1.

Handling rejection emotionally

A rejection says "not this team, not this day". It does not say "you cannot do this work". See Chapter 14 for feelings of self-doubt. Take a short break after three rejections in a row, then read your log and pick the next step.

Connection to the real job

Post-incident reviews work the same way: a blameless note of what happened, why, and what changes. You already know this habit.

Applications

- Job search, promotion cases, and any exam or certification attempt.

> **COMMON MISTAKE**
>
> Not logging, then repeating the same mistake and calling it bad luck. Log within the day.

Check yourself

1. What column in the log is the most useful? Answer: the reason or question you struggled with.
2. How do you decide what to study next? Answer: count reasons and fix the biggest one.
3. How often to update funnel numbers? Answer: about every two weeks.

Chapter recap

- A job search is a funnel; measure stage conversion rates and fix the weakest stage.
- Sixty applications at the sample rates gives about one offer; referrals can cut that number a lot.
- Referrals are the cheapest way to raise the first-stage rate; make requests short and easy to decline.
- Product, GCC, service and startup differ in tendencies; judge the team and score options using your own weights.
- An internal transfer uses a visible project, a shadow assignment and the official route.
- Research salary from at least three sources and compare total compensation, not CTC headlines; never invent or lie about numbers.
- Resign only after a signed offer; read your notice terms before applying; write a handover.
- Keep a rejection log, rank reasons by count and fix one weakness at a time.

Pattern cheat-sheet

Need	Pattern

Expected offers	applications x product of stage pass rates
Applications per offer	1 / overall conversion rate
Hike %	(new - old) / old x 100
Year-1 total	fixed + variable target x payout + joining + stock
Buyout (illustrative)	monthly pay / 30 x days
Referral ask	link + 2 lines of proof + easy exit
Rejection fix	count reasons, fix the top one, retest

Practice set

1. (Easy) 100 applications, screen 20%, technical 50%, final 50%, offer 50%. How many offers? **Answer:** 100 x 0.2 x 0.5 x 0.5 x 0.5 = 2.5 offers.
2. (Easy) Current fixed pay 10 lakh, new 13 lakh. Hike? **Answer:** 3 / 10 = 30%.
3. (Easy) What is the difference between CTC and in-hand? **Answer:** CTC is the total yearly cost to the company; in-hand is what reaches your bank after deductions and non-cash items.
4. (Medium) You send 25 messages and get 5 replies. Reply rate? **Answer:** 5 / 25 = 20%.
5. (Medium) Offer: fixed 15, variable target 3 paid at 70%, joining bonus 1.5, no stock. Year-1 total? **Answer:** 15 + 3 x 0.7 + 1.5 = 15 + 2.1 + 1.5 = 18.6 lakh.
6. (Medium) Find the bug in the plan: "I will resign today because the recruiter said the offer is coming." **Answer:** Resigning before a signed offer is risky; wait for the signed letter.
7. (Medium) The log shows 10 rejections: 5 system design, 3 SQL, 2 DSA. What do you study first and why? **Answer:** System design, because it caused 5 of 10 (50%) of rejections.
8. (Medium) Notice is 60 days, offer on 15 January 2027. Joining date if you resign that day? **Answer:** 15 Jan + 60 days = 16 March 2027 (January has 16 days left after the 15th, February has 28 in 2027, so 16 + 28 = 44, plus 16 in March = 60).
9. (Hard) Offer A: fixed 18, nothing else. Offer B: fixed 15, variable 3 at 100%, stock 4 per year valued at half. Which is higher in expected yearly value? **Answer:** B = 15 + 3 + 2 = 20 versus A = 18, so B is higher on paper; but B is less certain, so judge risk too.
10. (Interview) "What are your salary expectations?" Give a good answer. **Answer:** "I have researched market ranges for this role and level. I would like to understand the budget for

this role. Based on my research I am looking for a range of X to Y, and I am open to discussing the full package." (Use your own researched range.)

11. (Interview) "Why do you want to leave?" **Answer:** Speak about growth, not blame: "I have grown in my technical role and want to apply that analytical work to data systems. This role gives me that."
12. (Hard, scenario) You have two offers with 90-day and 30-day notice and the 90-day firm is slightly better. What do you check? **Answer:** Whether the better firm can wait or start remote work earlier, whether the buyout is possible, and whether the gap in growth justifies three more months of notice.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional at an Indian MNC moving into data roles. The topic is Chapter 12, the job search as a funnel. Teach me step by step how to compute expected offers from stage conversion rates, using small numbers first. Then ask me 5 questions one at a time and check my answers. Finish by helping me design my own weekly search routine with targets for applications, referral messages and tracking.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving into data roles in India. The topic is Chapter 12, referrals and outreach. Give me 8 graded exercises without solutions about writing referral requests, cold messages to hiring managers and following up politely. Then I will paste my answers, and you will review them for clarity, length and how easy it is to say yes. My draft: [PASTE HERE]
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional at an Indian MNC moving into data work. The topic is Chapter 12, comparing company types (product, GCC, service MNC, startup). Help me build a weighted decision score for my own priorities. Ask me for my weights, then guide me to research each company with free sources. Do not invent salaries or policies; tell me where to check them. Quiz me on the trade-offs at the end.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional in India preparing for offers. The topic is Chapter 12, salary research and negotiation. Run a mock negotiation: you are the recruiter, I am the candidate. First ask the salary expectation question, then present an offer with fixed, variable, stock and joining bonus, and respond to my counter. After the role-play, give me feedback on my script and my total-compensation math. Offer details: [PASTE HERE]
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data roles in India. The topic is Chapter 12, the rejection log and notice-period strategy. Give me a spreadsheet design for a rejection log, then review my entries and rank my weaknesses by count. Next, plan my notice-period timeline from my details and list the questions I should ask before resigning. My log and notice terms: [PASTE HERE]
```

CHAPTER 13

DOMAIN KNOWLEDGE AND BUSINESS SENSE

Two equally skilled people differ by how well they understand the business. Domain knowledge is the part of your skill set that tools do not replace.

A data model that is accurate but answers the wrong business question is useless. A manager at a bank does not ask for "a classifier with 0.91 AUC"; she asks "how many loans will go bad, and how much will we lose?" This chapter gives you the vocabulary and arithmetic of five big domains in India (banking and insurance, e-commerce, SaaS, telecom, healthcare), a method to learn any new domain in about a month, and worked unit-economics examples computed in Python. By the end you will be able to read a simple business, name its key numbers, and compute whether it makes money.

13.1 Why Domain Knowledge Is a Career Moat

INTUITION

A good mechanic can fix any car. A mechanic who knows rally cars knows which part fails at 200 km/h and what the driver cares about. Same skill, but far more valuable to the rally team. Domain knowledge is knowing what the "driver" cares about.

Words you need

Term	Meaning
Domain	An industry area, such as banking or telecom, with its own words, rules and numbers
KPI	Key Performance Indicator: a number a team watches to know if things are going well
Unit economics	Profit or loss for one "unit" (one order, one customer, one loan) before scaling up
Moat	Something that protects you from competition; here, a skill hard to copy

Regulation	Rules set by an authority (in India, for example RBI for banks, IRDAI for insurance, SEBI for markets, TRAI for telecom)

Check each regulator's current rules on its official site. Rules change; this book teaches you where to look, not the latest circular.

How a domain helps you in the job

Item	Example	Where it appears in the job
Right metric	Choosing "loss per loan" over "accuracy"	Model evaluation and dashboards
Right features	Days since last payment for credit risk	Feature engineering
Right constraints	Customer data must stay in India or be consented	Architecture and privacy
Right story	"Saves 12 lakh a month" instead of "AUC improved"	Stakeholder presentations (Chapter 5)

Pick **one primary and one secondary domain**. Any domain can work, but a strong choice connects to a place you could get a job: many GCCs and product firms in India are in finance, retail, telecom and health. Choose by interest, local job market, and any experience you already have (for example, you may have worked on banking or e-commerce applications. If so, you already know their real traffic patterns).

REAL JOB TIP

Think of the applications you have worked on or used. A UPI payment gateway, a retail checkout or a telecom billing system means you already know the system. Say that in interviews: "I know where this system breaks, and now I can analyse why."

Problem 13.1.1 Score three domains for yourself. Weights: interest 4, jobs nearby 3, prior knowledge 3. Scores 1 to 5. Banking: 3, 5, 4. E-commerce: 4, 4, 3. Healthcare: 5, 2, 1. Which is highest?

Formula	total = sum of weight x score
Banking	4x3 + 3x5 + 3x4 = 12 + 15 + 12 = 39
E-commerce	4x4 + 3x4 + 3x3 = 16 + 12 + 9 = 37
Healthcare	4x5 + 3x2 + 3x1 = 20 + 6 + 3 = 29
Answer	Banking 39, E-commerce 37, Healthcare 29. Pick banking first; e-commerce is a close second.
Check	The maximum is 10 x 5 = 50; every total is below it. Yes.
Why it matters	You are choosing with numbers, not with a mood of the day. Redo it with your own scores.

Problem 13.1.2 You study one domain for 1 hour a day, 5 days a week, for 4 weeks. How many hours?

Compute	1 x 5 x 4 = 20 hours
Answer	20 hours, enough to learn about 50 terms and build a small project (see 13.7).
Check	4 weeks x 5 days = 20 days; 20 days x 1 hour = 20. Yes.

COMMON MISTAKE

Trying to learn "all domains". Depth in one beats a list of six. Add a second domain after you finish a project in the first.

Check yourself

1. What is a KPI? Answer: a number a team watches to judge progress.
2. How many primary domains should you start with? Answer: one (and maybe a second later).
3. Why does domain knowledge protect your career? Answer: it is hard to copy and AI tools cannot easily supply business judgment.

13.2 BFSI: Banking, Financial Services and Insurance

INTUITION

A bank is a business that rents out money. It pays savers a little, lends to borrowers for more, and survives only if most borrowers repay. So the central data question is: who will not repay, and what will it cost?

Words you need

Term	Meaning
BFSI	Banking, financial services and insurance
UPI	Unified Payments Interface, India's real-time bank-to-bank payment system
NPA	Non-Performing Asset: a loan on which payment is overdue beyond a limit set by the regulator (check the current RBI rule for the exact period)
GNPA / NNPA	Gross NPA ratio = NPA / total loans; Net NPA ratio subtracts the money already set aside (provisions)
PD	Probability of Default: chance a borrower will fail to repay
LGD	Loss Given Default: share of the loan lost if the borrower defaults (after collateral recovered)
EAD	Exposure At Default: amount owed when default happens
KYC / AML	Know Your Customer / Anti-Money Laundering: identity checks and monitoring for suspicious money flows
Premium, claim	In insurance: the customer pays a premium; the insurer pays a claim when the insured event happens
Loss ratio	Claims paid / premium earned
Combined ratio	(Claims + expenses) / premium earned; below 100% means an underwriting profit

The credit lifecycle

```
Apply -> KYC check -> Credit score / underwriting -> Approve & disburse
   -> Repayment monitoring -> Early warning -> Collections -> Recovery /
write-off
```

Walk through it with one loan. A customer applies for a personal loan. The bank checks identity (KYC), looks at credit history (a score from a credit bureau) and income, and decides yes or no (underwriting). It then lends the money, collects monthly payments, watches for late payments (early warning), and if payments stop, tries to collect, then recovers what it can. Data science appears at each box: fraud detection at KYC, a PD model at underwriting, early-warning models during monitoring, collections prioritisation at the end.

Problem 13.2.1 A bank has loans of 5,00,000 (in lakh rupees) and 15,000 of them are NPA. It has set aside 9,000 as provisions. Compute the gross and net NPA ratios.

Formula	GNPA % = NPA / gross loans x 100; NNPA % = (NPA - provisions) / (gross loans - provisions) x 100
Plug in	GNPA = 15,000 / 5,00,000; NNPA = (15,000 - 9,000) / (5,00,000 - 9,000)
Compute	GNPA = 0.03 = 3.0%; NNPA = 6,000 / 4,91,000 = 1.22%
Answer	Gross NPA is 3.0% and net NPA is about 1.22%.
Check	Net must be smaller than gross because provisions reduce the exposed amount. 1.22 < 3.0. Yes.
Why it matters	In plain words: 3 of every 100 rupees lent are in trouble, but the bank has already prepared for part of it.

Problem 13.2.2 Expected loss on one loan with PD = 3%, LGD = 45%, EAD = 2,00,000 rupees. For a portfolio of 1,000 such loans, what is the expected loss?

Formula	expected loss (EL) = PD x LGD x EAD
Plug in	0.03 x 0.45 x 2,00,000
Compute	0.03 x 0.45 = 0.0135; 0.0135 x 2,00,000 = 2,700
Portfolio	1,000 x 2,700 = 27,00,000
Answer	Expected loss is 2,700 rupees per loan and 27 lakh for 1,000 loans.
Check	If 3% of 1,000 loans default, 30 loans default; each loses 45% of 2,00,000 = 90,000; 30 x 90,000 = 27,00,000. Yes.
Why it matters	The bank must charge enough interest to cover 2,700 per loan before it earns anything. A better PD model lowers that number.

Problem 13.2.3 An insurer earned 1,000 (in lakh rupees) of premium, paid 650 in claims and spent 280 on expenses. Compute loss ratio and combined ratio.

Compute	loss ratio = 650 / 1,000 = 0.65 = 65%; combined = (650 + 280) / 1,000 = 0.93 = 93%
Answer	Loss ratio 65%, combined ratio 93%, so an underwriting profit of 7% of premium.
Check	100% - 93% = 7%, and 7% of 1,000 = 70 lakh. Yes.
Why it matters	Pricing models try to keep the combined ratio below 100%.

Python check

```
gross_loans=500_000; npa=15_000; provision=9_000
print("GNPA %", npa/gross_loans*100)
print("NNPA %", (npa-provision)/(gross_loans-provision)*100)
pd_,lgd,ead=0.03,0.45,200_000
print("EL per loan", pd_*lgd*ead, "portfolio", 1000*pd_*lgd*ead)
print("loss ratio", 650/1000, "combined", (650+280)/1000)
```

```
GNPA % 3.0
NNPA % 1.2219959266802443
EL per loan 2700.0 portfolio 2700000.0
loss ratio 0.65 combined 0.93
```

The code line for NNPA matches our hand result 1.22%. Floating-point numbers sometimes show many decimals; round them when you present.

Other BFSI ideas

- **UPI and payments:** key numbers are transactions per second, success rate and fraud rate. A success rate of 98% means 2 in 100 payments fail, which at ten million payments a day is 200,000 failures. (Capacity thinking fits here: peak load, timeouts, retries.)
- **Fraud and AML:** rare events (very imbalanced classes), where precision and recall trade off with customer annoyance.
- **Regulation:** RBI rules on data and outsourcing; the DPDP Act (India's Digital Personal Data Protection Act) for personal data. Read the official texts or reliable summaries for current details.
- **Actuarial basics:** insurers estimate expected claims using probability and past data, which is statistics you already know (Math PDF).

Connection to the real job

Common ML use cases: credit scoring, fraud detection, collections prioritisation, customer churn, cross-sell, claims triage, document processing with LLMs.

Applications

- Data Science: PD models, with explainability needed by regulators.
- Data Engineering: reconciliation pipelines, daily settlement files, audit trails.
- Analytics: portfolio dashboards (GNPA by region, product).
- MLOps: model monitoring and drift for risk models.

COMMON MISTAKE

Optimising accuracy on default prediction. If only 3% default, a model that says "nobody defaults" is 97% accurate and worthless. Use recall, precision and expected loss in rupees.

Check yourself

1. What is EL = ? Answer: PD x LGD x EAD.
2. Is a combined ratio of 93% good or bad for an insurer? Answer: good, it means an underwriting profit.
3. What does KYC do? Answer: verifies customer identity.

13.3 E-commerce and Retail

INTUITION

An online shop is a funnel with a delivery van at the end. Visitors come, a few buy, and each order must pay for its product, its shipping, its returns and the advertising that brought the customer. If one order loses money and you sell a million, you lose a million times more.

Words you need

Term	Meaning
GMV	Gross Merchandise Value: total value of goods ordered
AOV	Average Order Value = GMV / number of orders
Conversion rate	Orders / sessions (visits)
CAC	Customer Acquisition Cost: marketing spend / new customers won
Contribution margin	What is left after variable costs of an order (product cost, shipping, payment fees, returns), before fixed costs like office and salaries
Return rate	Share of orders returned
Marketplace	A platform where many sellers sell; the platform earns commission and fees

Also know: catalog (the list of products and their attributes), pricing and promotions (discounts), logistics (warehouse, delivery, last mile), search and recommendation (the ML heart of the

shop). In India, quick commerce (delivery in minutes from small local warehouses) and open networks such as ONDC are new areas; read current news and company reports because the market changes quickly.

Running example: one shop, one month

Orders 2,000; average order 1,500 rupees; sessions 1,00,000; gross margin 20% of order value; shipping cost 60 per order; 10% of orders returned, and a return costs 120 (reverse shipping and handling); marketing spend 1,00,000; 60% of orders come from new customers.

Problem 13.3.1 Compute GMV, AOV and conversion rate.

Formula	GMV = orders x order value; AOV = GMV / orders; conversion = orders / sessions
Compute	GMV = 2,000 x 1,500 = 30,00,000; AOV = 30,00,000 / 2,000 = 1,500; conversion = 2,000 / 1,00,000 = 0.02
Answer	GMV 30 lakh, AOV 1,500, conversion 2%.
Check	1,500 x 2,000 = 3,000,000. Yes.
Why it matters	The funnel has two levers: more visitors, or more of them buying. A change from 2.0% to 2.2% conversion gives 10% more orders on the same traffic.

Problem 13.3.2 Compute the contribution per order and CAC. Then say if a new customer's first order covers the CAC.

Step 1	Gross margin per order = 20% x 1,500 = 300
Step 2	Expected return cost = 10% x 120 = 12
Step 3	Contribution = 300 - 60 (shipping) - 12 = 228
Step 4	New customers = 60% x 2,000 = 1,200; CAC = 1,00,000 / 1,200 = 83.33
Answer	Contribution is 228 per order and CAC is about 83. The first order alone covers the CAC, leaving about 145.
Check	228 - 83.33 = 144.67. Also total contribution 2,000 x 228 = 4,56,000, minus 1,00,000 marketing = 3,56,000, before fixed costs.
Why it matters	In plain words: each new customer pays back its advertising on the first order. If shipping were 200, contribution would be 300 - 200 - 12 = 88, and the CAC would barely pay back. Small cost changes decide the business.

```
orders=2000; gmv=orders*1500; print("AOV", gmv/orders)
sessions=100_000; print("conversion %", orders/sessions*100)
aov=1500; cm_pct=0.20; ship=60; ret=0.10; ret_cost=120
cm=aov*cm_pct; print("gross margin/order", cm)
print("expected return cost", ret*ret_cost)
print("contribution/order", cm-ship-ret*ret_cost)
print("CAC", 100_000/(orders*0.6))
```

```
AOV 1500.0
conversion % 2.0
gross margin/order 300.0
expected return cost 12.0
contribution/order 228.0
CAC 83.33333333333333
```

Connection to the real job

- **Search and recommendation:** ranking products to raise conversion; measured with click-through, add-to-cart and orders per session.
- **Demand forecasting:** how much stock for each warehouse.
- **Pricing and promotions:** uplift modelling (who buys only because of a discount).
- **Returns prediction** and **delivery-time estimation**.

Applications

- Data Analytics: funnel dashboards, cohort retention, category performance.
- Data Engineering: clickstream pipelines, order and inventory tables.
- ML: ranking, recommenders, demand forecast.
- GenAI: product-description generation, support chatbots.

COMMON MISTAKE

Looking at GMV and ignoring margin. Growing GMV while contribution per order is negative means growing losses.

Check yourself

1. AOV = ? Answer: GMV / number of orders.
2. If conversion goes from 2% to 3% with the same traffic, how many orders increase? Answer: 50% more.
3. What does contribution margin exclude? Answer: fixed costs.

13.4 SaaS and Product Companies

INTUITION

A SaaS (Software as a Service) company rents software by subscription. It is like a gym: the member pays every month, and the owner cares about how many join, how many stay, and how long it takes to earn back the cost of signing them up.

Words you need

Term	Meaning
MRR / ARR	Monthly / Annual Recurring Revenue (ARR = MRR x 12)
ARPA	Average Revenue Per Account per month
Churn	Share of customers (or revenue) lost in a period

Gross margin	(Revenue - cost to serve) / revenue
LTV	Lifetime Value: total gross profit from a customer over its life
CAC payback	Months of gross profit needed to earn back CAC
NRR	Net Revenue Retention: revenue from last period's customers now, including upgrades and losses, divided by their revenue then
GRR	Gross Revenue Retention: like NRR but ignoring upgrades
PLG	Product-Led Growth: users try the product free, then upgrade
Activation	The moment a new user first gets value (for example creates a first dashboard)

Problem 13.4.1 ARPA is 2,000 per month, gross margin 80%, monthly churn 4%, CAC 24,000. Compute LTV, LTV/CAC and CAC payback.

Formula	LTV = ARPA x gross margin / churn; payback months = CAC / (ARPA x gross margin)
Plug in	LTV = 2,000 x 0.80 / 0.04; payback = 24,000 / (2,000 x 0.80)
Compute	gross profit per month = 1,600; LTV = 1,600 / 0.04 = 40,000; payback = 24,000 / 1,600 = 15; LTV/CAC = 40,000 / 24,000 = 1.67
Answer	LTV is 40,000, payback is 15 months, LTV/CAC is 1.67.
Check	Average life is 1 / 0.04 = 25 months; 25 x 1,600 = 40,000. Yes.
Why it matters	Many teams look for LTV/CAC around 3 or more, but this is a rule of thumb, not a law. At 1.67 and a 15-month payback this business is weak: either lower CAC, lower churn or raise price.

Problem 13.4.2 A month's MRR starts at 1,00,000. New customers add 12,000; upgrades add 5,000; downgrades remove 2,000; churn removes 4,000. Compute ending MRR, ARR, NRR and GRR.

Step 1	Ending MRR = 1,00,000 + 12,000 + 5,000 - 2,000 - 4,000 = 1,11,000
Step 2	ARR = 1,11,000 x 12 = 13,32,000
Step 3	NRR excludes new customers: (1,00,000 + 5,000 - 2,000 - 4,000) / 1,00,000 = 99,000 / 1,00,000 = 99%
Step 4	GRR excludes upgrades: (1,00,000 - 2,000 - 4,000) / 1,00,000 = 94%
Answer	Ending MRR 1,11,000; ARR 13,32,000; NRR 99%; GRR 94%.
Check	NRR minus GRR = 5 points = the upgrades 5,000 / 1,00,000. Yes.
Why it matters	Growth looks fine (+11%) but NRR below 100% means existing customers shrink; growth depends entirely on new sales.

```
arpa=2000; churn=0.04; gm=0.80; cac=24000
print("LTV", arpa*gm/churn)
print("CAC payback months", cac/(arpa*gm), "LTV/CAC", arpa*gm/churn/cac)
start,new,exp,cont,churned=100_000,12_000,5_000,2_000,4_000
mrr=start+new+exp-cont-churned
print("MRR end",mrr,"ARR",mrr*12)
print("NRR %",(start+exp-cont-churned)/start*100,"GRR %",(start-cont-
churned)/start*100)
```

```
LTV 40000.0
CAC payback months 15.0 LTV/CAC 1.6666666666666667
MRR end 111000 ARR 1332000
NRR % 99.0 GRR % 94.0
```

Product thinking in one table

Idea	Meaning	Example
North-star metric	One number that best shows value delivered	Weekly active teams running a report
Input metrics	Things you can move that drive the north star	New-user activation rate
Guardrail	Things you must not harm	Page load time, support tickets

metrics		
Leading vs lagging	Early signal vs late result	Activation (leading) vs churn (lagging)

Experiments (A/B tests) and prioritisation methods (RICE: Reach, Impact, Confidence, Effort) are standard tools; Chapter 10 covers experiment design.

Connection to the real job

Product analytics roles compute funnels, retention curves and experiment results. Performance data (page speed) connects directly: slower pages lower activation, and you can measure how much.

Applications

- Analytics: activation and retention dashboards.
- DS: churn prediction, usage-based upsell scores.
- DE: event tracking pipelines, billing data.
- AI Engineering: in-product copilots, search.

COMMON MISTAKE

Using total churn to compute LTV when churn differs by segment. A small group of leaving customers can hide very loyal large ones. Compute per segment.

Check yourself

1. ARR = ? Answer: MRR x 12.
2. NRR below 100% means what? Answer: existing customers are shrinking in revenue.
3. What does CAC payback measure? Answer: months of gross profit needed to earn back the CAC.

13.5 Telecom and Healthcare

INTUITION

Telecom is a subscription business with a network that must work all day: customers leave if calls drop. Healthcare is a business of care where an error hurts a person and privacy is the law. The same data skills apply, but the cost of a mistake differs.

Telecom words

Term	Meaning
ARPU	Average Revenue Per User per period (usually month)
Churn	Subscribers leaving per period
Network KPIs	Call drop rate, call setup success, data throughput, latency, tower load
Billing / CDR	Charging records; a CDR (Call Detail Record) is one row per call or data session
Prepaid vs postpaid	Pay before use vs bill after use

Problem 13.5.1 A telecom has revenue of 300 crore over 3 months from 1 crore (10 million) subscribers. Churn is 2% a month. Compute monthly ARPU, expected customer lifetime in months and lifetime revenue.

Formula	ARPU = revenue / subscribers / months; lifetime = 1 / monthly churn; lifetime revenue = ARPU x lifetime
Plug in	Revenue 3,000,000,000 rupees (300 crore = 3 x 10^9)
Compute	ARPU = 3,000,000,000 / 10,000,000 / 3 = 100; lifetime = 1 / 0.02 = 50 months; lifetime revenue = 100 x 50 = 5,000
Answer	ARPU 100 rupees a month, a customer stays 50 months on average and brings about 5,000 rupees.
Check	100 rupees x 3 months x 10 million subs = 3 x 10^9. Yes.
Why it matters	A churn change from 2% to 1.5% lifts lifetime from 50 to 66.7 months, a 33% increase in lifetime revenue.

Problem 13.5.2 A churn model flags 1,000 at-risk customers with ARPU 150. A retention offer costs 30 per customer and saves 20% of them for an extra 3 months of revenue each. Is it worth it?

Plug in	saved customers = 1,000 x 0.20 = 200; revenue gain = 200 x 150 x 3 = 90,000; cost = 1,000 x 30 = 30,000
Compute	net = 90,000 - 30,000 = 60,000
Answer	Net gain is 60,000 rupees, so the offer is worth it under these assumptions.
Check	Break-even save rate = cost / (150 x 3 x 1,000) = 30,000 / 4,50,000 = 6.67%. The offer works if it saves more than 6.7% of at-risk customers. At 20% we are well above. Yes.
Why it matters	A churn model is only valuable when linked to an action and its cost.

Healthcare words

Term	Meaning
EHR / EMR	Electronic Health/Medical Record: a patient's digital chart

Clinical data	Diagnoses, lab results, prescriptions, images
Outcome	The result for the patient (recovered, readmitted)
Sensitivity	Share of sick patients a test correctly flags
Specificity	Share of healthy people a test correctly clears
PPV	Positive Predictive Value: when the test says "sick", how often it is right
PHI	Protected Health Information; personal health data requiring strict handling
Regulatory	Approvals and rules for medical products and data (check the current rules; they differ by country)

Problem 13.5.3 A screening test for a condition with prevalence 1% has sensitivity 90% and specificity 95%. Among 100,000 people, what share of positive results are true positives?

Formula	PPV = TP / (TP + FP)
Step 1	Sick people = 1% x 1,00,000 = 1,000; healthy = 99,000
Step 2	TP = 0.90 x 1,000 = 900; FP = (1 - 0.95) x 99,000 = 4,950
Step 3	PPV = 900 / (900 + 4,950) = 900 / 5,850 = 0.1538
Answer	Only about 15.4% of positive results are real. Most positives are false alarms because the condition is rare.
Check	Python gave 0.1538. Also total positives = 5,850 of 1,00,000 = 5.85%.
Why it matters	A doctor needs a second test after a positive. Reporting only "90% sensitive" would mislead patients. This is the same base-rate effect as in fraud detection (Chapter 7, stats).

Privacy and ethics

Health data is highly sensitive. Use consent, minimise the data you keep, remove identifiers, control access, and follow the law (in India the DPDP Act; other countries have HIPAA or GDPR). Do not use real patient data in public projects. Use public or synthetic datasets only.

Applications

- Telecom: churn, network anomaly detection, tower planning, fraud (SIM-box, spam calls).
- Healthcare: medical imaging, readmission prediction, claims, appointment no-show, clinical notes with NLP (privacy rules apply).
- Pharma: trial data analysis, supply chain, adverse-event monitoring.

COMMON MISTAKE

Judging a rare-event test by accuracy or sensitivity alone. Always compute PPV with the real prevalence.

Check yourself

1. ARPU stands for? Answer: Average Revenue Per User.
2. If PPV is low, what does a positive result mean? Answer: many positives are false alarms.
3. Why avoid real patient data in public projects? Answer: privacy and legal risk.

13.6 Other Domains and the Indian Context

INTUITION

Every business that moves goods, money or attention produces data. The names change; the questions (who, how much, how fast, how well) stay the same.

Domain	Typical KPIs	Typical data science use
Logistics, supply chain	On-time delivery %, cost per parcel, inventory turns	Route optimisation, demand forecast
Manufacturing	Yield, downtime, defect rate	Predictive maintenance
Energy, utilities	Load, outage minutes, loss %	Load forecasting, anomaly detection
Ed-tech	Completion rate, engagement	Personalised learning
Ad-tech, media	CTR, CPM, watch time	Ranking, attribution

Gaming	DAU, retention day 1/7/30, ARPDAU	Matchmaking, churn
Government, public services	Coverage, cost per service	Forecasting, targeting benefits
Agritech	Yield, price, rainfall	Satellite crop monitoring, price forecast

Indian context advantages (verify current status in news and official sites): payments at very large scale (UPI), open networks such as ONDC, fast-growing e-commerce and quick commerce, Indian-language NLP (initiatives such as AI4Bharat and Bhashini), agritech, Global Capability Centers that hire for many domains, and data rules (the DPDP Act and data-localisation expectations). A candidate who understands, for example, UPI failure patterns or Indian-language text is very useful locally.

Problem 13.6.1 A warehouse delivers 9,200 of 10,000 parcels on time. A new routing change leads to 9,500 on time. What are the on-time rates and the improvement in percentage points and in late parcels?

Compute	before 92%; after 95%; improvement 3 percentage points; late parcels fall from 800 to 500, which is 37.5% fewer
Answer	On-time rate rose from 92% to 95% (3 points), and late parcels fell by 37.5%.
Check	(800 - 500) / 800 = 0.375. Yes.
Why it matters	A "3 point" rise sounds small; "37.5% fewer late parcels" sounds large. Both are true; state which you mean. Be careful to say percentage points versus percent.

Problem 13.6.2 Gaming: 10,000 players install. 4,000 return on day 1, 1,500 on day 7, 600 on day 30. Compute retention each day.

Compute	D1 = 4,000 / 10,000 = 40%; D7 = 1,500 / 10,000 = 15%; D30 = 600 / 10,000 = 6%
Answer	D1 40%, D7 15%, D30 6%.
Check	Retention is non-increasing over days (40 > 15 > 6). Yes.
Why it matters	The drop between day 1 and day 7 tells the team to improve the first week.

COMMON MISTAKE

Mixing "percent" and "percentage points". 92% to 95% is +3 points, not +3%.

Check yourself

1. Name one KPI for manufacturing. Answer: yield, downtime or defect rate.
2. 20% to 25% is how many percentage points? Answer: 5.
3. Name an Indian data context. Answer: UPI payments (or ONDC, Indian-language NLP).

13.7 How to Learn a Domain Fast

INTUITION

Learning a domain is like learning a city. You first memorise the main roads (vocabulary), then walk a few streets (a project), then ask locals (experts). You do not need to know every lane.

The four-week plan (1 hour a day)

Week	Do	Output
1	Read two annual reports or investor presentations; list unknown words	Glossary of 50 terms
2	Listen to or read two earnings-call transcripts; note the numbers management talks about most	One page of 10 KPIs and their formulas
3	Talk to two people in the domain; ask what data problems cost them money	Notes with 3 real problems
4	Build one small project on public data (UPI statistics, a retail dataset, a telecom churn dataset) with a business question and a rupee estimate	A repo with a README (Chapter 11)

Also: shadow a business meeting if you can; read industry reports; follow two newsletters; use *Lean Analytics*, *Inspired* by Marty Cagan and *Trustworthy Online Controlled Experiments* by

Kohavi and others for product thinking.

The glossary habit

Write each term as: **Term | plain meaning | formula (if any) | where it shows up in data**. Review with spaced repetition (Chapter 2).

Questions to ask a domain expert

1. What number do you check first every morning?
2. What decision costs the most when it is wrong?
3. What data do you wish you had?
4. What happened the last time something went badly wrong?

Problem 13.7.1 You learn 50 terms in 4 weeks, with the plan "10 new terms per day, 5 days". Is that realistic? What daily pace is better?

Compute	10 x 5 = 50 terms in 5 days, but memory fades; with spaced review, 5 new terms a day for 10 days (50 terms) leaves time to review.
Answer	50 terms in 4 weeks is 2.5 new terms per working day (20 days), a comfortable pace with review.
Check	50 / 20 = 2.5. Yes.
Why it matters	Slow with review beats fast without it.

Problem 13.7.2 Write a domain project question with a rupee estimate. Example: "A bank sends 10,000 reminders a month. If a model targets the 20% likely to miss payment, how much does it save?" Suppose each message costs 2 rupees and recovers 150 rupees on 25% of targeted messages.

Step 1	Send to 2,000 (20% of 10,000); cost = 2,000 x 2 = 4,000
Step 2	Recoveries = 2,000 x 0.25 = 500; recovered value = 500 x 150 = 75,000
Step 3	Net = 75,000 - 4,000 = 71,000
Answer	Under the assumptions, net benefit is 71,000 rupees a month. State assumptions clearly, since the 25% and 150 are guesses to be tested.
Check	Cost per recovery = 4,000 / 500 = 8 rupees for 150 recovered. Yes.

Connection to the real job

In interviews, show domain curiosity: "I read your annual report; I noticed cost-to-serve is falling. How does the data team contribute to that?"

COMMON MISTAKE

Reading about a domain for months with no output. Build the small project in week 4.

Check yourself

1. What is the first output of week 1? Answer: a glossary of about 50 terms.
2. Name two sources for KPIs. Answer: annual reports and earnings calls.
3. Why talk to experts? Answer: they show the real costly problems.

Chapter recap

- Pick one primary domain by interest, jobs nearby and prior knowledge; use a score, not a mood.
- BFSI: NPA ratios, expected loss = PD x LGD x EAD, combined ratio; accuracy is a trap with rare defaults.
- E-commerce: GMV, AOV, conversion, CAC and contribution margin per order; growth with negative contribution is growing loss.
- SaaS: MRR/ARR, churn, LTV, CAC payback, NRR vs GRR; NRR below 100% means existing customers shrink.
- Telecom: ARPU and churn give lifetime value; a churn model needs an action with a cost.
- Healthcare: PPV depends on prevalence; privacy and consent are legal duties.

- Distinguish percent from percentage points; state assumptions with every rupee estimate.
- Learn a domain in four weeks: glossary, KPIs, experts, one small project.

Pattern cheat-sheet

Need	Formula
Gross / net NPA	NPA / loans; (NPA - provisions) / (loans - provisions)
Expected loss	PD x LGD x EAD
Combined ratio	(claims + expenses) / premium
AOV, conversion	GMV / orders; orders / sessions
Contribution per order	margin - shipping - expected return cost
CAC	marketing spend / new customers
LTV, payback	ARPA x GM / churn; CAC / (ARPA x GM)
NRR, GRR	(start + upgrades - downgrades - churn) / start; without upgrades
ARPU, lifetime	revenue / subs / months; 1 / churn
PPV	TP / (TP + FP)

Practice set

1. (Easy) Loans 2,000, NPA 60. GNPA %? **Answer:** 60 / 2,000 = 3%.
2. (Easy) 500 orders from 20,000 sessions. Conversion? **Answer:** 2.5%.
3. (Easy) MRR 50,000. ARR? **Answer:** 6,00,000.
4. (Medium) PD 2%, LGD 40%, EAD 1,00,000. EL? **Answer:** 0.02 x 0.4 x 1,00,000 = 800.
5. (Medium) ARPA 1,000, gross margin 70%, churn 5%. LTV? **Answer:** 700 / 0.05 = 14,000.
6. (Medium) Find the bug: a student computes lifetime revenue for ARPU 150 and churn 2% with `print(round(150/(1000*0.02),1))` and gets 7.5. **Answer:** The formula is ARPU / churn = 150 / 0.02 = 7,500. The extra `1000*` in the code is the bug (it printed 7.5 when run).
7. (Medium) Find the bug: "Sales rose from 10% to 12%, a 2% rise." **Answer:** It is 2

percentage points, or a 20% relative rise.

8. (Medium) Order value 1,000, margin 25%, shipping 50, return rate 8% at cost 100 each. Contribution? **Answer:** 250 - 50 - 8 = 192.
9. (Hard) Start MRR 2,00,000; new 20,000; upgrades 10,000; downgrades 4,000; churn 12,000. NRR and GRR? **Answer:** NRR = (2,00,000 + 10,000 - 4,000 - 12,000) / 2,00,000 = 1,94,000 / 2,00,000 = 97%. GRR = (2,00,000 - 16,000) / 2,00,000 = 92%.
10. (Hard) Prevalence 2%, sensitivity 95%, specificity 90%. PPV for 10,000 people? **Answer:** Sick 200, TP 190; healthy 9,800, FP 980; PPV = 190 / 1,170 = 16.2%.
11. (Interview) "A bank wants to reduce loan defaults. What would you ask first?" **Answer:** What counts as default, the current default rate and loss per default, what actions are possible (lower limit, early call), how much data history exists, and regulatory limits on features and explainability.
12. (Interview) "Our e-commerce conversion fell from 2.5% to 2.2%. How do you investigate?" **Answer:** Split by channel, device, new vs returning, page, and time (an issue tree, Chapter 6); check site speed and errors (your performance skills), stock-outs, price changes and tracking bugs before blaming demand.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional at an Indian MNC moving into data roles. The topic is Chapter 13, BFSI domain knowledge: NPA ratios, PD, LGD, EAD and expected loss, and the credit lifecycle. Explain each with tiny numbers first, then a realistic example. Then give me 8 graded exercises without solutions and check my answers one at a time. Also tell me where to read current RBI rules myself, without quoting rules from memory.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving into data roles in India. The topic is Chapter 13, e-commerce unit economics: GMV, AOV, conversion, CAC and contribution margin per order. Teach me step by step with a running example of one shop for one month. Then quiz me one question at a time, and make me find the bug in two wrong calculations. Finish by asking me to design a funnel dashboard with five KPIs.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional at an Indian MNC moving into data work. The topic is Chapter 13, SaaS metrics: MRR, ARR, churn, LTV, CAC payback, NRR and GRR. Explain the difference between NRR and GRR with an example. Then give me three scenarios with numbers and make me compute the metrics and say what a manager should do. Review my work here: [PASTE HERE]
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional in India. The topic is Chapter 13, how to learn a new domain fast (annual reports, earnings calls, a glossary of 50 terms, expert conversations, one small project). I want to learn the domain of [PASTE YOUR DOMAIN HERE]. Build a four-week one-hour-a-day plan, list 15 starter glossary terms with plain meanings, and suggest three public datasets to build a project. Do not invent company numbers.
```

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional moving into data roles in India. The topic is Chapter 13, business-sense interview questions in telecom and healthcare: ARPU, churn value of a retention offer, break-even save rate, and PPV with base rates. Run a mock interview with 6 case questions, one at a time. After each answer, tell me what a strong answer would add, especially assumptions, rupee impact and privacy.
```

CHAPTER 14

CERTIFICATIONS, OPEN SOURCE, KAGGLE AND LONG-TERM GROWTH

Choose credentials and side-projects on purpose, grow deep and wide, and look after the person who does all the work: you.

You now have skills, a plan and a job-search method. This chapter answers the questions that come after: Should I buy a certification? Should I do a degree? How do I start with open source? Is Kaggle worth my time? Should I aim to become a manager? And the quiet question under all of them: why do I sometimes feel I am not good enough? You will leave with decision rules you can compute, a contribution path you can start this month, honest facts about Kaggle scores, a picture of long-term growth (IC, manager, T-shaped), and tools for handling self-doubt without pretending it does not exist.

14.1 Certifications and Degrees: A Decision Guide

INTUITION

A certificate is a receipt for knowledge. A project is the meal itself. Some people (and software filters) ask for the receipt first, but nobody stays full on a receipt. Buy a receipt only when someone is going to check it.

Words you need

Term	Meaning
Certification	An exam, from a company or vendor, that proves you know a product or skill
Associate / Professional / Foundational	Common levels, from basic to advanced
Online degree	A university degree taught mostly online, often part-time
Opportunity cost	What you give up (time, other learning) when you choose something

Signal	Evidence a hiring person uses to guess your skill

Where certificates help

They help most when: (1) a job post or a company filter asks for one, (2) you have no experience on that tool and need structured learning, (3) your employer is a partner of that vendor and values them. They help least when you already have real projects with the same tool.

The certification map (pick only what matches your target job)

Target	Possible certificates (check current names and codes on vendor sites)
Analyst	Power BI Data Analyst (PL-300), Tableau Desktop Specialist. SQL needs none.
Data Engineer	AWS Cloud Practitioner then AWS Data Engineer Associate; Azure DP-900 then DP-700; Google Professional Data Engineer; Databricks Associate; Snowflake SnowPro Core
ML / AI Engineer	AWS ML Engineer Associate; Azure AI-102; Google Professional ML Engineer
Platform / MLOps	Terraform Associate; CKA / CKAD (Kubernetes)

Vendors rename and retire exams often. Before you spend money, open the official page, confirm the exam is current, read the objectives list and fee, and compare with the job posts you are targeting. Do not trust a list from a video or a book (including this one) for current names.

Free structured programs

DataTalks.Club Zoomcamps (data engineering, machine learning, MLOps, LLMs), Harvard CS50, Stanford CS229, CS231n and CS224n course material, fast.ai, Google Machine Learning Crash Course and Kaggle Learn. Free programs give the learning and a project. They do not give an exam badge, but a public project can be a stronger signal.

Degrees

Option	When it makes sense
Part-time or online degree (for example IIT Madras BS in Data Science and Applications, other Indian PG programs, Georgia Tech OMSCS or OMSA, UT Austin online programs)	You want long-term depth, a recognised degree for visa or role filters, and can commit hours for years
Research roles and PhD-heavy teams	A master's or PhD often matters
Most analyst, data engineer and ML engineer roles	Projects and experience usually win

Details like fees, duration, entry rules and recognition change. Check each programme page, ask alumni on LinkedIn, and read one honest review before committing.

Problem 14.1.1 A certificate needs 80 hours of study. You have 7 hours a week. When will you be ready, and what do you give up?

Formula	weeks = hours needed / hours per week
Plug in	80 / 7
Compute	80 / 7 = 11.43 weeks
Answer	About 11.4 weeks, so allow 12 weeks, plus a buffer for missed days.
Check	7 x 11.43 = 80. Also 12 weeks x 7 = 84 hours, enough.
Why it matters	These 12 weeks of study time could build one portfolio project instead. The decision depends on which one a recruiter will reward for your target role.

Problem 14.1.2 A degree needs about 1,300 study hours in total (a made-up example; check your programme). At 7 hours a week how many years? At 15 hours a week?

Compute	1,300 / 7 = 185.7 weeks = 185.7 / 52 = 3.57 years; 1,300 / 15 = 86.7 weeks = 1.67 years
Answer	About 3.6 years at 7 hours a week, and 1.7 years at 15 hours a week.
Check	Real output below.
Why it matters	For a working learner the time cost is years, not months. Be sure you have the energy (see 14.6).

```
for h in (60,80,120):
    print("cert",h,"h at 7h/week ->",round(h/7,1),"weeks")
print("degree 1300h at 7h/wk ->",round(1300/7/52,2),"years; at 15h/wk -
>",round(1300/15/52,2))
```

```
cert 60 h at 7h/week -> 8.6 weeks
cert 80 h at 7h/week -> 11.4 weeks
cert 120 h at 7h/week -> 17.1 weeks
degree 1300h at 7h/wk -> 3.57 years; at 15h/wk -> 1.67
```

The decision rule (five questions)

1. Does a job I want **ask** for it? Check ten real job posts.
2. Can a project show the same skill? If yes, build the project first.
3. Is the total cost (fee + hours + the project I did not build) small compared with the expected gain?
4. Will my employer pay or give study time?
5. Is it current, and does the exam test hands-on skill rather than memorised facts?

If you answer yes to two or more, go ahead. Otherwise skip it. Avoid expensive bootcamps that promise guaranteed jobs, and avoid collecting certificates without projects.

REAL JOB TIP

Count in 10 job posts for your target role how many name a certificate as required and how many as "preferred". If 0 or 1, skip it; if 6 or more, plan one.

Connection to the real job

A cloud certificate is useful if you will work on that cloud. It also forces you to see parts of the platform you may skip while building.

Applications

- Career changer without data job experience: one cloud data certificate plus two projects.
- Experienced engineer: skip entry-level certificates; they say little.
- Consultancy and service firms: certificates may count toward client requirements.

COMMON MISTAKE

Collecting five certificates and no project. Recruiters ask "what have you built?" and a list of badges cannot answer that.

Check yourself

1. When do certificates help most? Answer: when a job or filter asks for one, or you lack other proof.
2. What should you check before paying for an exam? Answer: that it is current, its objectives, its fee and whether jobs ask for it.
3. Is a degree needed for most DA/DE/MLE roles? Answer: usually no; projects and experience matter more.

14.2 The Open-Source Contribution Path

INTUITION

Open source is a public workshop. You can walk in, watch, hand someone a tool, then fix a small thing, then help build a big thing. Nobody starts by building the engine.

Words you need

Term	Meaning
Open source	Software whose code is public and can be used and improved by anyone under a licence

Repository (repo)	A project's folder of code and history, usually on GitHub
Issue	A written report of a bug or a feature request
Pull request (PR)	A proposal to change the code, submitted for review
Maintainer	A person who manages the project and approves changes
CONTRIBUTING.md	A file explaining how to contribute to the project
Good first issue	A label for tasks meant for new contributors

The ladder

```
Level 1  Use the library
Level 2  Read its code
Level 3  File a good bug report
Level 4  Fix docs and typos
Level 5  Fix a "good first issue"
Level 6  Add a small feature
Level 7  Review other people's PRs; maintain a part
```

Each step needs only the one before. Most people spend months at levels 1 to 4, and that is fine.

Beginner-friendly places to look

pandas, scikit-learn, Apache Airflow, dbt, Hugging Face libraries, Great Expectations, Evidently, LangChain and LlamaIndex, plus testing tools such as Locust, where your current skill gives you a real advantage. Check each project's labels and recent activity: pick a project with maintainers who respond in days, not years.

How to contribute (the practice)

1. Read CONTRIBUTING.md and the code of conduct.
2. Search existing issues so you do not duplicate.
3. Comment on an issue: "I would like to work on this. Is it still open?" Wait for a reply.
4. Make a small, focused change; add or update a test; run the project's test suite.
5. Write a clear PR description: what, why, how you tested.

6. Be patient with review. Say thanks. Make requested changes without argument.

A good bug report (template)

```
Title: DataFrame.groupby().agg() raises KeyError with duplicate column
names (version X)

What I did: <5 lines of code that reproduce it>
What I expected: <one line>
What happened: <error message pasted>
Environment: Python version, library version, OS
```

A bug report that can be reproduced in 5 lines is already a contribution.

Problem 14.2.1 Plan 8 weeks at 2 hours a week. Week 1 to 2: read CONTRIBUTING and code. Week 3: file or improve one bug report. Week 4: one docs fix. Weeks 5 to 8: one good-first-issue PR. How many hours in total, and how many on the PR?

Compute	total 2 x 8 = 16 hours; PR weeks 5 to 8 = 4 weeks x 2 = 8 hours
Answer	16 hours in total, 8 on the PR (half).
Check	Weeks 1-2: 4 h, week 3: 2 h, week 4: 2 h, PR weeks: 8 h. Then 4 + 2 + 2 + 8 = 16. Yes.
Why it matters	Sixteen hours gives you a public PR link, a stronger proof than most certificates.

Problem 14.2.2 What does a first PR to fix a typo cost and give? A typo PR takes about 30 minutes. If 5 such PRs are accepted in different projects, what is the value?

Step 1	Time: 5 x 0.5 = 2.5 hours.
Step 2	Value: you learn the fork, branch, commit and PR flow five times, and you get five accepted contributions.
Answer	2.5 hours for real practice with the workflow. Do not spam: make only real fixes. Many projects dislike low-value PRs made only for badges.
Check	The skill gained is the workflow, not the typo.

Your own small open-source tool

Publish one tool from your own work: a small loader that moves CSV files into a warehouse, or a simple anomaly detector (Chapter 11 shows the README). Add a licence, tests and a clear README. A small, tidy tool used by a few people is worth more than a big, messy one.

Programs and events

Hacktoberfest, Google Summer of Code (if you are eligible), Outreachy. Rules and eligibility change each year; read the official page.

Connection to the real job

Reviewers read your code and your messages. That is practice for code review at work (Chapter 4). Also hiring managers can read your PRs.

Applications

- Data Engineering: Airflow, dbt, Great Expectations contributions.
- ML: scikit-learn docs and examples, Hugging Face.
- Traffic testing: Locust, k6.

COMMON MISTAKE

Opening a PR with a big change without talking first. Comment on an issue first, keep the PR small, and add a test.

Check yourself

1. What is a PR? Answer: a proposal to change the code, submitted for review.
2. What is the first step before contributing? Answer: read CONTRIBUTING.md.
3. Why start with docs and bug reports? Answer: they are small, useful, and teach the process.

14.3 Kaggle Strategy

INTUITION

Kaggle is a driving school with an exam track. You learn handling in clean conditions and compare with others, but the real road has traffic and bad weather: messy data, vague goals, and deployment.

Words you need

Term	Meaning
Kaggle	A website with datasets, notebooks and machine-learning competitions
Public leaderboard (LB)	Scores computed on part of the hidden test data, visible during the contest
Private leaderboard	Scores on the rest of the hidden data, revealed at the end; decides the final rank
CV (cross-validation)	Testing on held-out parts of your training data, repeated across splits
Shake-up	A big change of ranks between public and private leaderboards
Ensembling / stacking	Combining several models' predictions
GBM	Gradient Boosting Machine, such as XGBoost or LightGBM

The progression

```
Kaggle Learn courses -> Titanic / House Prices -> Playground Series
(monthly)
    -> featured and research competitions
```

Playground Series competitions are made for learning. Other platforms: DrivenData, Zindi, Analytics Vidhya, AIcrowd, and Hugging Face challenges.

Skills Kaggle trains

A solid local cross-validation that tracks the leaderboard, feature engineering, tuning of gradient-boosted trees, ensembling, error analysis, and reading winning solutions after the contest. Goals: Notebooks Expert, then Competitions Expert (medals) over time. Rules and tier names can change; see the site.

Why trusting the public leaderboard is a trap

The public leaderboard uses only part of the test data. Small samples are noisy.

Problem 14.3.1 A model has a true accuracy of 80%. Compute the standard error of its measured accuracy on 200, 1,000 and 5,000 rows.

Formula	standard error = √(p x (1 - p) / n)
Plug in	p = 0.80, so p x (1 - p) = 0.16
Compute	n = 200: √(0.16 / 200) = √0.0008 = 0.0283; n = 1,000: √0.00016 = 0.0126; n = 5,000: √0.000032 = 0.0057
Answer	The measured accuracy wobbles by about ±2.8 points on 200 rows (95% band ±5.5 points), ±1.3 on 1,000 rows, and ±0.6 on 5,000 rows.
Check	Increasing n 25 times (200 to 5,000) should cut error by √25 = 5 times: 0.0283 / 5 = 0.0057. Yes.
Why it matters	Plain words: if the public leaderboard uses 200 rows, a difference of 2 points between two models is just noise.

Problem 14.3.2 Simulate 50 identical models (true accuracy 80%) scored on a public 200-row set. What does the best-looking one score? What does a model of the same quality score on a 5,000-row private set?

```
import random
random.seed(1)
best=max(sum(random.random()<0.8 for _ in range(200))/200 for _ in
range(50))
print("best of 50 identical models on 200 rows:",best)
random.seed(2)
priv=sum(random.random()<0.8 for _ in range(5000))/5000
print("same quality model on 5000 private rows:",priv)
```

Predict first: will the best of 50 be above or below 0.80? Real output:

```
best of 50 identical models on 200 rows: 0.855
same quality model on 5000 private rows: 0.802
```

- **Step 1** Each model gets a random score around 0.80 with spread about 0.028. The maximum of 50 draws is high by luck, about two standard deviations above the mean.
- **Step 2** The private set is large; the score settles near the true 0.80.
- **Answer** The "winner" scored 0.855 on the small public set purely by chance, but the same quality gives 0.802 on private data.
- **Check** 0.80 + 2 x 0.0283 = 0.857, close to 0.855. Yes.
- **Why it matters** If you chase the public leaderboard you overfit to noise and fall in the final ranks (the shake-up). Trust a solid local CV with many rows.

A strategy for a working learner

1. Pick **one** Playground competition per month, 4 to 6 hours a week.
2. Week 1: baseline model and a clean CV. Week 2: features. Week 3: tune and try one other model. Week 4: ensemble, then write up.
3. After the end, **read the top solutions** and note two ideas you did not use.
4. Share a public notebook with a short explanation (Chapter 4: learning in public).
5. Do not let Kaggle replace end-to-end projects. Competitions give clean data; jobs give messy data, deployment, monitoring and stakeholders (Chapters 9 and 11).

Connection to the real job

Kaggle builds feature engineering, validation discipline and error analysis. Those transfer to any ML job. Gradient-boosted trees are strong on tabular data, which is the kind of data most companies have.

Applications

- ML Engineer and Data Scientist: validation habits, baselines, ensembling.
- Data Analyst: Kaggle datasets and notebooks for portfolio practice.
- Career changers: a public notebook shows you can finish something.

COMMON MISTAKE

Optimising the public leaderboard score. Build a CV that matches the private result and trust it.

Check yourself

1. Which leaderboard decides final rank? Answer: the private one.
2. Does more data in the test set make the score noisier or steadier? Answer: steadier.
3. Name one thing Kaggle does not teach well. Answer: deployment and messy business problems.

14.4 Long-Term Growth: IC vs Manager, T-Shaped Skills

INTUITION

A career is a long road with forks. At some fork you choose: become a master craftsperson, or become the person who builds a team of craftspeople. Neither is "up". Both are different jobs. A T-shaped person is like a doctor with one deep specialty and enough general knowledge to talk to every other doctor.

Words you need

Term	Meaning
IC (Individual Contributor)	A person whose main value is their own technical work
Staff / Principal	Senior IC levels, with influence across teams
Manager	A person whose main value is the growth and output of a team

Director	A manager of managers
Specialist	A deep expert in a niche (MLOps, data architecture, applied science)
T-shaped	Broad knowledge across many areas (the top bar of the T) and deep skill in one or two (the stem)

Level names differ by company. Read each company's career ladder when available.

The paths

```
IC track:          Analyst -> Senior -> Staff -> Principal
Management track: Team lead -> Manager -> Director
Specialist:        MLOps, Data Architect, Applied Scientist
Other:             Product management, consulting, entrepreneurship,
freelancing
```

Choosing is not permanent. Many people try a lead role for a year and return to IC, and that is normal.

Problem 14.4.1 Compare IC and manager using your own weights (sum 10): craft 4, people 2, pay growth 2, flexibility 2. Scores 1 to 5: IC: craft 5, people 2, pay growth 3, flexibility 4. Manager: craft 2, people 5, pay growth 4, flexibility 2.

Formula	total = sum of weight x score
IC	4x5 + 2x2 + 2x3 + 2x4 = 20 + 4 + 6 + 8 = 38
Manager	4x2 + 2x5 + 2x4 + 2x2 = 8 + 10 + 8 + 4 = 30
Answer	IC 38, manager 30. With these weights, the IC path fits better.
Check	If you raise the weight of "people" and reduce "craft", the result flips. The score reveals your values, it does not decide for you.
Why it matters	Check the weights honestly. Do not choose management only for title or pay: ask what a manager does all day (meetings, feedback, hiring, conflict) and whether you would enjoy it.

T-shaped skills

The stem: one or two deep areas, for example data engineering with strong SQL and pipelines, or ML with strong evaluation. The bar: statistics, cloud, system design, communication, domain (Chapter 13).

Problem 14.4.2 You have about 365 study hours a year (1 hour a day). Split 50% depth (SQL and data engineering), 30% breadth (cloud, stats, ML basics), 20% communication. How many hours each?

Compute	365 x 0.50 = 182.5; 365 x 0.30 = 109.5; 365 x 0.20 = 73
Answer	182.5 hours depth, 109.5 hours breadth, 73 hours communication.
Check	182.5 + 109.5 + 73 = 365. Yes.
Why it matters	Without a plan the stem stays thin. Any earlier technical work already gives depth in measurement, a head start on the stem.

Choosing teams and companies for learning

- Choose a manager who gives feedback and a team with people better than you.
- Ask "what will I learn here in two years?" in interviews.
- Revisit your plan every 3 months: what did I learn, what is blocking me, what next?

Managing up and showing evidence

Keep a **brag document**: a running list of what you did, with numbers (hours saved, errors reduced, reports delivered). In reviews, bring this and the market data from Chapter 12. Ask your manager for specific feedback every month: "What is one thing I should do more or less?"

Mentors, networking and community

- Find one mentor. Ask for a specific 30-minute talk, not "be my mentor".
- Make about 5 genuine new connections a month by sharing useful posts and asking thoughtful questions. Never contact strangers only to ask for a job.
- Teamwork: give and receive code-review feedback kindly, disagree respectfully, own mistakes quickly.

Freelancing and side income (optional)

Small analytics or automation gigs, content writing and tutoring can bring income and sharpen

skills (teaching deepens learning). First check your employment contract for moonlighting and confidentiality rules, and never use company data or code.

Connection to the real job

Promotion is usually a case built from evidence of impact and scope. Start collecting that evidence from day one.

Applications

- IC path: senior and staff roles in data platform, ML platform, performance and reliability.
- Management path: team lead for analytics or data engineering.
- Specialist: MLOps, data architecture, applied scientist.

COMMON MISTAKE

Choosing management because it looks like the only way up. Staff and principal IC roles exist at many companies, and unhappy managers are common.

Check yourself

1. What does the stem of a T represent? Answer: deep skill in one or two areas.
2. Is the IC path "lower" than management? Answer: no, it is a different path.
3. What is a brag document? Answer: a running list of your achievements with numbers.

14.5 Wellbeing, Self-Doubt and Imposter Feelings

INTUITION

Your brain is like a shop that never closes. If you never allow idle time and never check the logs, it fails in a strange way at a bad moment. Rest and honest checking are part of the engineering.

I want to say this plainly. Learning a new field while working full time is hard. Feeling lost, slow or "not smart enough" is common and does not tell you your real ability. Many skilled people have felt it. I am not a doctor, and this is not medical advice. It is practical, ordinary advice, and 14.6 says when to talk to a professional.

Words you need

Term	Meaning
Imposter feelings	A belief that you are a fraud and will be "found out", despite evidence of ability
Comparison trap	Judging your behind-the-scenes against other people's highlights
Burnout	Deep tiredness and loss of interest from long stress
Evidence log	A written list of things you did and learned, used as proof against negative thoughts

Why it happens in this transition

You moved from being an expert (in your earlier role) to being a beginner (data and ML). The experience gap feels sharp because you remember how it felt to be good. Also social media shows only wins. Also interviews measure a narrow skill set under pressure, and a rejection feels personal when it is not.

Tools you can use

1. **Evidence log.** Each week write three things you can do now that you could not 4 weeks ago. Read it when you feel stuck.
2. **Compare with last month's you.** Not with a senior engineer with ten years' head start.
3. **Name the thought.** "I am a fraud" is a thought, not a fact. Replace with a testable statement: "I could not write this window function today. I will practise it for a week."
4. **Ask for help early.** A 10-minute question to a peer saves a day of silent struggle.
5. **Share the confusion.** Posting "what confused me this week" helps others and shows you are normal.
6. **Celebrate small wins.** A finished project README is a win.

Problem 14.5.1 Build an evidence log for 12 weeks. You record 3 items per week. How many items by the end? If you read the log whenever you feel stuck, what does it show?

Compute	3 x 12 = 36 items
Answer	36 items. Each is a concrete fact ("wrote a LAG query that finds day-over-day change", "explained percentiles to my team") that contradicts "I am learning nothing".
Check	Count is 3 per week for 12 weeks. Yes.
Why it matters	Feelings are loud; a written log is a calm second witness.

Problem 14.5.2 A week has 168 hours. You sleep 7.5 hours a night, work 45 hours, commute 1.5 hours on each of 5 days, and study 7 hours. How many hours remain for everything else (family, rest, meals, chores, exercise)?

Step 1	Sleep 7.5 x 7 = 52.5; commute 1.5 x 5 = 7.5
Step 2	Used = 52.5 + 45 + 7.5 + 7 = 112
Step 3	Remaining = 168 - 112 = 56
Answer	56 hours, about 8 hours a day, are left, and they must also cover meals, chores and family.
Check	Real output below.
Why it matters	Adding another 7 hours of study would leave 49 hours. Be honest about what you can carry; a plan that ignores sleep fails.

```
sleep=7.5*7; work=45; commute=5*1.5; study=7
print("left", 168-sleep-work-commute-study)
```

```
left 56.0
```

Energy habits that help a working learner

- A fixed daily slot (Chapter 2); one rest day a week.
- Seven or more hours of sleep, movement and breaks (for example 25 minutes of work, 5 minutes break).
- One project and one main resource per topic; finish before starting new courses.

- Financial resilience: keep an emergency fund, avoid debt for courses, and stay in your current role until a signed offer.

PROFESSOR'S NOTE

I cannot see your life, so I will not tell you how many hours you should study. I will say that the learner who studies one hour for 300 days beats the learner who studies six hours for ten days and quits.

Connection to the real job

The same habits keep you effective on call, in incidents and in hard projects. Teams value people who stay calm and ask for help early.

Applications

- Interview periods: rest before the day, sleep, do a light review.
- Notice period: take a few days off before the new job.
- Long projects: schedule review points, not only deadlines.

COMMON MISTAKE

Treating exhaustion as laziness and adding more hours. Fix sleep and the plan first.

Check yourself

1. What is an evidence log? Answer: a written list of what you did and learned, used as proof against self-doubt.
2. Whom should you compare yourself with? Answer: your own past self.
3. Why ask for help early? Answer: it saves time and prevents silent struggle.

14.6 When to Seek Help

INTUITION

When a bridge shows cracks you call an engineer. You do not wait for it to fall. Your mind deserves the same maintenance.

Ordinary doubt comes and goes. Talk to someone if you notice that:

- self-criticism is heavy or constant for weeks;
- you cannot sleep, eat or focus for several days;
- you have lost interest in things you usually enjoy;
- you feel hopeless, or have thoughts of harming yourself.

What to do: talk to a trusted friend or family member, your doctor, or a qualified counsellor or therapist. Many companies offer an employee assistance programme, so check your HR portal. If you are in danger or may harm yourself, contact local emergency services or a crisis helpline in your country right away, and tell someone near you. Mental health support is part of a sustainable career, not a detour from it.

Mentors and peers

A study buddy, a community such as DataTalks.Club Slack, a local meetup, or a mentor makes the road less lonely. Meet a buddy weekly for 30 minutes: each explains one concept to the other (teaching back, Chapter 2).

Problem 14.6.1 You plan 12 weekly 30-minute study-buddy sessions. How many hours? Compare with explaining alone: why does it help?

Compute	12 x 0.5 = 6 hours
Answer	Six hours in total, and each session has you explain a topic aloud, which exposes gaps (teaching back).
Check	12 weeks x 30 minutes = 360 minutes = 6 hours. Yes.
Why it matters	A small, regular, social habit protects motivation.

COMMON MISTAKE

Waiting until things are very bad before telling anyone.

Check yourself

1. Name one warning sign to talk to someone. Answer: heavy constant self-criticism, poor sleep for days, or loss of interest.

2. Who can you talk to? Answer: a friend, family, doctor, counsellor or employer assistance programme.
3. Is seeking help a career detour? Answer: no, it is part of a sustainable career.

Chapter recap

- Certificates are receipts; projects are the meal. Buy a certificate only when a job or filter asks for it.
- Check certificate names, fees and objectives on the vendor site; they change often.
- Free structured programs (Zoomcamps, CS50, fast.ai, Kaggle Learn) teach as much as paid ones.
- A degree matters for research roles, some visa routes and long-term depth; compute the hours before committing.
- Open source ladder: use, read, report bugs, fix docs, good first issues, features. Read CONTRIBUTING.md and keep PRs small with tests.
- On Kaggle, trust cross-validation over a small public leaderboard; the best of many equal models looks lucky.
- IC and management are different jobs, not higher and lower; use a T-shaped plan with evidence in a brag document.
- Self-doubt is common: keep an evidence log, compare with your past self, plan around 168 hours, and ask for professional help when it gets heavy.

Pattern cheat-sheet

Need	Pattern
Weeks to certify	hours needed / hours per week
Certificate rule	2 or more "yes" on the five questions, else build a project
Open-source first PR	read CONTRIBUTING, claim issue, small change, test, clear PR text
Standard error of accuracy	$\sqrt{(p(1 - p) / n)}$
Kaggle rule	trust CV, not a small public LB

Weighted choice	sum of weight x score
Week budget	168 - sleep - work - commute - study
Self-doubt tools	evidence log, compare with past self, ask early

Practice set

1. (Easy) A cert needs 60 hours; you have 5 hours a week. How many weeks? **Answer:** 60 / 5 = 12 weeks.
2. (Easy) What is a pull request? **Answer:** A proposal to change a project's code, submitted for the maintainers to review.
3. (Easy) Which leaderboard decides the final Kaggle rank? **Answer:** the private leaderboard.
4. (Medium) Standard error for p = 0.9 and n = 400? **Answer:** √(0.9 x 0.1 / 400) = √0.000225 = 0.015 (1.5 points).
5. (Medium) You scored 38 for IC and 30 for manager but you doubt it. What do you do? **Answer:** Check the weights and scores with a mentor, try one stretch task (mentoring a junior) before deciding, and rerun the scoring.
6. (Medium) Find the bug: "I will submit a 2,000-line PR to a project I have never used, without opening an issue." **Answer:** Large unannounced PRs are usually rejected. Open an issue, discuss, start small, add tests.
7. (Medium) You study 6 hours a week for 40 weeks. Hours? What is a T-shaped split of 50/30/20 depth/breadth/communication? **Answer:** 240 hours; 120, 72 and 48.
8. (Medium) A week: sleep 8 x 7, work 40, commute 5, study 10. Hours left? **Answer:** 168 - 56 - 40 - 5 - 10 = 57.
9. (Hard) Public LB uses 500 rows; a model scores 0.84 and another 0.82. Is the first better? **Answer:** Standard error at p about 0.83 is √(0.83 x 0.17 / 500) = 0.0168, so a 2-point difference is about 1.2 standard errors. It is not reliable; compare with CV.
10. (Interview) "Do you have any certifications? Why or why not?" **Answer:** Be honest: name the one you hold or plan, say you picked it because of the target job and the cloud used, and describe the project you built with it. If none, say you chose projects and name two with results.
11. (Interview) "Where do you see yourself in five years: manager or senior engineer?" **Answer:** State a path with a reason and openness: "I want to go deep in data platform

engineering, lead through design and mentoring, and I am open to leading a team if it grows naturally."

12. (Hard, scenario) You feel like a fraud after three rejections in a row. What do you do this week? **Answer:** Read your evidence log, review your rejection log (Chapter 12) for one fixable weakness, talk to a peer or mentor, rest one day, and if the heavy feeling lasts for weeks, talk to a doctor or counsellor.

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional at an Indian MNC moving into data roles. The topic is Chapter 14, deciding on certifications and degrees. Ask me my target role and my time per week, then help me apply the five-question decision rule and compute the hours. Tell me how to check current exam names, fees and objectives myself rather than quoting them. Then quiz me with 5 scenarios, one at a time.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional moving into data roles in India. The topic is Chapter 14, the open-source contribution path (use, read, bug report, docs fix, good first issue, feature). Give me an 8-week plan at 2 hours a week for a project I choose: [PASTE PROJECT HERE]. Explain how to read CONTRIBUTING.md, how to write a reproducible bug report, and how to write a clear PR description. Then review my draft issue or PR text: [PASTE HERE]
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional at an Indian MNC moving into data work. The topic is Chapter 14, Kaggle strategy and why the public leaderboard is noisy. Explain standard error of accuracy step by step with small numbers, then give me 8 graded exercises without solutions about cross-validation, leaderboard noise and shake-ups. Finally help me plan a one-month Playground Series routine at 5 hours a week.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in
India moving into data roles. The topic is Chapter 14, long-term growth:
individual contributor versus manager, and T-shaped skills. Ask me what I
enjoy and dislike at work, build a weighted comparison with my own
weights, and plan a yearly 365-hour learning split for depth, breadth and
communication. Then run a mock interview on "where do you see yourself in
five years" and give feedback.
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional in
India studying while working full time, and I sometimes feel like an
imposter. The topic is Chapter 14, wellbeing and self-doubt. Be kind and
practical, not clinical. Help me build an evidence log, plan a realistic
168-hour week, and write a short script to ask a peer for help. Remind me
that you are not a doctor and tell me when to talk to a trusted person or
a counsellor. My week and worries: [PASTE HERE]
```

CHAPTER 15

PROJECT LAB — PORTFOLIO SITE AND FIRST OPEN-SOURCE CONTRIBUTION

Two projects that turn private skill into public proof, plus the working method that makes any project finishable.

Chapter 11 told you that six strong projects beat thirty half-done ones, and showed what a strong project looks like. This chapter is the workshop. First you learn how to run any project from first idea to published write-up: how to set a question, lay out a repo, commit in small steps, decide when you are done, and talk about it in public. Then you build two projects end to end. Project 43 is a portfolio website made by a small Python script, hosted free on GitHub Pages, with three blog posts. Project 44 is your first open-source contribution; you rehearse the full workflow on your own laptop first, so the real thing feels familiar. Every code block here was run, and the output shown is the real output. Chapter 16 adds two more projects: a mock-interview kit and a resume sprint.

15.1 How to Run a Project in This Area

INTUITION

Building a project is like cooking a meal for guests. You decide the menu first (the question), buy ingredients (data and tools), cook in steps and taste as you go (small commits and tests), and you only serve when the plate is ready (definition of done). Guests also need a menu card (the README) and a photo of the dish (the demo and the post).

In this area (career proof: sites, contributions, interview material, resumes) the "data" is often text and the "tests" are checks on links, files and wording. The method is the same as for a data project.

Words you need

Term	Meaning
Repository (repo)	A folder tracked by git, holding your files and their full history

Commit	A saved snapshot of your files with a message
Branch	A separate line of work, so experiments do not break the main version
Static site	A website made of plain HTML files; no server code runs when someone visits
Fork	Your own copy of someone else's repository on a hosting site
Pull request (PR)	A request to the owner: "please take my change into your project"
Definition of done	A written list of checks; when all pass, the project is finished

Plan the question and the success criteria

Write one sentence: "I will build X so that Y can do Z." Then write three **measurable** success criteria before coding. Example for Project 43: "A visitor can open my site, read three posts and reach my GitHub in two clicks; the build script runs in under one second; a link check finds 0 broken links." If you cannot measure a criterion, rewrite it until you can. Then time-box: choose hours (Chapter 11.1 suggests about 25 hours per project) and stop when they run out, shipping what works.

Repo layout (a pattern that fits every project here)

```
project-name/
├── README.md            # the menu card: what, why, how to run, result
├── LICENSE              # MIT is a common, simple choice
├── .gitignore           # keeps junk (__pycache__/, .env, data dumps) out
├── src/ or scripts/     # the code you run
├── data/                # small sample or generated data (never private
data)
├── tests/               # checks that prove it works
├── docs/                # output or notes (GitHub Pages can serve this
folder)
└── notes/decisions.md   # why you chose A over B (gold in interviews)
```

Environment and dependencies

Use a virtual environment (an isolated set of Python packages for one project): `python3 -m venv .venv`, then `source .venv/bin/activate`. Prefer the Python standard library when it is enough. Every project in this chapter uses only the standard library, so there is nothing to install and nothing to break. When you do add packages, list them in `requirements.txt`, and state your Python version in the README (the examples here ran on Python 3.14; any recent Python 3 should work).

Data sourcing or generation

Never use company data, client names or private messages. Good sources: your own writing, public datasets, open-source repositories, or data you generate with a short script and a fixed random seed (so anyone gets the same numbers). Write the source and the licence in the README.

Commit rhythm and good commit messages

Commit when one small thing works, at least once per work session. A good message says what changed, in the imperative mood, in under about 60 characters, with a longer body only if the reason is not obvious.

Weak	Strong
`update`	`Add link checker for built pages`
`fix stuff`	`Close open list before code block in builder`
`final final v2`	`Write README with run steps and result`

Why it matters: your commit history is a diary a reviewer can read in 30 seconds. A tidy history says "careful engineer".

Definition of done

A project is done when all of these are true: it runs from a clean clone using only the README steps; your success criteria are met and measured; at least three checks or tests pass; the README has a result with a number; known limits are written down; and one public write-up exists. "Done" is not "perfect".

README template

```
# Project title: one-line result

**Problem.** One sentence a non-expert understands.
**Result.** The headline number or screenshot.

## Demo
Link to the live page or a 2-minute video.

## How to run
git clone <url> && cd project-name
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt   # if you have one
python3 scripts/main.py

## Data
Source, licence, size, how it was generated.

## Method
3-5 bullets: what you did and why.

## Results
Table or numbers. State what the numbers mean.

## Limits and next steps
What does not work yet; what you would do next.
```

Demo and deploy options

Option	Good for	Cost and effort
GitHub Pages	Static sites, reports, docs	Free; a few clicks
A screen recording (2 to 3 minutes)	Anything with a terminal	Free; use any screen recorder
Screenshots in the README	Charts, dashboards	Free; fastest
A hosted notebook or app service	Interactive demos	Free tiers exist; limits change, check current terms

How to write the LinkedIn or blog post

Use five short paragraphs: (1) the problem in one sentence; (2) why you cared; (3) what you built, with one picture; (4) the result with a number; (5) one mistake you made and what you learned, then the link. Chapter 4 covered tone, and Chapter 11.5 covered posting rhythm. Post within a week of finishing, while the details are fresh.

Testing and debugging checklist for this area

- Does the project run from a **fresh clone** in an empty folder? (Most "works on my machine" failures show here.)
- Do all links work? Run a link checker (you will write one in 15.2).
- Open the output in a different browser and on a phone-sized window.
- Are there secrets, private names or personal phone numbers in any file or in the git history? Search before you push.
- Is `.gitignore` in place before the first commit? Files added by mistake stay in history.
- Spell-check the README and the posts; read them aloud once (Chapter 3).
- When something breaks: reproduce it, shrink it to the smallest case, read the full error message from the bottom up, change one thing at a time, and write the fix as a test.

REAL JOB TIP

Keep a `notes/decisions.md` file with one line per decision ("Used plain HTML, not a framework, because the site is small and I wanted zero dependencies"). In interviews you will be asked "why did you choose that?" and the file answers for you.

Check yourself

1. Why write success criteria before coding? Answer: so "done" is measurable and you do not keep polishing forever.
2. What goes in `.gitignore`? Answer: files that should never be committed, such as `__pycache__/`, `.env`, `.venv/` and big data dumps.
3. Name two things in a strong commit message. Answer: what changed, in the imperative mood, short; plus a reason in the body when it is not obvious.

15.2 Project: Portfolio Website and Three Blog Posts (Project #43)

Goal and why employers care

You will build a small website of your own, generated by a Python script you wrote, with a home page and three blog posts, and publish it free on GitHub Pages. Recruiters often search your name. A clean site with three clear posts shows that you can build, explain and finish. Writing the generator yourself proves you can handle files, text and automation, which are everyday data skills.

Skills used

- Python file handling and string processing (the Python study document).
- Clear writing, headline and structure (Chapter 4, Written Communication).
- Explaining results with numbers (Chapter 5, Data Storytelling).
- README, profile and project selection (Chapter 11, Portfolio, GitHub, LinkedIn and Resume).
- Git basics: commit, push (15.1 above).

Data

Your data is your writing. For this lab the three posts are tiny placeholders about neutral topics (loading shop orders into SQLite, chart mistakes, a first pull request). Replace them with your real posts. Do not copy other people's text.

Architecture and folder structure

```
portfolio/
├── posts/                  # you write these (plain text with a tiny
header)
│   ├── 01-first-pipeline.md
│   ├── 02-chart-lessons.md
│   └── 03-first-pr.md
├── build_site.py           # reads posts/, writes docs/
├── check_site.py           # finds broken links in docs/
└── docs/                   # generated HTML; GitHub Pages serves this
    ├── index.html
    ├── first-pipeline.html
    ├── chart-lessons.html
    ├── first-pr.html
    └── .nojekyll
```

Flow: `posts/*.md` → `build_site.py` → `docs/*.html` → `check_site.py` → git push → public site.

Step-by-step build

1. **Create the folder and the posts.** Each post starts with `title:` and `date:` lines, then a line containing only `---` , then the body. Before you run anything, predict: what will the file name of `01-first-pipeline.md` become in `docs/` ? (Answer below.)

```
title: What I learned loading shop orders into SQLite
date: 2026-10-01
---
# What I learned loading shop orders into SQLite
I loaded 1,000 fake orders into a database.
## The problem
Two rows had a missing city.
- I counted them first
- I then decided to label them "Unknown"
```

1. **Write the builder.** It reads each post, converts a small subset of Markdown (headings, paragraphs, bullets, code blocks) into HTML, escapes special characters so text cannot break the page, and makes an index page sorted newest first.

```
"""Tiny static site builder. Usage: python3 build_site.py posts docs"""
import html, re, sys
from pathlib import Path
```

```
PAGE = """<!doctype html><html lang="en"><head><meta charset="utf-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>{title}</title>
<style>body{{font-family:sans-serif;max-width:42rem;margin:2rem
auto;padding:0 1rem;line-height:1.6}}
pre{{background:#f4f4f4;padding:.8rem;overflow:auto}}</style></head>
<body><nav><a href="index.html">Home</a></nav>{body}</body></html>"""

def parse_post(text):
    head, _, body = text.partition("\n---\n")
    meta = dict(line.split(": ", 1) for line in head.splitlines())
    return meta, body

def md_to_html(md):
    out, in_code, in_list = [], False, False
    for line in md.splitlines():
        if in_list and not in_code and not line.startswith("- "):
            out.append("</ul>"); in_list = False
        if line.startswith("```"):
            out.append("</pre>" if in_code else "<pre>")
            in_code = not in_code
            continue
        if in_code:
            out.append(html.escape(line))
            continue
        if line.startswith("## "):
            out.append(f"<h2>{html.escape(line[3:])}</h2>")
        elif line.startswith("# "):
            out.append(f"<h1>{html.escape(line[2:])}</h1>")
        elif line.startswith("- "):
            if not in_list:
                out.append("<ul>"); in_list = True
            out.append(f"<li>{html.escape(line[2:])}</li>")
        elif line.strip():
            out.append(f"<p>{html.escape(line)}</p>")
    if in_list: out.append("</ul>")
    return "\n".join(out)

def main(src, dst):
    src, dst = Path(src), Path(dst)
    dst.mkdir(exist_ok=True)
    posts = []
    for f in sorted(src.glob("*.md")):
        meta, body = parse_post(f.read_text())
        slug = re.sub(r"^\d+-", "", f.stem) + ".html"
        (dst / slug).write_text(PAGE.format(title=meta["title"],
body=md_to_html(body)))
```

```
        posts.append((meta["date"], meta["title"], slug))
    items = "\n".join(f'<li>{d} — <a href="{s}">{html.escape(t)}</a></li>'
                      for d, t, s in sorted(posts, reverse=True))
    index = f"<h1>Asha Rao: Data Portfolio</h1><p>Short bio here.</p>
<h2>Posts</h2><ul>{items}</ul>"
    (dst / "index.html").write_text(PAGE.format(title="Portfolio",
body=index))
    (dst / ".nojekyll").write_text("")
    print(f"built {len(posts)} posts + index into {dst}/")

if __name__ == "__main__":
    main(sys.argv[1], sys.argv[2])
```

How it works, line by line: `parse_post` splits the file at the first `---` line and turns the header lines into a dictionary. `md_to_html` walks line by line and keeps two flags, `in_code` and `in_list`, so it knows when to open and close `<pre>` and `<ul>`. `html.escape` turns `"` and `<` into safe codes. The slug removes the leading number: `01-first-pipeline` becomes `first-pipeline.html`. The empty `.nojekyll` file tells GitHub Pages to serve the files as they are.

1. **Run it.**

```
python3 build_site.py posts docs
ls -a docs
```

```
built 3 posts + index into docs/
.
..
.nojekyll
chart-lessons.html
first-pipeline.html
first-pr.html
index.html
```

The prediction answer: `first-pipeline.html`. Open `docs/index.html` in a browser by double-clicking it.

1. **Check the generated HTML.** Here is the end of the first post.

```
<p>Two rows had a missing city.</p>
<ul>
<li>I counted them first</li>
<li>I then decided to label them &amp;quot;Unknown&amp;quot;</li>
</ul>
<pre>
print(&amp;quot;rows:&amp;quot;, 1000)
</pre></body></html>
```

The first version of my builder had a bug: the list was not closed before the code block, so `<pre>` ended up inside `<ul>`. The fix was to close the list check *before* the code-fence check. This is a typical lesson: render, read the output, and fix what you see.

1. **Write the link checker.** It reads every page, finds each `href`, and complains if a local target file does not exist.

```
import re, sys
from pathlib import Path
docs = Path(sys.argv[1]); bad = 0
for f in docs.glob("*.html"):
    for href in re.findall(r'href="([^"#]+)"', f.read_text()):
        if not href.startswith("http") and not (docs / href).exists():
            print("BROKEN", f.name, "->", href); bad += 1
print("pages:", len(list(docs.glob('*.html'))), "broken links:", bad)
sys.exit(1 if bad else 0)
```

```
python3 check_site.py docs
```

```
pages: 4 broken links: 0
```

To prove the checker works, I added `<a href="missing.html">x</a>` to the index and ran it again:

```
BROKEN index.html -> missing.html
pages: 4 broken links: 1
```

It also exits with code 1, so a script or CI job can fail on it. A check you have never seen fail is a check you cannot trust.

1. **Publish on GitHub Pages (not run here: it needs your account).** Steps: create a new

public repository; push the folder; open the repository's Settings, then Pages; under "Build and deployment" choose "Deploy from a branch", branch `main`, folder `/docs`; save. After a minute the site appears at `https://<your-username>.github.io/<repo-name>/`. Menu names change over time, so follow the current GitHub Pages documentation if a label differs. Put the link in your README, LinkedIn Featured section and resume.

1. **Write the three posts.** Use outlines, not blank pages.

Post	Working title	Outline (5 parts each)
1. A project write-up	"I loaded 1,000 shop orders into SQLite and found 2 bad rows"	problem; data; method; result with a number; one mistake
2. A lesson learned	"Three chart mistakes I stopped making"	the mistake; a before picture; an after picture; the rule; a one-line checklist
3. A contribution story	"My first open-source pull request"	why I chose that project; what I changed; what the maintainer said; what I learned; next step

Evaluation and acceptance criteria

- `build_site.py` finishes with no error and writes 4 HTML pages plus `.nojekyll`.
- `check_site.py` reports `broken links: 0` and exits 0.
- Index lists 3 posts, newest first (dates 2026-11-01, 2026-10-15, 2026-10-01 in this lab).
- Each post has a title, a result with at least one number, and a closing "what I would do next".
- Site loads on a phone-sized window with readable text.
- The public URL works from a private browser window.
- A friend can reach your GitHub link within 2 clicks.

Common pitfalls

1. **Raw characters break the page.** A `<` in your text vanishes or breaks a tag. Detect it: view the page source; fix it with `html.escape`.
2. **Broken links after moving to Pages.** Links starting with `/` point to the site root, not your repo folder. Use relative links like `first-pr.html`; the link checker will catch missing files.

3. **Stale output.** You edit a post but forget to rebuild, so the live site is old. Make rebuilding part of your commit routine.
4. **Pages shows a 404.** Usually the wrong branch or folder is selected, or the repository is private on a plan that does not allow it. Re-check the Pages settings.
5. **Posts with no numbers.** A post that says "improved a lot" persuades nobody. Detect it by searching each post for digits.

Three extensions

1. Add an RSS feed file (`feed.xml`) generated from the same list of posts.
2. Add a tags field to the header and generate one page per tag.
3. Add a GitHub Actions workflow that runs the builder and link checker on every push (not run here).

Interview talking points

Q1. Why did you write your own site generator instead of using a template? Model answer: "I wanted to learn how static sites work and keep zero dependencies. The generator is 59 lines. I also learned to test it: I wrote a link checker and made it fail on purpose to prove it worked. For a bigger site I would use a ready tool."

Q2. What was the hardest bug? Model answer: "My bullet list stayed open when a code block started, so the HTML was invalid. I found it by reading the generated file, not the source. I fixed the order of two checks and I now always inspect the output."

Q3. How do you know the site works? Model answer: "The build prints a count, the link checker returns zero broken links, and I open the public link in a private window. I do not rely on 'it looked fine on my laptop'."

README outline

Title with live link; one-sentence purpose; screenshot; how to run (`python3 build_site.py posts docs`); folder tree; how to add a post (header format); how to publish; checks (`check_site.py`); limits (only headings, lists, code supported); next steps.

15.3 Project: First Open-Source Contribution (Project #44)

Goal and why employers care

You will find a beginner-friendly issue in a real open-source project, understand the repository, make a small correct change, and open a good pull request. Before that you rehearse the whole workflow locally with git, so the commands are familiar. Employers care because a merged pull request is public proof that you can read other people's code, follow a team's rules, communicate in writing, and take review feedback calmly.

Skills used

- Git: branch, commit, remote, rebase (15.1 and this section).
- Reading unfamiliar code and tests (Chapter 6, Problem Solving).
- Writing clearly: issue comments and PR descriptions (Chapter 4, Written Communication).
- Python basics and testing (the Python study document).
- Professional presence on GitHub (Chapter 11).

Data

No dataset. Your "data" is a practice repository you create in `/tmp`, containing a tiny Python helper library, plus (later) a real project you choose. A small but real project beats a famous giant one: maintainers answer faster and the codebase is readable in an afternoon.

Architecture: who owns which copy

```
upstream.git  (the maintainer's original repo)
     |   ^
fork |   | pull request (you ask them to merge)
     v   |
fork.git      (your copy on the server)
     |   ^
clone|   | push
     v   |
myfork/       (your laptop)  -- remote "origin" = fork.git
                             -- remote "upstream" = upstream.git
```

Read it as follows. You cannot push to the maintainer's repo. You push to your fork, then ask the maintainer to pull from it. You add `upstream` as a second remote so you can fetch their newest changes.

Step-by-step build

1. **Find a good first issue (not run here: needs the internet).** On GitHub, open a project you actually use, go to Issues, and filter by labels such as `good first issue` or `help wanted`. Look for: a recent maintainer reply (the project is alive), a clear description, no one already assigned, and a fix that touches one or two files. Read `README.md`, `CONTRIBUTING.md` and the code of conduct. Comment on the issue: "I would like to work on this. My plan is X. Is that okay?" and wait for a yes. Good first changes: a documentation typo, a missing test, a clearer error message.

1. **Read the repository in this order:** README (what it is) → CONTRIBUTING (the rules) → the test folder (how it should behave) → the file named in the issue → `git log --oneline -10` (the style of past changes). Run the tests before you change anything, to know what "passing" looks like.

1. **Rehearse locally.** The script below creates a maintainer's repo with a small bug (`mean([])` raises a confusing `ZeroDivisionError`), forks and clones it, makes the fix on a branch with a test, lets the maintainer push a new commit while you work, and then rebases your branch. Before you run it, predict: after the rebase, will your commit come before or after the maintainer's "license note" commit?

```
set -e
export GIT_AUTHOR_NAME=Asha GIT_AUTHOR_EMAIL=asha@example.com
GIT_COMMITTER_NAME=Asha GIT_COMMITTER_EMAIL=asha@example.com
export PYTHONDONTWRITEBYTECODE=1
cd /tmp/lab15/oss
# --- setup: the maintainer's project (the "upstream") ---
git init -q --bare -b main upstream.git
git clone -q upstream.git maintainer 2>/dev/null
cd maintainer; git checkout -q -b main
mkdir src tests
printf 'def mean(values):\n    return sum(values) / len(values)\n' >
src/stats_utils.py
printf 'from src.stats_utils import mean\nassert mean([2, 4, 6]) ==
4\nprint("tests passed")\n' > tests/test_stats.py
printf '__pycache__/\n' > .gitignore
printf '# stats-utils\nSmall helpers.\n\n## Contributing\nOpen an issue
first. Add a test.\n' > README.md
git add . && git commit -qm "Initial version" && git push -q origin main
cd ..
echo "=== 1-3: fork, clone, add upstream remote"
git clone -q --bare upstream.git fork.git
git clone -q fork.git myfork
```

```
cd myfork
git remote add upstream ../upstream.git
git remote -v
echo "=== 4-5: branch, change, test"
git checkout -q -b fix-mean-empty-list
cat > src/stats_utils.py <<'PY'
def mean(values):
    """Return the average of a list of numbers.

    Raises ValueError with a clear message if the list is empty.
    """
    if not values:
        raise ValueError("mean() needs at least one value")
    return sum(values) / len(values)
PY
cat >> tests/test_stats.py <<'PY'
try:
    mean([])
except ValueError as e:
    print("empty list ->", e)
else:
    raise AssertionError("expected ValueError")
PY
python3 -m tests.test_stats
git diff --stat
git add -A
git commit -q -m "Raise ValueError for empty input in mean()" -m
"Previously mean([]) raised ZeroDivisionError, which hides the cause. Add
a clear message and a test. Fixes #1."
echo "=== 6: maintainer moves on while you wait"
cd ..; git clone -q upstream.git m2 2>/dev/null; cd m2
printf '\n## License\nMIT\n' >> README.md
GIT_AUTHOR_NAME=Maint GIT_COMMITTER_NAME=Maint git commit -qam "Add
license note"; git push -q origin main
cd ../myfork
echo "=== 7: sync with upstream and rebase"
git fetch -q upstream
git rebase upstream/main
git push -q origin fix-mean-empty-list
git log --oneline --graph --decorate -4
```

Output (the commit hashes differ every time you run it):

```
=== 1-3: fork, clone, add upstream remote
origin  /tmp/lab15/oss/fork.git (fetch)
origin  /tmp/lab15/oss/fork.git (push)
upstream        ../upstream.git (fetch)
upstream        ../upstream.git (push)
=== 4-5: branch, change, test
tests passed
empty list -> mean() needs at least one value
 src/stats_utils.py  | 6 ++++++
 tests/test_stats.py | 6 ++++++
 2 files changed, 12 insertions(+)
=== 6: maintainer moves on while you wait
=== 7: sync with upstream and rebase
Rebasing (1/1)Successfully rebased and updated refs/heads/fix-mean-empty-
list.
* 7e064f5 (HEAD -> fix-mean-empty-list, origin/fix-mean-empty-list) Raise
ValueError for empty input in mean()
* f49f058 (upstream/main, upstream/HEAD) Add license note
* 7553e48 (origin/main, origin/HEAD, main) Initial version
```

Answer to the prediction: your commit sits on top, after the maintainer's. Rebase replays your work on top of the newest upstream, giving a straight line that is easy to review.

1. **Understand what each command did.**

Command	What it does	Why
`git clone --bare upstream.git fork.git`	Makes a server-side copy	This stands in for the "Fork" button
`git remote add upstream ...`	Names the original repo	So you can fetch new changes
`git checkout -b fix-mean-empty-list`	New branch	One change per branch keeps PRs small
`git diff --stat`	Summary of changes	Review your own work before committing
`git fetch upstream` + `git rebase upstream/main`	Update your branch	Avoids conflicts in the review
`git push origin <branch>`	Send to your fork	The PR is created from this

branch

1. **Make the real contribution.** Repeat the same steps with the real project: click Fork, clone your fork, add the `upstream` remote, branch, change, run the project's tests, commit, push, and open the PR on GitHub (not run here). If the project has a formatter or linter, run it.

1. **Write a good pull request.** Use this shape.

```
Title: Raise ValueError for empty input in mean()

What: mean([]) now raises ValueError("mean() needs at least one value").
Why: it used to raise ZeroDivisionError, which hides the real cause. Fixes
#1.
How tested: added a test in tests/test_stats.py; all tests pass locally.
Notes: no behaviour change for non-empty lists. Happy to adjust the
message.
```

Keep it small (under about 100 changed lines), link the issue, say how you tested, and be polite. When the maintainer asks for changes, reply "Thanks, done" and push to the same branch; the PR updates by itself.

1. **After the merge.** Delete the branch, sync your fork, and add the PR link to your resume and LinkedIn. Then write post 3 from Project 43 ("My first open-source pull request").

Evaluation and acceptance criteria

- The local rehearsal runs without error and ends with your commit on top of `upstream/main`.
- Your fix has at least one test that fails before the change and passes after.
- The PR is under about 100 changed lines, has a title, a "what and why" and a "how tested" part, and links an issue.
- You received a response (merged, or review comments answered within two days).
- `git status` is clean and no unrelated files (`__pycache__`, editor settings) are in the diff.
- For the real PR: the project's own checks pass.

Common pitfalls

1. **Committing junk files.** In my first attempt a `__pycache__` file got staged because the

repo had no `.gitignore`. Detect with `git status` and `git diff --stat` before every commit. The final script puts `__pycache__/` into `.gitignore` first.

2. **Missing git identity.** `git commit` fails with "Author identity unknown". Set `git config user.name` and `user.email` for the repo.
3. **A giant PR.** Maintainers avoid reviewing 800 lines from a stranger. Split by topic; one PR, one idea.
4. **Working without asking.** You may build something the maintainers do not want. Comment on the issue first.
5. **Force-pushing carelessly.** After a rebase you may need to force-push your own branch; prefer `git push --force-with-lease`, which refuses if someone else pushed. Never force-push the shared main branch.

Three extensions

1. Fix a second issue in the same project and ask to be added to a review rota (or simply review someone else's PR politely).
2. Improve the project's documentation with a "how to run the tests" section.
3. Add a small GitHub Actions test workflow to your practice repo (not run here).

Interview talking points

Q1. Tell me about your open-source contribution. Model answer: "I chose a small library I use. I commented on a good-first-issue to confirm my plan, read the tests, fixed an unclear error and added a test. The change was 12 lines. The maintainer asked me to reword the message; I did, and it was merged. I learned to keep changes small."

Q2. What is the difference between merge and rebase? Model answer: "Both bring in new upstream commits. Merge keeps both histories and adds a merge commit. Rebase replays my commits on top of the newest upstream, so history is a straight line. I rebase my own unpublished branch and never rewrite shared history."

Q3. How did you handle review feedback? Model answer: "I read every comment as help, not criticism. I answered each one, pushed the fix to the same branch, and asked when something was unclear. I did not argue about style the project had already chosen."

README outline

For the practice repo: purpose; the diagram of upstream, fork and clone; the commands (the

script); a table of what each does; lessons learned; a link to your real merged PR, with a one-line description.

Chapter recap

- A project is finishable when you write measurable success criteria and a definition of done before you start.
- A repo should run from a fresh clone using only README steps; add `.gitignore` before the first commit.
- Commit small and often with imperative messages; the history is your diary.
- Use public or generated data; never company or private data.
- A static site is plain HTML; a 59-line Python script can build one from text posts.
- Always test your checker by making it fail once (the broken-link test).
- Open-source flow: fork, clone, add `upstream`, branch, change, test, push to your fork, open a PR.
- Rebase your own branch onto `upstream/main` to keep a straight history; use `--force-with-lease` if you must force-push.
- A good PR is small, linked to an issue, explains what, why and how tested.

Pattern	Command or rule
Build the site	`python3 build_site.py posts docs`
Check links	`python3 check_site.py docs` (exit 1 if broken)
Pages setup	Settings, Pages, branch `main`, folder `/docs`
Add original repo	`git remote add upstream <url>`
Update your branch	`git fetch upstream` then `git rebase upstream/main`
Safe force push	`git push --force-with-lease origin <branch>`
PR body	what, why, how tested, issue link

Practice set

1. (Easy) Rewrite the commit message `update` for a change that adds a link checker. **Answer:** `Add link checker for built pages` (imperative, specific, short).

2. (Easy) What does `01-first-pipeline.md` become after `build_site.py` runs? **Answer:** `first-pipeline.html`, because the regex removes the leading digits and dash.

3. (Easy) List three things that belong in `.gitignore` for a Python project. **Answer:** `__pycache__/`, `.venv/`, `.env` (secrets). Data dumps are also common.

4. (Medium) What is the output of `check_site.py` if `index.html` links to `about.html` and no such file exists, with 4 pages in `docs/`? **Answer:** `BROKEN index.html -> about.html` then `pages: 4 broken links: 1`, exit code 1.

5. (Medium) Find the bug: a builder opens `<ul>` for each bullet line and never closes it before a heading. What does the browser show and how do you detect it? **Answer:** following content is indented inside the list or the layout breaks; detect by reading the generated HTML, and fix by closing the list when a non-bullet line arrives.

6. (Medium) You ran `git rebase upstream/main` and then `git push origin my-branch` was rejected. Why, and what do you do? **Answer:** rebase rewrote your commits, so the remote branch has different history. Use `git push --force-with-lease origin my-branch` on your own branch only.

7. (Interview) "Why do you keep an `upstream` remote?" **Answer:** `origin` is my fork; `upstream` is the original. I fetch from upstream to get new changes and rebase my branch, so my PR has no conflicts.

8. (Scenario) You found an issue labelled good first issue that is 2 years old with no replies. Do you start? **Answer:** Not yet. Comment and ask if it is still wanted; if no answer after a few days, pick a project with recent activity. A dead project will not review your PR.

Prompt pack

✨ **AI prompt: copy and paste into any AI**

```
You are my patient, practical professor. I am a working professional
moving into data roles. The topic is Chapter 15, "How to run a project"
(success criteria, repo layout, commit rhythm, definition of done). I will
paste my plan for a portfolio project. Ask me what is unclear, then
rewrite my plan with three measurable success criteria, a time box, a repo
tree and a definition of done. Explain each choice in simple English and
```

```
check my understanding with two questions. My plan: [PASTE HERE]
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Review my static portfolio site project from
Chapter 15 (a Python script that builds HTML pages from text posts and
publishes on GitHub Pages). I will paste my build script and one post.
Find bugs, security problems such as unescaped text, and missing checks.
Rank issues by importance, explain each in simple English, and show a fix
for the top two. My files: [PASTE HERE]
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Help me debug a git problem from my first open-
source contribution workflow (fork, clone, branch, rebase, push). I will
paste the commands I ran and the error message. Explain in simple English
what each command did, why the error appears, and give the safest fix.
Warn me about commands that could lose work and tell me how to back up
first. Problem: [PASTE HERE]
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Help me choose a first open-source issue. I will
paste three candidate issues with the project name, labels, last activity
date and description. Score each on clarity, size, project health and my
skill fit, pick one, and write the short comment I should post to ask
permission. Explain your reasoning simply. Candidates: [PASTE HERE]
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Review my pull request description and my blog
post about it, from Chapter 15. Check that the PR says what, why, and how
it was tested, is small and polite, and that the post follows five short
paragraphs with one number and one mistake. Give a score out of 10 for
each, three specific edits, and a rewritten first paragraph. Drafts:
[PASTE HERE]
```

CHAPTER 16

PROJECT LAB 2 — MOCK-INTERVIEW KIT AND RESUME SPRINT

Two small tools that make interview practice measurable and resume bullets honest, specific and strong.

Chapter 15 gave you the project method and two public-proof projects. This chapter adds two projects that sharpen the last step of the journey: being chosen. In Project A you build a mock-interview kit: a question tracker in SQLite, a STAR story template, a scoring sheet, and a script that quizzes you on your weakest questions first. In Project B you run a resume and LinkedIn refresh sprint: you rewrite weak bullets into bullets with impact numbers, and a small script flags the weak ones for you. Both projects use only the Python standard library, and every output shown below was produced by running the code. Chapter 7 (Technical Interview Preparation by Role) gives the question types; Chapter 11 (Portfolio, GitHub, LinkedIn and Resume) gives the bullet rules. Here you turn both into working tools. Use the method of Chapter 15.1: success criteria first, small commits, definition of done.

16.1 Project: Mock-Interview Kit and Answer Bank

Goal and why employers care

You will build a personal practice system: a database of questions with model answers, a way to score every practice attempt, a report of strengths and gaps, and a quiz that picks weak questions more often. Employers never see the tool itself, but they see the result: calm, structured answers. As a portfolio item, the repository shows that you can design a small database, write a command-line tool and use data to improve your own performance, which is exactly what analysts do for a business.

Skills used

- Interview round types and what each tests (Chapter 7, Technical Interview Preparation by Role).
- Spoken delivery and structure (Chapter 3.3, Spoken English; Chapter 2.1, Learning How to Learn, for spaced practice).
- SQL tables, joins, `GROUP BY` and `HAVING` (the SQL study document) and Python's

`sqlite3` module.

- Weighted random choice (a gentle use of probability).
- Project method (Chapter 15.1).

Data

You generate it yourself. The script seeds six starter questions across four topics (behavioral, SQL, statistics, project walk-through). You add your own with one command. Never paste confidential interview questions that you signed an agreement about, and never store real company or client details in your STAR stories.

Architecture and folder structure

```
interview-kit/
├── kit.py              # the command-line tool (init, add, quiz, log,
report)
├── kit.db              # created on first run (add to .gitignore)
├── stories/
│   └── star_template.md
├── scoring_sheet.md
├── README.md
└── .gitignore
```

Flow: `init` creates two tables, `questions` and `attempts` → you practise → `quiz` shows a question and the model answer → you score yourself 1 to 5 on clarity, structure and evidence → `report` shows the average by topic.

Step-by-step build

1. **Design the tables.** One row per question; one row per practice attempt, with a `CHECK` rule so a score outside 1 to 5 is rejected.

1. **Write the tool.** Before you read it, predict: why does the quiz use a weight such as `6 - average score` instead of picking at random?

```
"""Mock-interview kit. Commands: init | add | quiz | log | report"""
import argparse, random, sqlite3, sys
from datetime import date

DB = "kit.db"
```

```
SCHEMA = """
CREATE TABLE IF NOT EXISTS questions(
  id INTEGER PRIMARY KEY, topic TEXT NOT NULL, question TEXT NOT NULL,
  model_answer TEXT NOT NULL);
CREATE TABLE IF NOT EXISTS attempts(
  id INTEGER PRIMARY KEY, question_id INTEGER NOT NULL REFERENCES
questions(id),
  day TEXT NOT NULL, clarity INTEGER, structure INTEGER, evidence INTEGER,
  CHECK (clarity BETWEEN 1 AND 5 AND structure BETWEEN 1 AND 5 AND
evidence BETWEEN 1 AND 5));
"""
SEED = [
 ("behavioral", "Tell me about a time you disagreed with a teammate.",
"STAR: situation, task, action (listened, tested both ideas), result (with
a number)."),
 ("behavioral", "Describe a mistake you made and what you changed.", "Own
the mistake early, show the fix, show the habit that now prevents it."),
 ("sql", "What is the difference between WHERE and HAVING?", "WHERE
filters rows before grouping; HAVING filters groups after aggregation."),
 ("sql", "How do you find duplicate order ids?", "GROUP BY order_id HAVING
COUNT(*) > 1."),
 ("stats", "What does a p-value of 0.03 mean?", "If there were no real
effect, we would see a result this extreme about 3 percent of the time."),
 ("project", "Walk me through your best project in two minutes.",
"Problem, data, method, result with a number, one limit, one next step."),
]

def connect():
    con = sqlite3.connect(DB); con.row_factory = sqlite3.Row
    con.executescript(SCHEMA); return con

def cmd_init(a):
    con = connect()
    if con.execute("SELECT COUNT(*) FROM questions").fetchone()[0] == 0:
        con.executemany("INSERT INTO
questions(topic,question,model_answer) VALUES(?,?,?)", SEED)
        con.commit()
    print("questions:", con.execute("SELECT COUNT(*) FROM
questions").fetchone()[0])

def cmd_add(a):
    con = connect()
    con.execute("INSERT INTO questions(topic,question,model_answer)
VALUES(?,?,?)", (a.topic, a.question, a.answer))
    con.commit(); print("added")

def pick(con, n, rng):
```

```
    """Weakest-first: questions never tried or with a low average score
are likelier."""
    rows = con.execute("""
      SELECT q.id, q.topic, q.question, q.model_answer,
             COALESCE(AVG((at.clarity+at.structure+at.evidence)/3.0), 0)
AS avg_score
      FROM questions q LEFT JOIN attempts at ON at.question_id = q.id
      GROUP BY q.id""").fetchall()
    weights = [6 - r["avg_score"] if r["avg_score"] else 5 for r in rows]
    chosen = []
    pool = list(zip(rows, weights))
    for _ in range(min(n, len(pool))):
        r = rng.choices([p[0] for p in pool], [p[1] for p in pool])[0]
        chosen.append(r); pool = [p for p in pool if p[0]["id"] !=
r["id"]]
    return chosen

def cmd_quiz(a):
    con = connect(); rng = random.Random(a.seed)
    for r in pick(con, a.n, rng):
        print(f"\n[{r['topic']}] Q{r['id']}: {r['question']}")
        if a.demo:
            print("  (demo mode: answer aloud, then compare)")
            print("  Model answer:", r["model_answer"])
            continue
        input("  Answer aloud, then press Enter to see the model answer...
")
        print("  Model answer:", r["model_answer"])
        s = [int(input(f"  Score {k} 1-5: ")) for k in ("clarity",
"structure", "evidence")]
        con.execute("INSERT INTO
attempts(question_id,day,clarity,structure,evidence) VALUES(?,?,?,?,?)",
                    (r["id"], str(date.today()), *s)); con.commit()

def cmd_log(a):
    con = connect()
    con.execute("INSERT INTO
attempts(question_id,day,clarity,structure,evidence) VALUES(?,?,?,?,?)",
                (a.qid, a.day, a.clarity, a.structure, a.evidence));
con.commit(); print("logged")

def cmd_report(a):
    con = connect()
    print(f"{'topic':<11}{'tries':>6}{'avg':>6}")
    for r in con.execute("""SELECT q.topic, COUNT(at.id) n,
        ROUND(AVG((at.clarity+at.structure+at.evidence)/3.0),2) avg
        FROM questions q LEFT JOIN attempts at ON at.question_id=q.id
```

```
        GROUP BY q.topic ORDER BY avg IS NULL, avg"""):
        print(f"{r['topic']:<11}{r['n']:>6}{(r['avg'] if r['avg'] is not
None else '-'):>6}")
    ready = con.execute("""SELECT COUNT(*) FROM (SELECT question_id FROM
attempts
        GROUP BY question_id HAVING AVG((clarity+structure+evidence)/3.0)
>= 4)""").fetchone()[0]
    total = con.execute("SELECT COUNT(*) FROM questions").fetchone()[0]
    print(f"interview-ready questions (avg >= 4): {ready} of {total}")

p = argparse.ArgumentParser(); sub = p.add_subparsers(dest="cmd",
required=True)
sub.add_parser("init").set_defaults(f=cmd_init)
s = sub.add_parser("add"); [s.add_argument(x) for x in ("topic",
"question", "answer")]; s.set_defaults(f=cmd_add)
s = sub.add_parser("quiz"); s.add_argument("-n", type=int, default=3);
s.add_argument("--seed", type=int)
s.add_argument("--demo", action="store_true"); s.set_defaults(f=cmd_quiz)
s = sub.add_parser("log"); s.add_argument("qid", type=int);
s.add_argument("day")
[s.add_argument(x, type=int) for x in ("clarity", "structure",
"evidence")]; s.set_defaults(f=cmd_log)
sub.add_parser("report").set_defaults(f=cmd_report)
a = p.parse_args(); a.f(a)
```

How it works: `pick` joins every question to its attempts and computes the average of the three scores. A question never tried gets weight 5 (high). Otherwise the weight is `6 - avg`, so an average of 2 gives weight 4 and an average of 5 gives weight 1: weak questions come up about four times as often as mastered ones. `random.Random(seed)` makes a quiz repeatable, which is useful for testing.

1. **Initialise and add your own question.**

```
python3 kit.py init
python3 kit.py add stats "What is overfitting?" "The model memorises
training data and fails on new data; check with a held-out test set."
```

```
questions: 6
added
```

1. **Log some attempts and read the report.** `log qid day clarity structure evidence` records one attempt.

```
python3 kit.py log 3 2026-11-01 3 2 2
python3 kit.py log 3 2026-11-05 4 4 4
python3 kit.py log 1 2026-11-02 2 3 2
python3 kit.py log 5 2026-11-03 5 4 4
python3 kit.py report
```

```
logged
logged
logged
logged
topic       tries   avg
behavioral      1  2.33
sql             2  3.17
stats           1  4.33
project         0     -
interview-ready questions (avg >= 4): 1 of 7
```

Check by hand: SQL question 3 has attempts averaging (3+2+2)/3 = 2.33 and (4+4+4)/3 = 4.00, so the topic average is (2.33 + 4.00) / 2 = 3.17. The `project` row shows `-` because it has no attempts yet, and it sorts last because the query puts empty averages at the end. Plain meaning: behavioral is weakest (2.33), so practise it next; project walk-through is untouched.

1. **Quiz yourself.** Use `--demo` to preview without scoring, or run without it to be asked for scores.

```
python3 kit.py quiz -n 3 --seed 7 --demo
```

```
[sql] Q3: What is the difference between WHERE and HAVING?
  (demo mode: answer aloud, then compare)
  Model answer: WHERE filters rows before grouping; HAVING filters groups
after aggregation.

[behavioral] Q2: Describe a mistake you made and what you changed.
  (demo mode: answer aloud, then compare)
  Model answer: Own the mistake early, show the fix, show the habit that
now prevents it.

[project] Q6: Walk me through your best project in two minutes.
  (demo mode: answer aloud, then compare)
  Model answer: Problem, data, method, result with a number, one limit,
one next step.
```

The weighted pick is random, so the order depends on the seed (and on your Python version); the weak and untried topics appear. In interactive mode, answer out loud first (record yourself if you can; Chapter 3), then compare, then enter three scores.

1. **Test the guard rail.** A bad score is rejected by the `CHECK` rule:

```
python3 kit.py log 3 2026-11-05 9 1 1
```

```
sqlite3.IntegrityError: CHECK constraint failed: clarity BETWEEN 1 AND 5
AND structure BETWEEN 1 AND 5 AND evidence BETWEEN 1 AND 5
```

The raw traceback is fine for a personal tool; a production tool would catch it and print a friendly message.

1. **Add the STAR story template.** Save as `stories/star_template.md` and write eight stories (Chapter 7.3 explains the eight stories; Chapter 3.3 explains STAR).

```
Title:      (6 words, e.g. "Fixed the wrong monthly report")
Situation:  (1–2 sentences. Where, when, what was at stake?)
Task:       (1 sentence. What was YOUR responsibility?)
Action:     (3–4 bullets. What did YOU do? Use "I", not "we".)
Result:     (1–2 sentences WITH A NUMBER. What changed?)
Lesson:     (1 sentence. What do you do differently now?)
Fits questions: (conflict / mistake / leadership / deadline / ambiguity)
Time to tell:   (target 90 seconds, about 200 words)
```

1. **Add the scoring sheet.** Save as `scoring_sheet.md` and use it after every mock.

Dimension	1 (weak)	3 (okay)	5 (strong)
Clarity	Rambles; jargon undefined	Mostly clear; one muddy part	One main idea per sentence; terms defined
Structure	No order	Has a start and end	Clear frame, e.g. STAR or "answer, reason, example"
Evidence	Opinions only	One example	Specific example with a number

Rule of thumb: total of 12 or more out of 15 (an average of 4) counts as interview-ready, which is the threshold the `report` command uses.

1. **Schedule it.** Spaced practice beats cramming: 3 questions a day, 5 days a week, one full 45-minute mock every Sunday with a friend or an AI interviewer (see the Prompt pack). Review the report every Sunday and choose next week's weakest topic.

Evaluation and acceptance criteria

- `init` creates both tables and seeds 6 questions; running it twice does not duplicate them.
- `add`, `log`, `quiz` and `report` all run without error.
- A score outside 1 to 5 is rejected by the database.
- After 10 attempts, `report` shows a topic average that matches a hand calculation for one topic.
- Weak questions are picked more often: run `quiz` 200 times with different seeds in a test and check that a question with average 2 appears more often than one with average 5.
- The question bank has at least 40 questions and 8 written STAR stories by the end of week 4.
- After 4 weeks, at least 60% of questions have an average of 4 or more.

Common pitfalls

1. **Scoring yourself too kindly.** Everything becomes a 4. Detect it: if your average never goes below 3.5, record yourself and ask a friend to score three answers; compare.
2. **Reading the model answer first.** You then recognise rather than recall. Say the answer aloud before pressing Enter.
3. **A bank full of easy questions.** Count questions per topic with `report`; add questions from the topics with the lowest averages.
4. **Memorised scripts.** A word-for-word story sounds robotic. Memorise the five STAR beats, not the sentences.
5. **Leaking secrets in stories.** Remove client names, internal tool names and exact figures that belong to a company; use rounded or relative numbers ("cut by about a third").

Three extensions

1. Add a `due` column and a rule to show a question again after 1, 3, 7 and 14 days (spaced repetition).
2. Plot your weekly average score with matplotlib and add the chart to the README.
3. Add a `timer` command that gives 90 seconds per answer and records the answer length.

Interview talking points

Q1. Tell me about a tool you built to improve yourself. Model answer: "I built a small interview tracker. Questions and attempts live in two SQLite tables, and a report shows my average score by topic. The quiz picks weak questions more often. In four weeks my average on behavioral questions went from about 2.3 to above 4. The data told me where to spend time."

Q2. Why two tables and not one? Model answer: "One question has many attempts. If I stored attempts inside the question row, I would repeat the question text or lose history. Two tables linked by a question id avoids repetition and lets me use a join and group by."

Q3. How would you check that the weighting works? Model answer: "I would seed a random generator, run the picker many times, count how often each question appears, and check that low-score questions appear more. Fixed seeds make the test repeatable."

README outline

Title and one-line result; why (practice is measured); how to run (the five commands); the schema in a small table; sample report output; the STAR template and scoring sheet links; limits (self-scoring is biased); next steps.

16.2 Project: Resume and LinkedIn Refresh Sprint

Goal and why employers care

In one focused week you will rewrite your resume bullets so each one shows an action and a measurable result, refresh your LinkedIn, and use a script to flag weak bullets. Recruiters scan a resume in seconds (Chapter 11), and bullets like "worked on data pipelines" give them nothing to remember. A bullet with a verb and a number gives them a reason to call you.

Skills used

- The bullet formula and impact numbers (Chapter 11.3, Resume Bullets With Impact Numbers).
- Percentages and before/after comparison (the learner's Math PDF; Chapter 5, Data Storytelling).
- LinkedIn headline, About and Featured (Chapter 11.4).

- Python strings, regular expressions and exit codes (the Python study document).
- Honest reporting: never invent numbers, because interviewers will ask how you counted (Chapter 7.4 on resume alignment).

Data

Your own bullets, in a plain text file, one per line. For this lab, `bullets.txt` holds six realistic example bullets built on neutral shop-order work. Use your real work, but remove client names and anything confidential.

Architecture and folder structure

```
resume-sprint/
├── bullets.txt            # before: your current bullets, one per line
├── after.txt              # after: your rewritten bullets
├── check_bullets.py       # flags weak bullets
├── linkedin_checklist.md  # headline, About, Featured, skills
└── README.md
```

Flow: `bullets.txt` → `check_bullets.py` → read flags → rewrite → `after.txt` → check again until all pass.

Step-by-step build

1. **Collect the before bullets.** Write down everything you did in the last 2 to 3 years, one line each. Do not edit yet.

```
Responsible for testing of applications and reporting issues
Worked on data pipelines using Python and SQL
Built a nightly Python job that loads 50,000 shop orders into SQLite,
cutting manual report time from 3 hours to 10 minutes
Helped the team with dashboards
Wrote SQL checks that caught 14 duplicate-order bugs before release,
reducing refund requests by 22%
Participated in meetings and was involved in various activities
```

1. **Write the checker.** Predict first: which of these six bullets will pass? The rules are: a strong first verb, at least one digit, no filler words, and 8 to 30 words.

```
"""Flag weak resume bullets. Usage: python3 check_bullets.py
bullets.txt"""
import re, sys

WEAK_STARTS = ("responsible for", "worked on", "helped", "participated",
"involved in",
               "assisted", "duties included", "tasked with")
FILLER = ("various", "etc", "multiple things", "many", "several", "hard-
working")
STRONG_VERBS = {"built", "wrote", "cut", "reduced", "automated",
"designed", "launched",
                "analysed", "analyzed", "migrated", "created", "improved",
"led", "fixed", "tested"}

def check(b):
    low, issues = b.lower().strip(), []
    if low.startswith(WEAK_STARTS):
        issues.append("weak opening phrase")
    elif low.split()[0] not in STRONG_VERBS:
        issues.append("does not start with a strong verb")
    if not re.search(r"\d", b):
        issues.append("no number")
    if any(re.search(rf"\b{re.escape(w)}\b", low) for w in FILLER):
        issues.append("filler word")
    n = len(b.split())
    if n < 8: issues.append(f"too short ({n} words)")
    if n > 30: issues.append(f"too long ({n} words)")
    return issues

bullets = [l.strip() for l in open(sys.argv[1]) if l.strip()]
flagged = 0
for i, b in enumerate(bullets, 1):
    issues = check(b)
    flagged += bool(issues)
    print(f"{i}. [{'FIX' if issues else 'OK '}] {b[:60]}{'...' if len(b) >
60 else ''}")
    for x in issues: print(f"      - {x}")
print(f"\n{len(bullets) - flagged} of {len(bullets)} bullets pass")
sys.exit(1 if flagged else 0)
```

How it works: `low.startswith(WEAK_STARTS)` accepts a tuple, so one call tests all weak openers. The `elif` checks the first word against a set of strong verbs. `re.search(r"\d", b)` finds any digit. The regex `\b...\b` matches whole words, so "many" does not match inside "company". The script exits with code 1 if any bullet fails, so it can run in an automatic

check.

1. **Run it on the before file.**

```
python3 check_bullets.py bullets.txt
```

```
1. [FIX] Responsible for testing of applications and reporting issues
      - weak opening phrase
      - no number
2. [FIX] Worked on data pipelines using Python and SQL
      - weak opening phrase
      - no number
3. [OK ] Built a nightly Python job that loads 50,000 shop orders int...
4. [FIX] Helped the team with dashboards
      - weak opening phrase
      - no number
      - too short (5 words)
5. [OK ] Wrote SQL checks that caught 14 duplicate-order bugs before ...
6. [FIX] Participated in meetings and was involved in various activit...
      - weak opening phrase
      - no number
      - filler word

2 of 6 bullets pass
```

Two of six pass. The exit code is 1.

1. **Rewrite using the formula** `Action verb + what you did + how/with what + result with a number` . A number can be a count, a percentage, a time saved, a money amount or a team size. If you truly have no number, estimate honestly ("about", "roughly") and be ready to explain how you counted. Never invent one.

Before	After	Impact numbers used
Responsible for testing of applications and reporting issues	Tested 12 order-processing screens and logged 31 defects, 9 of them critical, before the release	12 screens, 31 defects, 9 critical
Worked on data pipelines using Python and SQL	Built a Python and SQL pipeline that loads 50,000 daily orders and flags 2 percent bad	50,000 rows, 2 percent

	rows	
Helped the team with dashboards	Wrote 4 dashboard views used weekly by a 6-person team, cutting report requests by 40%	4 views, 6 people, 40%

1. **Check the after bullets.**

```
python3 check_bullets.py after.txt
```

```
1. [OK ] Tested 12 order-processing screens and logged 31 defects, 9 ...
2. [OK ] Built a Python and SQL pipeline that loads 50,000 daily orde...
3. [OK ] Wrote 4 dashboard views used weekly by a 6-person team, cutt...

3 of 3 bullets pass
```

An honest note: my first run flagged the first after-bullet, "Tested 12 order-processing...", with "does not start with a strong verb", because "tested" was not in my verb list. I added it. The checker is a **heuristic** (a rough rule of thumb), not a judge. It cannot tell whether a number is real or whether a bullet matters. Use it to catch the obvious weaknesses, then read each bullet as a stranger would.

Impact arithmetic is worth practising. For the third bullet of the earlier file, "cutting manual report time from 3 hours to 10 minutes":

- **Formula** time saved % = (old − new) / old × 100
- **Plug in** old = 3 hours = 180 minutes; new = 10 minutes. (180 − 10) / 180 × 100
- **Compute** 170 / 180 = 0.9444, so 94.4%
- **Answer** The job went from 180 minutes to 10, about 94% faster, so you can write "cut report time by 94%".
- **Check** 180 × (1 − 0.944) = 10.1 minutes, which matches the new time.

1. **Refresh LinkedIn with a checklist.** Save as `linkedin_checklist.md` and tick each item.

Section	Rule	Example
Headline	Target role + 2 skills + proof	"Data analyst

About	3 short paragraphs: who, proof with a number, what you want	First line works without clicking "see more"
Featured	Link to your site (Project 43), best repo, best post	3 items, newest proof first
Experience	Reuse the after bullets (same numbers on both)	Consistent with the resume
Skills	Top 10 match the job posts you want	Pin the 3 most wanted
Photo and URL	Clear face photo; custom profile URL	`linkedin.com/in/your-name`

1. **Run a 7-day sprint.** Day 1 collect bullets; Day 2 run the checker and mark weak ones; Day 3 rewrite with numbers; Day 4 resume layout (one page, plain format that scanners read well); Day 5 LinkedIn; Day 6 ask one friend to read it for 30 seconds and tell you what they remember; Day 7 fix and publish.

Evaluation and acceptance criteria

- At least 80% of bullets pass the checker (for example, 8 of 10), and the remaining ones have a written reason.
- Every bullet has a number you can defend in two sentences; keep a note of how each was counted.
- Resume is one page for under 8 years of experience (a common guideline, not a law).
- LinkedIn headline, first About line and Featured links are updated and match the resume numbers.
- A friend can recall two of your strongest facts after a 30-second read.
- No confidential names or exact internal figures appear.

Common pitfalls

1. **Inventing or inflating numbers.** It collapses in interviews. Detect it by asking yourself "How did I measure this?" for each bullet; if you cannot answer in two sentences, change it.
2. **Gaming the checker.** Adding a random digit ("worked on 3 things") passes the rule but not a human. Detect it by reading each passing bullet aloud: does it say what changed?
3. **One resume for every role.** Match the keywords in the job post (honestly). Keep a master

file and trim copies per role.

4. **Tools list instead of results.** "Python, SQL, Excel" lines tell nothing about depth. Attach each tool to an outcome.
5. **Resume and LinkedIn disagree.** Different dates or numbers look careless. Compare the two side by side before sending.

Three extensions

1. Add a rule that flags passive voice ("was done by") and bullets longer than 2 lines.
2. Add a keyword-match score between your resume and a pasted job post (count shared skill words) and print the missing ones.
3. Generate a one-page resume as HTML with a print stylesheet, using your Project 43 builder.

Interview talking points

Q1. How did you quantify your impact? Model answer: "I listed each task, then asked three questions: how many, how long, how much better. For example, a nightly job that took 3 hours by hand now takes 10 minutes, a 94% cut. When I did not have an exact number I gave an honest range and noted how I counted."

Q2. Tell me about a time you improved something with data. Model answer: "I reviewed my own resume as a dataset. A script flagged four of six bullets as weak because they had no number and a weak opening. I rewrote them and reached a pass rate of 100%. The lesson was that a quick automatic check finds the obvious problems so I can spend my time on the hard ones."

Q3. What are the limits of your checker? Model answer: "It is a rule of thumb. It checks for a strong verb, a digit, filler words and length. It cannot tell if a number is true or meaningful, and a clever writer can fool it. So it supports a human review; it does not replace one."

README outline

Title and one-line result (pass rate before and after); why (recruiters scan quickly); how to run; the rules the checker uses; a before/after table; LinkedIn checklist; limits (heuristic, English only); next steps.

Chapter recap

- A practice system needs two things: questions with model answers, and scores over time.
- Two linked SQLite tables (questions, attempts) give you history without repeating text.
- A `CHECK` rule in the table stops bad data (scores outside 1 to 5) from being saved.
- Weighted picking (`6 - average score`) makes weak questions come up more often.
- Score yourself on clarity, structure and evidence; an average of 4 or more means interview-ready.
- A strong bullet is: action verb + what + how + result with a number.
- Percent saved = (old − new) / old × 100; 180 to 10 minutes is 94.4%.
- A checker script is a heuristic: it finds obvious weaknesses, not truth.
- Resume, LinkedIn and your projects should tell one consistent story.

Pattern	Command or rule
Create bank	`python3 kit.py init`
Add question	`python3 kit.py add topic "question" "answer"`
Log attempt	`python3 kit.py log qid day clarity structure evidence`
Weekly review	`python3 kit.py report`
Quiz weak first	`python3 kit.py quiz -n 3`
Bullet formula	Verb + what + how + number
Check bullets	`python3 check_bullets.py file.txt` (exit 1 if any weak)
Percent change	(old − new) / old × 100

Practice set

1. (Easy) Why does the `attempts` table store `question_id` instead of the question text? **Answer:** to avoid repeating the text and to link many attempts to one question; a join brings the text back when needed.

2. (Easy) A question has attempts scoring (3,3,3) and (5,5,5). What is its average and what quiz weight does it get? **Answer:** average = (3 + 5) / 2 = 4.0; weight = 6 − 4.0 = 2.0.

3. (Easy) Does "Helped the team with dashboards" pass `check_bullets.py`? **Answer:** No: weak opening phrase, no number, and too short (5 words).

4. (Medium) Calculate the percent change: a report took 45 minutes and now takes 9. **Answer:** (45 − 9) / 45 × 100 = 36 / 45 × 100 = 80% faster. Check: 45 × 0.2 = 9.

5. (Medium) Rewrite: "Responsible for monthly sales reports." You know it covered 5 regions and took you 2 hours a month, down from 6. **Answer:** "Automated the monthly sales report for 5 regions, cutting prep time from 6 hours to 2 (a 67% saving)." Check: (6 − 2) / 6 = 0.667.

6. (Medium) What is the output of the checker for a bullet "Led many projects"? **Answer:** It starts with a strong verb, but it has no number, contains the filler word "many", and is too short (3 words), so three flags.

7. (Interview) "Tell me about your weakest area and what you did about it." **Answer:** a model structure: name one real weakness (behavioral answers rambling), show the data (average 2.3), the action (daily STAR practice, recorded), and the result (average above 4 in four weeks). Keep it under 90 seconds.

8. (Scenario) Your checker passes every bullet but a friend says your resume is still unclear. What do you do? **Answer:** trust the human. Read the bullets aloud, look for vague nouns ("things", "stuff", "systems"), remove numbers that have no meaning, and add the missing outcome (who benefited and how).

Prompt pack

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. The topic is the Chapter 16 mock-interview kit: a
question bank, STAR stories, a scoring sheet and a quiz script. Act as my
interviewer for a 45-minute mock for a data role. Ask one question at a
time, wait for my answer, then score clarity, structure and evidence from
1 to 5 with one sentence of advice. At the end, summarise my averages and
name my weakest area. My target role: [PASTE HERE]
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Review my STAR story from the Chapter 16 template.
I will paste it. Check that the Situation is short, the Task is my own
```

```
responsibility, the Action uses "I" with specific steps, and the Result
contains a real number. Tell me what the interviewer will remember, what
will confuse them, and rewrite it in about 200 words, keeping every fact.
Story: [PASTE HERE]
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Help me debug my interview-kit project from
Chapter 16 (Python, SQLite, two tables: questions and attempts). I will
paste my code and the error message. Explain in simple English what the
error means, show the smallest fix, and suggest one test I can add so it
does not return. Also point out any SQL mistakes in grouping or joins.
Code and error: [PASTE HERE]
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Rewrite my resume bullets using the Chapter 16
formula: action verb, what I did, how or with what, and a result with a
number. I will paste my bullets and the facts I know. Do not invent any
number; if a number is missing, ask me a question to find an honest one.
Show before and after, and mark which numbers still need proof. Bullets
and facts: [PASTE HERE]
```

✨ AI prompt: copy and paste into any AI

```
You are my patient, practical professor. I am a working professional
moving into data roles. Review my resume sprint from Chapter 16: my bullet
checker script, the pass rates before and after, and my LinkedIn headline
and About text. I will paste them. List the checker's blind spots, suggest
three new rules with code, and critique my headline and first About line
for clarity and proof. Rank your advice from most to least useful.
Material: [PASTE HERE]
```

APPENDIX

CHEAT SHEET

One page per chapter of patterns, formulas and rules. Use it to revise, not to learn. If a line feels unclear, go back to the chapter.

Chapter 1: The Career Map

Need	Pattern
Read a job description	Count tools, ignore the title
Six roles	Analyst, Data Engineer, Analytics Engineer, Data Scientist, ML Engineer, AI Engineer
Percent improvement	`(old – new) / old x 100`
Study hours per year	`hours per week x weeks x (1 – missed share)`
Transition story	Where I was, what I noticed, what I built, where I am going

Chapter 2: Learning How to Learn

Need	Pattern
Review schedule	Gaps of 1, 3, 7, 14, 30 days
New cards per day	3 to 5 maximum
One-hour session	10 recall, 30 build, 10 interleave, 10 teach back
Video to practice	20 minutes video : 40 minutes doing
Focus blocks	25 on, 5 off

Chapter 3: Spoken English

Need	Pattern
Explain a concept	PREP: Point, Reason, Example, Point
Behavioral answer	STAR with a 15 / 60 / 25 time split
Stand-up	Did, result, next, blockers
Do not know	"I do not know yet. My guess is... I will find out by..."
Answer length	words = minutes x 130

Chapter 4: Written Communication

Need	Pattern
Email	Subject = ask; line 1 = conclusion; bullets; next steps
README	One sentence, result, how to run, how it works, limits
ADR	Status, context, decision, alternatives, consequences
Good question	Goal, tried, expected vs actual, minimal example
Weekly post	Problem, what I did, mistake, next

Chapter 5: Data Storytelling

Need	Pattern
Percent change	`(new − old) / old x 100`
Conversion rate	`converted / users`
Chart title	One full sentence stating the message
Executive summary	Headline, evidence, recommendation, risk, ask

Q&A	Pause, restate, brief answer, honest "not yet"

Chapter 6: Problem Solving

Need	Pattern
Coding steps	Clarify, restate, examples, brute force, optimise, code, test, Big-O
KPI drop	Check data first, then revenue = orders x AOV
Bisect	`ceil(log2(n))` steps
Events per second	`events per day / 86,400`
Concurrent users	`throughput x response time` (Little's law)

Chapter 7: Interview Preparation

Need	Pattern
Time split	`hours x weight / sum of weights`
Second highest per group	`dense_rank() over (partition by g order by v desc)`, then `r = 2`
Precision / recall / F1	`TP / flagged`; `TP / actual`; `2PR / (P + R)`
Case steps	Clarify, structure, prioritise, calculate, recommend
Eight STAR stories	Reuse across questions; keep a number in each

Chapter 8: System Design I

Need	Pattern
Requests in flight	`L = λ x W`

Servers needed	`L / concurrency per server`, plus spare
Latency with cache	`h x t_cache + (1 − h) x t_db`
Queue wait	`service / (1 − utilisation)`
Availability	series: multiply; parallel: $1 - (1 - a)^n$
Framework	Requirements, estimate, API, data, design, deep dive, trade-offs

Chapter 9: System Design II (Data and ML)

Need	Pattern
Precision from base rate	$r \cdot b \,/\, (r \cdot b + FPR \cdot (1 - b))$
Net value of a model	`caught x loss − flags x review cost`
Point-in-time join	`feature.ts <= label.ts`, latest one
Vector index size	`chunks x dims x bytes per number`
Ranking quality	precision@k, recall@k

Chapter 10: Research Skills

Need	Pattern
Three passes	10 min scan, 1 h understand, hours to rebuild
Standard error of a mean difference	$\sqrt{(s_1^2/n + s_2^2/n)}$
False positives over m tests	$1 - 0.95^m$; Bonferroni $0.05 / m$
Ablation	Compare drop with noise
Abstract	Problem, method, result with number, limit

Chapter 11: Portfolio, GitHub, LinkedIn, Resume

Need	Pattern
Resume bullet	`Verb + what + how + result with number + scale`
README order	Promise, result, demo, architecture, run, data, method, limits, next
Headline	Target role + skills + niche
First post	Hook, problem, what I did, result, lesson, question

Chapter 12: Job Search in India

Need	Pattern
Expected offers	applications x product of stage pass rates
Applications per offer	`1 / overall conversion rate`
Hike %	`(new − old) / old x 100`
Year-1 total pay	`fixed + variable target x payout + joining + stock`
Referral ask	Link + two lines of proof + easy exit
Notice rules	Read contract, resign after a signed offer, write a handover
Rejection fix	Count reasons, fix the top one, retest

Chapter 13: Domain Knowledge and Business Sense

Need	Formula
Gross / net NPA	`NPA / loans` ; `(NPA − provisions) / (loans − provisions)`
Expected loss	`PD x LGD x EAD`
Combined ratio	`(claims + expenses) / premium`

AOV, conversion	`GMV / orders`; `orders / sessions`
Contribution per order	`margin − shipping − expected return cost`
CAC	`marketing spend / new customers`
LTV, payback	`ARPA x GM / churn`; `CAC / (ARPA x GM)`
NRR, GRR	`(start + upgrades − downgrades − churn) / start`; same without upgrades
ARPU, lifetime	`revenue / subs / months`; `1 / churn`
PPV	`TP / (TP + FP)`
Learn a domain	Glossary of 50 terms, KPIs, experts, one small project

Chapter 14: Certifications, Growth, Wellbeing

Need	Pattern
Weeks to certify	`hours needed / hours per week`
Certificate rule	Does a job ask for it? Can a project replace it? Is it current?
First open-source PR	Read CONTRIBUTING, claim issue, small change, test, clear PR text
Standard error of accuracy	`√(p(1 − p) / n)`
Kaggle rule	Trust cross-validation, not a small public leaderboard
Growth	IC and manager are different paths; build a T shape; keep a brag document
Self-doubt	Evidence log, compare with your past self, plan the 168-hour week, ask early

An Honest Closing Note

If you worked through these chapters with the book closed, you can now do real things. You can read a job description and name the true role, learn with recall and spacing instead of re-watching, explain a result aloud and in writing, break a problem into parts, prepare for a loop by role, sketch a software, data or ML system with numbers, read a paper in three passes, and present a portfolio. You can also run a job search as a measured funnel, compare offers on total pay, learn a business domain in a month, choose credentials with a rule, and plan your own growth without pretending to feel confident every day.

This document does not give you the technical skills themselves. SQL, Python, statistics, machine learning, data engineering and cloud live in the other study documents. It does not give you current salary figures, exam codes, company policies or regulation texts, because they go stale; I showed where to check them. It does not replace practice with real people: mock interviews, real code reviews, real conversations. And it is not medical or legal advice. For tax, contracts or health, ask a qualified person.

What to practise next, in order: (1) pick one target role and one domain; (2) finish one portfolio project with a README and a number (Chapter 11); (3) do one mock interview a week for the next four weeks and keep a rejection log (Chapters 7 and 12); (4) write one short public post every week; (5) review your plan every three months.

Some days you will feel behind. That feeling is common among people who are changing fields, and it is not a measurement of your ability. Keep the daily hour, keep the evidence log, and stay in your current role until you hold a signed offer. Slow, solid steps add up faster than you expect. I am confident in the method, and the work is yours.